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Biologic drugs for induction and maintenance of remission in Crohn's disease: a network meta-analysis (Review)

Gordon M, Sinopoulou V, Akobeng AK, Freeman SC, Moran GW, Cherayath R, Masitara M, Sherafat A, Khan MI, Alqusous F, Elleithy A, Phlananthachai S

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Cochrane Database of Systematic Reviews 2026, Issue 6. Art. No.: CD012751.

DOI: [10.1002/14651858.CD012751.pub2](https://doi.org/10.1002/14651858.CD012751.pub2).

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[Intervention Review]

Biologic drugs for induction and maintenance of remission in Crohn's disease: a network meta-analysis

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Editorial group: Cochrane Central Editorial Service.

Publication status and date: New, published in Issue 6, 2026.

Citation: Gordon M, Sinopoulou V, Akobeng AK, Freeman SC, Moran GW, Cherayath R, Masitara M, Sherafat A, Khan MI, Alqusous F, Elleithy A, Phlananthachai S. Biologic drugs for induction and maintenance of remission in Crohn's disease: a network meta-analysis. *Cochrane Database of Systematic Reviews* 2026, Issue 6. Art. No.: CD012751. DOI: [10.1002/14651858.CD012751.pub2](https://doi.org/10.1002/14651858.CD012751.pub2).

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ABSTRACT

Rationale

Crohn's disease is an inflammatory bowel disease (IBD) that can extend to any part of the gut. Many people with Crohn's disease fail to reach and maintain remission using conventional therapies. Biologic therapies with different mechanisms have advanced rapidly over recent decades. Comparative data for efficacy and safety are needed to clarify the relative position of these agents and guide decision-making.

Objectives

To assess the effects of biologic and advanced treatments for induction and maintenance of remission in Crohn's disease.

A secondary objective was to rank the included interventions.

Search methods

In June 2025, we searched CENTRAL, MEDLINE, Embase, ClinicalTrials.gov and WHO ICTRP.

Eligibility criteria

Randomised controlled trials (RCTs) comparing biologics with any other active comparator, placebo or no treatment for induction or maintenance of remission in adults with Crohn's disease.

Outcomes

Our critical outcome was clinical remission/relapse. Important outcomes were clinical response/loss of response, endoscopic remission/relapse, withdrawals due to adverse events, serious adverse events and total adverse events.

Risk of bias

We assessed the risk of bias using the Cochrane tool (RoB 1).

Synthesis methods

We expressed outcomes as risk ratios (RR) with 95% confidence intervals (CI). We conducted network meta-analyses for a network of induction studies and a network of maintenance studies. We assessed the certainty of the evidence using GRADE.

Biologic drugs for induction and maintenance of remission in Crohn's disease: a network meta-analysis (Review)

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Included studies

We included 94 RCTs, of which 66 were on induction ($n = 20,653$) and 22 on maintenance ($n = 6823$) of remission.

Synthesis of results

Induction

For clinical remission, there is high-certainty evidence that ustekinumab is more effective (RR 2.01, 95% CI 1.67 to 2.43, number needed to treat for an additional beneficial outcome (NNTB) = 6, small effect size) and moderate-certainty evidence that adalimumab combined with purine analogues (RR 2.6, 95% CI 1.71 to 3.96, NNTB = 3, moderate effect size), guselkumab (RR 2.49, 95% CI 1.95 to 3.18, NNTB = 4, moderate effect size), adalimumab (RR 2.36, 95% CI 1.75 to 3.18, NNTB = 4, moderate effect size), upadacitinib (RR 1.76, 95% CI 1.34 to 2.3, NNTB = 7, small effect size), vedolizumab (RR 1.57, 95% CI 1.15 to 2.13, NNTB = 10, trivial effect size) and natalizumab (RR 1.37, 95% CI 1.08 to 1.73, NNTB = 15, trivial effect size) are probably more effective than placebo. There is low-certainty evidence that risankizumab (RR 2.26, 95% CI 1.8 to 2.84, moderate effect), BI695501 (adalimumab biosimilar) (RR 2.09, 95% CI 1.31 to 3.33, small effect), a combination of infliximab with purine analogues (RR 2.04, 95% CI 1.14 to 3.64, small effect) and filgotinib (RR 1.55, 95% CI 1.18 to 2.04, trivial effect) may be more effective.

The evidence for humicade, a combination of natalizumab with infliximab and purine analogues, CTP13 (infliximab biosimilar) with purine analogues, infliximab, mirikizumab, etrolizumab, certolizumab, tesnatilimab, onercept, andecaliximab and etanercept was of very low certainty and no conclusions can be drawn.

There is moderate-certainty evidence that vedolizumab (RR 0.71, 95% CI 0.41 to 1.23), ustekinumab (RR 0.72, 95% CI 0.52 to 0.99) and upadacitinib (RR 0.8, 95% CI 0.46 to 1.39) probably lead to a similar number of withdrawals due to adverse events. There is low-certainty evidence that risankizumab may lead to fewer withdrawals due to adverse events (RR 0.36, 95% CI 0.22 to 0.59, small effect size) and low-certainty evidence that humicade (RR 0.81, 95% CI 0.39 to 1.68), natalizumab (RR 0.79, 95% CI 0.47 to 1.33), apilimod mesylate (RR 0.84, 95% CI 0.36 to 1.95), certolizumab (RR 0.98, 95% CI 0.65 to 1.49), filgotinib (RR 1.06, 95% CI 0.64 to 1.78) and tesnatilimab (RR 1.51, 95% CI 0.82 to 2.8) may lead to a similar number of withdrawals due to adverse events.

Maintenance

For preventing clinical relapse, there is moderate-certainty evidence that adalimumab (RR 0.69, 95% CI 0.58 to 0.82, NNTB = 2, moderate effect size) is probably more effective than placebo. There is low-certainty evidence that CTP13 (RR 0.55, 95% CI 0.43 to 0.71, large effect size), infliximab (RR 0.68, 95% CI 0.57 to 0.81, moderate effect size) and upadacitinib (RR 0.72, 95% CI 0.56 to 0.94, small effect size) may be more effective than placebo. Vedolizumab (RR 0.78, 95% CI 0.6 to 1.01), ustekinumab (RR 0.83, 95% CI 0.67 to 1.03), etrolizumab (RR 0.89, 95% CI 0.67 to 1.17) and tofacitinib (RR 0.9, 95% CI 0.67 to 1.21) may be similar to placebo.

The certainty of the evidence ranged between high and very low, with the main limitations being imprecision and risk of bias.

There is moderate-certainty evidence that vedolizumab probably leads to a similar number of withdrawals due to adverse events during maintenance (RR 0.84, 95% CI 0.63 to 1.11). There is low-certainty evidence that tofacitinib (RR 0.78, 95% CI 0.47 to 1.31), ustekinumab (RR 0.74, 95% CI 0.4 to 1.38), etrolizumab (RR 1.42, 95% CI 0.73 to 2.77) and upadacitinib (RR 1.51, 95% CI 0.73 to 3.11) may be no different to placebo in withdrawals due to adverse events.

The evidence for certolizumab, natalizumab, filgotinib, risankizumab, infliximab combined with purine analogues, and vedolizumab combined with purine analogues was of very low certainty and no conclusions can be drawn.

Authors' conclusions

This review has supported evidence generation, but head-to-head comparative trials for key interventional subclasses or specific agents may be needed, as guided by key stakeholders; for example, on the use of biosimilar versions of medications out of patent.

The role of concomitant purine analogues is a key confounding factor and future studies may not only want to investigate specifically the role of combinations, but also consider stratifying populations or purposefully excluding participants based on this key class of therapy to avoid such heterogeneity.

Given the majority of evidence focusses on the outcomes of clinical remission and relapse, future studies may want to further consider other outcomes of clear interest to clinicians and patients, particularly endoscopic remission.

Finally, long-term safety data are limited throughout the networks. Whilst future studies with longer follow-ups could provide increased data, it may be that study designs outside of randomised trials are employed for such outcomes.

Funding

This project is funded by an NIHR Evidence Synthesis Programme Grant (NIHR132748).

Registration

A protocol for this review was published in 2017: doi.org/10.1002/14651858.CD012751.

PLAIN LANGUAGE SUMMARY

Do 'biologics' (medications that target specific parts of the immune system) help manage Crohn's disease?

Key messages

- For making symptoms go away (remission), ustekinumab works slightly better than placebo (sham treatment). Adalimumab plus immune system drugs, guselkumab and upadacitinib probably work slightly better than placebo, while vedolizumab and natalizumab probably work only a little better than placebo.
- For stopping symptoms coming back (preventing active disease) within one to two years of remission, adalimumab probably works slightly better than placebo.
- In the short term (about one year), compared to placebo, upadacitinib, ustekinumab and vedolizumab probably lead to fewer people stopping treatment due to unwanted effects. We can't be sure about unwanted effects of biologics in the long term based on the existing evidence.

What is Crohn's disease?

Crohn's disease is a life-long (chronic) inflammatory disease that can affect any part of the gut. Common symptoms include bloody poo (stool or faeces), diarrhoea, stomach ache, fever, weight loss, fatigue and others. When someone is experiencing symptoms of Crohn's, they are said to have 'active' disease, and when their symptoms are under control, then they are said to be 'in remission'. We don't know what exactly causes Crohn's, but it could be related to a combination of genes, immune system malfunction, 'bad' gut bacteria and environmental reasons. There is no known cure, but the symptoms are usually managed with medications, such as steroids, immune system medications and, if necessary, surgery.

What are biologics?

At the turn of the century, a new category of medications called biologics, or advanced treatments, became available and started being widely used for the treatment of Crohn's. These are new and different treatments that target parts of the immune system more specifically than the older treatments, and they are thought to be better and have fewer unwanted effects. They can be delivered through a vein (intravenous), under the skin or sometimes orally.

What did we want to find out?

We wanted to find out whether all the biologics (medications that target specific parts of the immune system) or advanced treatments that are being used (or have been tested) can treat Crohn's, whether they have unwanted effects and how they compare to each other. The things we looked for were whether a medication can make symptoms go away, improve symptoms or make the gut look healed during a colonoscopy (examination inside the colon), as well as whether these medications have unwanted effects in the short term (up to one to two years) and long term (longer than two years).

What did we do?

We searched for studies that compared biologics with any other medical treatment or a placebo (a sham treatment) for adults with Crohn's disease. People in the studies could be men or women. They could have had Crohn's disease for any length of time and have used any medication for their Crohn's disease. We summarised the findings and assessed how confident we could be in the results, based on how the studies were designed and carried out.

What did we find?

We included 94 studies with 27,476 people, with about equal numbers of men and women. Most people were around 30 to 40 years old, but they ranged from older adolescents to people in their sixties. Severity in the induction studies was moderate to severe. In the maintenance studies, some people were in remission and some had responded to previous therapy and still had some symptoms. Most induction studies followed people for three to six months and maintenance studies for about one year.

Main results

- There were 66 induction studies, involving 20,653 people, and 22 maintenance studies, involving 6823 people.
- For making symptoms go away (remission), ustekinumab works slightly better than placebo (sham treatment). Adalimumab plus immune system drugs, guselkumab and upadacitinib probably work slightly better than placebo, while vedolizumab and natalizumab probably work only a little better than placebo. Risankizumab, BI695501 (a medication that resembles adalimumab), a combination of infliximab with immune system medications, and filgotinib may also work, but we are less certain about them.
- We are uncertain whether other medications work or not, compared to placebo.

- For stopping symptoms coming back (preventing active disease) within one to two years of remission, adalimumab probably works slightly better than placebo.
- In the short term (about one year), compared to placebo, upadacitinib, ustekinumab and vedolizumab probably lead to fewer people stopping treatment due to unwanted effects. Other medications like risankizumab, certolizumab and filgotinib could be similar to placebo too, but we're less certain about them.
- We can't be sure about unwanted effects of biologics in the long term based on the existing evidence.

What are the limitations of the evidence?

The included studies had people with very different characteristics between them, such as mixes of people who have or have not used biologics or advanced treatments before, people with more or less serious disease, and people using or not using other treatments at the same time. When these characteristics of people are not similar across the studies, it can make it difficult to be sure of the results.

The information we have is mainly around clinical symptoms, and information from colonoscopies or microscope examination was limited. There were also some issues with the way the included studies were carried out and the quality of the methods they used.

How up to date is this review?

This review is up to date to June 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Biologics for induction in Crohn's disease: clinical remission

Clinical remission (59 studies, n = 18,765 participants)

Patient or population: adult people with active Crohn's disease

Settings: hospital setting

Intervention: biologics/advanced therapies

Comparison: placebo (47 studies, n = 5089)

Intervention vs placebo (in order of GRADE certainty)	Network evidence		Anticipated absolute effects for network estimate			Notes
	RR (95% CI)	Certainty	Risk with placebo ^a	Risk with biologic ^b (95% CI)	Absolute risk difference with biologic ^c (95% CI)	
Ustekinumab (9 studies, n = 1912)	2.01 (1.67 to 2.43)	High ⊕⊕⊕⊕	179 per 1000	360 per 1000 (299 to 435)	18.2% more (12.1% more to 25.7% more)	Ustekinumab is better than placebo, with a small effect size (ranging from small to moderate).
Adalimumab and purine analogues (3 studies n = 353)	2.6 (1.71 to 3.96)	Moderate ^d ⊕⊕⊕⊖	179 per 1000	465 per 1000 (306 to 709)	28.9% more (12.8% more to 53.3% more)	Adalimumab and purine analogues are probably better than placebo, with a moderate effect size (ranging from small to large).
Guselkumab (4 studies, n = 1045)	2.49 (1.95 to 3.18)	Moderate ^d ⊕⊕⊕⊖	179 per 1000	446 per 1000 (350 to 569)	26.9% more (17.2% more to 39.2% more)	Guselkumab is probably better than placebo, with a moderate effect size (ranging from small to large).
Adalimumab (5 studies, n = 646)	2.36 (1.75 to 3.18)	Moderate ^d ⊕⊕⊕⊖	179 per 1000	422 per 1000 (313 to 569)	24.4% more (13.5% more to 39.2% more)	Adalimumab is probably better than placebo, with a moderate effect size (ranging from small to large).
Upadacitinib (3 studies, n = 857)	1.76 (1.34 to 2.3)	Moderate ^e ⊕⊕⊕⊖	179 per 1000	315 per 1000 (240 to 412)	13.6% more (6.1% more to 23.5% more)	Upadacitinib is probably better than placebo, with a small effect size (ranging from trivial to moderate).
Vedolizumab (5 studies, n = 779)	1.57 (1.15 to 2.13)	Moderate ^e ⊕⊕⊕⊖	179 per 1000	281 per 1000 (206 to 381)	10.2% more (2.7% more to 20.4% more)	Vedolizumab is probably better than placebo, with a trivial effect size (ranging from trivial to moderate).

Natalizumab (4 studies, n = 1186)	1.37 (1.08 to 1.73)	Moderate ^e ⊕⊕⊕⊕	179 per 1000	245 per 1000 (193 to 310)	6.6% more (1.4% more to 13.2% more)	Natalizumab is probably better than placebo, with a trivial effect size (ranging from trivial to small).
Risankizumab (4 studies, n = 1267)	2.26 (1.8 to 2.84)	Low ^f ⊕⊕⊕⊕	179 per 1000	405 per 1000 (322 to 508)	22.6% more (14.3% more to 33.1% more)	Risankizumab may be better than placebo, with a moderate effect size (ranging from small to large).
BI695501 (adalimumab biosimilar) (1 study, n = 72)	2.09 (1.31 to 3.33)	Low ^g ⊕⊕⊕⊕	179 per 1000	374 per 1000 (235 to 596)	19.7% more (5.6% more to 42% more)	BI695501 may be better than placebo, with a small effect size (ranging from trivial to large).
Infliximab with purine analogues (6 studies, n = 437)	2.04 (1.14 to 3.64)	Low ^g ⊕⊕⊕⊕	179 per 1000	365 per 1000 (204 to 651)	18.6% more (2.5% more to 47.6% more)	Infliximab with purine analogues may be better than placebo, with a small effect size (ranging from trivial to large).
Filgotinib (3 studies, n = 1032)	1.55 (1.18 to 2.04)	Low ^g ⊕⊕⊕⊕	179 per 1000	277 per 1000 (211 to 365)	9.9% more (3.3% more to 18.7% more)	Filgotinib may be better than placebo with a trivial effect size (ranging from trivial to small).
Tofacitinib (2 studies, n = 293)	1.18 (0.81 to 1.72)	Low ^h ⊕⊕⊕⊕	179 per 1000	211 per 1000 (145 to 308)	3.3% more (3.4% less to 12.9% more)	Tofacitinib may be the same as placebo, with an effect size that can range from trivially less than placebo to a small effect more than placebo.
Apilimod mesylate (1 study, n = 147)	0.67 (0.34 to 1.34)	Low ^h ⊕⊕⊕⊕	179 per 1000	120 per 1000 (61 to 240)	5.9% less (11.9% less to 6.2% more)	Apilimod mesylate may be the same as placebo, with an effect size that can range from a small effect less than placebo to a trivial effect more than placebo.
Humicade (1 study, n = 111)	3.14 (0.71 to 13.9)	Very low ⁱ ⊕⊕⊕⊕	179 per 1000	562 per 1000 (127 to 1000)	38.4% more (5.3% less to 100% more)	The evidence is very uncertain.
Natalizumab with infliximab and purine analogues (1 study, n = 52)	2.51 (0.98 to 6.42)	Very low ⁱ ⊕⊕⊕⊕	179 per 1000	449 per 1000 (175 to 1000)	27.2% more (0.3% less to 97.6% more)	The evidence is very uncertain.
CTP13 (infliximab biosimilar) and purine analogues (1 study, n = 111)	1.92 (0.94 to 3.91)	Very low ⁱ ⊕⊕⊕⊕	179 per 1000	344 per 1000 (168 to 700)	16.5% more (1.1% less to 52.3% more)	The evidence is very uncertain.

Infliximab (4 studies, n = 323)	1.77 (1.01 to 3.1)	Very low ^j ⊕⊕⊕⊕	179 per 1000	317 per 1000 (181 to 555)	13.9% more (0.2% more to 37.7% more)	The evidence is very uncertain.
Mirikizumab (2 studies, n = 758)	1.65 (1.16 to 2.34)	Very low ^k ⊕⊕⊕⊕	179 per 1000	295 per 1000 (208 to 419)	11.6% more (2.9% more to 24.1% more)	The evidence is very uncertain.
Etrolizumab (2 studies, n = 529)	1.3 (0.87 to 1.93)	Very low ^l ⊕⊕⊕⊕	179 per 1000	232 per 1000 (156 to 345)	5.3% more (2.3% less to 16.7% more)	The evidence is very uncertain.
Certolizumab (3 studies, n = 623)	1.17 (0.88 to 1.56)	Very low ⁱ ⊕⊕⊕⊕	179 per 1000	209 per 1000 (157 to 279)	3.1% more (2.2% less to 10.1% more)	The evidence is very uncertain.
Tesnatilimab (2 studies, n = 221)	1.17 (0.76 to 1.79)	Very low ^m ⊕⊕⊕⊕	179 per 1000	210 per 1000 (136 to 320)	3.1% more (4.2% less to 14.3% more)	The evidence is very uncertain.
Onercept (1 study, n = 169)	1 (0.5 to 1.99)	Very low ⁱ ⊕⊕⊕⊕	179 per 1000	179 per 1000 (89 to 356)	0% (9% less to 17.9% more)	The evidence is very uncertain.
Andecaliximab (1 study, n = 159)	0.76 (0.33 to 1.77)	Very low ⁱ ⊕⊕⊕⊕	179 per 1000	136 per 1000 (59 to 317)	4.3% less (12.1% less to 13.8% more)	The evidence is very uncertain.
Etanercept (1 study, n = 23)	0.43 (0.09 to 2.18)	Very low ⁱ ⊕⊕⊕⊕	179 per 1000	77 per 1000 (16 to 390)	10.2% less (16.4% less to 21.2% more)	The evidence is very uncertain.

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

CI: confidence interval; RR: risk ratio

^aThe risk with placebo has been calculated based on the cumulative placebo rates of all studies with a placebo arm.

^bThe risk with treatment has been calculated by multiplying the risk with control with the RR (95% CI). If the calculation results in more than 1000 per 1000 people, the number has been capped at 1000. Numbers have been rounded up to the closest whole number.

^cThe % risk difference has been calculated by subtracting the risk with control from the risk with treatment (95% CI) and dividing by 10. If the calculation results in more than 100%, the number has been capped at 100%. Numbers have been rounded up to the closest whole number.

- ^dDowngraded once due to risk of bias.
^eDowngraded once due to imprecision.
^fDowngraded once due to risk of bias and once due to incoherence.
^gDowngraded once due to risk of bias and once due to imprecision.
^hDowngraded twice due to serious imprecision.
ⁱDowngraded once due to risk of bias and twice due to serious imprecision.
^jDowngraded once due to risk of bias, once due to imprecision and once due to incoherence.
^kDowngraded once due to risk of bias, once due to imprecision and once due to inconsistency.
^lDowngraded once due to inconsistency and twice due to serious imprecision.
^mDowngraded once due to inconsistency and twice due to serious imprecision.

Summary of findings 2. Biologics for induction in Crohn's disease: withdrawals due to adverse events

Withdrawals due to adverse events (59 studies, n = 19,594 participants)

Patient or population: adult people with active Crohn's disease

Settings: hospital setting

Intervention: biologics/advanced therapies

Comparison: placebo (46 studies, n = 5097)

Comparisons (in order of GRADE certainty)	Network evidence		Anticipated absolute effects for network estimate			Notes
	RR (95% CI)	Certainty	Risk with placebo ^a	Risk with biologic ^b (95% CI)	Absolute risk difference with biologic ^c (95% CI)	
Vedolizumab (5 studies, n = 779)	0.71 (0.41 to 1.23)	Moderate ^e ⊕⊕⊕⊖	142 per 1000	101 per 1000 (58 to 175)	4.2% less (8.5% less to 3.3 % more)	Vedolizumab probably leads to similar withdrawal events compared to placebo (ranging from a small effect size less to a trivial effect size more).
Ustekinumab (11 studies, n = 2490)	0.72 (0.52 to 0.99)	Moderate ^d ⊕⊕⊕⊖	142 per 1000	102 per 1000 (74 to 141)	4% less (6.8% less to 0.2% less)	Ustekinumab probably leads to fewer withdrawal events with a trivial effect size compared to placebo.
Upadacitinib (3 studies, n = 857)	0.8 (0.46 to 1.39)	Moderate ^e ⊕⊕⊕⊖	142 per 1000	114 per 1000 (65 to 197)	2.9% less (7.1% less to 5.5% more)	Upadacitinib probably leads to similar withdrawal events compared to placebo (ranging from a small effect size less to a trivial effect size more).
Risankizumab (4 studies, n = 1267)	0.36 (0.22 to 0.59)	Low ^f ⊕⊕⊖⊖	142 per 1000	51 per 1000 (31 to 84)	9.2% less (11.2% less to 5.9% less)	Risankizumab may lead to fewer withdrawal events, with a small effect size compared to placebo (ranging from a

						small effect size less to a trivial effect size less).
Humicade (1 study, n = 111)	0.81 (0.39 to 1.68)	Low ^f ⊕⊕⊕⊕	142 per 1000	115 per 1000 (55 to 239)	2.7% less (8.7% less to 9.7% more)	Humicade may lead to similar withdrawal events compared to placebo (ranging from a small effect size less to a small effect size more).
Natalizumab (4 studies, n = 1186)	0.79 (0.47 to 1.33)	Low ^f ⊕⊕⊕⊕	142 per 1000	112 per 1000 (66 to 189)	3% less (7.6% less to 4.7% more)	Natalizumab may lead to similar withdrawal events compared to placebo (ranging from a small effect size less to a trivial effect size more).
Apilimod mesylate (1 study, n = 147)	0.84 (0.36 to 1.95)	Low ^g ⊕⊕⊕⊕	142 per 1000	119 per 1000 (51 to 277)	2.3% less (9.2% less to 13.6% more)	Apilimod mesylate may lead to similar withdrawal events compared to placebo (ranging from a small effect size less to a small effect size more).
Certolizumab (4 studies, n = 842)	0.98 (0.65 to 1.49)	Low ^h ⊕⊕⊕⊕	142 per 1000	139 per 1000 (92 to 212)	0.3% less (5% less to 7% more)	Certolizumab may lead to similar withdrawal events compared to placebo (ranging from a trivial effect size less to a small effect size more).
Filgotinib (n = 3 studies, n = 1049)	1.06 (0.64 to 1.78)	Low ^f ⊕⊕⊕⊕	142 per 1000	277 per 1000 (211 to 365)	0.9% more (5.2% less to 11.1% more)	Filgotinib may lead to similar withdrawal events compared to placebo (ranging from a trivial effect size less to a small effect size more).
Tesnatilimab (2 studies, n = 221)	1.51 (0.82 to 2.8)	Low ⁱ ⊕⊕⊕⊕	142 per 1000	214 per 1000 (116 to 398)	7.3% more (2.6% less to 25.7% more)	Tesnatilimab may lead to similar withdrawal events compared to placebo (ranging from a trivial effect size less to a large effect size more).
Mirikizumab (2 studies, n = 706)	0.39 (0.15 to 1.03)	Very low ^j ⊕⊕⊕⊕	142 per 1000	55 per 1000 (21 to 146)	8.7% less (12.2% less to 0.4% more)	The evidence is very uncertain.
Tofacitinib (2 studies, n = 293)	0.53 (0.26 to 1.08)	Very low ^k ⊕⊕⊕⊕	142 per 1000	75 per 1000 (37 to 153)	6.7% less (10.6% less to 1.1% more)	The evidence is very uncertain.
Onercept (1 study, n = 169)	0.52 (0.12 to 2.3)	Very low ^k ⊕⊕⊕⊕	142 per 1000	74 per 1000 (17 to 327)	6.8% less (12.6% less to 18.5% more)	The evidence is very uncertain.

Adalimumab (5 studies, n = 646)	0.88 (0.42 to 1.88)	Very low ^j ⊕⊕⊕⊕	142 per 1000	125 per 1000 (60 to 267)	1.7% less (8.4% less to 12.6% more)	The evidence is very uncertain.
Etrolizumab (2 studies, n = 529)	0.87 (0.41 to 1.85)	Very low ^l ⊕⊕⊕⊕	142 per 1000	124 per 1000 (58 to 262)	1.9% less (8.4% less to 12.1% more)	The evidence is very uncertain.
BI695501 (adalimumab biosimilar) (1 study, n = 72)	0.86 (0.26 to 2.86)	Very low ^k ⊕⊕⊕⊕	142 per 1000	122 per 1000 (37 to 406)	2% less (10.6% less to 26.6% more)	The evidence is very uncertain.
Guselkumab (3 studies, n = 1031)	0.85 (0.52 to 1.4)	Very low ^m ⊕⊕⊕⊕	142 per 1000	121 per 1000 (74 to 199)	2.1% less (6.8% less to 5.7% more)	The evidence is very uncertain.
Adalimumab and purine analogues (3 studies, n = 353)	1 (0.33 to 3.01)	Very low ^k ⊕⊕⊕⊕	142 per 1000	142 per 1000 (47 to 427)	0% (9.6% less to 28.7% more)	The evidence is very uncertain.
CTP13 (infliximab biosimilar) and purine analogues (1 study, n = 111)	1.06 (0.17 to 6.48)	Very low ^k ⊕⊕⊕⊕	142 per 1000	151 per 1000 (24 to 920)	0.9% more (11.8% less to 78.4% more)	The evidence is very uncertain.
Natalizumab with infliximab and purine analogues (1 study, n = 52)	1.22 (0.16 to 9.49)	Very low ^k ⊕⊕⊕⊕	142 per 1000	173 per 1000 (23 to 1000)	3.1% more (12.1% more to 100% more)	The evidence is very uncertain.
Duvakitug (1 study, n = 92)	1.16 (0.54 to 2.51)	Very low ^k ⊕⊕⊕⊕	142 per 1000	164 per 1000 (77 to 356)	2.3% more (6.6% less to 21.6% more)	The evidence is very uncertain.
Infliximab (3 studies, n = 240)	1.25 (0.27 to 5.8)	Very low ^k ⊕⊕⊕⊕	142 per 1000	177 per 1000 (38 to 824)	3.5% more (10.5% less to 68.6% more)	The evidence is very uncertain.
Infliximab with purine analogues (6 studies, n = 437)	1.34 (0.28 to 6.41)	Very low ^k ⊕⊕⊕⊕	142 per 1000	190 per 1000 (38 to 910)	4.9% more (10.3% less to 77.4% more)	The evidence is very uncertain.
Etanercept (1 study, n = 23)	1.35 (0.55 to 3.34)	Very low ^k ⊕⊕⊕⊕	142 per 1000	192 per 1000 (78 to 474)	5% more (6.5% less to 33.4% more)	The evidence is very uncertain.

Andecaliximab (1 study, n = 159)	2.11 (0.25 to 17.52)	Very low ^k ⊕⊕⊕⊕	142 per 1000	300 per 1000 (35 to 1000)	15.9% more (10.7% less to 100% more)	The evidence is very uncertain.
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GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

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CI: confidence interval; RR: risk ratio

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^bThe risk with treatment has been calculated by multiplying the risk with control with the RR (95% CI). If the calculation results in more than 1000 per 1000 people, the number has been capped at 1000. Numbers have been rounded up to the closest whole number.

^cThe % risk difference has been calculated by subtracting the risk with control from the risk with treatment (95% CI) and dividing by 10. If the calculation results in more than 100%, the number has been capped at 100%. Numbers have been rounded up to the closest whole number.

^dDowngraded once due to incoherence.

^eDowngraded once due to imprecision.

^fDowngraded once due to risk of bias and once due to imprecision.

^gDowngraded twice due to serious imprecision.

^hDowngraded once due to risk of bias and once due to inconsistency.

ⁱDowngraded twice due to serious imprecision.

^jDowngraded once due to risk of bias, once due to imprecision and once due to incoherence.

^kDowngraded once due to risk of bias and twice due to serious imprecision.

^lDowngraded once due to inconsistency and twice due to serious imprecision.

^mDowngraded twice due to serious inconsistency and once due to risk of bias.

Summary of findings 3. Biologics for maintenance in Crohn's disease: clinical relapse

Clinical relapse (23 studies, n = 6543 participants)

Patient or population: adult people with inactive Crohn's disease or with response to biologic treatment

Settings: hospital setting

Intervention: biologics and advanced therapies

Comparison: placebo (19 studies, n = 2151)

Comparisons (in order of GRADE certainty)	Network evidence		Anticipated absolute effects for network estimate			Notes
	RR (95% CI)	Certainty	Risk with placebo ^a	Risk with biologic ^b	Absolute risk difference with biologic ^c (95% CI)	

Adalimumab (4 studies, n = 455)	0.69 (0.58 to 0.82)	Moderate ^d ⊕⊕⊕⊖	714 per 1000	492 per 1000 (414 to 585)	22.3% less (30.1% less to 13.1% less)	Adalimumab is probably better than placebo, with a moderate effect size (ranging from small to moderate).
CTP13 (3 studies, n = 359)	0.55 (0.43 to 0.71)	Low ^e ⊕⊕⊕⊖	714 per 1000	392 per 1000 (307 to 507)	32.2% less (40.9% less to 21% less)	CTP13 may be better than placebo, with a large effect size (ranging from moderate to large).
Infliximab (6 studies, n = 557)	0.68 (0.57 to 0.81)	Low ^e ⊕⊕⊕⊖	714 per 1000	485 per 1000 (407 to 578)	23% less (30.9% less to 13.5% less)	Infliximab may be better than placebo, with a moderate effect size (ranging from small to moderate).
Upadacitinib (1 study, n = 337)	0.72 (0.56 to 0.94)	Low ^f ⊕⊕⊕⊖	714 per 1000	514 per 1000 (400 to 671)	19.7% less (31.6% less to 4.3% less)	Upadacitinib may be better than placebo, with a small effect size (ranging from trivial to large).
Vedolizumab (2 studies, n = 287)	0.78 (0.6 to 1.01)	Low ^f ⊕⊕⊕⊖	714 per 1000	557 per 1000 (428 to 721)	15.9% less (28.6% less to 0.7% more)	Vedolizumab may be the same as placebo, with an effect size that can range from a moderate effect size less than placebo to a trivial effect size more than placebo.
Ustekinumab (2 studies, n = 336)	0.83 (0.67 to 1.03)	Low ^f ⊕⊕⊕⊖	714 per 1000	592 per 1000 (478 to 735)	12.1% less (23.5% less to 1.9% more)	Ustekinumab may be the same as placebo, with an effect size that can range from a moderate effect size less than placebo to a trivial effect size more than placebo.
Etrolizumab (1 study, n = 217)	0.89 (0.67 to 1.17)	Low ^f ⊕⊕⊕⊖	714 per 1000	635 per 1000 (478 to 835)	8% less (23.2% less to 12% more)	Etrolizumab may be the same as placebo, with an effect size that can range from a moderate effect size less than placebo to a small effect size more than placebo.
Tofacitinib (1 study, n = 121)	0.9 (0.67 to 1.21)	Low ^f ⊕⊕⊕⊖	714 per 1000	642 per 1000 (478 to 864)	7% less (23.3% less to 15% more)	Tofacitinib may be the same as placebo, with an effect size that can range from a moderate effect size less than placebo to a small effect size more than placebo.
Certolizumab (1 study, n = 216)	0.73 (0.55 to 0.97)	Very low ^g ⊕⊕⊕⊖	714 per 1000	521 per 1000 (393 to 693)	19.3% less (32.1% less to 2.3% less)	The evidence is very uncertain.
Natalizumab (1 study, n = 214)	0.82 (0.58 to 1.15)	Very low ^g ⊕⊕⊕⊖	714 per 1000	585 per 1000 (414 to 821)	12.8% less (29.8% less to 11% more)	The evidence is very uncertain.

Filgotinib (1 study, n = 217)	0.92 (0.70 to 1.21)	Very low ^g ⊕⊕⊕⊕	714 per 1000	657 per 1000 (500 to 864)	5.5% less (21.3% less to 15.3% more)	The evidence is very uncertain.
Risankizumab (1 study, n = 358)	0.93 (0.7 to 1.23)	Very low ^g ⊕⊕⊕⊕	714 per 1000	664 per 1000 (500 to 878)	5.2% less (21.4% less to 16.2% more)	The evidence is very uncertain.
Infliximab with purine analogues (2 studies, n = 130)	1 (0.4 to 2.55)	Very low ^h ⊕⊕⊕⊕	714 per 1000	714 per 1000 (286 to 1000)	0.3% more (43.2% less to 100% more)	The evidence is very uncertain.
Vedolizumab with purine analogues (1 study, n = 308)	2.46 (0.99 to 6.11)	Very low ^g ⊕⊕⊕⊕	714 per 1000	1000 per 1000 (707 to 1000)	100% more (0.9% less to 100% more)	The evidence is very uncertain.

GRADE Working Group grades of evidence

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^dDowngraded once due to risk of bias.

^eDowngraded once due to risk of bias and once due to incoherence.

^fDowngraded twice due to very serious imprecision.

^gDowngraded once due to risk of bias and twice due to very serious imprecision.

^hDowngraded once due to inconsistency and twice due to very serious imprecision.

Summary of findings 4. Biologics for maintenance in Crohn's disease: withdrawals due to adverse events

Withdrawals due to adverse events (21 studies, n = 6149 participants)

Patient or population: adult people with inactive Crohn's disease or with response to biologic treatment

Settings: hospital setting

Intervention: biologics and advanced therapies

Comparison: placebo (17 studies, n = 1912)

Comparisons (in order of GRADE certainty)	Network evidence		Anticipated absolute effects for network estimate			Notes
	RR (95% CI)	Certainty	Risk with placebo ^a	Risk with biologic ^b	Absolute risk difference with biologic ^c (95% CI)	
Vedolizumab (3 studies, n = 595)	0.84 (0.63 to 1.11)	Moderate ^d ⊕⊕⊕⊖	221 per 1000	186 per 1000 (139 to 245)	4% less (9.1% less to 2.8% more)	Vedolizumab is probably the same as placebo, with an effect size that can range from a small effect size less than placebo to a trivial effect size more than placebo.
Tofacitinib (1 study, n = 121)	0.78 (0.47 to 1.31)	Low ^e ⊕⊕⊖⊖	221 per 1000	393 per 1000 (104 to 289)	5.3% less (13% less to 7.5% more)	Tofacitinib may be the same as placebo, with an effect size that can range from a small effect size less than placebo to a small effect size more than placebo.
Ustekinumab (2 studies, n = 336)	0.74 (0.40 to 1.38)	Low ^e ⊕⊕⊖⊖	221 per 1000	163 per 1000 (88 to 305)	6.3% less (14.6% less to 9.2% more)	Ustekinumab may be the same as placebo, with an effect size that can range from a moderate effect size less than placebo to a small effect size more than placebo.
Etrolizumab (1 study, n = 217)	1.42 (0.73 to 2.77)	Low ^e ⊕⊕⊖⊖	221 per 1000	314 per 1000 (161 to 612)	10.3% more (6.6% less to 43.3% more)	Etrolizumab may be the same as placebo, with an effect size that can range from a trivial effect size less than placebo to a large effect size more than placebo.
Upadacitinib (1 study, n = 337)	1.51 (0.73 to 3.11)	Low ^e ⊕⊕⊖⊖	221 per 1000	334 per 1000 (161 to 687)	12.5% more (6.6% less to 51.8% more)	Upadacitinib may be the same as placebo, with an effect size that can range from a trivial effect size less than placebo to a large effect size more than placebo.
Natalizumab (1 study, n = 214)	0.48 (0.28 to 0.82)	Very low ^f ⊕⊖⊖⊖	221 per 1000	106 per 1000 (62 to 181)	12.9% less (17.7% less to 4.5% less)	The evidence is very uncertain.
Infliximab (6 studies, n = 369)	0.54 (0.29 to 0.99)	Very low ^f ⊕⊖⊖⊖	221 per 1000	119 per 1000 (64 to 219)	11.3% less (17.3% less to 0.2% less)	The evidence is very uncertain.
Certolizumab (1 study, n = 216)	0.58 (0.38 to 0.90)	Very low ^f ⊕⊖⊖⊖	221 per 1000	128 per 1000 (84 to 199)	10.2% less (15.3% less to 2.3% less)	The evidence is very uncertain.

Adalimumab (4 studies, n = 162)	0.84 (0.54 to 1.30)	Very low ^f ⊕⊕⊕⊕	221 per 1000	186 per 1000 (119 to 287)	3.9% less (11.2% less to 7.4% more)	The evidence is very uncertain.
Risankizumab (1 study, n = 358)	0.87 (0.38 to 2.03)	Very low ^f ⊕⊕⊕⊕	221 per 1000	192 per 1000 (84 to 449)	3.1% less (15.3% less to 25.3% more)	The evidence is very uncertain.
CTP13 (2 studies, n = 146)	0.72 (0.21 to 2.45)	Very low ^f ⊕⊕⊕⊕	221 per 1000	159 per 1000 (46 to 541)	6.8% less (19.3% less to 35.6% more)	The evidence is very uncertain.
Infliximab with purine analogues (1 study, n = 71)	0.92 (0.31 to 2.68)	Very low ^f ⊕⊕⊕⊕	221 per 1000	203 per 1000 (69 to 592)	2.1% less (16.8% less to 41.1% more)	The evidence is very uncertain.
Filgotinib (1 study, n = 217)	2.58 (1.09 to 6.10)	Very low ^f ⊕⊕⊕⊕	221 per 1000	570 per 1000 (241 to 1000)	38.7% more (2.2% more to 100% more)	The evidence is very uncertain.

GRADE Working Group grades of evidence

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BACKGROUND

Description of the condition

Crohn's disease is a chronic inflammatory bowel disease (IBD), characterised by transmural inflammation of the gastrointestinal tract. Its pathophysiology is thought to involve a complex interplay between genetic susceptibility, immune and environmental factors [1]. Worldwide, the incidence of Crohn's disease is increasing, with the highest incidence in westernised nations and even within these nations the incidence continues to increase [2, 3].

Symptoms depend on the area of bowel involved but often include diarrhoea, abdominal pain, gastrointestinal bleeding and weight loss. Complications may further arise with the development of stricturing or fistulising disease and surgery is a relatively common occurrence for patients.

Until recently, the most common therapeutic approach was the use of corticosteroids followed by an immunomodulator such as a thiopurine (azathioprine, 6-mercaptopurine), which are a subset of purine analogues that act as anti-metabolites, methotrexate or a tumour necrosis factor (TNF) inhibitor (infliximab, adalimumab, certolizumab). However, this approach results in up to 55% to 60% of patients failing to achieve remission at one year following diagnosis [4, 5].

Despite the advent of TNF inhibitors, many patients have either primary non-response or secondary loss of response and, as such, new therapies with different mechanisms have advanced rapidly over the last decade.

Description of the intervention and how it might work

Biologic medications (monoclonal antibodies such as infliximab) and small molecules (such as JAK (Janus associated kinase) inhibitors and S1P (sphingosine-1-phosphate) modulators) have revolutionised the treatment of IBD, by targeting specific parts of the immune system. There are currently three tumour necrosis factor- α (TNF- α) antagonists approved for the treatment of Crohn's disease: infliximab, adalimumab and certolizumab pegol. In the last few years, numerous new biologic drug classes have become actively available for the treatment of Crohn's disease, with many more in various stages of approval.

Natalizumab and vedolizumab are humanised monoclonal antibodies against the human α 4 (α 4) integrin subunit and the α 4 β 7 (α 4 β 7) integrin on leukocytes, respectively.

Ustekinumab, a fully human monoclonal antibody against the p40 subunit that is shared by interleukin-12 and interleukin-23, is currently available for the treatment of Crohn's disease. Guselkumab and risankizumab are IL23-binding antibodies that block the activation of pro-inflammatory cytokines. Janus kinase inhibitors, such as tofacitinib and filgotinib, are small molecules that are orally administered.

Several other integrin and specific IL23 antagonists are under investigation for Crohn's disease, with some in late-stage clinical development [6, 7].

TNF- α antagonists are monoclonal antibodies directed towards TNF- α . Although TNF- α antagonists have been the benchmark biologic therapies for more than a decade, the exact mechanism

of action is still incompletely understood [8, 9]. These drugs specifically and avidly bind to TNF- α , a cytokine with pleiotropic functions that is relevant to the pathogenesis of IBD. However, a decrease in the concentration of inflammatory markers such as serum C-reactive protein (CRP) following treatment with a TNF- α antagonist is not consistently associated with improvement in endoscopic disease activity. Although direct comparisons are lacking, available data suggest that not all TNF- α antagonists are equally effective in IBD. Specifically, soluble (decoy) TNF- α receptors such as etanercept and onercept seem to be ineffective [10] (Rutgeerts 2006 [11]).

In addition, TNF- α antagonists that lack the Fc region (such as certolizumab pegol), or have an Fc region with strongly reduced affinity towards the Fc receptor (such as CDP571), demonstrate disappointing results in terms of mucosal healing in Crohn's disease patients [12]. Collectively, these observations suggest that additional factors beyond the simple neutralisation of soluble TNF- α are relevant to the mechanism of action of this class of drug. In particular, TNF- α antagonists appear to induce lamina propria T-cell apoptosis and potentiation of M2-type wound-healing macrophages, both of which are dependent on the presence of a functional Fc region and binding of membrane-bound TNF- α [8].

Integrin antagonists block the recruitment of lymphocytes from the circulation to the gut by preventing the interaction between the integrins on these cells with endothelial adhesion molecules on intestinal microvascular endothelial cells [13]. Natalizumab blocks the interaction between α 4-expressing lymphocytes and the endothelial adhesion molecules VCAM-1 and MAdCAM-1, whereas vedolizumab only blocks the interaction between α 4 β 7-expressing lymphocytes and MAdCAM-1. Unlike VCAM-1, MAdCAM-1 is exclusively expressed by intestinal microvascular endothelial cells, resulting in the gut-selective action of vedolizumab but not of natalizumab [13]. A drug exposure-response relationship is seen for vedolizumab despite maximal saturation of the target α 4 β 7 at the lowest concentrations, suggesting that other, as yet unidentified, mechanisms of actions are involved [14, 15].

Ustekinumab is a fully human monoclonal antibody directed towards p40, a shared subunit of interleukin-12 and interleukin-23, which are involved in T helper 1 and T helper 17 cell responses, respectively [16].

Large phase III trials have demonstrated the efficacy of all three drug classes as induction and maintenance agents in Crohn's disease, which has led to the market authorisation of these drugs [16, 17].

A major innovation has been the development of small molecules that can be delivered orally as biological therapy. Janus kinase inhibitors act by competitively blocking the adenosine triphosphate-binding site in the JH1 domain through non-covalent interactions [18]. Sphingosine-1-phosphate works mainly through S1PR1 receptor binding [19].

Conventional therapy can be combined with biologics, known as combination therapy, with increasing numbers of patients using it.

Why it is important to do this review

Recent Cochrane meta-analyses have investigated the role of individual biologics in the induction and maintenance of remission

in Crohn's disease [20, 21, 22, 23, 24, 25, 26]. However, comparative efficacy data are needed in order to clarify the relative position of the increasingly complex portfolio of biologic agents and to guide decision-making regarding Crohn's disease therapeutic algorithms. Specifically, clarity is needed as to whether agents should be positioned as first-line therapies for patients who are anti-TNF- α naive [16, 27, 28], or be used as second-line therapies if patients are refractory to the more established biological classes.

These data are best achieved by head-to-head trials. However, such trials take time and will likely compare only two biologic drugs within a specific population, if these trials are conducted at all.

Thus far, there are limited active head-to-head trials comparing biologic treatments for Crohn's disease. An alternative strategy for obtaining comparative data is to conduct a network meta-analysis in which multiple treatments are compared using both direct comparisons of interventions within randomised controlled trials (RCTs) and indirect comparisons across trials based on a common comparator (i.e. placebo). In other words, if compound A is compared with compound B in one trial, and the same compound B is compared with compound C in another trial, indirect information can be obtained from compound A versus compound C under the assumption of transitivity.

OBJECTIVES

To assess the effects of biologic and advanced treatments for induction and maintenance of remission in Crohn's disease.

A secondary objective was to rank the included interventions.

METHODS

The protocol for this review was published by a different author team in 2017 [29]. We have updated the methods section to adhere to current Cochrane methodology.

Compared to the original protocol, in this review we have included any dosages of any biologic and advanced treatment, irrespective of whether they have received approval by regulatory authorities.

We have considered studies with more than 50% of participants with concomitant purine analogue use as combination studies ([Types of interventions](#)). Purine analogues are an active treatment commonly used for the treatment of Crohn's and their combination with biologics and advanced agents is likely to have different efficacy and safety profiles to monotherapy.

We have removed histologic and endoscopic response outcomes, as we did not consider them to be critical. We have amended the pre-planned subgroup analyses, specifically those around dosage and administration route, exposure to immunosuppressives and biological naivety, and we have deemed those around disease location and clinical and objective measures to determine patient eligibility to not be achievable with the currently available data. The subgroup analysis on disease activity has now been listed as a sensitivity analysis; the sensitivity analysis based on 'relevant loss to follow-up' has been removed because attrition is taken into account in the risk of bias assessment; and the sensitivity analysis based on 'sample size of less than 50% participants' has been replaced with comparison-adjusted funnel plots assessments. We have also updated the network meta-analysis and GRADE assessment methodology according to the latest standards.

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review and PRISMA 2020 for the reporting.

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs) (phase 2b or above) comparing biologics or advanced therapies with any other active comparator, placebo or no treatment for induction or maintenance of remission in Crohn's disease. We excluded studies comparing different regimens of the same biologic.

We considered cluster-RCTs and the pre-cross-over phase of cross-over RCTs for inclusion.

We considered all relevant RCTs published as full text, abstract, press release or as results posted in a trial register for inclusion.

We excluded non-randomised or quasi-randomised trials (such as sequential allocation, allocation by birthdate or medical record number) or parts of trials, such as non-randomised induction or maintenance phases, long-term follow-ups or non-randomised control groups.

Types of participants

We considered trials in adult participants (> 18 years of age) with Crohn's disease as diagnosed by conventional clinical, radiological or endoscopic criteria for inclusion.

There were no limits on follow-up time, language, setting, gender, disease activity, disease duration, disease location or any other factors.

We included participants with active Crohn's disease, as defined by the included studies, in the network for induction of remission. We included participants with inactive Crohn's disease, or with response to biologics at the beginning of the RCT, as defined by the included studies, in the network for maintenance of remission.

We included studies with only a subset of eligible participants and planned sensitivity analyses around them.

Types of interventions

We considered all biologic and advanced therapies for inclusion, including TNF- α inhibitors, anti-integrins, IL23/IL12 antagonists, JAK inhibitors, S1P receptor modulators and biosimilars. We considered all types of administration routes for inclusion. We considered all tested dose regimens.

After discussion with the editorial group and the Cochrane methods unit, it was decided that a threshold would be set for populations with mixed exposure to purine analogues. Thus, for studies where more than 50% of participants were receiving concomitant purine analogues, we considered the purine analogues as part of the intervention. That is, if a comparison was defined by the authors as 'biologic against placebo', however more than 50% of the participants were on purine analogues during the trial, for our analysis, the comparison would be 'biologic and purine analogues against purine analogues'.

Purine analogues and corticosteroids as standalone treatments were ineligible for this review, as they are not biologic treatments.

Our search strategy was not designed to comprehensively include all RCTs for these treatments on this review's topic. However, trials comparing them to biologics were eligible and included in the analyses. As the evidence for these treatments in this review is incomplete because trials of purine analogues or corticosteroids versus placebo or other non-biologic interventions were ineligible, we have not assessed or discussed the results we generated for them.

The above criteria for studies, participants and interventions meant that potential effect modifiers would be evenly distributed among the included studies, ensuring the transitivity assumption would not be violated.

Outcome measures

We only considered dichotomous outcome data for inclusion, as we expected continuous outcomes to be scarcely reported and less clinically relevant for the context of this review.

Critical outcomes

The critical outcome for induction of remission studies (I.a) was the proportion of participants who entered clinical remission, as defined by the original study (e.g. a Crohn's Disease Activity Index score < 150).

The critical outcome for maintenance of remission studies (M.a) was the proportion of participants who relapsed (failed to maintain clinical remission), as defined in the original study.

If the outcomes were reported at more than one time point, we used the one defined as primary by the original studies.

Important outcomes

For the induction studies, the important outcome measures were the proportion of participants with the following:

1. Clinical response, as defined by the original study (using the strictest definition for analysis if more than one available) (I.b)
2. Endoscopic remission, as defined by the original study (I.c)
3. Withdrawals due to adverse events (I.d)
4. Serious adverse events (I.e)
5. Total adverse events (I.f)

For the maintenance studies, the important outcome measures were the proportion of participants with the following:

1. Clinical loss of response, as defined by the original study (using the strictest definition if more than one available) (M.b)
2. Endoscopic relapse, as defined by the original study (M.c)
3. Withdrawals due to adverse events (M.d)
4. Serious adverse events (M.e)
5. Total adverse events (M.f)

Search methods for identification of studies

Electronic searches

In June 2025, an Information Specialist searched the following sources:

- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 6) via the Cochrane Library;

- MEDLINE via Ovid SP (1946 to 2025);
- Embase via Ovid SP (1974 to 2025);
- ClinicalTrials.gov;
- World Health Organization International Clinical Trials Registry Platform (WHO ICTRP).

We followed the latest guidelines from the *Cochrane Handbook for Systematic Reviews of Interventions* in designing and running the searches [30]. We also used the Cochrane highly sensitive search strategy for identifying randomised trials in MEDLINE (sensitivity-maximising version, 2008 revision, Ovid format) and Cochrane's RCT search filter for Embase [31] to identify randomised controlled trials. The MEDLINE search strategy was adapted and translated into the syntax of other sources. We did not apply any date, language, document type or publication status limitations to this search [32]. The search strategies for all sources are listed in [Supplementary material 1](#).

Searching other resources

We performed manual searches of reference lists from included studies and potentially relevant publications to ensure that studies that may have been missed by the computer-assisted search were identified.

We also performed handsearches of the conference proceedings of major gastroenterology meetings (e.g. Digestive Disease Week; the annual meeting of the European Crohn's and Colitis Organisation; and United European Gastroenterology Week) from the last five years.

To identify potentially problematic studies, we considered any post-publication amendments, including expressions of concern, errata, corrigenda and retractions on included or eligible studies. Retracted studies would have been excluded, while errata and corrigenda would have been considered when extracting data and for our risk of bias assessments. Expressions of concern would have been discussed on a case by case basis between authors and our decisions on whether to include them or not would have been captured in the review.

Data collection and analysis

Selection of studies

Based on the inclusion criteria above, at least two authors independently screened the titles and abstracts identified by the search for eligibility. Disagreements were resolved by discussion and consensus. A third author (MG) resolved cases in which consensus was not reached. Full-text screening followed the same method.

Data extraction and management

Pairs of review authors independently extracted outcome data and data on effect modifiers. Any disagreements were resolved by discussion with a third author. We used a piloted data extraction form. We extracted the following data from the included studies:

1. General information (study design, study arms, single or multi-centre, countries, study dates, setting).
2. Participant characteristics (disease activity at inclusion; number and percentage of TNF- α naive versus TNF- α experienced

participants; disease duration and location; luminal or fistulating disease, age, sex, smoking status).

3. Critical and important outcome data.
4. Sources of funding and conflicts of interest of study authors.

We entered data into Review Manager (RevMan) [33]. We attempted to contact the authors or sponsors of the studies for unclear or missing data.

Risk of bias assessment in included studies

The methodological quality of each included study was independently assessed by pairs of authors using the Cochrane risk of bias tool (RoB 1) [34]. The domains assessed were:

1. random sequence generation;
2. allocation concealment;
3. blinding of participants, outcome assessors and investigators;
4. incomplete outcome data;
5. selective outcome reporting; and
6. other potential sources of bias.

We judged studies to be at low, high or unclear risk of bias based on these domains. We assessed the likely magnitude and direction of the bias and whether we considered it likely to impact on the results of our analysis. Any disagreements regarding risk of bias assessment between the pairs of authors were resolved through discussion until consensus was reached. If consensus could not be reached, a third author (MG) was consulted to resolve the disagreement. We attempted to contact the authors or sponsors of the studies for unclear or missing information regarding any unclear risk of bias judgements.

Measures of treatment effect

We summarised the treatment effects as risk ratio (RR) estimates and associated two-sided 95% confidence intervals (CI). We also calculated absolute risk difference percentages (RD%) and number needed to treat (NNT), which is the number of patients you need to treat to prevent one additional bad outcome. We only calculated the NNT for treatments with high and moderate certainty of evidence for our critical outcomes.

Unit of analysis issues

The participant was the unit of the analysis. Correlated effect estimates arising from multiple-arm studies were accounted for using network meta-analysis methodology, and there was no need for splitting the placebo group evenly in order to avoid a unit of analysis error [35]. In multi-arm studies where different dosages of the same drug were compared, we combined the different dosage arms for the purpose of our analysis.

Where outcomes were reported at several time points, we undertook analyses for the time point defined as primary by the original trial. If that was not defined, then we used the single time point that was consistently reported by the trials.

To avoid carry-over effects, we only included the first part of the study (i.e. before cross-over) for any included cross-over studies.

No cluster-RCTs were eligible for inclusion. If cluster-randomised trials are identified in future updates, these will be included and, if unit of analysis issues are identified (e.g. randomisation and

analysis at different units), we will adjust the sample sizes or standard errors appropriately [36]

Dealing with missing data

We based our analysis on the data made available by the authors. We contacted authors or sponsors for missing data, and data that were not reported in sufficient detail or unclear.

For all outcomes, we performed analyses on a modified intention-to-treat basis; that is, we attempted to include all participants randomised to each group in the analyses, and we analysed all participants in the group to which they were allocated, regardless of whether they received the allocated intervention or not.

For safety outcomes, we considered participants with missing or unclear withdrawal data as withdrawals due to adverse events. The denominators used for this outcome were as reported by the authors. For serious and total adverse events, we used the numbers of participants with events, as reported by the authors. We discarded outcome data reported for mixes of randomised and non-randomised participants and did not use such data for analysis.

In maintenance trials, relapse and loss of response rates were rarely reported directly, while remission and response were reported more often. We accounted for that by inverting the reported remission and response rates to infer relapse and loss of response, for the studies where this was the case. Withdrawals have been added to the inverted rates, as all withdrawals have been counted as treatment failures.

Reporting bias assessment

To assess the presence of small-study effects in the network meta-analysis, we used comparison adjusted funnel plots for active treatments versus placebo and used Egger's test for asymmetry [37]. We assumed that small studies may show that the active treatments are more effective than placebo. This accounts for the fact that each set of studies estimates a different summary effect [38]. We investigated publication bias via visual inspection of funnel plots for each pairwise analysis if there were at least 10 studies [39].

Synthesis methods

Pairwise meta-analysis

Direct comparisons of treatment effects

We combined data for meta-analysis from individual studies when the interventions, participant groups and outcomes were similar, as deemed by review author consensus. We used a random-effects model to pool data [40]. We used RevMan for the analyses [33].

Network meta-analysis

Direct, indirect and network comparisons

We used the direct pairwise comparisons to perform the network meta-analysis (NMA). Through the NMA, direct effect estimates (direct evidence) can also be obtained (adjusted for multi-arm studies). In such cases, effect estimates derived from pairwise meta-analysis may slightly differ from the corresponding direct estimates derived from NMA. This is due to multi-arm studies being included, as the NMA model used makes adjustments for multi-arm studies.

We initially generated and assessed the network diagrams to determine if a network meta-analysis was feasible. A network diagram represents graphically the connections of the interventions included in the network. Nodes are represented by a point. The lines connecting the points are the connections between the interventions and referred to as edges. Two points connected via a single edge represent a direct comparison, while points connected via more than one edge represent indirect comparisons. A minimum of 10 connected studies was required to perform the network meta-analysis. In the case of disconnected networks, we would analyse the largest network. If the network did not consist of at least 10 studies, we did not perform the analysis.

We performed the network meta-analysis on all outcomes within a frequentist framework using multivariate meta-analysis. We used the R statistical package [41] and the netmeta library [42].

Relative treatment ranking

We estimated the cumulative probabilities for each biologic being at each possible rank and obtained a treatment hierarchy using the surface under the cumulative ranking curve (SUCRA) based on 100 simulations; the larger the SUCRA the higher its rank among all available agents [43]. The probabilities to rank the treatments are estimated under a Bayesian model with flat priors, assuming that the posterior distribution of the parameter estimates is approximated by a normal distribution with mean and variance equal to the frequentist estimates and variance-covariance matrix. Rankings are constructed by drawing 100 samples from their approximate posterior density. For each draw, the linear predictor is evaluated for each trial, and the largest linear predictor is noted [42].

Investigation of heterogeneity and subgroup analysis

Clinical and methodological heterogeneity

To evaluate the presence of clinical heterogeneity, we examined trial and trial population characteristics across all eligible trials that compared each pair of interventions. We assessed the presence of clinical heterogeneity within each direct comparison by comparing these characteristics.

Assessment of statistical heterogeneity

We assessed statistically the presence of heterogeneity within each direct comparison using the I^2 statistic and its 95% CI [36]. We based the assessment of statistical heterogeneity on the magnitude of the heterogeneity variance parameter (τ^2). The heterogeneity estimator was DerSimonian-Laird (DL) both for network and pairwise meta-analyses.

The I^2 statistic quantifies heterogeneity and reflects the percentage of the variability in effect estimates that is due to heterogeneity rather than sampling error (chance). We interpreted I^2 values as follows, as described in the *Cochrane Handbook for Systematic Reviews of Interventions*:

- 0% to 40%: might not be important;
- 30% to 60%: may represent moderate heterogeneity;
- 50% to 90%: may represent substantial heterogeneity;
- 75% to 100%: considerable heterogeneity.

Assessment of transitivity across treatment comparisons

The assumption of transitivity refers to the notion that all studies included in a network are similar in all important aspects other than the interventions compared [44]. These important aspects can also be referred to as potential effect modifiers, since if they are not evenly distributed between studies, they have the potential of modifying the true effect estimates. The potential effect modifiers we considered were setting, age, sex, disease location, disease duration, disease phenotype, disease activity and concomitant steroid use.

We assessed the assumption of transitivity by comparing the distribution of potential effect modifiers listed above across the different pairwise comparisons. In this context, we expect that the transitivity assumption will hold assuming the common treatment used to compare different biologic drugs indirectly is similar when it appears in different trials, and all pairwise comparisons do not differ with respect to the distribution of effect modifiers.

We evaluated the assumption of intransitivity epidemiologically by comparing the clinical and methodological characteristics of sets of studies from the various treatment comparisons.

Assessment of statistical inconsistency/incoherence at the network level

Consistency (or coherence) is the statistical manifestation of transitivity. Intransitivity may result in differences between direct and indirect evidence. We assumed a common estimate for the heterogeneity variance across the different comparisons. We used global and local approaches to evaluate the statistical agreement between the various sources of evidence in a network of interventions (coherence). To evaluate the presence of inconsistency locally, we used the loop-specific SIDE approach [45]. This method evaluates the consistency assumption in each closed loop of the network separately as the difference between direct and indirect estimates for a specific comparison in the loop (inconsistency factor). Then, the magnitude of the inconsistency factors and their 95% CIs can be used to infer the presence of inconsistency in each loop. To check the assumption of coherence in the entire network, we used the 'design-by-treatment' model as described by Higgins et al. [35]. This method accounts for different sources of inconsistency that can occur when studies with different designs, among different sets of interventions, give different results as well as disagreement between direct and indirect evidence. Using this approach, we inferred the presence of inconsistency from any source in the entire network based on a χ^2 test. We ran all analyses with the R statistical package [41] and the netmeta library [42].

Subgroup analyses

Our pre-planned NMA subgroup analyses were as follows:

1. Participants who were biologically naive or not (whether they had been exposed to a biologic prior to randomisation; if mixed, the threshold of 50% exposure was set to define whether a group was classified as naive or exposed).
2. Participants who were biologically naive or not (any study with populations with mixed exposure was removed).
3. We had originally planned a subgroup analysis on different dosages (not considering dosages used in phase 2a studies; dosages only considered if licenced for patient use). However, as

most of the biologics included in the review have not yet been licenced, we did not think it would be clinically appropriate to perform an analysis whereby only the non-licenced dosages of the licenced drugs would be removed.

The assessment of the subgroup analyses was based on visual inspection for overlap between the estimates and confidence intervals for the interventions that had evidence for all subgroups. As the evidence used for the subgroup analyses could only come from the included studies that reported the required data, the interventions lacking data were not represented or connected within each subgroup, therefore these interventions are not presented.

Equity-related assessment

No equity-related assessment was planned for this review.

Sensitivity analysis

We planned to perform the following NMA sensitivity analyses to examine the impact of the following variables on the pooled effect estimate:

1. Random-effects versus fixed-effect modelling.
2. Only studies with low risk of bias (removing studies with more than two unclear domains or any high risk of bias domains).
3. Removal of studies where the population is mixed regarding the use of immunosuppressives (purine analogues) (> 20%).
4. In the context of maintenance studies, addition of studies with mixed populations of participants in a state of both clinical remission and clinical response at baseline randomisation.
5. Removal of studies with participants with more severe forms of disease, such as fistulating disease.
6. Removal of non-peer-reviewed full-text publications (e.g. abstracts, press releases, etc.).

Certainty of the evidence assessment

The summary of findings tables (SoF) present evidence comparing all methods with a reference comparator (placebo). Each table describes key features of the evidence relating to a single outcome. We have presented SoF tables for each critical outcome (clinical remission for induction and clinical relapse for maintenance) and the safety outcome of 'withdrawals due to adverse events' in accordance with the GRADE approach. Our critical outcomes are presented in the abstract and plain language summary. These outcomes were 'prioritised' for presentation in the summary versions of the review because they are the most clinically important outcomes for decision-makers.

We assessed the certainty of the evidence using the GRADE approach as outlined in the GRADE Handbook in order to assess the certainty of the body of evidence relating to each outcome for all comparisons [46].

In order to create summary of findings tables, we used GRADEpro GDT [47] to import data from Review Manager [33]. We used the GRADE working group's approach for rating the certainty of the network meta-analysis effect estimates for all the comparisons and all outcomes [48, 49]. We appraised the certainty of the direct, indirect and network evidence sequentially.

First, we assessed the certainty of the direct evidence (where available) for a given outcome, and rated the evidence using the standard GRADE approach based on consideration of trial design limitations (risk of bias), inconsistency, imprecision, indirectness and publication bias [50]. In this approach, the direct estimates are rated for risk of bias, inconsistency, indirectness and publication bias, then the indirect estimates are rated based on the lowest ratings of the direct comparisons forming the most dominant loop and intransitivity; finally the network estimates are rated based on imprecision, incoherence and the rating of the direct or indirect estimate that contributes the most.

We then rated the certainty of the indirect evidence for the same given outcomes, based on the lower of the certainty ratings of the two direct arms forming the dominant 'first-order' loop in the network diagram for this outcome.

Finally, we determined the certainty of network evidence based on the following:

- The higher certainty rating of the direct and indirect evidence.
- Whether the relevant network exhibited 'transitivity', that is, whether all the comparisons contributing data to the estimate were directly consistent with the PICO question.
- Consideration of coherence between direct and indirect effect estimates (visual inspection of how much the direct and indirect evidence estimates agree and their confidence intervals overlap, and the statistical tests described earlier). We considered whether both the direct and indirect estimates contributed importantly to the network estimate, and whether incoherence was associated with intransitivity, imprecision and indirectness [51].
- Precision of the network effect estimate. Imprecision judgements were based on the following thresholds of risk difference [52]:
 - Clinical remission/relapse: < 10%: trivial difference; 10% to 20%: small difference; 20% to 31%: moderate difference; > 31% large difference.
 - Clinical response/loss of response: < 13%: trivial difference; 13% to 24%: small difference; 24% to 35%: moderate difference; > 35% large difference.
 - Endoscopic remission/relapse: < 9%: trivial difference; 9% to 17%: small difference; 17% to 28%: moderate difference; > 28% large difference.
 - Withdrawals due to adverse events: < 7%: trivial difference; 7% to 14%: small difference; 14% to 23%: moderate difference; > 23% large difference.
 - Serious adverse events: < 6%: trivial difference; 6% to 10%: small difference; 10% to 17%: moderate difference; > 17% large difference.
 - Total adverse events: < 8.5%: trivial difference; 8.5% to 15.5%: small difference; 15.5% to 24%: moderate difference; > 24% large difference.

At each of these stages, two review authors (MG, VS) independently appraised the certainty ratings for the direct, indirect and network evidence. We resolved disagreements between authors through discussion and consultation. We rated the certainty of network evidence for each outcome as 'high', 'moderate', 'low' or 'very low' in accordance with the GRADE approach.

- High certainty: we are very confident that the true effect lies close to that of the effect.
- Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.
- Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

Consumer involvement

Consumers were not involved in this review due to limited resources.

RESULTS

Description of studies

Results of the search

Our search, conducted up to June 2025, identified 24,290 records. After removing duplicates, 22,424 records underwent title and abstract screening to assess eligibility, of which 21,691 were excluded. The remaining 733 records, which comprised 154 studies, underwent full-text review, of which 21 studies were excluded with reasons. There are 24 studies awaiting assessment and 15 ongoing studies. The remaining records comprised a total of included 94 RCTs.

A total of 66 RCTs (n = 20653) on induction of remission were included (Allez 2023 (TRIDENT- Part 1) [53, 54, 55]; Allez 2023 (TRIDENT- Part 2) [56]; Chen 2020 [57, 58, 59]; Colombel 2007 (CHARM) - Induction (non-responders) [60, 61, 62]; Colombel 2010 (SONIC) [63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73]; D'Haens 1999 (refractory CD) [74, 75, 76]; D'Haens 2008 [77]; D'Haens 2022 (ADVANCE) [78, 79, 80]; D'Haens 2022 (SERENE - Induction) [81, 82, 83]; Duan 2013 [84]; Feagan 2008c [85, 86]; Feagan 2015a (UNITI-1) [87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110]; Feagan 2015b (UNITI-2) [111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134]; Feagan 2017 [135, 136, 137]; Ghosh 2003 [138, 139, 140, 141]; Gordon 2001 [142, 143, 144, 145]; Hanauer 2002 (ACCENT I) - Induction (non-response) [146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163]; Hanauer 2006 (CLASSIC I) [164, 165, 166]; Hanauer 2021 (VOLTAGE-CD) [167, 168, 169, 170, 171, 172]; Lemann 2006 [173]; Loftus 2023 (U-EXCEED) [174]; Panaccione 2022 (MOTIVATE) [175, 176, 177]; Panes 2017 Induction [178, 179]; Present 1999 [180, 181, 182]; Sandborn 2001a [183]; Sandborn 2001b [184]; Sandborn 2005a (ENACT 1) [185, 186, 187, 188, 189, 190, 191, 192, 193]; Sandborn 2007a [194, 195, 196]; Sandborn 2007c (PRECISE 1) [197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207]; Sandborn 2008 (Population 1) [208, 209, 210, 211, 212]; Sandborn 2008 (Population 2) [213]; Sandborn 2011 [214, 215, 216, 217]; Sandborn 2012 (CERTIFI) - Induction [218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228]; Sandborn 2012 (CERTIFI) - Induction (non-response) [229]; Sandborn 2014 [230]; Sandborn 2023 - BERGAMOT (cohort 1) [231, 232, 233, 234]; Sandborn 2023 - BERGAMOT (cohort 3) [235]; Sandborn 2020d (CELEST) - Induction [236, 237]; Sandborn 2022 - GALAXI 1 [238, 239, 240, 241, 242,

243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254]; Sands 2004 (ACCENT II) Induction (Non-responders) [255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265]; Sands 2007 [266, 267]; Sands 2010 [268]; Sands 2014 (GEMINI III) [269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280]; Sands 2022 (SEAVUE) [281, 282, 283, 284, 285, 286, 287]; Sands 2022 (SERENITY) [288]; Chen 2025 [289, 290]; Ferrante 2024 (VIVID-1) [291, 292]; Hart 2025 (GRAVITI) [293, 294]; Jairath 2025 (RELIEVE UCCD) [295]; Loftus 2023 (U-EXCEL) [296, 297, 298]; Matsumoto 2016 (DIAMOND) [299, 300]; Panaccione 2024 (GALAXI-2) [301, 302]; Panaccione 2024 (GALAXI-3) [303, 304]; Peyrin-Biroulet 2024 - SEQUENCE [305, 306, 307, 308]; Schreiber 2005 [309, 310, 311, 312]; Schreiber 2018a [313, 314]; Schwartz 2021 - ENTERPRISE [315, 316, 317]; Targan 1997 [318, 319, 320, 321]; Targan 2007 (ENCORE) [322, 323, 324, 325, 326, 327, 328, 329, 330]; Vermeire 2017 FITZROY - induction [331]; Vermeire 2017 FITZROY - induction non-responders [332]; Vermeire 2025 - DIVERSITY A [333, 334]; Vermeire 2025 - DIVERSITY B [335, 336]; Watanabe 2012 - Induction [337, 338]; Watanabe 2020 - Induction [339, 340, 341]; Winter 2004 [342]; Ye 2019 [343, 344, 345, 346, 347, 348, 349]).

A total of 28 RCTs (n = 6823) on maintenance of remission were included (Buhl 2022 (STOP-IT) [350, 351, 352, 353, 354]; Colombel 2007 (CHARM) - Maintenance (responders) [355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386]; D'Haens 2022 (SERENE - Maintenance) [387, 388]; Feagan 2015c (IM-UNITI) [389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419]; Ferrante 2022 - FORTIFY [420, 421, 422]; Hanauer 2002 (ACCENT I) - Maintenance (Responders) [423]; Jorgensen 2017 (NORSWITCH) [424, 425, 426, 427, 428]; Louis 2023 - SPARE [429, 430, 431, 432]; Panes 2017 Maintenance [433]; Rutgeerts 1999 [434]; Rutgeerts 2012 (EXTEND) [435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461]; Sandborn 2005a (ENACT 2) [462, 463, 464, 465, 466, 467, 468]; Sandborn 2007b (CLASSIC II) [469, 470, 471, 472]; Sandborn 2010b (WELCOME) [473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483]; Sandborn 2012 (CERTIFI) - Maintenance [484]; Sandborn 2013 (GEMINI II) - Maintenance [485, 486]; Sandborn 2020d (CELEST) - Maintenance [487]; Sands 2004 (ACCENT II) Maintenance (Responders) [488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499]; Schreiber 2007 (PRECISE 2) [500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518]; Schreiber 2021 [519, 520, 521, 522]; VanAssche 2012 (SWITCH) [523, 524, 525]; Vermeire 2017 FITZROY - maintenance responders [526]; Vermeire 2021 (VISIBLE 2) [527, 528, 529, 530, 531, 532, 533, 534, 535]; Vermeire 2025 - DIVERSITY (maintenance) [536, 537]; Volkers 2017 - SIMILAR [538, 539, 540]; Watanabe 2012 - Maintenance [541, 542]; Watanabe 2020 - Maintenance [543]; Young 2024 - iBaSS [544, 545]).

There are 24 records of studies awaiting classification (Supplementary material 4) (Baima 2016 [546]; EUCTR2005-004104-37 [547]; EUCTR2007-001913-41 [548]; EUCTR2007-003239-21 [549]; EUCTR2008-001131-35 (AZORIX) [550, 551]; EUCTR2008-004926-18 (CHAMP) [552]; EUCTR2012-001723-12 [553]; EUCTR2020-002701-26 [554]; Huang 2012 [555]; JPRN-UMIN000005689 [556]; JPRN-UMIN000010524 (OASIS) [557]; NCT00349752 (COSPAR1) [558, 559, 560]; NCT00796250 [561]; NCT01674413 (CADHUM) [562]; NCT01696942 [563]; NCT02771457 [564]; NCT03221166 [565]; NCT03370601 (STAMP) [566]; NCT03393247 [567]; NCT04131322 (ADA-SWITCH) [568];

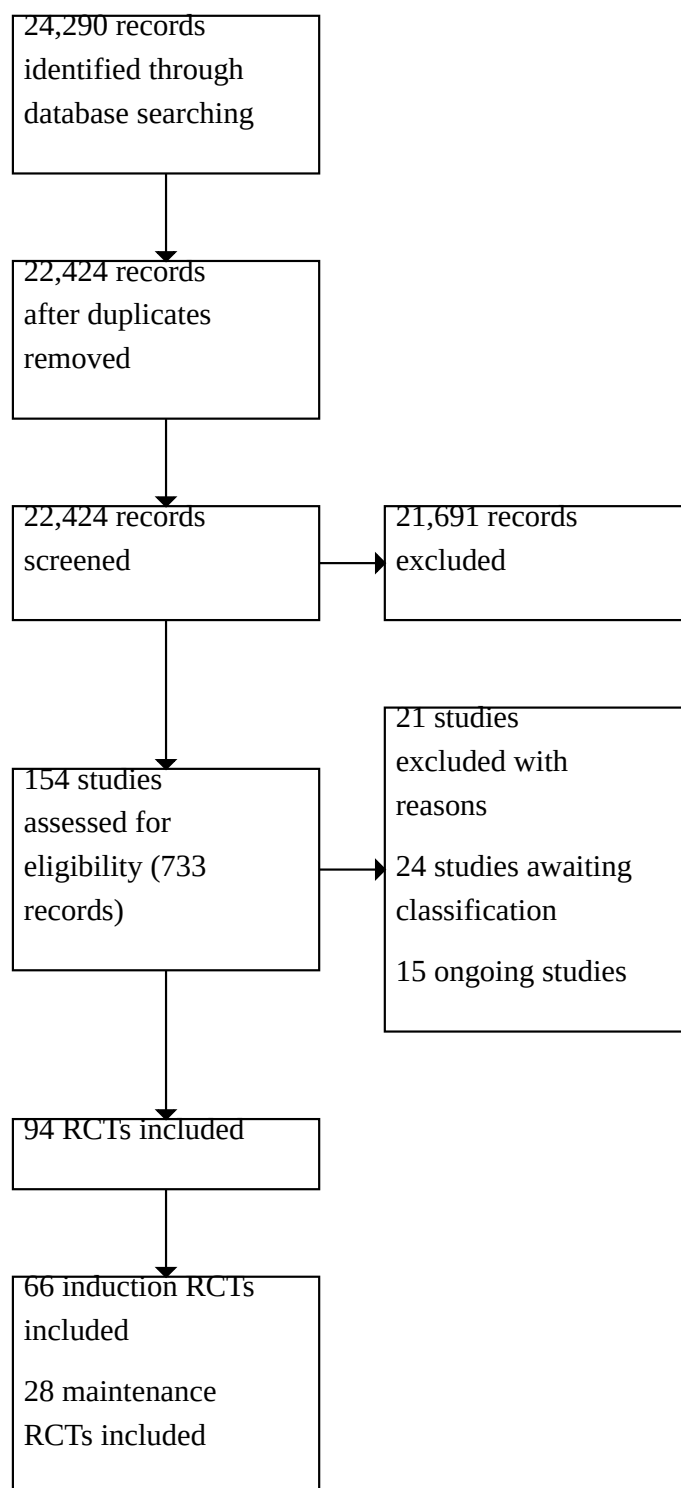
NCT04655807 [569]; NCT05057273 [570]; Rogler 2014 [571]; Wu 2016 [572, 573]).

There are 15 records of ongoing studies ([Supplementary material 5](#)) (ACTRN12621001498886 [574]; ACTRN12622001458729 (DISCUS-IBD) [575]; Chaparro 2019 (GETECCU EXIT) [576, 577, 578]; ChiCTR2200060604 [579]; CTRI/2020/12/029571 [580, 581]; EUCTR2017-001226-18 [582]; EUCTR2018-004614-18 [583];

EUCTR2020-004527-16 (InCharge) [584]; EUCTR2021-002869-18 (AIM-CD) [585]; Jonkers 2021 - CoCros [586]; JRCT2031200406 - DUET [587]; KCT0007470 - CHAMELEON [588]; NCT03759288 - INTREPID [589]; NCT05837897 [590]; NCT06227910 (VICTRIVIA) [591, 592]).

The results of the search are presented in the study flow diagram ([Figure 1](#)).

Figure 1. Study flow diagram



Included studies

A summary of key characteristics and intervention details for the included induction studies is shown in [Table 1](#) and [Table 2](#), and for the maintenance studies in [Table 3](#) and [Table 4](#). The included studies characteristics tables are in [Supplementary material 2](#).

The approved dose and route of administration of the treatments that have been approved so far is shown in [Supplementary material 8](#).

Study design

All studies except Duan 2013 and VanAssche 2012 (SWITCH) were multi-centre RCTs.

Forty-one RCTs were published as papers describing studies where more than one RCT was carried out, such as studies with randomised induction phases followed by randomised maintenance phases of responders, and randomised follow-up phases of non-responders (which we classified under induction) (Allez 2023 (TRIDENT- Part 1); Allez 2023 (TRIDENT- Part 2); Colombel 2007 (CHARM) - Maintenance (responders); D'Haens 2022 (SERENE - Maintenance); Feagan 2015a (UNITI-1); Feagan 2015b (UNITI-2); Feagan 2015c (IM-UNITI); Hanauer 2002 (ACCENT I) - Induction (non-response); Hanauer 2002 (ACCENT I) - Maintenance (Responders); Loftus 2023 (U-ENDURE) [593]; Loftus 2023 (U-EXCEED); Loftus 2023 (U-EXCEL) Panes 2017 Induction; Panes 2017 Maintenance; Sandborn 2005a (ENACT 1); Sandborn 2005a (ENACT 2); Sandborn 2008 (Population 1); Sandborn 2012 (CERTIFI) - Induction; Sandborn 2012 (CERTIFI) - Induction (non-response); Sandborn 2012 (CERTIFI) - Maintenance; Sandborn 2013 (GEMINI II) - Induction [594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646]; Sandborn 2013 (GEMINI II) - Maintenance; Sandborn 2020d (CELEST) - Induction; Sandborn 2020d (CELEST) - Maintenance; Sandborn 2023 - BERGAMOT (cohort 1); Sandborn 2023 - BERGAMOT (cohort 3); Sandborn 2023 - BERGAMOT Maintenance [647] Panaccione 2024 (GALAXI-2); Panaccione 2024 (GALAXI-3); Sands 2004 (ACCENT II) Induction (Non-responders); Sands 2004 (ACCENT II) Maintenance (Responders); Vermeire 2017 FITZROY - induction; Vermeire 2017 FITZROY - induction non-responders; Vermeire 2017 FITZROY - maintenance responders; Vermeire 2025 - DIVERSITY (maintenance); Vermeire 2025 - DIVERSITY A; Vermeire 2025 - DIVERSITY B; Watanabe 2012 - Induction; Watanabe 2012 - Maintenance; Watanabe 2020 - Induction; Watanabe 2020 - Maintenance).

Induction studies

Of the induction studies, 27 were two-armed (Allez 2023 (TRIDENT- Part 1); Chen 2020; Lemann 2006; Loftus 2023 (U-ENDURE); Loftus 2023 (U-EXCEED); Sandborn 2023 - BERGAMOT (cohort 3); Sands 2022 (SERENITY) Allez 2023 (TRIDENT- Part 1); Chen 2025; Loftus 2023 (U-EXCEL); Matsumoto 2016 (DIAMOND); Peyrin-Biroulet 2024 - SEQUENCE; Sandborn 2001b; Sandborn 2005a (ENACT 1); Sandborn 2007a; Sandborn 2007c (PRECISE 1); Sandborn 2008 (Population 1); Sandborn 2012 (CERTIFI) - Induction; Sandborn 2013 (GEMINI II) - Induction; Sands 2004 (ACCENT II) Induction (Non-responders); Sands 2007; Sands 2014 (GEMINI III); Sands 2022 (SEAVUE); Targan 2007 (ENCORE); Vermeire 2017 FITZROY - induction; Watanabe 2020 - Induction; Ye 2019); 20 were three-

armed (Colombel 2007 (CHARM) - Induction (non-responders); Colombel 2010 (SONIC); D'Haens 2022 (ADVANCE); Duan 2013; Feagan 2008c; Feagan 2015a (UNITI-1); Feagan 2015b (UNITI-2); Feagan 2017; Ferrante 2024 (VIVID-1); Hanauer 2002 (ACCENT I) - Induction (non-response); Hart 2025 (GRAVITI); Jairath 2025 (RELIEVE UCCD); Panaccione 2022 (MOTIVATE); Present 1999; Sandborn 2023 - BERGAMOT (cohort 1); Sands 2010; Vermeire 2017 FITZROY - induction non-responders; Vermeire 2025 - DIVERSITY A; Vermeire 2025 - DIVERSITY B; Watanabe 2012 - Induction); 12 were four-armed (D'Haens 1999 (refractory CD); Ghosh 2003; Hanauer 2006 (CLASSIC I); Panaccione 2024 (GALAXI-2); Panaccione 2024 (GALAXI-3); Panes 2017 Induction; Sandborn 2008 (Population 1); Sandborn 2012 (CERTIFI) - Induction; Sandborn 2014; Schreiber 2005; Schwartz 2021 - ENTERPRISE; Targan 1997); three were five-armed (Allez 2023 (TRIDENT- Part 2); Sandborn 2022 - GALAXI 1; Winter 2004); and two were six-armed (Sandborn 2001a; Sandborn 2020d (CELEST) - Induction).

Five studies had more than 50% of their participants receiving concomitant purine analogues during the trial, without considering them part of their interventions. We considered them as part of their interventions for the purposes of our review (Chen 2020; Sandborn 2007a; Sands 2004 (ACCENT II) Induction (Non-responders); Sands 2007; Ye 2019).

Seventeen studies were in participants naive to biologics (Chen 2020; Colombel 2010 (SONIC); D'Haens 1999 (refractory CD); D'Haens 2008; Feagan 2008c; Ghosh 2003; Gordon 2001; Hanauer 2002 (ACCENT I) - Induction (non-response); Hanauer 2006 (CLASSIC I); Lemann 2006; Present 1999; Sandborn 2001a; Sandborn 2011; Sands 2004 (ACCENT II) Induction (Non-responders); Sands 2022 (SEAVUE); Targan 1997; Ye 2019) and 22 studies were in participants not naive to biologics (Allez 2023 (TRIDENT- Part 1); Allez 2023 (TRIDENT- Part 2); Chen 2025; D'Haens 2022 (ADVANCE); Feagan 2015a (UNITI-1); Feagan 2017; Ferrante 2024 (VIVID-1); Hart 2025 (GRAVITI); Jairath 2025 (RELIEVE UCCD); Loftus 2023 (U-EXCEL); Panaccione 2024 (GALAXI-2); Panaccione 2024 (GALAXI-3); Peyrin-Biroulet 2024 - SEQUENCE; Sandborn 2007a; Sandborn 2012 (CERTIFI) - Induction; Sandborn 2012 (CERTIFI) - Induction (non-response); Sandborn 2020d (CELEST) - Induction; Sands 2007; Sands 2014 (GEMINI III); Schreiber 2018a; Vermeire 2025 - DIVERSITY A; Vermeire 2025 - DIVERSITY B). One study was unclear on this (Duan 2013) and the remaining studies were on mixed populations of naive and not naive patients.

One study was exclusively on participants with fistulating disease (Present 1999).

All studies took place in hospital settings. The median age and standard deviation (SD) in the included studies was 37.3 (12.95), but age at diagnosis was not reported. Participants were 47.3% male and 52.7% female. In 41% of participants, disease location was colonic, 38.6% ileocolonic and 20.6% ileal. Median disease duration was 8.4 years, with some studies including disease duration of up to 20 years. Disease phenotype was not well reported. Disease activity was moderate to severe in all studies, with only two studies also including people with mild disease (Gordon 2001; Present 1999). All studies that reported baseline clinical disease activity reported it on the Crohn's Disease Activity Index (CDAI) scale, and some also reported it as Inflammatory Bowel Disease Questionnaire (IBDQ) scores. The median CDAI score was 300 (150 to 350) and for the IBDQ was 124 (100 to 160). Endoscopic baseline activity was only reported in about 40% of studies. Simple Endoscopic

Score for Crohn's Disease (SES-CD) scores ranged between 9 and 16 and Crohn's Disease Endoscopic Index of Severity (CDEIS) scores between 8 and 15. Only one study required systemic steroids to be excluded (Feagan 2008c). In the rest of the studies that reported steroid use, about 40% of participants were receiving steroids at baseline and were allowed to keep using them at stable or tapering dosages, but reporting varied.

Details of interventions, length of treatment and outcome definitions can be found in [Synthesis of results](#) and details per study in [Table 1](#) and [Table 2](#).

Based on the available data and distributions, effect modifiers were similarly distributed or assessed between the included studies to support the assumption of transitivity.

Maintenance studies

Twenty maintenance studies were two-armed (Buhl 2022 (STOP-IT); Colombel 2007 (CHARM) - Maintenance (responders); Feagan 2015c (IM-UNITI); Hanauer 2002 (ACCENT I) - Maintenance (Responders); Jorgensen 2017 (NORSWITCH); Panes 2017 Maintenance; Rutgeerts 1999; Rutgeerts 2012 (EXTEND); Sandborn 2005a (ENACT 2); Sandborn 2007b (CLASSIC II); Sandborn 2012 (CERTIFI) - Maintenance; Sandborn 2013 (GEMINI II) - Maintenance; Sands 2004 (ACCENT II) Maintenance (Responders); Schreiber 2007 (PRECISE 2); VanAssche 2012 (SWITCH); Vermeire 2021 (VISIBLE 2); Volkers 2017 - SIMILAR; Watanabe 2012 - Maintenance; Watanabe 2020 - Maintenance; Young 2024 - iBaSS); two studies were three-armed (Ferrante 2022 - FORTIFY; Louis 2023 - SPARE; Vermeire 2025 - DIVERSITY (maintenance)) and one was four-armed (Vermeire 2017 FITZROY - maintenance responders).

One study had more than 50% of their participants receiving concomitant purine analogues during the trial, without considering them part of their interventions. We considered them as part of their interventions for the purposes of our review (Sandborn 2013 (GEMINI II) - Maintenance).

Two studies were exclusively in participants on remission at the point of randomisation (Buhl 2022 (STOP-IT); Sandborn 2007b (CLASSIC II)). All other maintenance studies were in mixed populations of people in remission and response to a biologic, or only in people who had responded. Mean or median disease activity at baseline could not be meaningfully established as many studies reported activity before remission or response to treatment was reached (induction phases). Five studies reported endoscopic activity, four on the SES-CD (Buhl 2022 (STOP-IT); Ferrante 2022 - FORTIFY; Rutgeerts 2012 (EXTEND); Sandborn 2023 - BERGAMOT Maintenance, mean range: 0 to 13.2) and one on the CDEIS (Lemann 2006, range: 2.8 to 3.5).

No studies were in participants naive to biologics.

One study had not been published as a full paper by the time this review was conducted (Volkers 2017 - SIMILAR).

One study was exclusively on fistulating disease (Sands 2004 (ACCENT II) Maintenance (Responders)).

All studies took place in hospital settings. The median age ranged between 30 and 42, but age at diagnosis was not reported. Participants were 48% male and 52% female. In 24% of participants, disease location was colonic, 38% ileocolonic and 30% ileal.

Median disease duration was 7.5 years, with some studies including disease duration of up to 15 years. Disease phenotype was not well reported. Two studies required steroids to be excluded (Buhl 2022 (STOP-IT); Louis 2023 - SPARE). In the remaining studies that reported steroid use, about 45% to 55% of participants were receiving steroids at baseline and were allowed to keep using them at stable or tapering dosages, but reporting varied.

Details of interventions, length of treatment and outcome definitions can be found in [Synthesis of results](#) and details per study in [Table 3](#) and [Table 4](#).

Based on the available data and distributions, effect modifiers were similarly distributed or assessed between the included studies to support the assumption of transitivity.

Funding

Three studies were publicly funded (Buhl 2022 (STOP-IT); Jorgensen 2017 (NORSWITCH)) or did not receive financial support (Matsumoto 2016 (DIAMOND)). Two studies had unclear funding (Duan 2013; Volkers 2017 - SIMILAR).

The remaining studies were fully or partly funded by pharmaceutical companies. Twelve studies were funded by AbbVie (Chen 2020; D'Haens 2022 (ADVANCE); D'Haens 2022 (SERENE - Induction); D'Haens 2022 (SERENE - Maintenance); Feagan 2017; Ferrante 2022 - FORTIFY; Loftus 2023 (U-EXCEL); Panaccione 2022 (MOTIVATE); Peyrin-Biroulet 2024 - SEQUENCE; Sandborn 2020d (CELEST) - Induction; Sandborn 2020d (CELEST) - Maintenance; Wu 2016). Nine studies were funded by Abbott (Colombel 2007 (CHARM) - Induction (non-responders); Colombel 2007 (CHARM) - Maintenance (responders); Hanauer 2006 (CLASSIC I); Rutgeerts 2012 (EXTEND); Sandborn 2007a; Sandborn 2007b (CLASSIC II); VanAssche 2012 (SWITCH); Watanabe 2012 - Induction; Watanabe 2012 - Maintenance). Twelve studies were funded by Centocor (Colombel 2010 (SONIC); D'Haens 1999 (refractory CD); D'Haens 2008; Hanauer 2002 (ACCENT I) - Induction (non-response); Hanauer 2002 (ACCENT I) - Maintenance (Responders); Present 1999; Rutgeerts 1999; Sandborn 2008 (Population 1); Sandborn 2008 (Population 2); Sands 2004 (ACCENT II) Induction (Non-responders); Sands 2004 (ACCENT II) Maintenance (Responders); Targan 1997). Thirteen studies were funded by Janssen/Johnson & Johnson (Allez 2023 (TRIDENT- Part 1); Allez 2023 (TRIDENT- Part 2); Feagan 2015a (UNITI-1); Feagan 2015b (UNITI-2); Feagan 2015c (IM-UNITI); Hart 2025 (GRAVITI); Panaccione 2024 (GALAXI-2); Panaccione 2024 (GALAXI-3); Sandborn 2012 (CERTIFI) - Induction; Sandborn 2012 (CERTIFI) - Induction (non-response); Sandborn 2012 (CERTIFI) - Maintenance; Sandborn 2022 - GALAXI 1; Sands 2022 (SEAVUE)). Three studies were funded by Millennium (Feagan 2008c; Sandborn 2013 (GEMINI II) - Induction; Sandborn 2013 (GEMINI II) - Maintenance). Six studies were funded by Takeda (Chen 2025; Sands 2014 (GEMINI III); Schwartz 2021 - ENTERPRISE; Vermeire 2021 (VISIBLE 2); Watanabe 2020 - Induction; Watanabe 2020 - Maintenance). Six studies were funded by Elan (Ghosh 2003; Gordon 2001; Sandborn 2005a (ENACT 1); Sandborn 2005a (ENACT 2); Sandborn 2007a; Targan 2007 (ENCORE)). One study was funded by Boehringer Ingelheim (Hanauer 2021 (VOLTAIRE-CD)). Two studies were funded by the GETAID group (Lemann 2006; Louis 2023 - SPARE). Three studies were funded by Pfizer (Panes 2017 Induction; Panes 2017 Maintenance; Sandborn 2014). Three studies were funded by Celltech (Sandborn 2001a; Schreiber

2005; Winter 2004). Four studies were funded by UCB (Sandborn 2007c (PRECISE 1); Sandborn 2010b (WELCOME); Sandborn 2011; Schreiber 2007 (PRECISE 2)). One study was funded by Immunex (Sandborn 2001b). One study was funded by Synta (Sands 2010). One study was funded by Gilead (Schreiber 2018a), three studies were funded by Galapagos (Vermeire 2017 FITZROY - induction; Vermeire 2017 FITZROY - induction non-responders; Vermeire 2017 FITZROY - maintenance responders), and three by both (Vermeire 2025 - DIVERSITY (maintenance); Vermeire 2025 - DIVERSITY A; Vermeire 2025 - DIVERSITY B). Two studies were funded by Celltrion (Schreiber 2021; Ye 2019). One study was funded by Hoffmann-La Roche (Sandborn 2023 - BERGAMOT (cohort 1)), one study was funded by Eli Lilly (Ferrante 2024 (VIVID-1)), one by Teva/Sanofi (Jairath 2025 (RELIEVE UCCD)) and one by Biogen (Young 2024 - iBaSS).

Conflicts of interest

No studies declared all their authors to be conflict free. All of them either had conflicted authors or did not clearly report conflicts.

We contacted the majority of authors for clarifications on their methods and data when the published information was unclear. Authors were not contacted when we had no questions for them.

Responses varied between some authors not responding at all and others answering our questions, redirecting us to others or not being able to answer.

A data package can be accessed in [Supplementary material 7](#).

Excluded studies

The reasons for exclusion of 21 example excluded studies are presented in the 'Characteristics of excluded studies' table in [Supplementary material 3](#).

The main reasons for exclusion were non-RCT designs and ineligible interventions, such as comparisons of pharmacokinetics, trough levels or treatment strategies, and all included participants receiving the same biologic, therefore not being able to use it as a node in the NMA.

Risk of bias in included studies

The risk of bias in the 94 included studies is summarised and presented graphically in [Figure 2](#) and [Figure 3](#). We have contacted the authors or study sponsors, or both, for clarification on unclear judgements.

Figure 2. Review authors' judgements about each risk of bias item presented as percentages across all included studies.

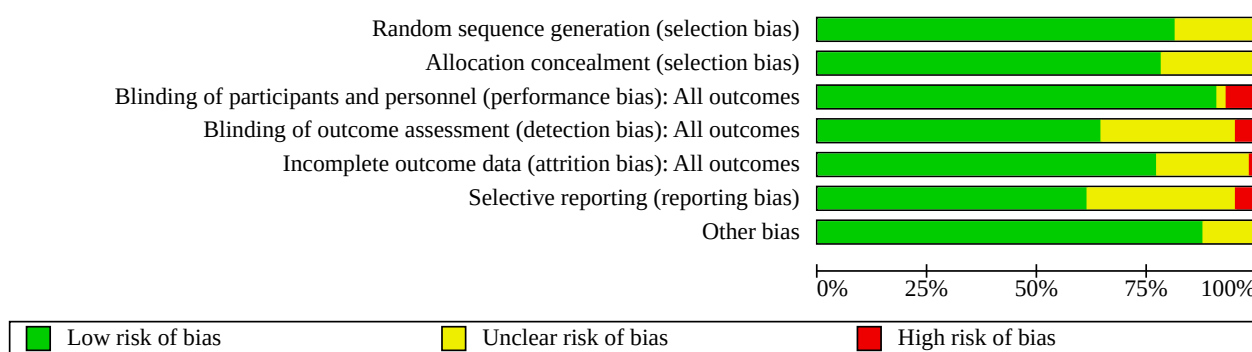


Figure 3. Review authors' judgements about each risk of bias item for each included study.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Allez 2023 (TRIDENT- Part 1)	+	+	+	+	?	+	+
Allez 2023 (TRIDENT- Part 2)	+	+	+	+	+	+	+
Buhl 2022 (STOP-IT)	+	+	+	+	+	?	+
Chen 2020	+	+	+	+	+	+	+
Chen 2025	?	+	?	?	?	+	?
Colombel 2007 (CHARM) - Induction (non-responders)	+	+	+	+	?	-	?
Colombel 2007 (CHARM) - Maintenance (responders)	+	+	+	+	?	+	?
Colombel 2010 (SONIC)	+	+	+	+	+	+	+
Colombel 2023 - LIBERTY-CD	?	?	+	?	?	+	?
D'Haens 1999 (refractory CD)	+	+	+	+	+	?	+
D'Haens 2008	+	+	-	-	+	+	+
D'Haens 2022 (ADVANCE)	+	+	+	+	+	+	+
Duan 2013	?	?	?	?	+	?	+
Feagan 2008c	?	?	+	?	+	+	+
Feagan 2015a (UNITI-1)	+	+	+	?	?	+	+
Feagan 2015b(UNITI-2)	+	+	+	?	?	+	+
Feagan 2015c (IM-UNITI)	+	+	+	?	+	+	+

Figure 3. (Continued)

Feagan 2015c (IM-UNITI)	+	+	+	?	+	+	+
Feagan 2017	+	+	+	+	+	+	+
Ferrante 2022 - FORTIFY	+	+	+	+	?	?	+
Ferrante 2024 (VIVID-1)	+	+	+	+	?	+	+
Ghosh 2003	+	+	+	+	+	?	+
Gordon 2001	?	+	+	+	+	?	+
Hanauer 2002 (ACCENT I) - Induction (non-response)	+	+	+	+	?	?	?
Hanauer 2002 (ACCENT I) - Maintenance (Responders)	+	+	+	+	+	?	?
Hanauer 2006 (CLASSIC I)	+	+	+	+	+	?	+
Hanauer 2021 (VOLTAIRE-CD)	+	+	+	+	+	+	+
Hart 2025 (GRAVITI)	+	+	+	+	?	+	+
Jairath 2025 (RELIEVE UCCD)	?	?	+	?	?	?	+
Jorgensen 2017 (NORSWITCH)	+	+	+	+	+	+	+
Lemann 2006	+	+	+	+	+	?	+
Loftus 2023 (U-ENDURE)	+	+	+	?	+	+	+
Loftus 2023 (U-EXCEED)	+	+	+	?	+	+	+
Loftus 2023 (U-EXCEL)	+	+	+	?	+	+	+
Louis 2023 - SPARE	+	+	-	-	+	+	+
Matsumoto 2016 (DIAMOND)	+	+	-	-	+	+	+
Panaccione 2022 (MOTIVATE)	+	+	+	+	+	+	+
Panaccione 2024 (GALAXI-2)	+	?	+	+	+	?	+
Panaccione 2024 (GALAXI-3)	+	?	+	+	+	?	+
Panes 2017 Induction	+	+	+	+	+	+	+
Panes 2017 Maintenance	+	+	+	+	+	+	+
Peyrin-Biroulet 2024 - SEQUENCE	?	?	-	+	?	?	+
Present 1999	?	+	+	?	+	-	+
Röder 2021	+	?	+	?	+	?	?
Rutgeerts 1999	+	?	+	+	+	?	+
Rutgeerts 2006	+	+	+	?	?	?	+
Rutgeerts 2012 (EXTEND)	?	?	+	+	+	+	+
Sandborn 2001a	+	+	+	+	+	?	+
Sandborn 2001b	+	+	+	?	+	?	+
Sandborn 2005a (ENACT 1)	+	+	+	+	+	?	+
Sandborn 2005a (ENACT 2)	+	+	+	+	?	?	+
Sandborn 2007a	+	+	+	+	+	+	+
Sandborn 2007b (CLASSIC II)	?	+	+	+	+	+	+

Figure 3. (Continued)

Sandborn 2007a	+	+	+	+	+	+	+
Sandborn 2007b (CLASSIC II)	?	+	+	+	+	+	+
Sandborn 2007c (PRECISE 1)	+	?	+	+	+	+	+
Sandborn 2008 (Population 1)	+	+	+	?	+	+	+
Sandborn 2011	+	+	+	?	+	?	+
Sandborn 2012 (CERTIFI) - Induction	+	+	+	+	+	+	+
Sandborn 2012 (CERTIFI) - Induction (non-response)	+	+	+	+	+	?	+
Sandborn 2012 (CERTIFI) - Maintenance	+	+	+	+	+	+	+
Sandborn 2013 (GEMINI II) - Induction	+	+	+	+	+	+	+
Sandborn 2013 (GEMINI II) - Maintenance	+	+	+	+	+	+	+
Sandborn 2014	?	+	+	+	+	+	+
Sandborn 2020d (CELEST) - Induction	+	+	+	+	+	+	+
Sandborn 2022 - GALAXI 1	+	+	+	+	+	+	+
Sandborn 2023 - BERGAMOT (cohort 1)	+	+	+	+	?	+	?
Sandborn 2023 - BERGAMOT (cohort 3)	+	+	+	+	+	+	+
Sandborn 2023 - BERGAMOT Maintenance	+	+	+	?	+	+	+
Sands 2004 (ACCENT II) Induction (Non-responders)	+	+	+	?	+	-	+
Sands 2004 (ACCENT II) Maintenance (Responders)	+	+	+	?	+	-	+
Sands 2007	?	?	+	?	+	?	+
Sands 2010	+	+	+	+	+	+	+
Sands 2014 (GEMINI III)	+	+	+	+	+	+	+
Sands 2022 (SEAVUE)	+	+	+	+	+	+	+
Sands 2022 (SERENITY)	+	+	+	?	+	+	+
Schreiber 2005	+	+	-	+	+	+	+
Schreiber 2007 (PRECISE 2)	+	+	+	+	-	+	+
Schreiber 2018a	?	?	+	+	+	+	+
Targan 1997	+	+	+	+	+	?	+
Targan 2007 (ENCORE)	+	+	+	+	+	?	+
VanAssche 2012 (SWITCH)	+	+	-	-	?	?	+
Vermeire 2017 FITZROY - induction	+	+	+	+	+	+	+
Vermeire 2017 FITZROY - induction non-responders	+	+	+	+	+	?	+
Vermeire 2017 FITZROY - maintenance responders	+	+	+	+	+	?	+
Vermeire 2021 (VISIBLE 2)	+	+	+	?	+	+	+
Vermeire 2025 - DIVERSITY (maintenance)	+	?	+	+	?	+	?
Vermeire 2025 - DIVERSITY A	+	?	+	+	+	+	+
Vermeire 2025 - DIVERSITY B	+	?	+	+	+	+	+

Figure 3. (Continued)

Vermeire 2025 - DIVERSITY B	+	?	+	+	+	+	+
Volkers 2017 - SIMILAR	?	?	+	?	?	?	?
Watanabe 2012 - Induction	?	?	+	?	+	+	+
Watanabe 2012 - Maintenance	?	?	+	?	+	+	+
Watanabe 2020 - Induction	+	+	+	?	+	+	+
Watanabe 2020 - Maintenance	+	+	+	?	+	+	+
Winter 2004	?	?	+	?	+	?	+
Ye 2019	+	+	+	+	+	+	+
Young 2024 - iBaSS	+	+	+	+	?	?	?

We judged 77 studies at low risk of bias for random sequence generation and the remaining 17 studies at unclear risk. We judged 74 studies at low risk of bias for allocation concealment and the remaining 20 studies at unclear risk.

We judged six studies at high risk of bias for blinding of participants/personnel and 86 studies at low risk. We judged two studies at unclear risk. We judged four studies at high risk of bias for blinding of outcome assessment and 61 studies at low risk. We judged 29 studies at unclear risk.

We judged one study at high risk of attrition bias and 73 at low risk. We judged 20 studies at unclear risk.

We judged four studies at high risk of bias for selective reporting and 58 studies at low risk. We judged 32 studies at unclear risk.

We judged 83 studies at low risk of other bias and 11 studies at unclear risk.

Synthesis of results

All analyses presented in [Supplementary material 6](#) relate to the pairwise evidence, and we used them to grade the evidence, as described in our methods. The following section presents the results as reported in the figures.

The figures present the results as network diagrams, forest plots with direct, indirect and network (combining direct and indirect) effect estimates, and SUCRA rankings, and GORDON plots which

combine numerical results, statistical ranking and GRADE certainty, for all the outcomes with available data.

The figures present the results for all biologics included in the review in comparison to placebo.

The treatments are ordered by GRADE certainty and whether the results showed difference with placebo or not (crossed the line of no effect). Within GRADE certainty treatments are ordered by their SUCRA rankings. However, SUCRA rankings are only mentioned if a treatment showed a difference from placebo with high, moderate or low GRADE certainty.

Results are based on random-effects model analyses.

Biologic drugs for induction of remission in Crohn's disease

The extracted data for the induction studies can be found in [Table 5](#). Outcome measurements ranged from 2 weeks to 26 weeks.

Critical outcome: Clinical remission

The results for this outcome are presented in [Summary of findings 1](#), with placebo as the reference treatment. A GORDON plot of the network estimates ordered by GRADE certainty and SUCRA ranking for the treatments not crossing the line of no effect with high, moderate and low certainty can be seen in [Figure 4](#). A comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all treatments compared to placebo, and SUCRA rankings for this outcome can be seen in [Figure 5](#), [Figure 6](#), [Figure 7](#), [Figure 8](#) and [Figure 9](#). The comparison-adjusted funnel plot did not reveal large imbalances.

Figure 4. Gordon plot for clinical remission for biologics for induction of remission in Crohn's Disease, with placebo as reference treatment.



Figure 4. (Continued)

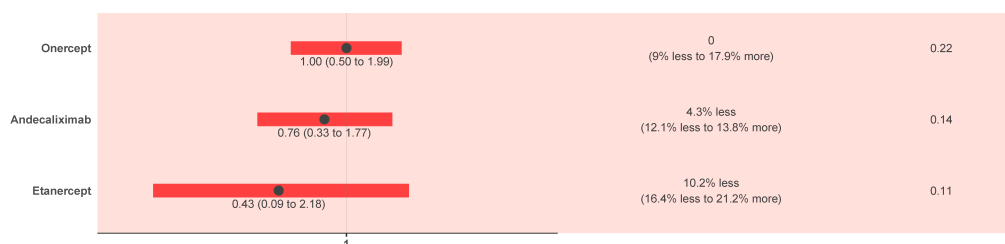


Figure 5. Comparison-adjusted funnel plot for induction of clinical remission

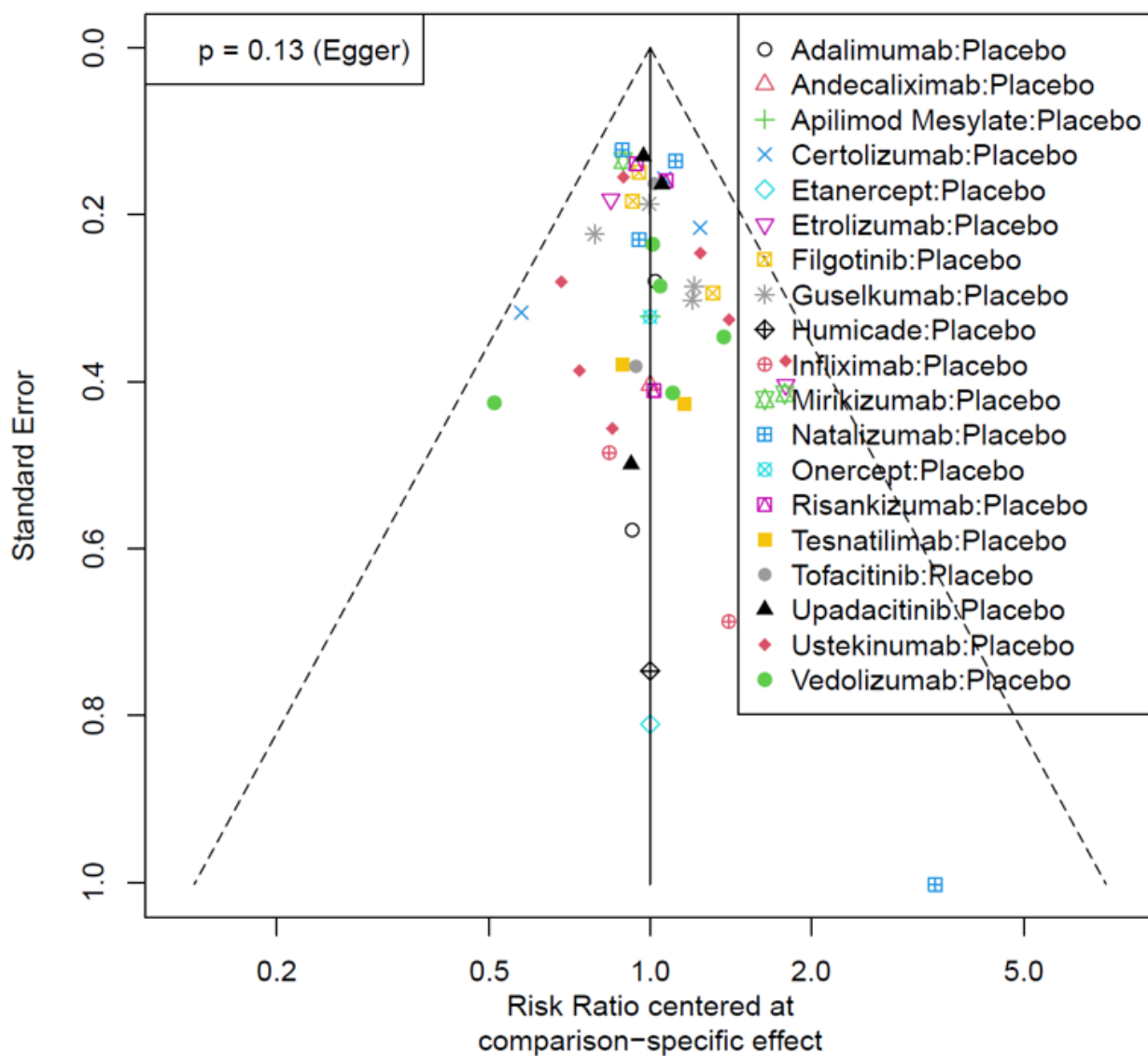


Figure 6. Network plot for induction of clinical remission (shaded regions denote multi-arm studies)

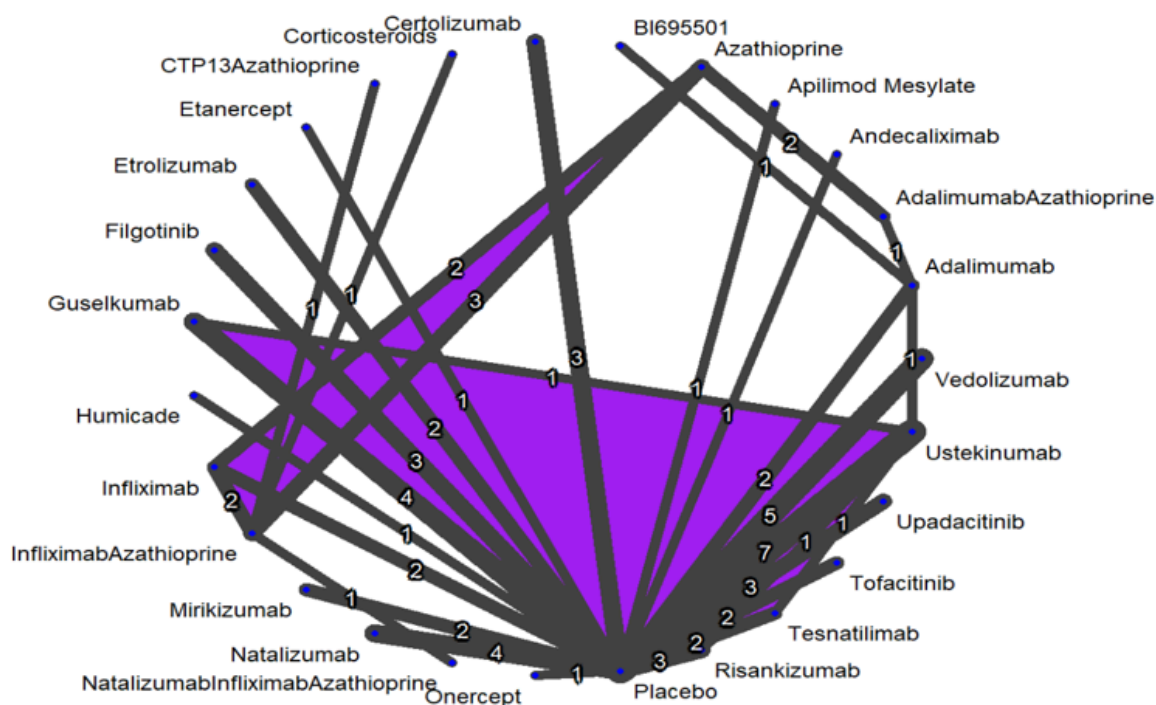


Figure 7. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for induction of clinical remission.

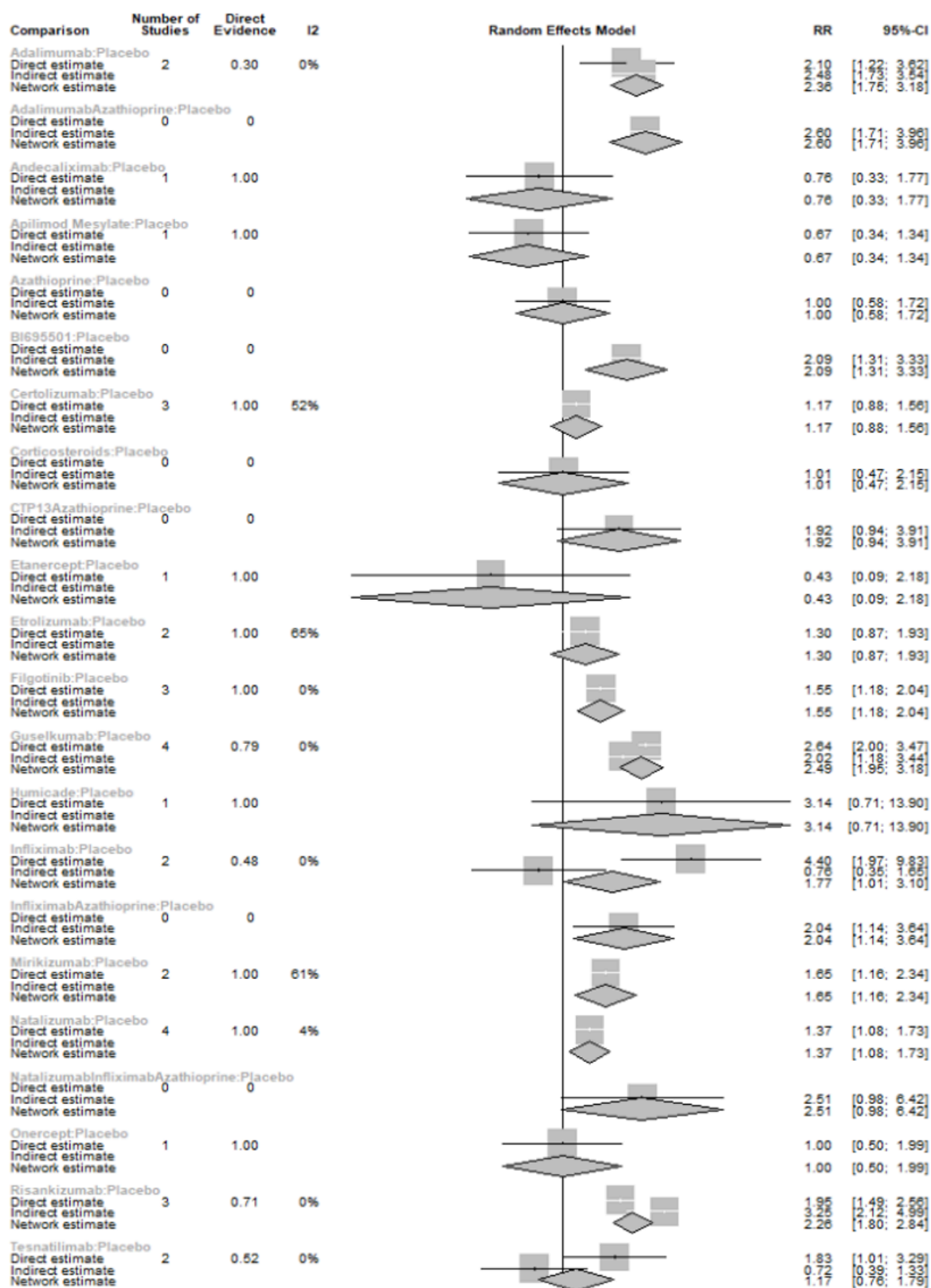


Figure 7. (Continued)

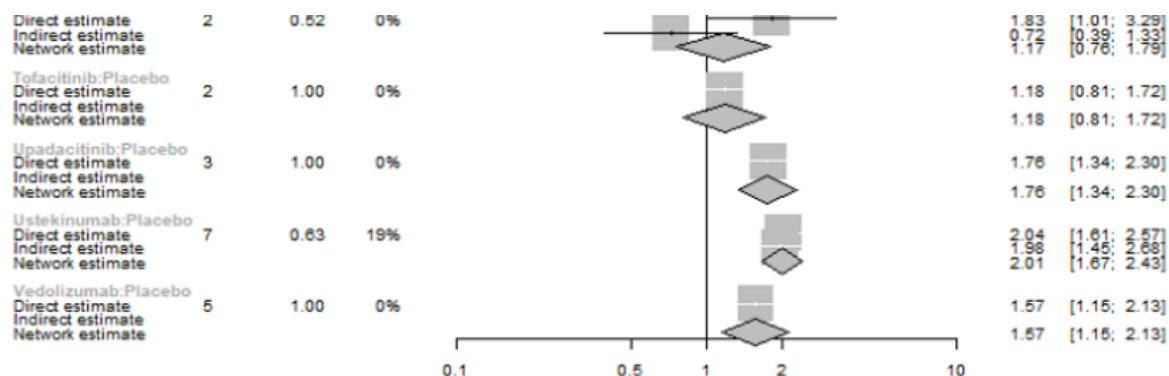


Figure 8. Forest plot for induction of clinical remission

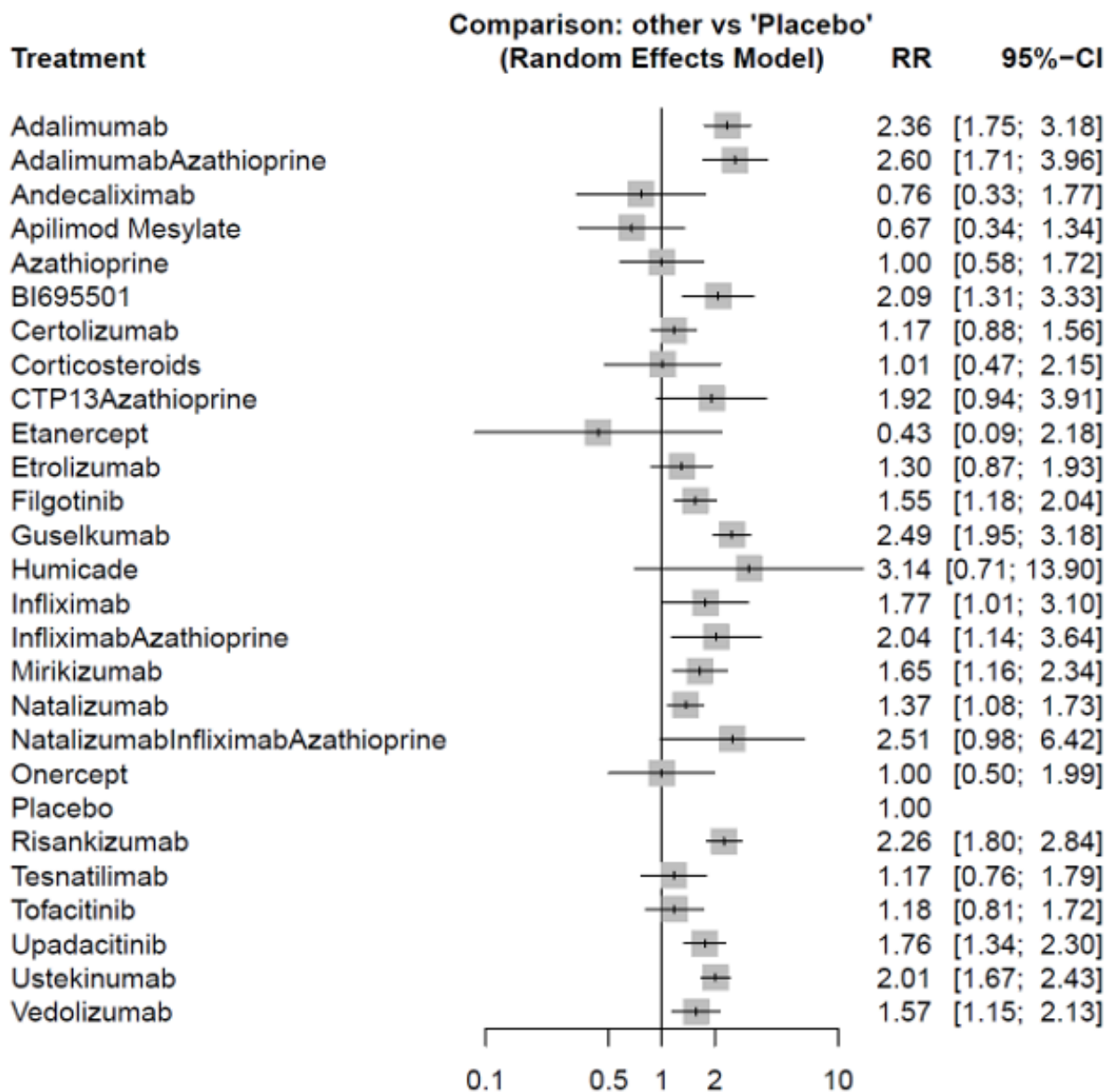


Figure 9. SUCRA rankings for induction of clinical remission

	SUCRA (common)	SUCRA (random)
AdalimumabAzathioprine	0.8323	0.8777
Guselkumab	0.9185	0.8673
Adalimumab	0.8508	0.8481
Humicade	0.8258	0.8415
Risankizumab	0.8858	0.8169
NatalizumabInfliximabAzathioprine	0.7215	0.8000
Ustekinumab	0.7519	0.7385
BI695501	0.7369	0.7250
InfliximabAzathioprine	0.6488	0.7100
CTP13Azathioprine	0.5950	0.6477
Infliximab	0.5000	0.5781
Upadacitinib	0.6785	0.5758
Mirikizumab	0.5827	0.5592
Filgotinib	0.5523	0.5158
Vedolizumab	0.5708	0.5042
Natalizumab	0.4331	0.4188
Etrolizumab	0.3665	0.3762
Tesnatilimab	0.2500	0.2992
Tofacitinib	0.3177	0.2985
Certolizumab	0.3450	0.2938
Corticosteroids	0.1819	0.2469
Onercept	0.2731	0.2362
Azathioprine	0.1492	0.2212
Placebo	0.2077	0.1727
Andecaliximab	0.1204	0.1258
Etanercept	0.0946	0.1150
Apilimod Mesylate	0.1092	0.0900

- based on 100 simulations

All studies included in the analysis defined remission as a CDAI score of more than 150, except Present 1999, which defined 'complete response' as the absence of any draining fistulas at two consecutive visits, and Sandborn 2020d (CELEST) - Induction, which defined it as daily stool frequency of ≤ 1.5 and an abdominal pain score of ≤ 1.0 , with neither worse than the baseline value.

A total of 27 treatments were included in this network from a total of 59 studies ($n = 18,765$ participants), and 67 pairwise comparisons were carried out.

Direct comparisons to placebo included two studies on infliximab ($n = 202$), two on mirikizumab ($n = 758$), one on humicade ($n = 169$), four on guselkumab ($n = 1045$), three on risankizumab ($n = 1551$), two on adalimumab ($n = 388$), seven on ustekinumab ($n = 2454$), three on filgotinib ($n = 1032$), three on upadacitinib ($n = 1241$), five on vedolizumab ($n = 1341$), four on natalizumab ($n = 1692$), two on etrolizumab ($n = 685$), three on certolizumab ($n = 1193$), two on tofacitinib ($n = 419$), two on tesnatilimab ($n = 341$), one on onercept

($n = 207$), one on andecaliximab ($n = 187$), one on etanercept ($n = 43$) and one on apilimod mesylate ($n = 220$).

There is high-certainty evidence that ustekinumab is more effective in inducing clinical remission in Crohn's disease compared to placebo (RR 2.01, 95% CI 1.67 to 2.43, SUCRA = 0.71, NNT = 6, small effect size).

There is moderate-certainty evidence that adalimumab combined with purine analogues (RR 2.6, 95% CI 1.71 to 3.96, SUCRA = 0.87, NNT = 3, moderate effect size), guselkumab (RR 2.49, 95% CI 1.95 to 3.18, SUCRA = 0.86, NNT = 4, moderate effect size), adalimumab (RR 2.36, 95% CI 1.75 to 3.18, SUCRA = 0.83, NNT = 4, moderate effect size), upadacitinib (RR 1.76, 95% CI 1.34 to 2.3, SUCRA = 0.63, NNT = 7, small effect size), vedolizumab (RR 1.57, 95% CI 1.15 to 2.13, SUCRA = 0.51, NNT = 10, trivial effect size) and natalizumab (RR 1.37, 95% CI 1.08 to 1.73, SUCRA = 0.41, NNT = 15, trivial effect size) are probably more effective in inducing clinical remission in Crohn's disease compared to placebo.

There is low-certainty evidence that risankizumab (RR 2.26, 95% CI 1.8 to 2.84, SUCRA = 0.82, moderate effect), BI695501 (RR 2.09, 95% CI 1.31 to 3.33, SUCRA = 0.74, small effect), a combination of infliximab with purine analogues (RR 2.04, 95% CI 1.14 to 3.64, SUCRA = 0.71, small effect) and filgotinib (RR 1.55, 95% CI 1.18 to 2.04, SUCRA = 0.5, trivial effect) may be more effective in inducing clinical remission in Crohn's disease compared to placebo.

There is low-certainty evidence that tofacitinib (RR 1.18, 95% CI 0.81 to 1.72) and apilimod mesylate (RR 0.67, 95% CI 0.34 to 1.34) may be equally effective to placebo in inducing clinical remission in Crohn's disease.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

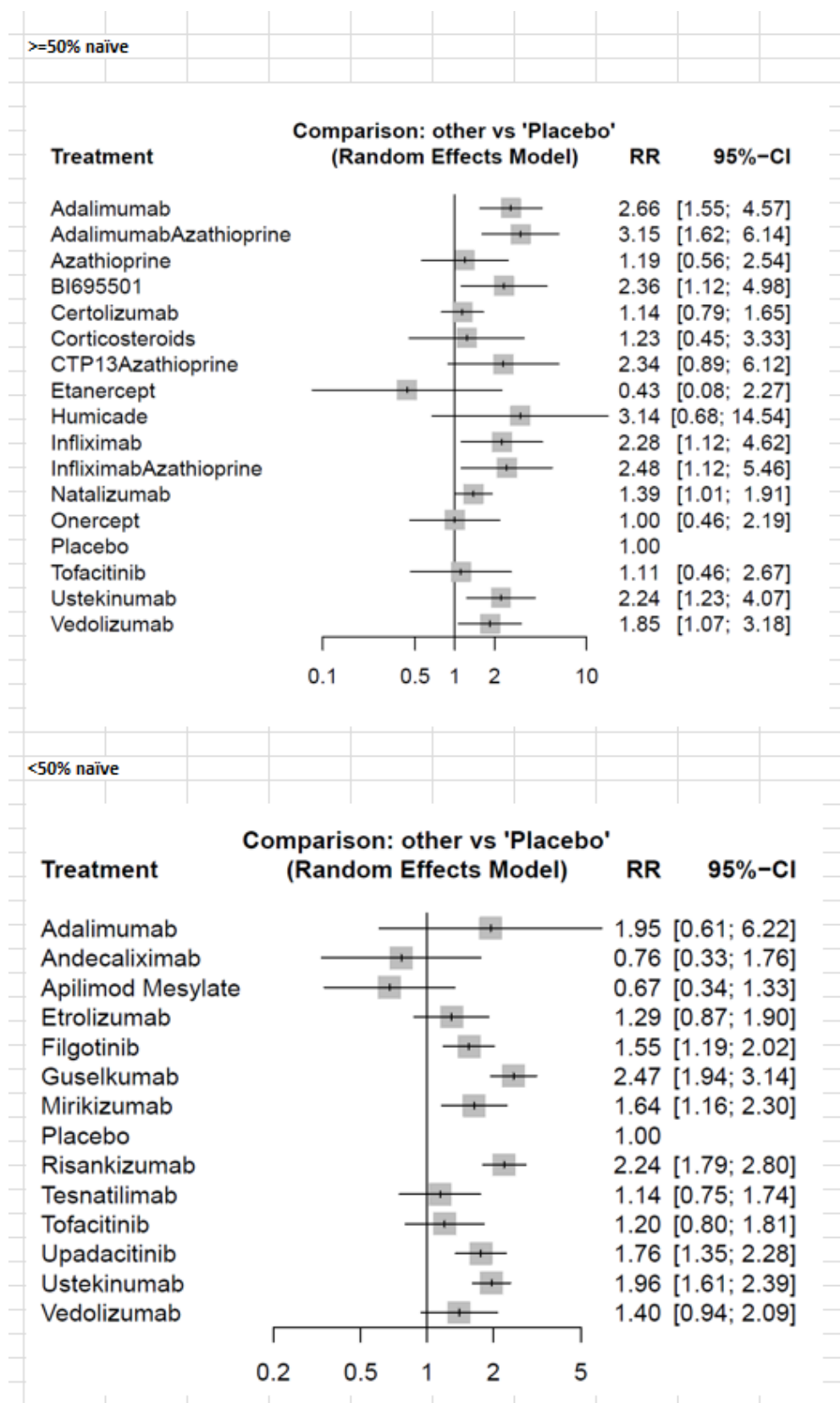
The τ^2 for this outcome was 0.0025 and the $I^2 = 4.3\%$ (total $Q = 50.07$, P value = 0.0742). The χ^2 test results and P values for assessing inconsistency within study designs were not statistically

significant ($Q = 24.28$, P value = 0.71). Between study designs ($Q = 25.78$, P value = 0.0011), the designs between tesnatilimab and ustekinumab ($Q = 19.62$, P value = 0.0032), tesnatilimab and placebo ($Q = 22.74$, P value = 0.0019), adalimumab and ustekinumab ($Q = 22.91$, P value = 0.0018), ustekinumab and placebo ($Q = 25.17$, P value = 0.0007), guselkumab and placebo ($Q = 25.56$, P value = 0.0006), infliximab with purine analogues and placebo ($Q = 25.72$, P value = 0.0006), and adalimumab and placebo ($Q = 25.77$, P value = 0.0006) were statistically significant.

Subgroup analyses

The forest plots for the subgroup analysis for studies with more and less than 50% of participants naive to biologics for induction of clinical remission can be seen in [Figure 10](#). For the treatments that could be compared (adalimumab, tofacitinib, ustekinumab, vedolizumab) no major differences were observed upon visual inspection of the plots.

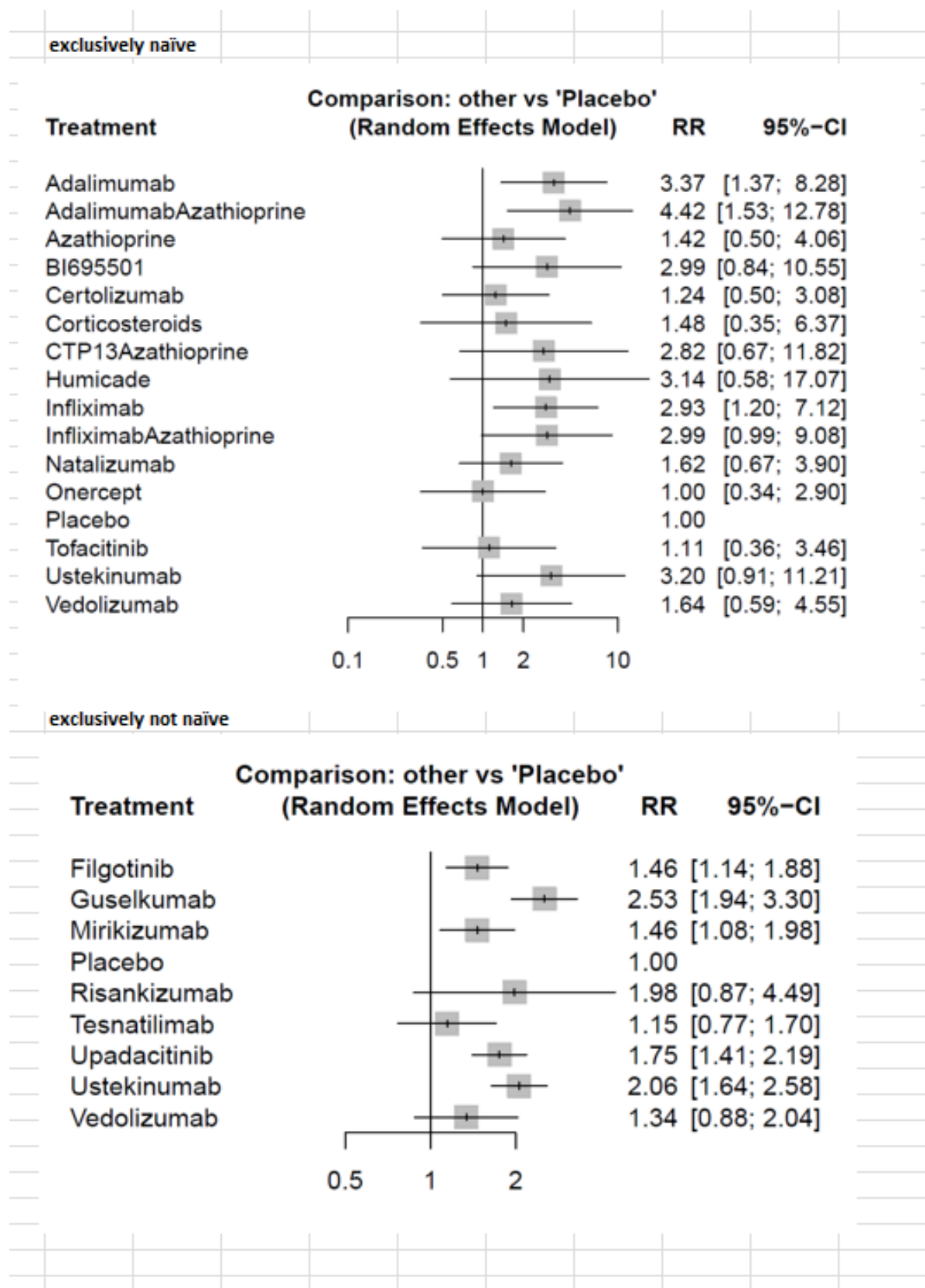
Figure 10. Subgroup analysis for studies with more and less than 50% of participants naive to biologics for induction of clinical remission.



The forest plots for the subgroup analysis for studies with participants exclusively naive or not to biologics for induction of clinical remission can be seen in [Figure 11](#). For the treatments

that could be compared (ustekinumab, vedolizumab), no major differences were observed upon visual inspection of the plots.

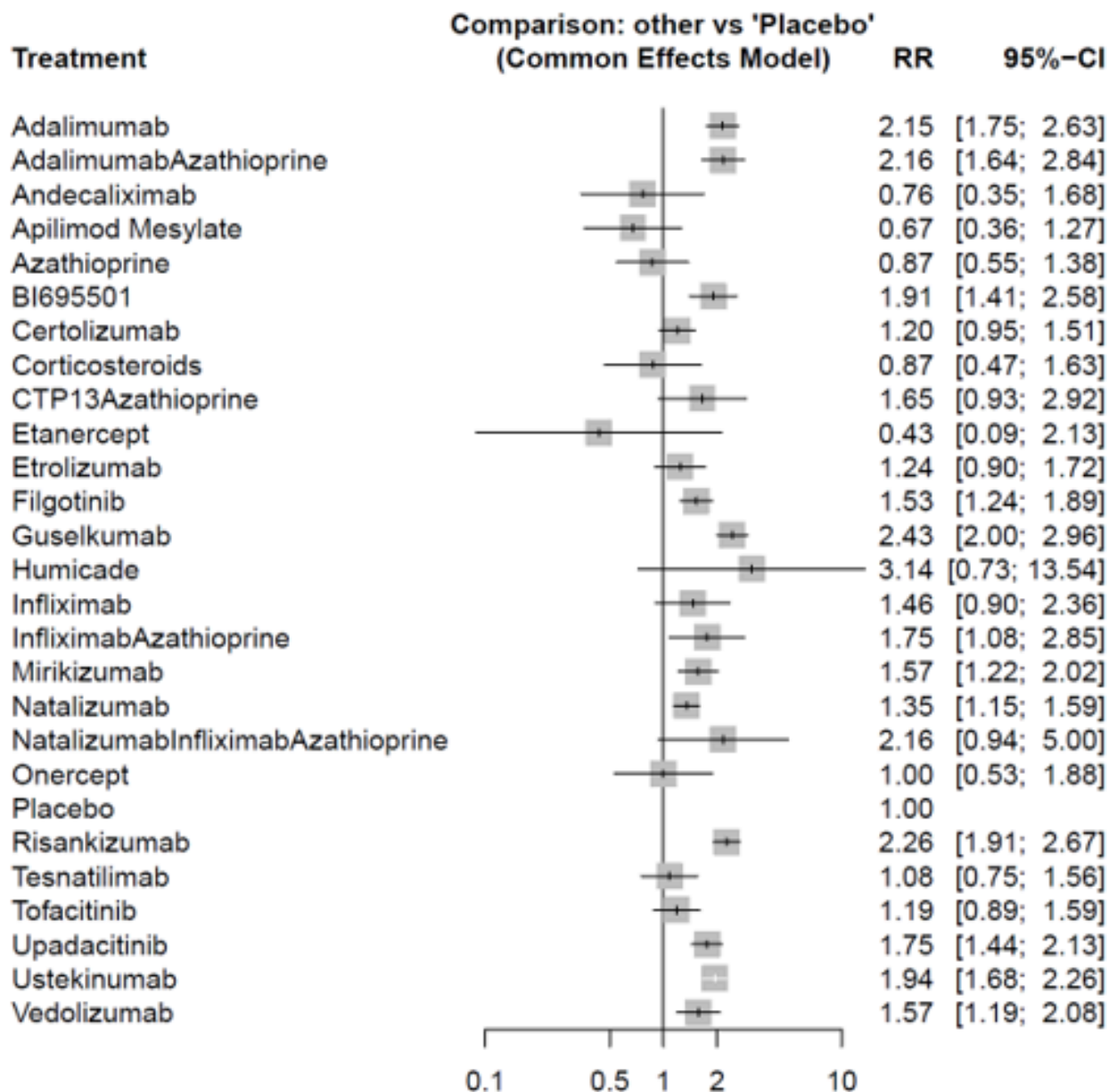
Figure 11. Subgroup analysis for studies with participants exclusively naïve or not naïve to biologics for induction of clinical remission.



Sensitivity analyses

The sensitivity analysis using a fixed-effect model revealed no differences to the random-effects model analysis (Figure 12).

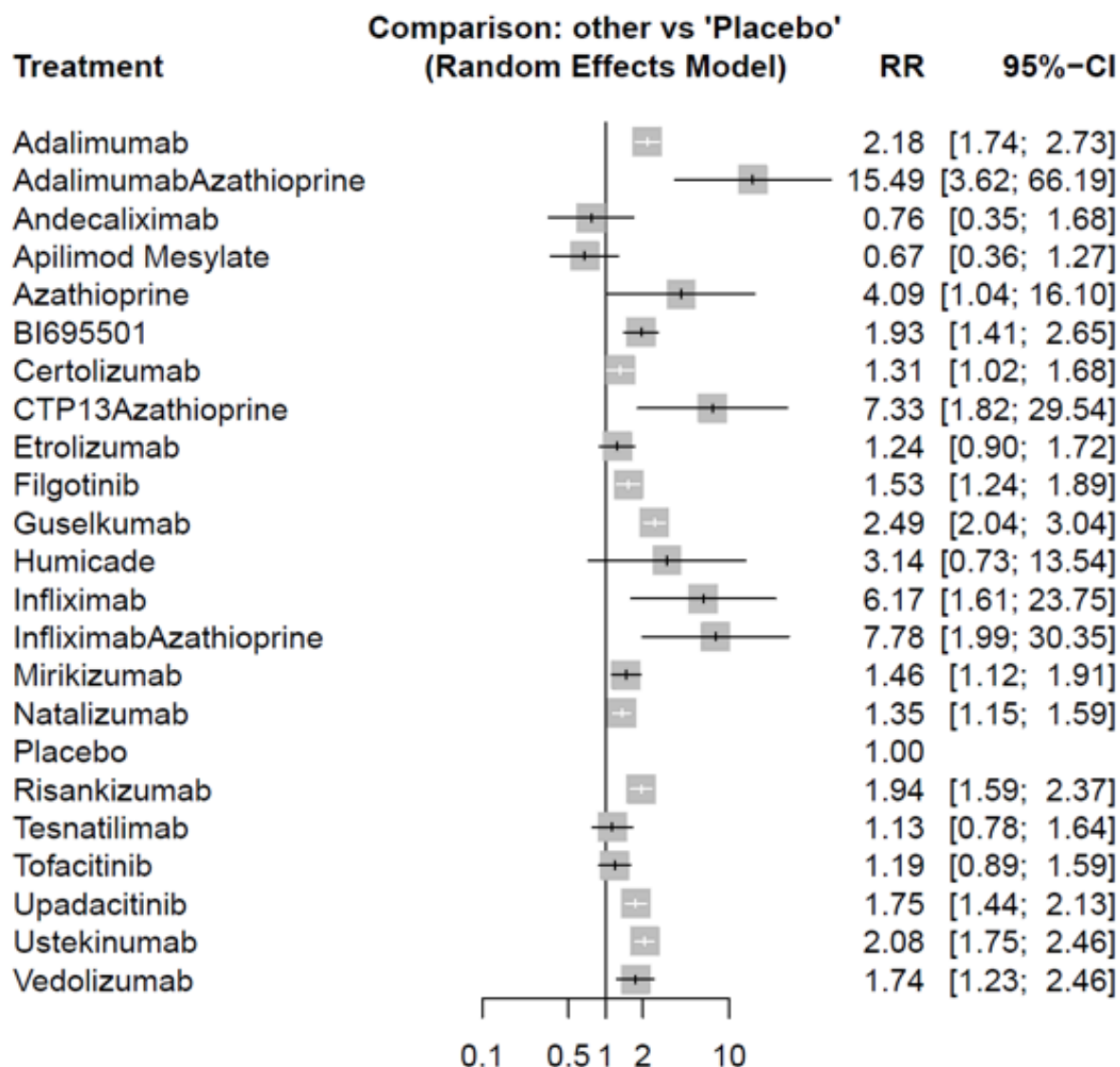
Figure 12. Common (fixed) effect sensitivity analysis for induction of remission



The forest plots for the sensitivity analysis for risk of bias can be seen in Figure 13. Differences were observed for infliximab and the combinations of infliximab, CTP13 and adalimumab with purine

analogues compared to the main analysis; however, due to the higher degree of imprecision in the sensitivity network no clear comments can be made.

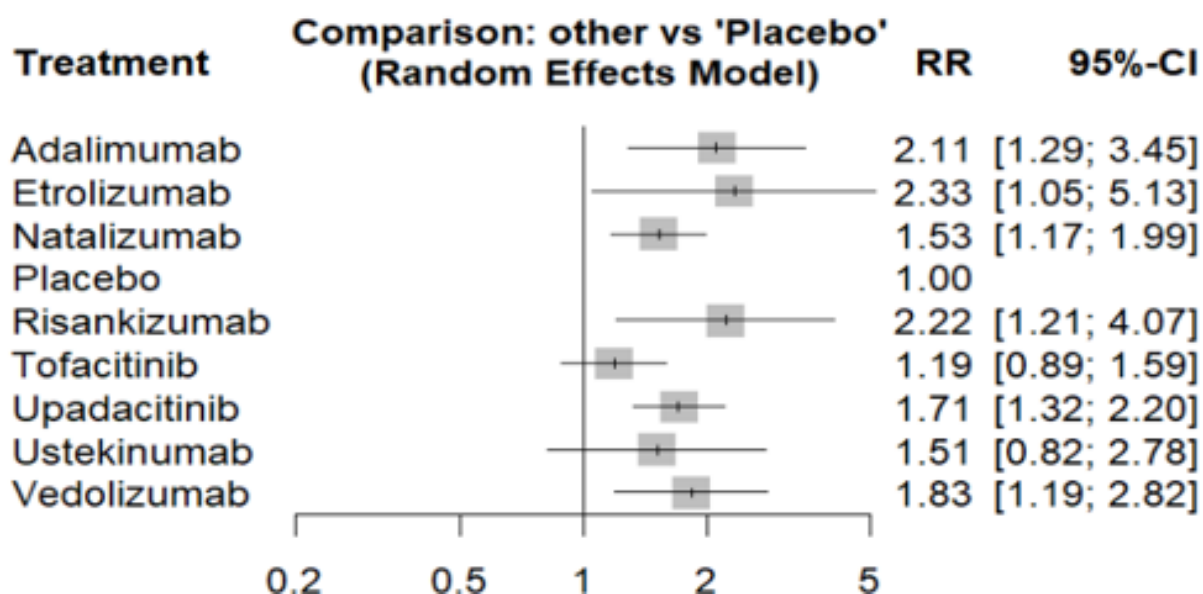
Figure 13. Sensitivity analysis for risk of bias (removal of high risk of bias studies)



The forest plots for the sensitivity analysis for studies with less or more than 20% of participants on concomitant purine analogues

can be seen in [Figure 14](#). No major differences to the main analysis were observed.

Figure 14. Sensitivity analysis for studies with less than 20% concomitant purine analogues for induction of remission (removal of studies where the population is mixed regarding the use of purine analogues)



The sensitivity analysis excluding studies exclusively on severe forms of Crohn's (fistulating disease) and the sensitivity analysis excluding studies not published as full papers could not be performed as no networks could be formed (one fistulating study and two studies not published in full).

treatments compared to placebo, and SUCRA rankings can be seen in [Figure 15](#), [Figure 16](#), [Figure 17](#), [Figure 18](#), [Figure 19](#) and [Figure 20](#). The comparison-adjusted funnel plot did not reveal large imbalances visually for the individual interventions, but the Egger test was statistically significant indicating the presence of publication bias in the network.

Important outcomes

Clinical response

A GORDON plot, comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all

Figure 15. Gordon plot for clinical response for biologics for induction of remission in Crohn's disease, with placebo as reference treatment.

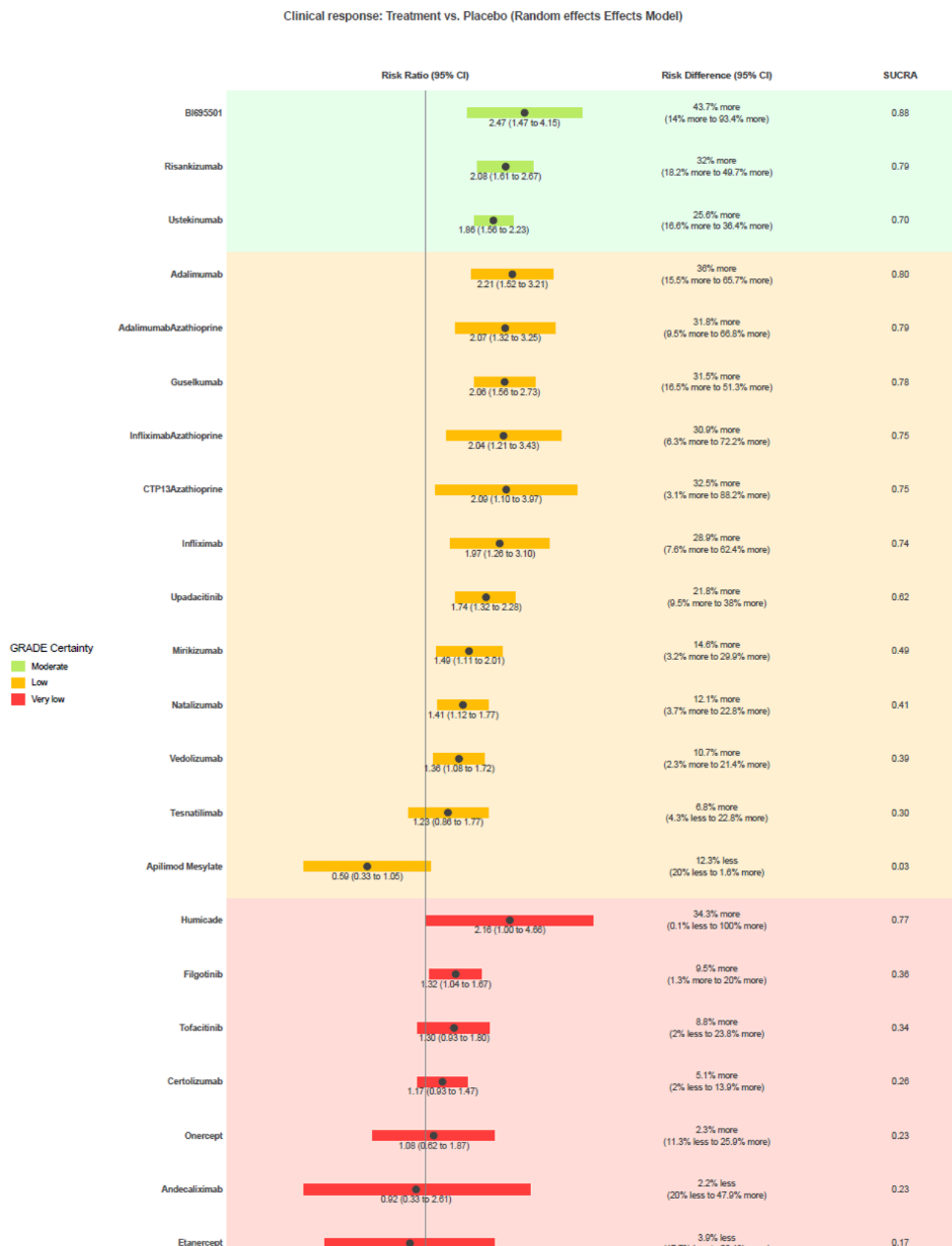


Figure 15. (Continued)



Figure 16. Comparison-adjusted funnel plot for induction of clinical response

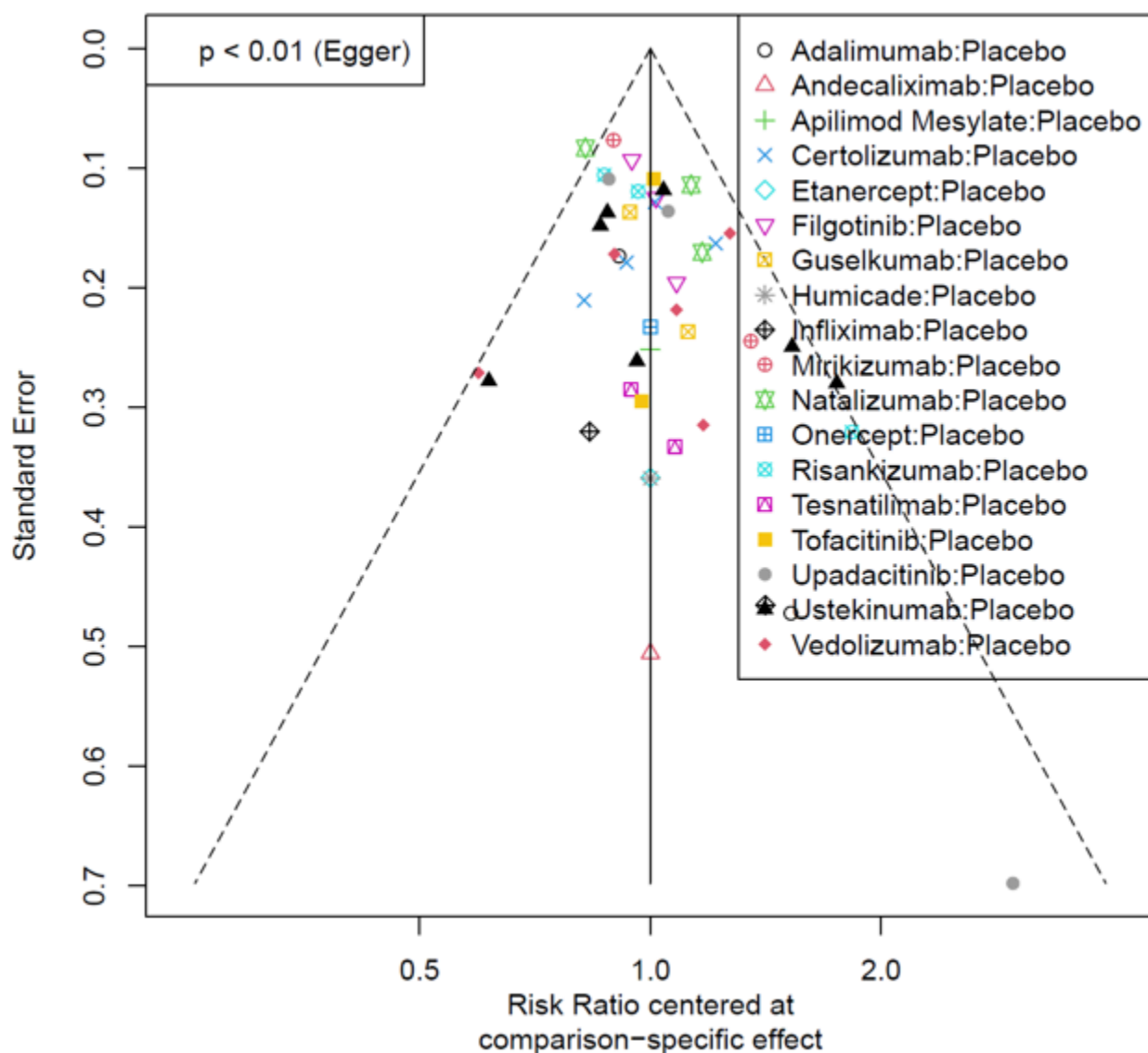


Figure 17. Network plot for induction of clinical response (shaded regions denote multi-arm studies)

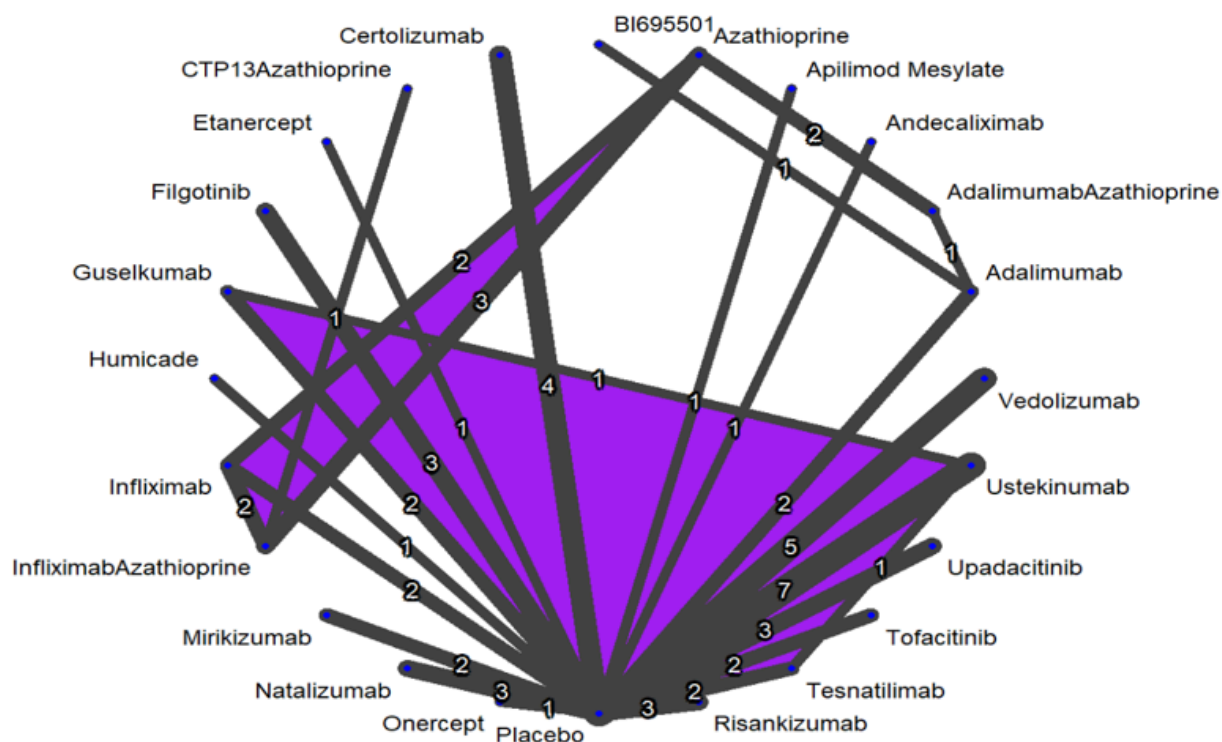


Figure 18. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for induction of clinical response.

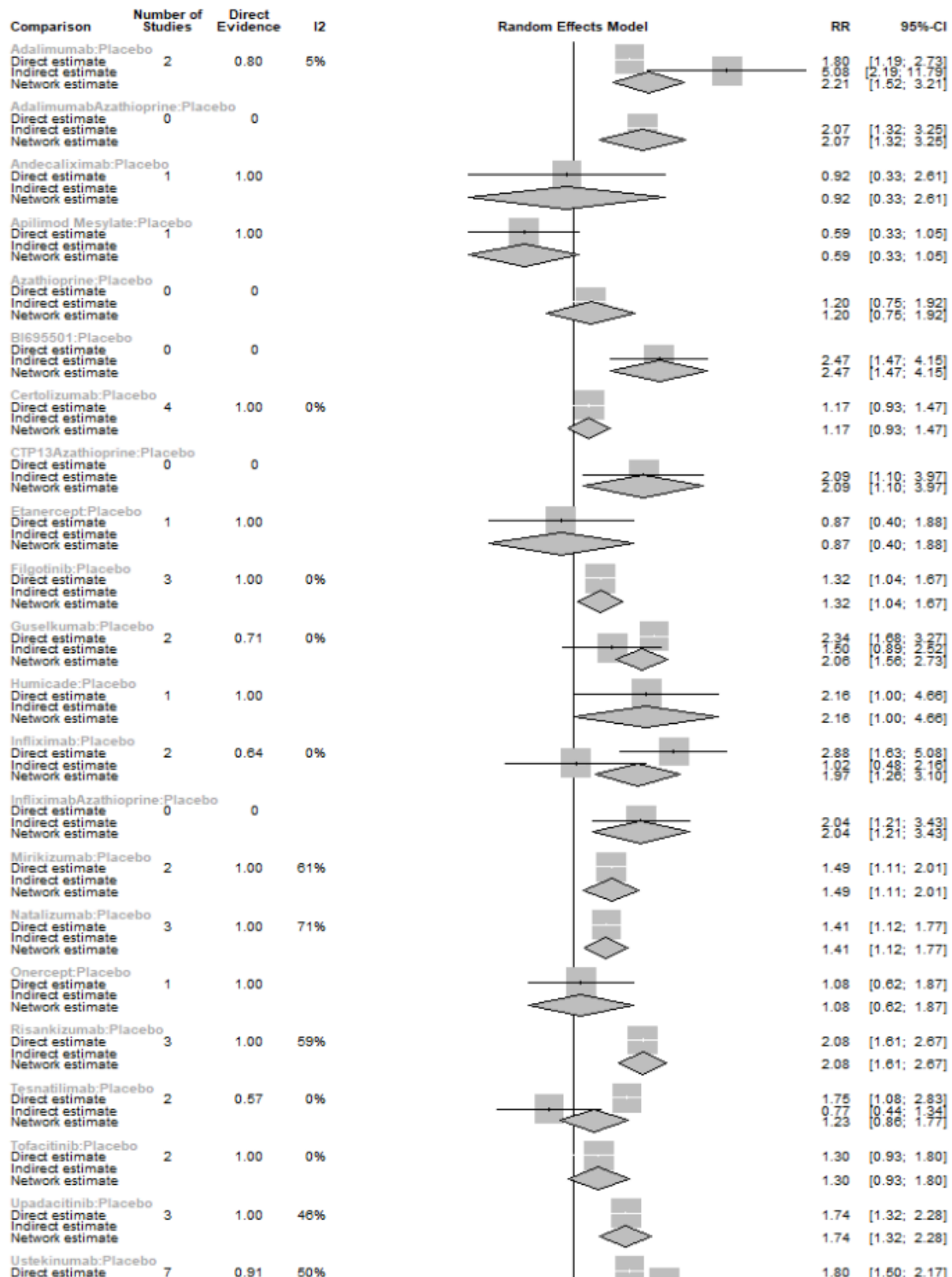


Figure 18. (Continued)

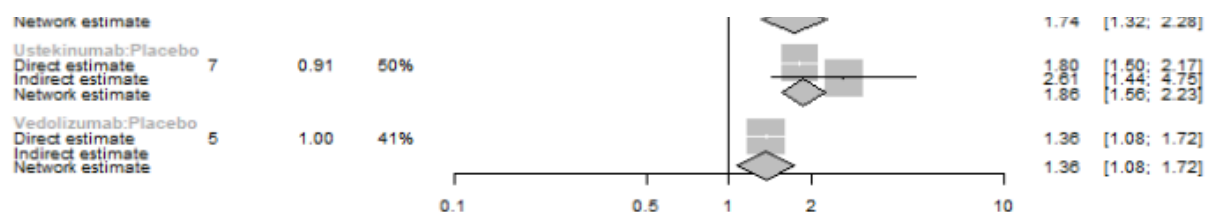


Figure 19. Forest plot for induction of clinical response

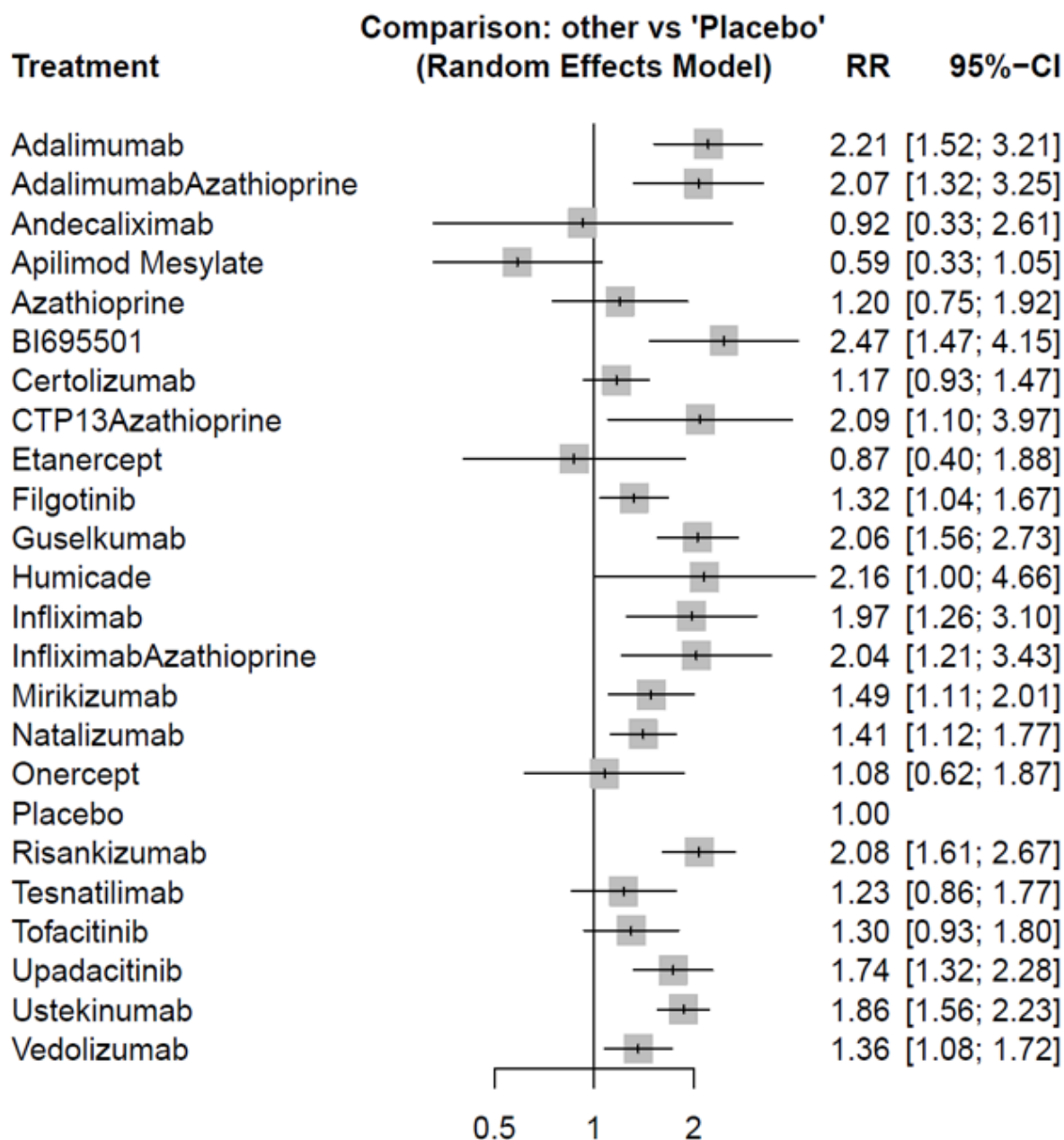


Figure 20. SUCRA rankings for induction of clinical response

	SUCRA (common)	SUCRA (random)
BI695501	0.9248	0.9083
Adalimumab	0.8783	0.8417
Risankizumab	0.7943	0.7783
AdalimumabAzathioprine	0.7391	0.7735
Humicade	0.7991	0.7683
Guselkumab	0.8330	0.7613
InfliximabAzathioprine	0.7291	0.7470
CTP13Azathioprine	0.7391	0.7348
Infliximab	0.6478	0.7257
Ustekinumab	0.7383	0.6970
Upadacitinib	0.6365	0.6274
Mirikizumab	0.4539	0.4965
Natalizumab	0.4130	0.4165
Vedolizumab	0.4417	0.3804
Filgotinib	0.3809	0.3522
Tofacitinib	0.3817	0.3235
Tesnatilimab	0.2191	0.3157
Azathioprine	0.2009	0.2896
Certolizumab	0.2639	0.2787
Onercept	0.2326	0.2235
Andecaliximab	0.2296	0.2204
Etanercept	0.1522	0.1652
Placebo	0.1457	0.1383
Apilimod Mesylate	0.0252	0.0365

- based on 100 simulations

All studies included in the analysis defined response as a reduction of 70 or 100 points on the CDAI scale, except Present 1999, which defined it as a reduction of 50% or more from baseline in the number of draining fistulas observed at two or more consecutive study visits, Sands 2004 (ACCENT II) Induction (Non-responders), which defined loss of response as the recrudescence of draining fistulas, the need for a change in medication for Crohn's disease or the need for additional therapy for persistent or worsening luminal disease activity, the need for a surgical procedure for Crohn's disease, or the discontinuation of the study medication owing to a perceived lack of efficacy, and Schreiber 2018a, which defined it as liquid or very soft stool frequency and abdominal pain composite score ≤ 8 at week 8.

A total of 24 treatments were included in this network from a total of 51 studies ($n = 16,702$ participants), and 59 pairwise comparisons were carried out.

Direct comparisons to placebo included two studies on infliximab ($n = 202$), two on mirikizumab ($n = 758$), one on humicade ($n = 169$), two on guselkumab ($n = 457$), three on risankizumab ($n = 1551$), two on adalimumab ($n = 388$), seven on ustekinumab ($n = 2454$), three on filgotinib ($n = 1074$), three on upadacitinib ($n = 1241$), five on vedolizumab ($n = 1341$), three on natalizumab ($n = 1662$), four on certolizumab ($n = 1485$), two on tofacitinib ($n = 419$), two on tesnatilimab ($n = 341$), one on onercept ($n = 207$), one on andecaliximab ($n = 187$), one on etanercept ($n = 43$) and one on apilimod mesylate ($n = 220$).

We did not judge the evidence for any treatments at high certainty using GRADE for this outcome.

There is moderate-certainty evidence that BI695501 (RR 2.47, 95% CI 1.47 to 4.15, SUCRA = 0.88, large effect size), risankizumab (RR 2.08, 95% CI 1.61 to 2.67, SUCRA = 0.79, moderate effect size) and ustekinumab (RR 1.86, 95% CI 1.56 to 2.23, SUCRA = 0.7, moderate effect size) are probably more effective in inducing clinical response in Crohn's disease compared to placebo.

There is low-certainty evidence that adalimumab (RR 2.21, 95% CI 1.52 to 3.21, SUCRA = 0.8, large effect size), a combination of adalimumab and purine analogues (RR 2.07, 95% CI 1.32 to 3.25, SUCRA = 0.79, moderate effect size), guselkumab (RR 2.06, 95% CI 1.56 to 2.73, SUCRA = 0.78, moderate effect size), a combination of infliximab with purine analogues (RR 2.04, 95% CI 1.21 to 3.43, SUCRA = 0.75, moderate effect size), a combination of CTP13 (infliximab biosimilar) and purine analogues (RR 2.09, 95% CI 1.1 to 3.97, SUCRA = 0.75, moderate effect size), infliximab (RR 1.97, 95% CI 1.26 to 3.1, SUCRA = 0.74, moderate effect size), upadacitinib (RR 1.74, 95% CI 1.32 to 2.28, SUCRA = 0.62, small effect size), mirikizumab (RR 1.49, 95% CI 1.11 to 2.01, SUCRA = 0.49, moderate effect), natalizumab (RR 1.41, 95% CI 1.12 to 1.77, SUCRA = 0.41, trivial effect size) and vedolizumab (RR 1.36, 95% CI 1.08 to 1.72, SUCRA = 0.39, trivial effect size) may be more effective in inducing clinical response in Crohn's disease compared to placebo.

There is low-certainty evidence that tesnatilimab (RR 1.23, 95% CI 0.86 to 1.77) and apilimod mesylate (RR 0.59, 95% CI 0.33 to 1.05) may be equally effective to placebo in inducing clinical response in Crohn's disease.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The τ^2 for this outcome was 0.0255 and the $I^2 = 44.8\%$ (total $Q = 57.97$, P value = 0.0033). The χ^2 test results and P values for assessing inconsistency within study designs were not statistically significant ($Q = 37.24$, P value = 0.0711). Between study designs ($Q = 20.73$, P value = 0.0021), the designs between tesnatilimab and ustekinumab ($Q = 13.6$, P value = 0.0087), tesnatilimab and placebo ($Q = 17.75$, P value = 0.0033), guselkumab and placebo ($Q = 19.73$, P value = 0.0014), infliximab with purine analogues and purine analogues ($Q = 20.35$, P value = 0.0011), and guselkumab, placebo and ustekinumab ($Q = 16.63$, P value = 0.0023) were statistically significant.

Endoscopic remission

A GORDON plot, comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all treatments compared to placebo, and SUCRA rankings can be seen in [Figure 21](#), [Figure 22](#), [Figure 23](#), [Figure 24](#), [Figure 25](#) and [Figure 26](#). The comparison-adjusted funnel plot did not reveal large imbalances.

Figure 21. Gordon plot for endoscopic remission for biologics for induction of remission in Crohn's disease, with placebo as reference treatment.

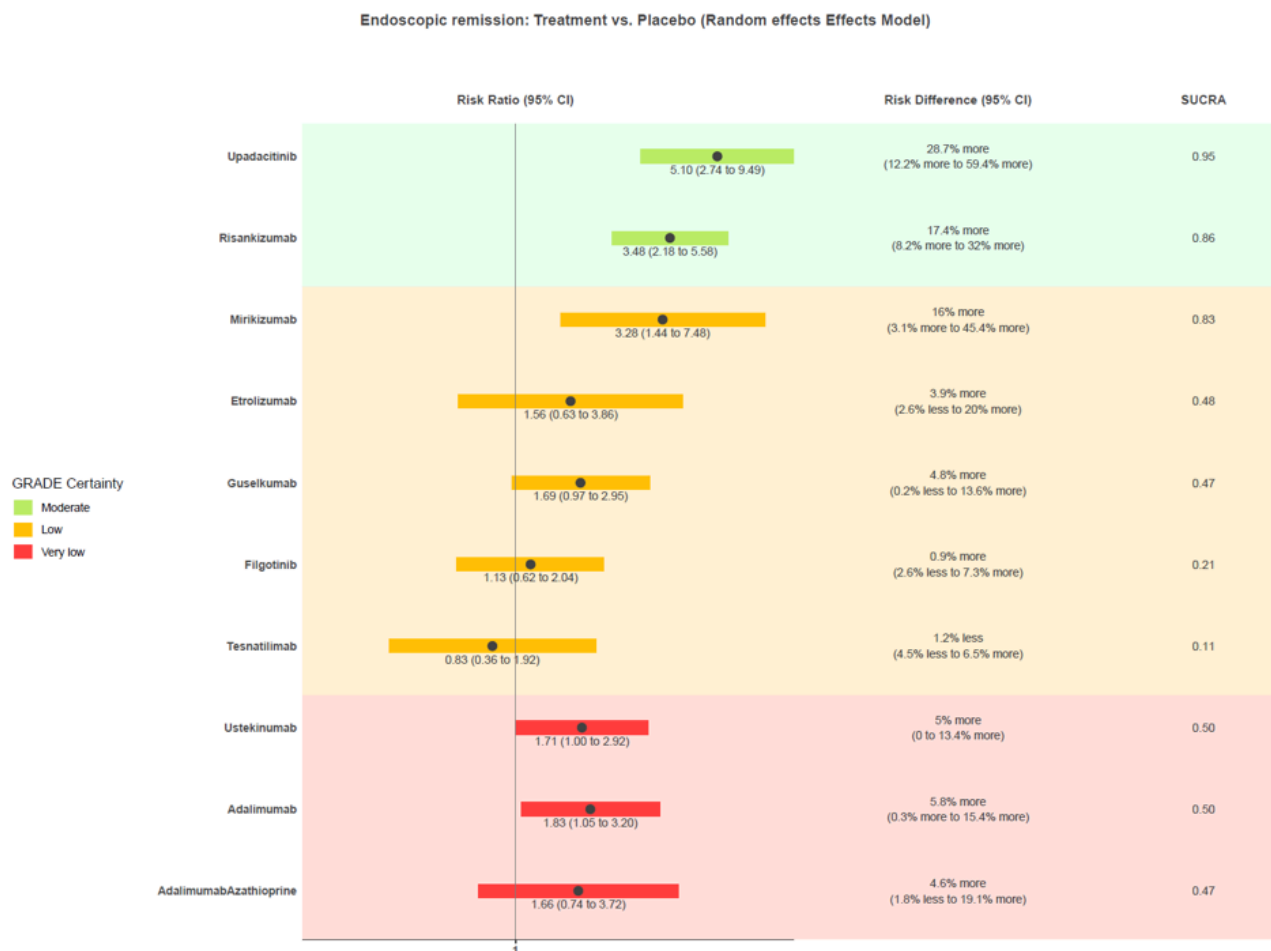


Figure 22. Comparison-adjusted funnel plot for induction of endoscopic remission

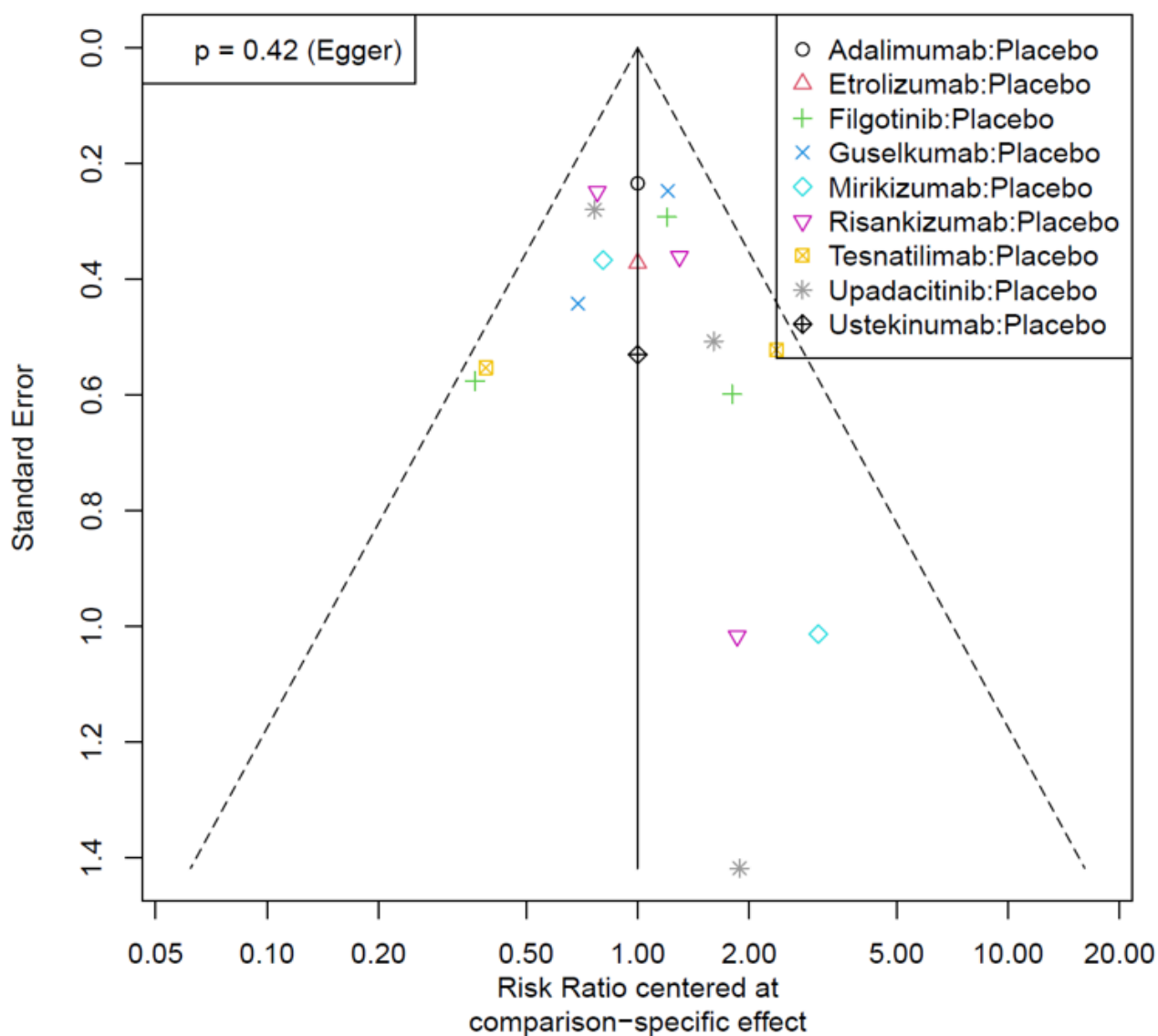


Figure 23. Network plot for induction of endoscopic remission (shaded regions denote multi-arm studies)

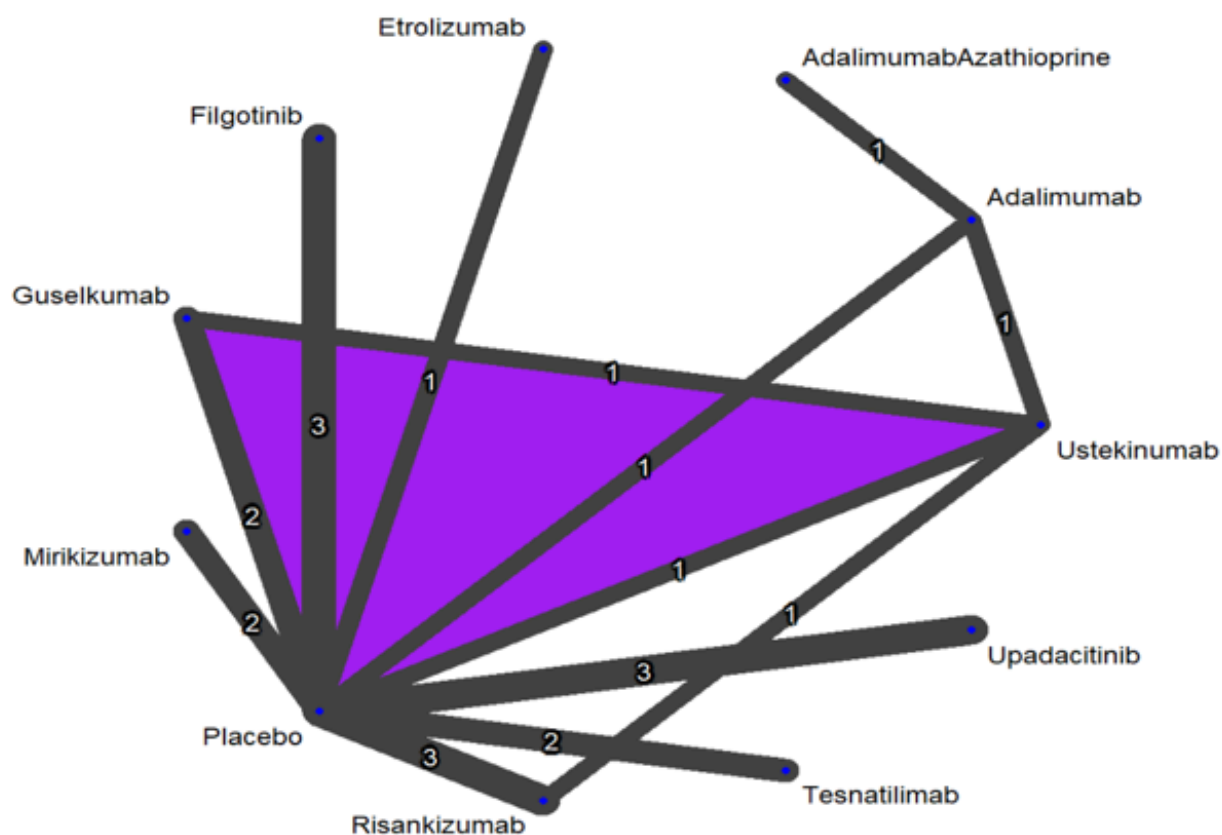


Figure 24. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for induction of endoscopic remission.

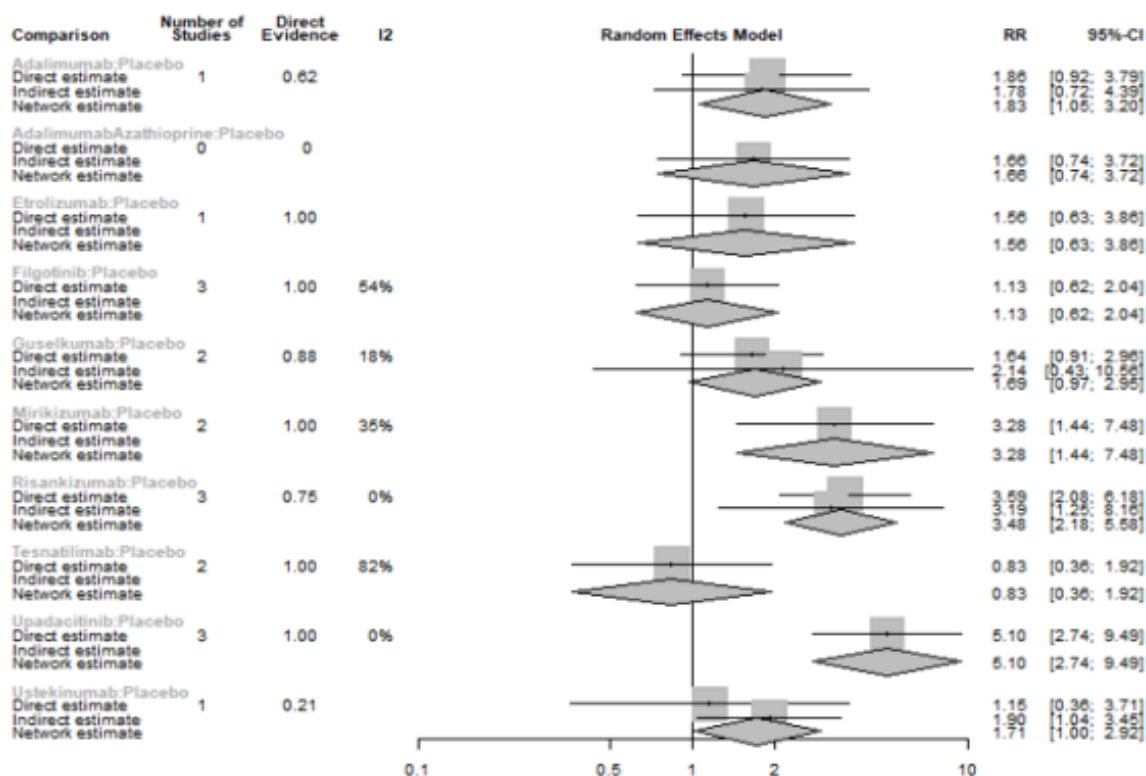


Figure 25. Forest plot for induction of endoscopic remission

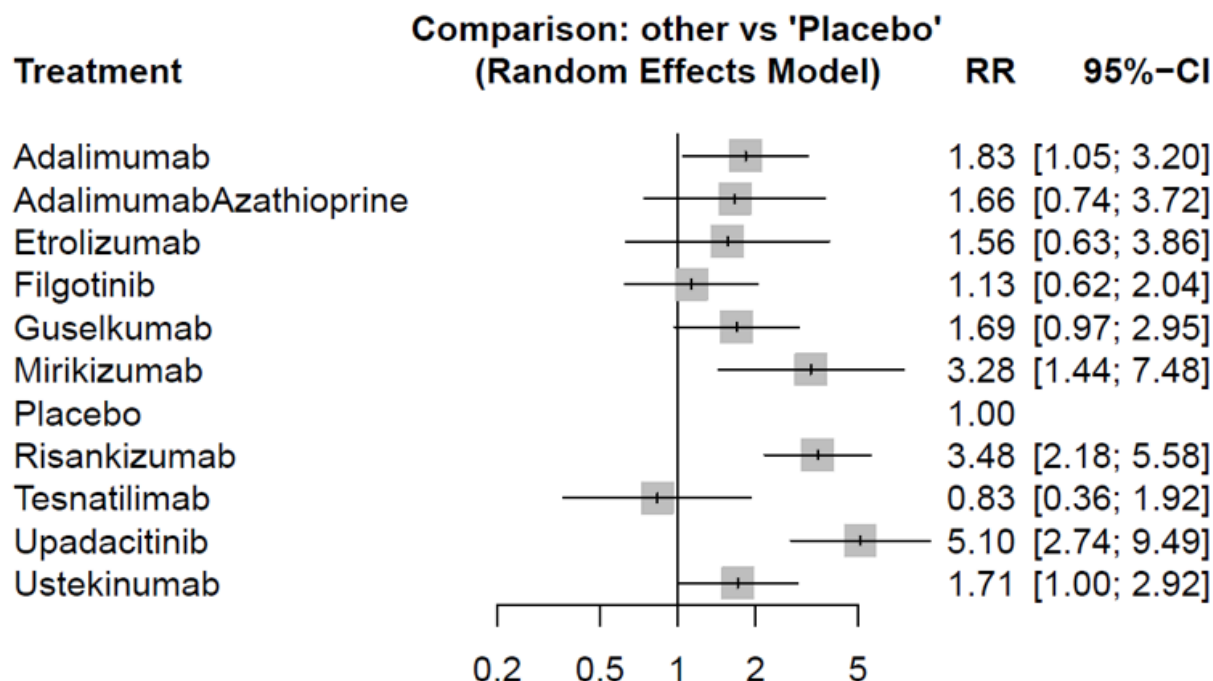


Figure 26. SUCRA rankings for induction of endoscopic remission

	SUCRA (common)	SUCRA (random)
Upadacitinib	0.978	0.948
Risankizumab	0.860	0.853
Mirikizumab	0.817	0.823
Adalimumab	0.571	0.556
Ustekinumab	0.506	0.510
AdalimumabAzathioprine	0.439	0.486
Guselkumab	0.485	0.461
Etrolizumab	0.432	0.413
Filgotinib	0.244	0.211
Placebo	0.104	0.132
Tesnatilimab	0.064	0.107

- based on 100 simulations

A total of 11 treatments were included in this network from a total of 20 studies (n = 7543 participants), and 22 pairwise comparisons were carried out.

Direct comparisons to placebo included two studies on mirikizumab (n = 822), two on guselkumab (n = 459), three on risankizumab (n = 1551), one on adalimumab (n = 129), one on ustekinumab (n = 141), three on filgotinib (n = 601), three on

upadacitinib (n = 1241), two on tesnatilimab (n = 341) and one on etrolizumab (n = 385).

We did not judge the evidence for any treatments at high certainty using GRADE for this outcome.

There is moderate-certainty evidence that upadacitinib (RR 5.1, 95% CI 2.74 to 9.49, SUCRA = 0.95, large effect size) and risankizumab (RR 3.48, 95% CI 2.18 to 5.58, SUCRA = 0.86, NNT = 6, moderate effect size) are probably more effective in inducing endoscopic remission in Crohn's disease compared to placebo.

There is low-certainty evidence that mirikizumab may be more effective in inducing endoscopic remission in Crohn's disease compared to placebo (RR 3.28, 95% CI 1.44 to 7.48, SUCRA = 0.83, small effect size).

There is low-certainty evidence that etrolizumab (RR 1.56, 95% CI 0.63 to 3.86), guselkumab (RR 1.69, 95% CI 0.97 to 2.95), filgotinib (RR 1.13, 95% CI 0.62 to 2.04) and tesnatilimab (RR 0.83, 95% CI 0.36 to 1.92) may be equally effective to placebo in inducing endoscopic remission in Crohn's disease.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The τ^2 for this outcome was 0.0762 and the $I^2 = 33.4\%$ (total $Q = 16.52$, P value = 0.1229). The χ^2 test results and P values for assessing inconsistency within study designs were not statistically significant ($Q = 15.27$, P value = 0.0541), but the design within tesnatilimab and placebo was statistically significant ($Q = 5.64$, P value = 0.0176). Between study designs ($Q = 1.25$, P value = 0.7407) the results were not statistically significant.

Withdrawals due to adverse events

A GORDON plot, comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all treatments compared to placebo, and SUCRA rankings can be seen in [Figure 27](#), [Figure 28](#), [Figure 29](#), [Figure 30](#), [Figure 31](#) and [Figure 32](#). The comparison-adjusted funnel plot did not reveal large imbalances.

Figure 27. Gordon plot for withdrawals due to adverse events for biologics for induction of remission in Crohn's disease, with placebo as reference treatment



Figure 27. (Continued)

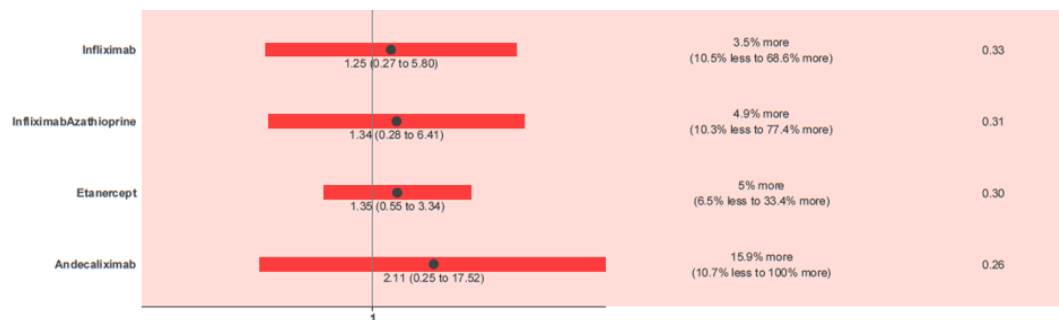


Figure 28. Comparison-adjusted funnel plot for withdrawals due to adverse events during induction.

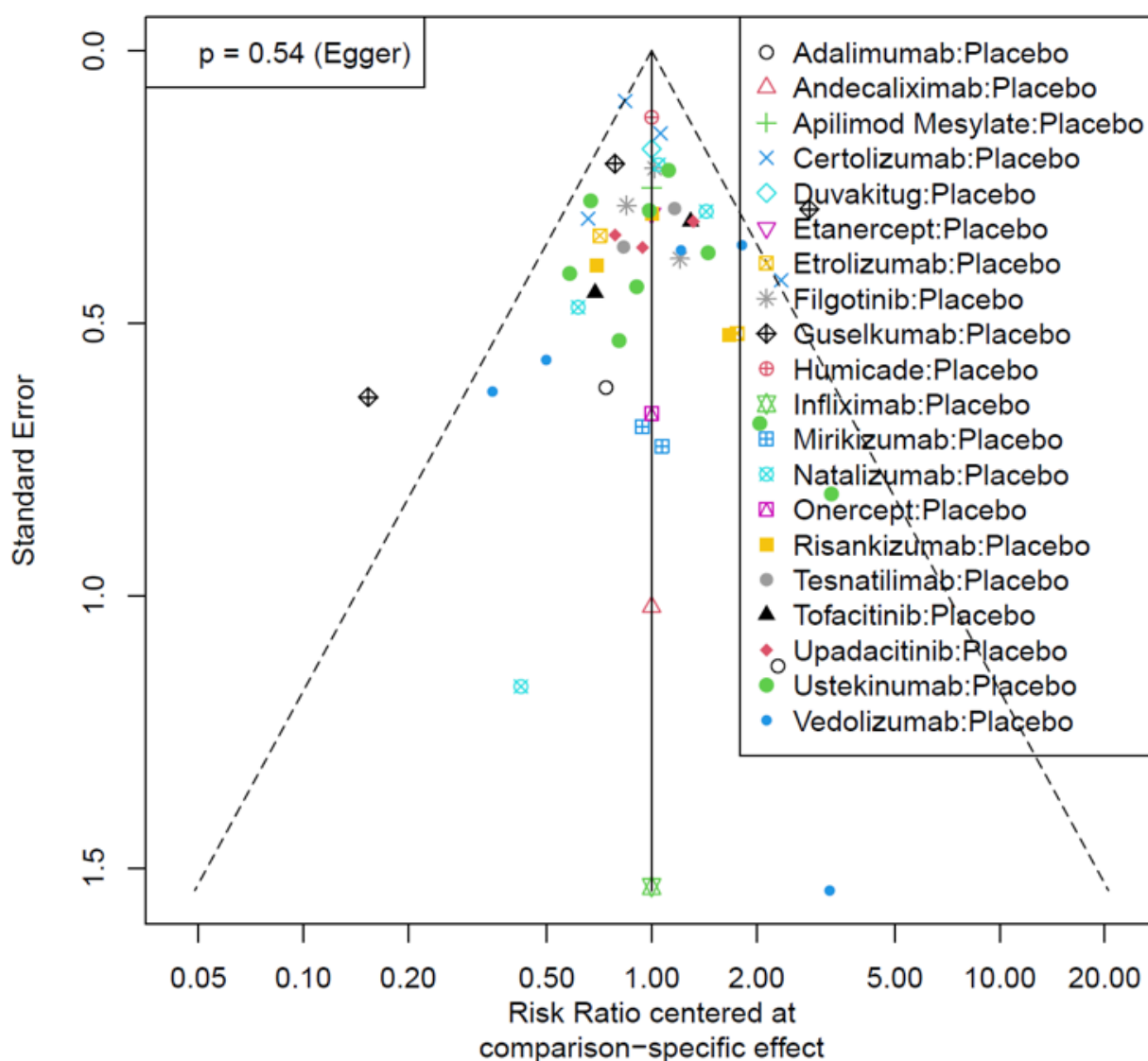


Figure 29. Network plot for withdrawals due to adverse events during induction (shaded regions denote multi-arm studies)

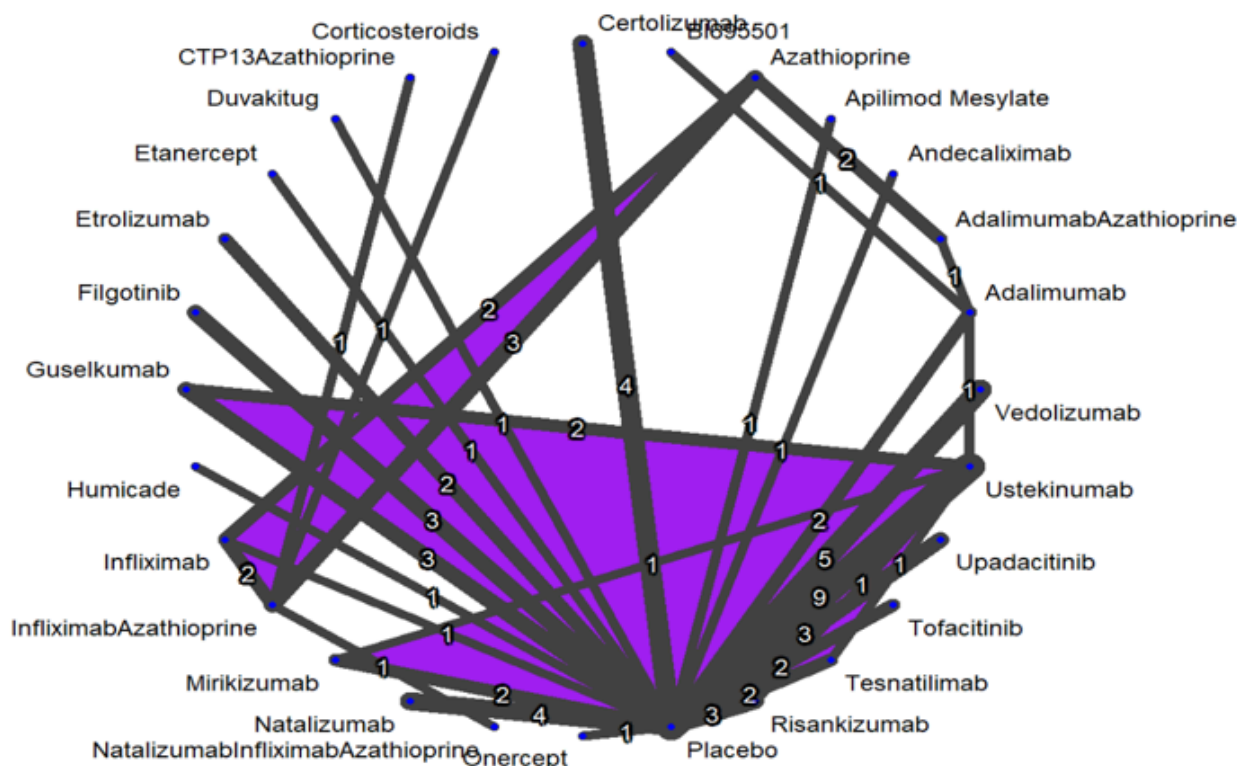


Figure 30. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for withdrawals due to adverse events during induction.

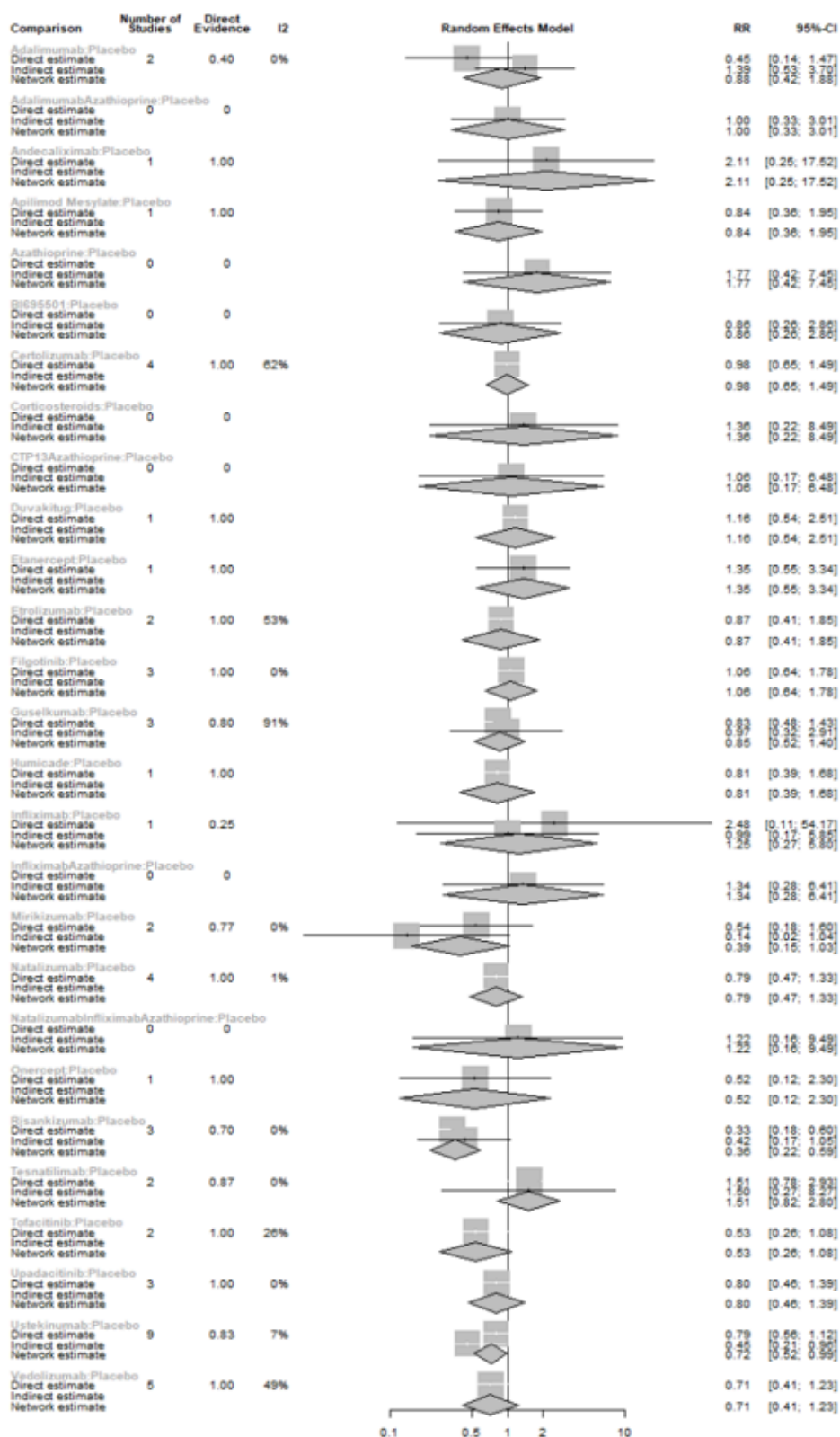


Figure 31. Forest plot for withdrawals due to adverse events during induction

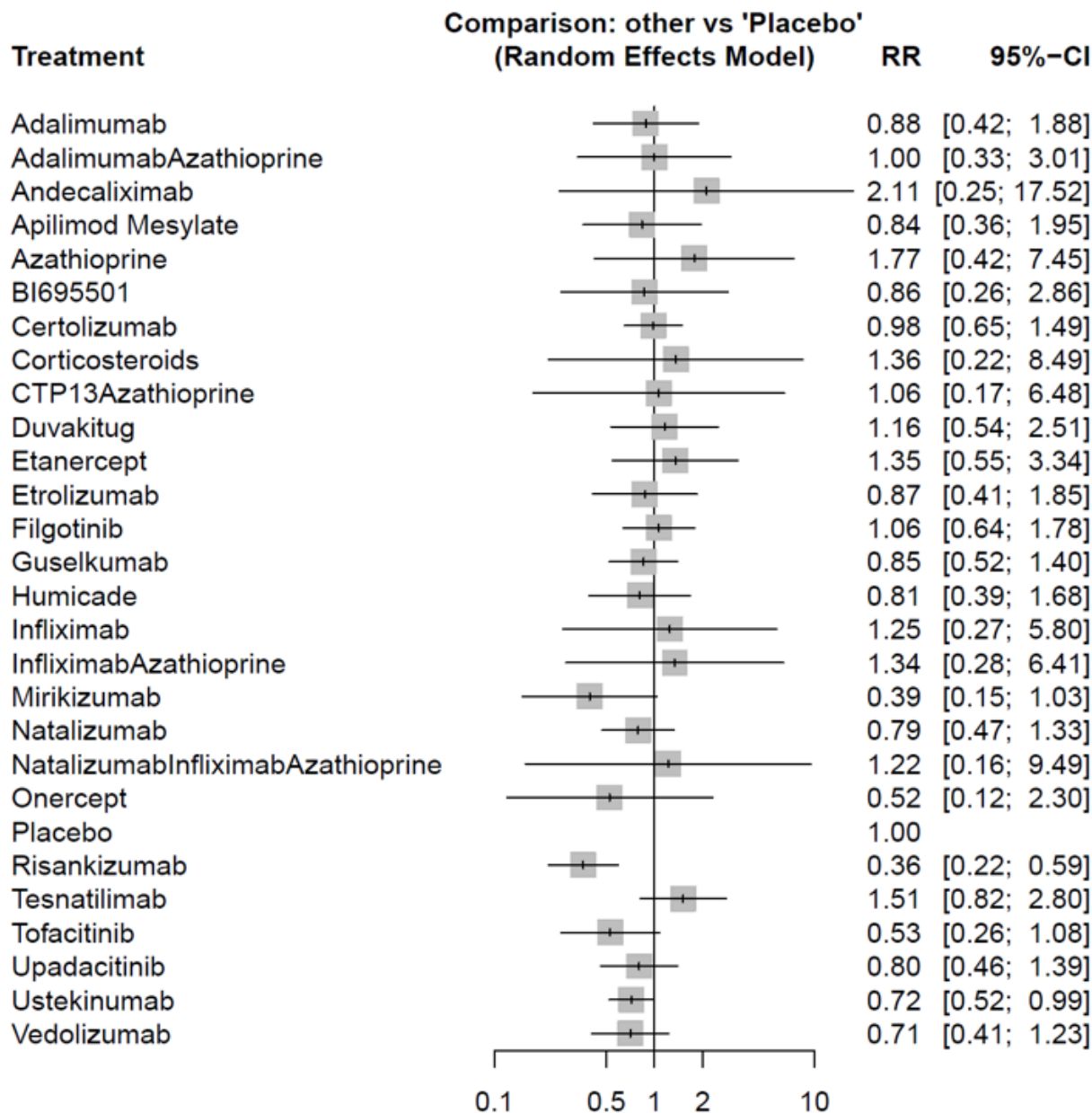


Figure 32. SUCRA rankings for withdrawals due to adverse events during induction

	SUCRA (common)	SUCRA (random)
Risankizumab	0.9626	0.9252
Mirikizumab	0.9007	0.8411
Tofacitinib	0.8278	0.7863
Onercept	0.7385	0.7593
Ustekinumab	0.7148	0.6663
Vedolizumab	0.6570	0.6485
Natalizumab	0.5811	0.5985
Upadacitinib	0.6033	0.5707
Adalimumab	0.4656	0.5604
Humicade	0.6311	0.5589
Guselkumab	0.5937	0.5419
Apilimod Mesylate	0.5548	0.5374
Etrolizumab	0.6004	0.5344
BI695501	0.4667	0.5159
AdalimumabAzathioprine	0.4085	0.5119
Certolizumab	0.5922	0.4670
Placebo	0.4293	0.4400
CTP13Azathioprine	0.4496	0.4011
Filgotinib	0.4078	0.3981
NatalizumabInfliximabAzathioprine	0.3419	0.3807
InfliximabAzathioprine	0.2107	0.3626
Infliximab	0.3167	0.3507
Duvakitug	0.3937	0.3470
Etanercept	0.2922	0.3000
Corticosteroids	0.2581	0.2959
Andecaliximab	0.2493	0.2733
Tesnatilimab	0.1996	0.2541
Azathioprine	0.1522	0.1726

- based on 100 simulations

A total of 28 treatments were included in this network from a total of 59 studies (n = 19,594 participants), and 71 pairwise comparisons were carried out.

Direct comparisons to placebo included one study on infliximab (n = 94), two on mirikizumab (n = 770), one on humicade (n = 169), three on guselkumab (n = 1041), three on risankizumab (n = 1551), two on adalimumab (n = 388), nine on ustekinumab (n = 3032), three on filgotinib (n = 1071), three on upadacitinib (n = 1241), five on vedolizumab (n = 1341), four on natalizumab (n = 1692), two on etrolizumab (n = 685), four on certolizumab (n = 1485), two on tofacitinib (n = 419), two on tesnatilimab (n = 341), one on onercept (n = 207), one on andecaliximab (n = 187), one on etanercept (n = 43), one on apilimod mesylate (n = 220) and one on duvakitug (n = 92).

We did not judge the evidence for any treatments at high certainty using GRADE for this outcome.

There is moderate-certainty evidence that vedolizumab (RR 0.71, 95% CI 0.41 to 1.23), ustekinumab (RR 0.72, 95% CI 0.52 to 0.99) and

upadacitinib (RR 0.8, 95% CI 0.46 to 1.39) probably lead to a similar number of withdrawals due to adverse events during induction compared to placebo.

There is low-certainty evidence that risankizumab may lead to fewer withdrawals due to adverse events compared to placebo (RR 0.36, 95% CI 0.22 to 0.59, small effect size).

There is low-certainty evidence that humicade (RR 0.81, 95% CI 0.39 to 1.68), natalizumab (RR 0.79, 95% CI 0.47 to 1.33), apilimod mesylate (RR 0.84, 95% CI 0.36 to 1.95), certolizumab (RR 0.98, 95% CI 0.65 to 1.49), filgotinib (RR 1.06, 95% CI 0.64 to 1.78) and tesnatilimab (RR 1.51, 95% CI 0.82 to 2.8) may lead to a similar number of withdrawals due to adverse events during induction compared to placebo.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The τ^2 for this outcome was 0.123 and the $I^2 = 50.4\%$ (total $Q = 76.58$, P value = 0.0002). The χ^2 test results and P values for assessing inconsistency within study designs ($Q = 58.41$, P value = 0.0006) showed the designs within guselkumab, placebo and ustekinumab ($Q = 24.25$, P value < 0.0001), and certolizumab and placebo ($Q = 7.99$, P value = 0.0461) to be statistically significant. Between study designs ($Q = 18.17$, P value = 0.0522), the designs between purine analogues and infliximab with purine analogues ($Q = 17.37$, P value = 0.0433), risankizumab and ustekinumab ($Q = 17.38$, P value = 0.0432), placebo and risankizumab ($Q = 17.38$, P value = 0.0432), placebo and tesnatilimab ($Q = 17.71$, P value = 0.0387), mirikizumab and placebo ($Q = 17.94$, P value = 0.0359), adalimumab, purine analogues and adalimumab with purine analogues ($Q = 17.95$, P value = 0.0358), infliximab and

placebo ($Q = 17.95$, P value = 0.0358), placebo and ustekinumab ($Q = 18.02$, P value = 0.035), mirikizumab, placebo and ustekinumab ($Q = 16.6$, P value = 0.0345), purine analogues, infliximab and infliximab with purine analogues ($Q = 17.16$, P value = 0.0285), and placebo, tesnatilimab and ustekinumab ($Q = 17.63$, P value = 0.0242) were statistically significant.

Serious adverse events

A GORDON plot, comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all treatments compared to placebo, and SUCRA rankings can be seen in [Figure 33](#), [Figure 34](#), [Figure 35](#), [Figure 36](#), [Figure 37](#) and [Figure 38](#). The comparison-adjusted funnel plot did not reveal large imbalances.

Figure 33. Gordon plot for serious adverse events for biologics for induction of remission in Crohn's disease, with placebo as reference treatment

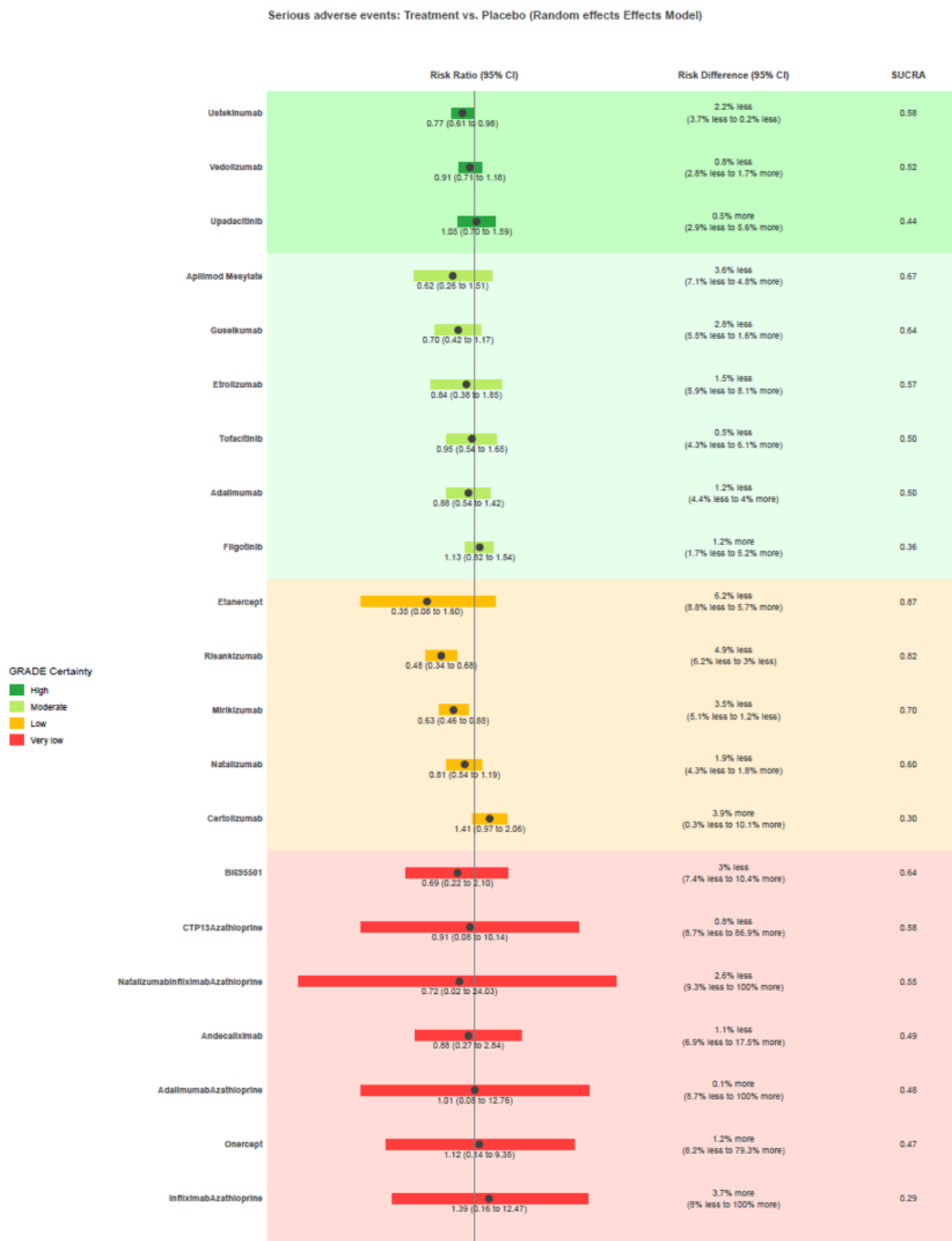


Figure 33. (Continued)

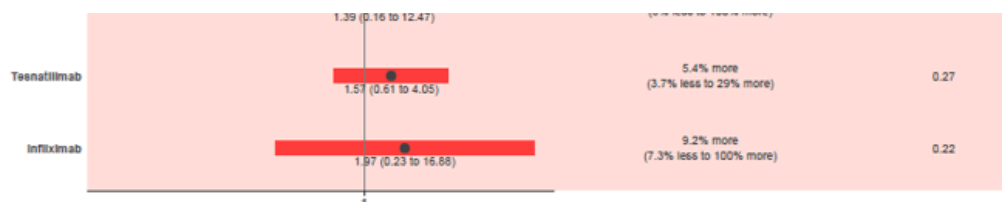


Figure 34. Comparison-adjusted funnel plot for serious adverse events during induction.

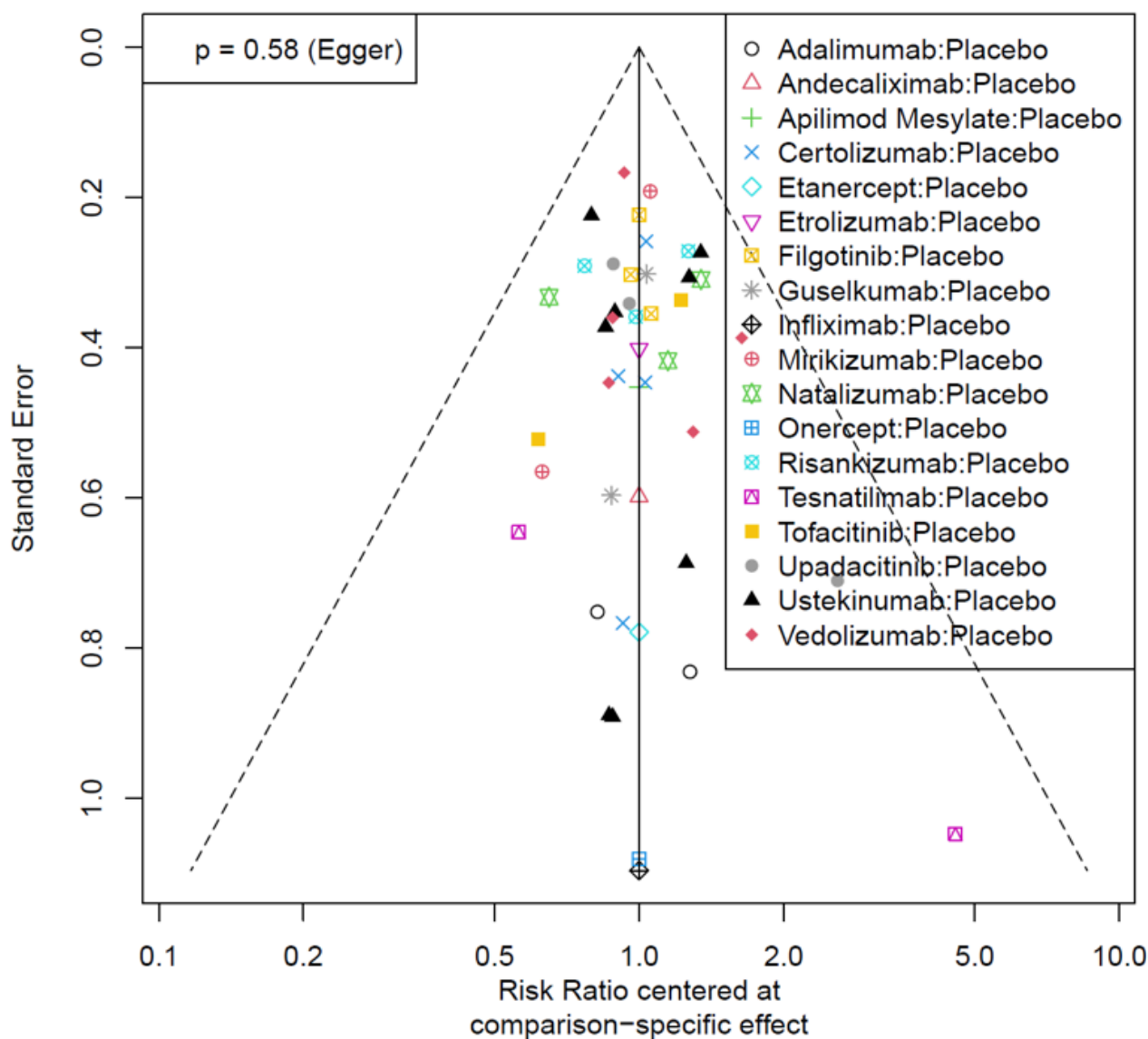


Figure 35. Network plot for serious adverse events during induction (shaded regions denote multi-arm studies)

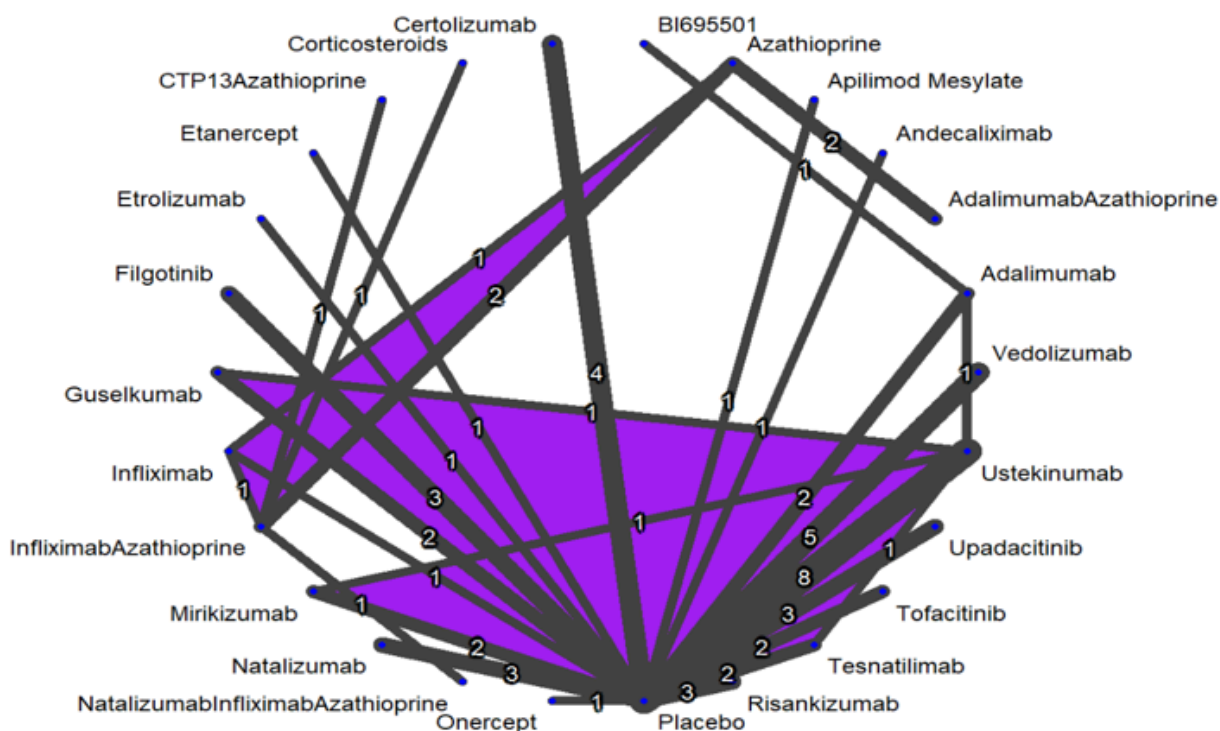


Figure 36. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for serious adverse events during induction.

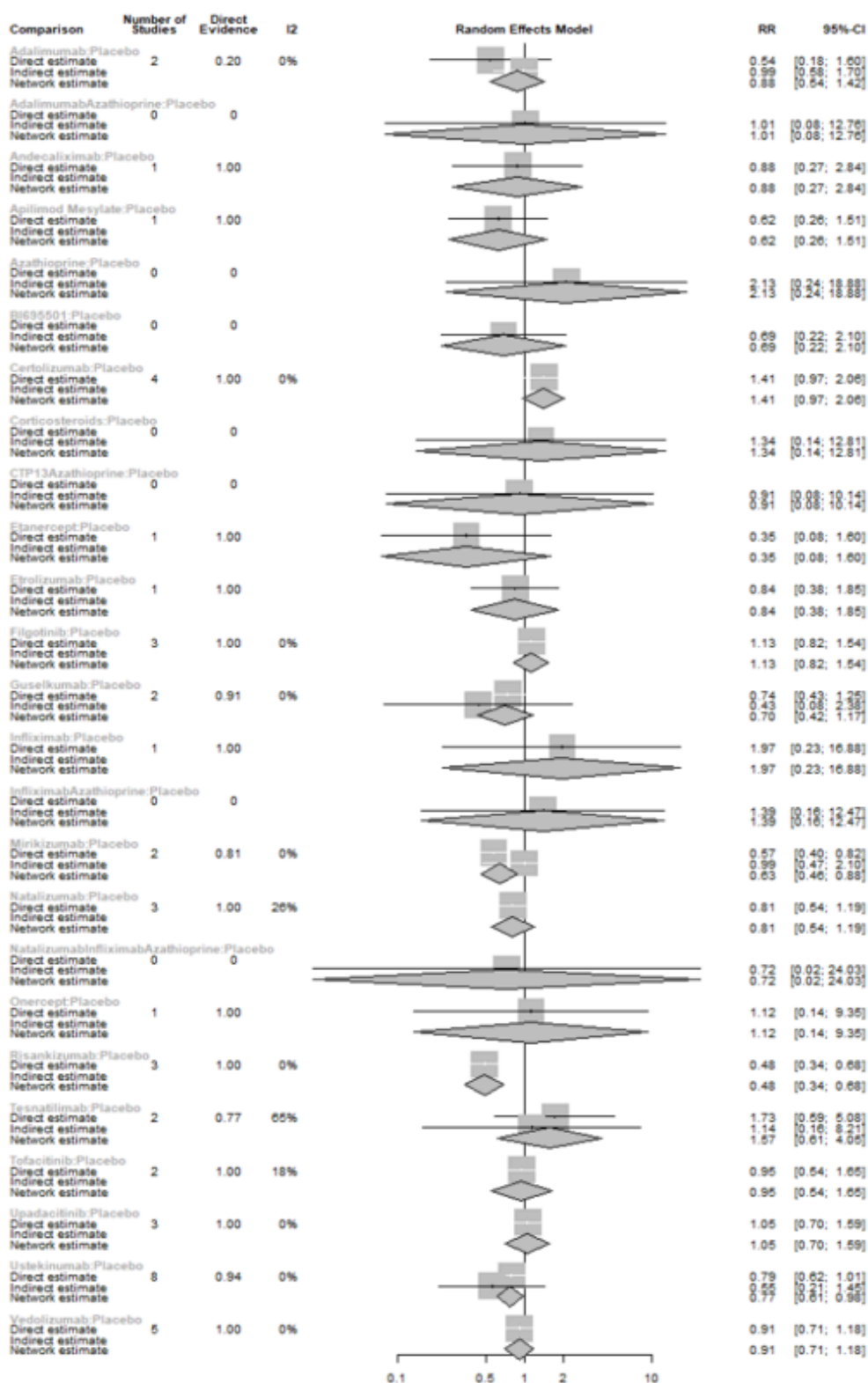


Figure 37. Forest plot for serious adverse events during induction

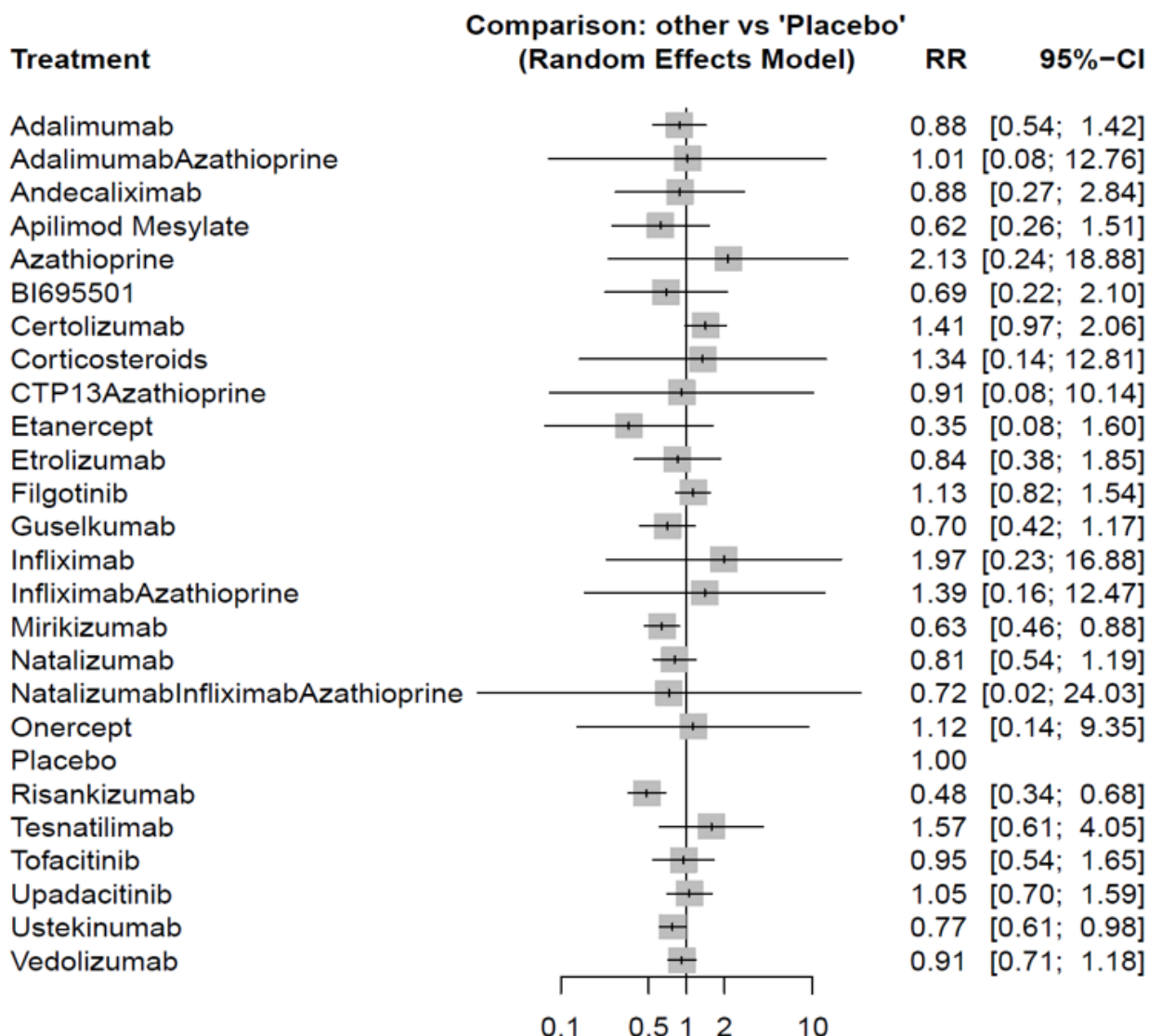


Figure 38. SUCRA rankings for serious adverse events during induction

	SUCRA (common)	SUCRA (random)
Risankizumab	0.8496	0.8340
Etanercept	0.8220	0.7688
Mirikizumab	0.6856	0.7172
Guselkumab	0.6352	0.6508
Apilimod Mesylate	0.6712	0.6424
Ustekinumab	0.5960	0.6052
BI695501	0.6264	0.5924
Natalizumab	0.5888	0.5640
NatalizumabInfliximabAzathioprine	0.5148	0.5492
Etrolizumab	0.5264	0.5212
CTP13Azathioprine	0.4820	0.5156
Vedolizumab	0.5104	0.5096
Adalimumab	0.5108	0.5036
Tofacitinib	0.5192	0.4988
Andecaliximab	0.5392	0.4848
Onercept	0.4116	0.4744
AdalimumabAzathioprine	0.4148	0.4452
Placebo	0.4396	0.4356
Filgotinib	0.4288	0.4244
Upadacitinib	0.4556	0.4228
InfliximabAzathioprine	0.3612	0.4076
Corticosteroids	0.4024	0.3832
Tesnatilimab	0.2380	0.2932
Certolizumab	0.2828	0.2688
Infliximab	0.2584	0.2500
Azathioprine	0.2292	0.2372

- based on 100 simulations

A total of 26 treatments were included in this network from a total of 52 studies (n = 17,789 participants), and 60 pairwise comparisons were carried out.

Direct comparisons to placebo included one study on infliximab (n = 94), two on mirikizumab (n = 821), two on guselkumab (n = 457), three on risankizumab (n = 1551), two on adalimumab (n = 388), eight on ustekinumab (n = 2763), three on filgotinib (n = 1071), three on upadacitinib (n = 1241), five on vedolizumab (n = 1341), three on natalizumab (n = 1662), one on etrolizumab (n = 385), four on certolizumab (n = 1485), two on tofacitinib (n = 419), two on tesnatilimab (n = 341), one on onercept (n = 207), one on andecaliximab (n = 187), one on etanercept (n = 43) and one on apilimod mesylate (n = 220).

There is high-certainty evidence that ustekinumab (RR 0.77, 95% CI 0.61 to 0.98), vedolizumab (RR 0.91, 95% CI 0.71 to 1.18) and upadacitinib (RR 1.05, 95% CI 0.7 to 1.59) lead to a similar number of serious adverse events during induction compared to placebo.

There is moderate-certainty evidence that apilimod mesylate (RR 0.62, 95% CI 0.26 to 1.51), guselkumab (RR 0.7, 95% CI 0.42 to 1.17), etrolizumab (RR 0.84, 95% CI 0.38 to 1.85), tofacitinib (RR 0.95, 95% CI 0.54 to 1.65), adalimumab (RR 0.88, 95% CI 0.54 to 1.42) and filgotinib (RR 1.13, 95% CI 0.82 to 1.54) are probably no different to placebo for serious adverse events.

There is low-certainty evidence that etanercept (RR 0.35, 95% CI 0.08 to 1.6), risankizumab (RR 0.48, 95% CI 0.34 to 0.68), mirikizumab (RR 0.63, 95% CI 0.46 to 0.88), natalizumab (RR 0.81, 95% CI 0.54 to 1.19) and certolizumab (RR 1.41, 95% CI 0.97 to 2.06) may lead to a similar number of serious adverse events during induction compared to placebo.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The tau² for this outcome was 0 and the I² = 0% (total Q = 20.98, P value = 0.89). The Chi² test results and P values for assessing inconsistency within study designs (Q = 13.72, P value = 0.91) were

not statistically significant. Between study designs ($Q = 7.26$, P value = 0.51) there were no statistically significant results.

Total adverse events

A GORDON plot, comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all

treatments compared to placebo, and SUCRA rankings can be seen in [Figure 39](#), [Figure 40](#), [Figure 41](#), [Figure 42](#), [Figure 43](#) and [Figure 44](#). The comparison-adjusted funnel plot did not reveal large imbalances.

Figure 39. Gordon plot for total adverse events for biologics for induction of remission in Crohn's disease, with placebo as reference treatment



Figure 39. (Continued)

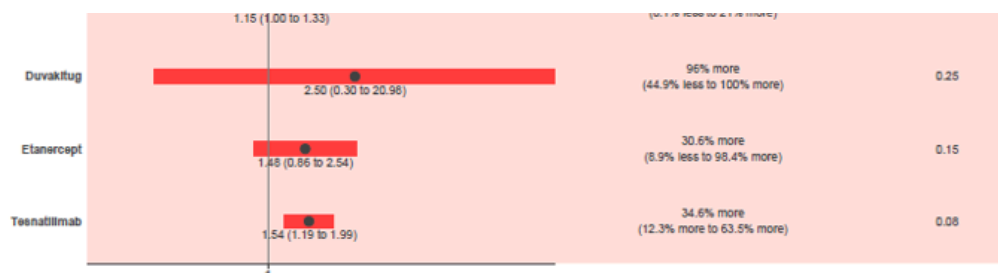


Figure 40. Comparison-adjusted funnel plot for total adverse events during induction.

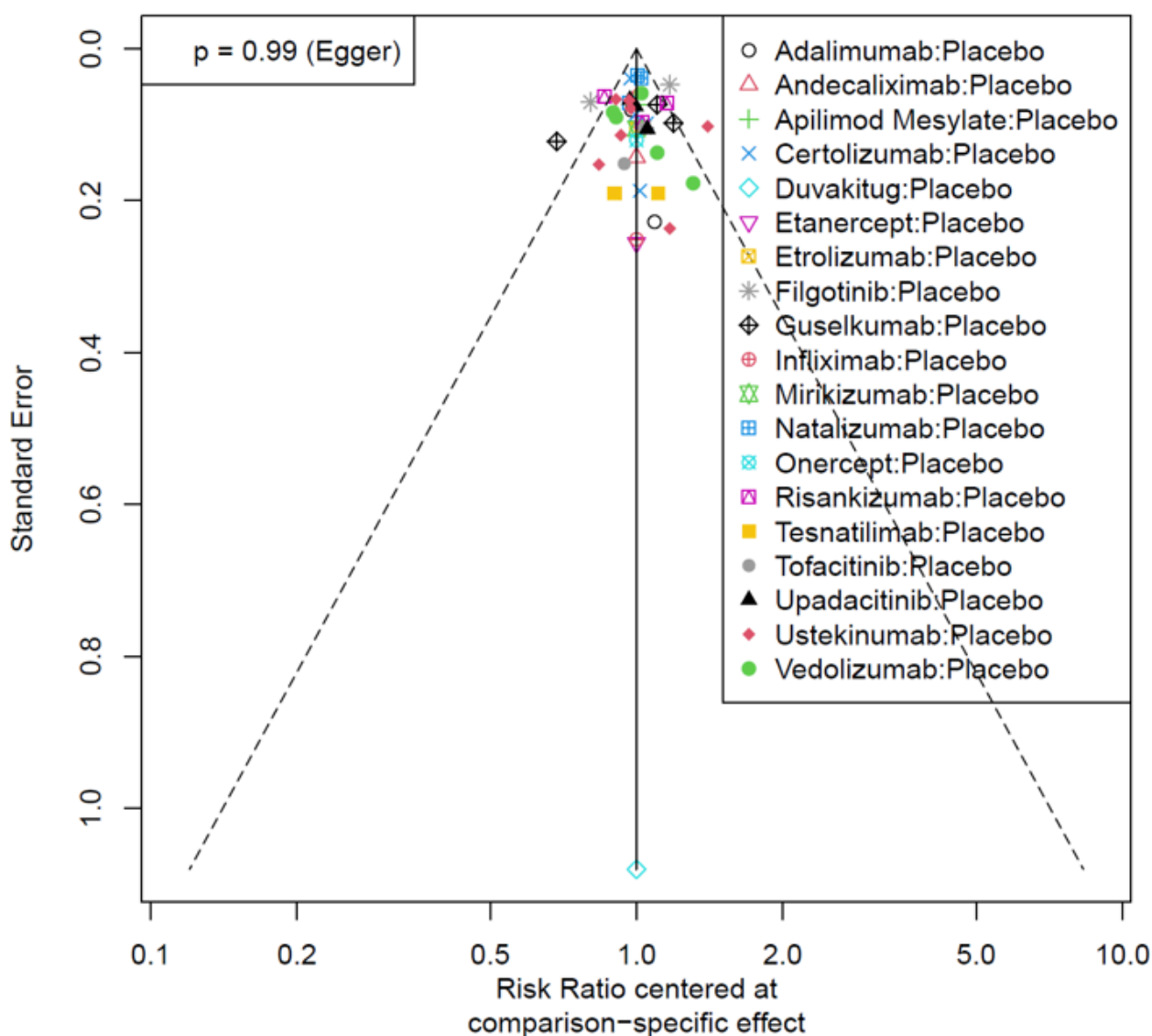


Figure 41. Network plot for total adverse events during induction (shaded regions denote multi-arm studies)

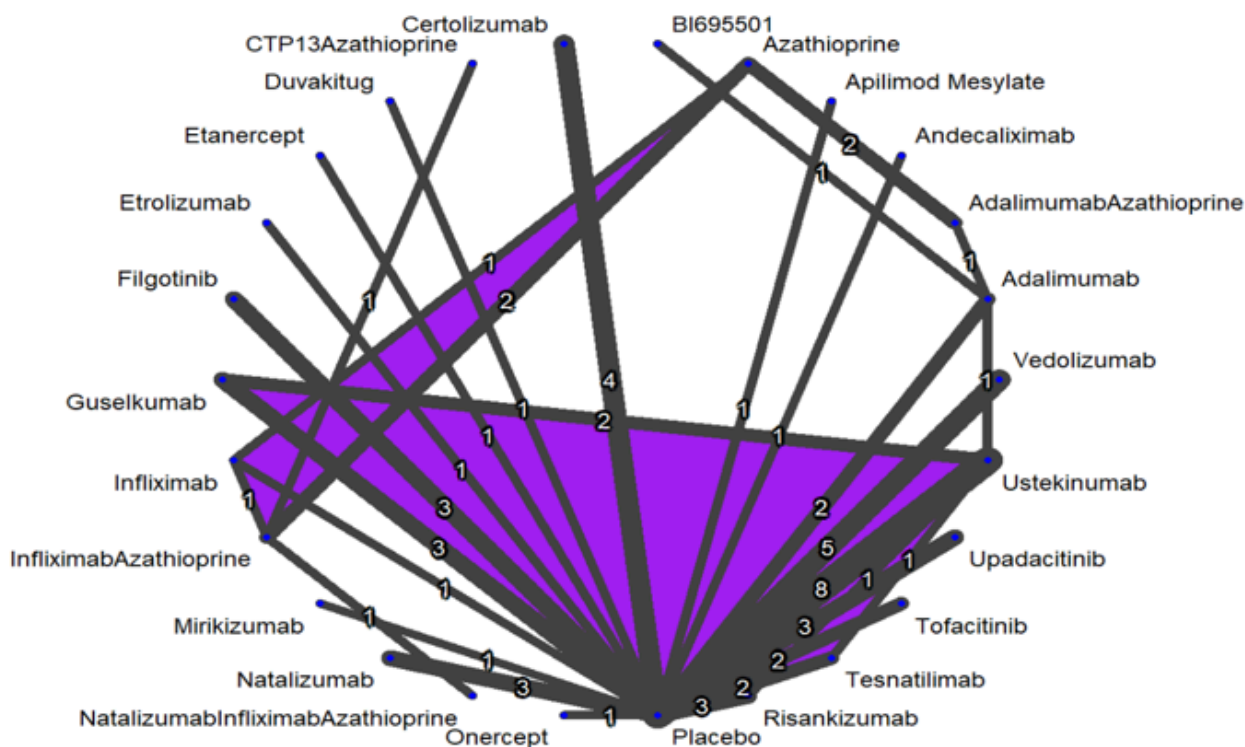


Figure 42. Network plot for total adverse events during induction.

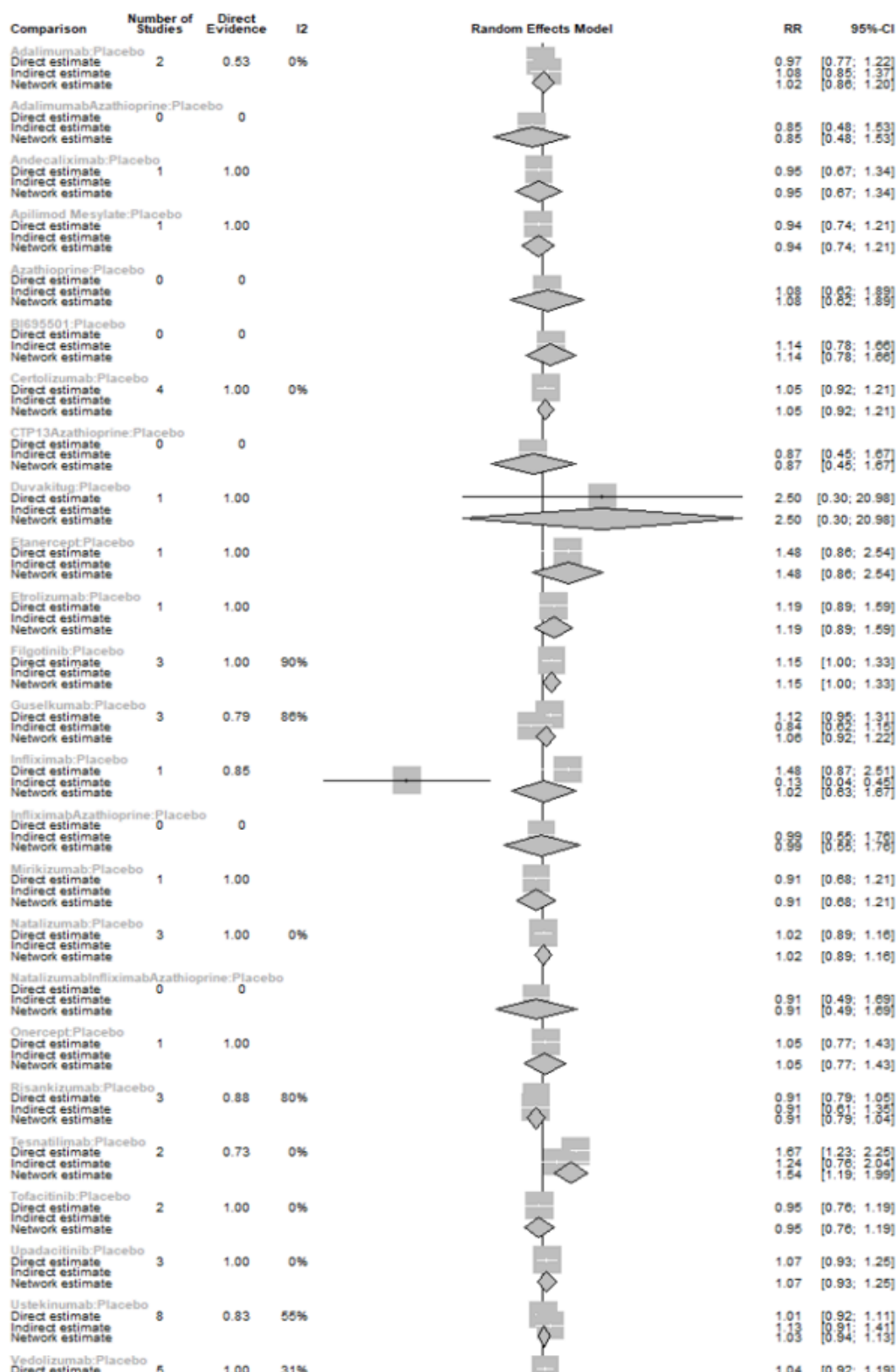


Figure 42. (Continued)

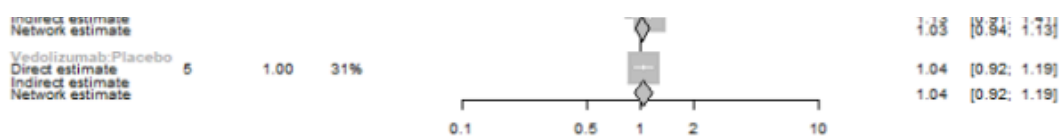


Figure 43. Forest plot for total adverse events during induction

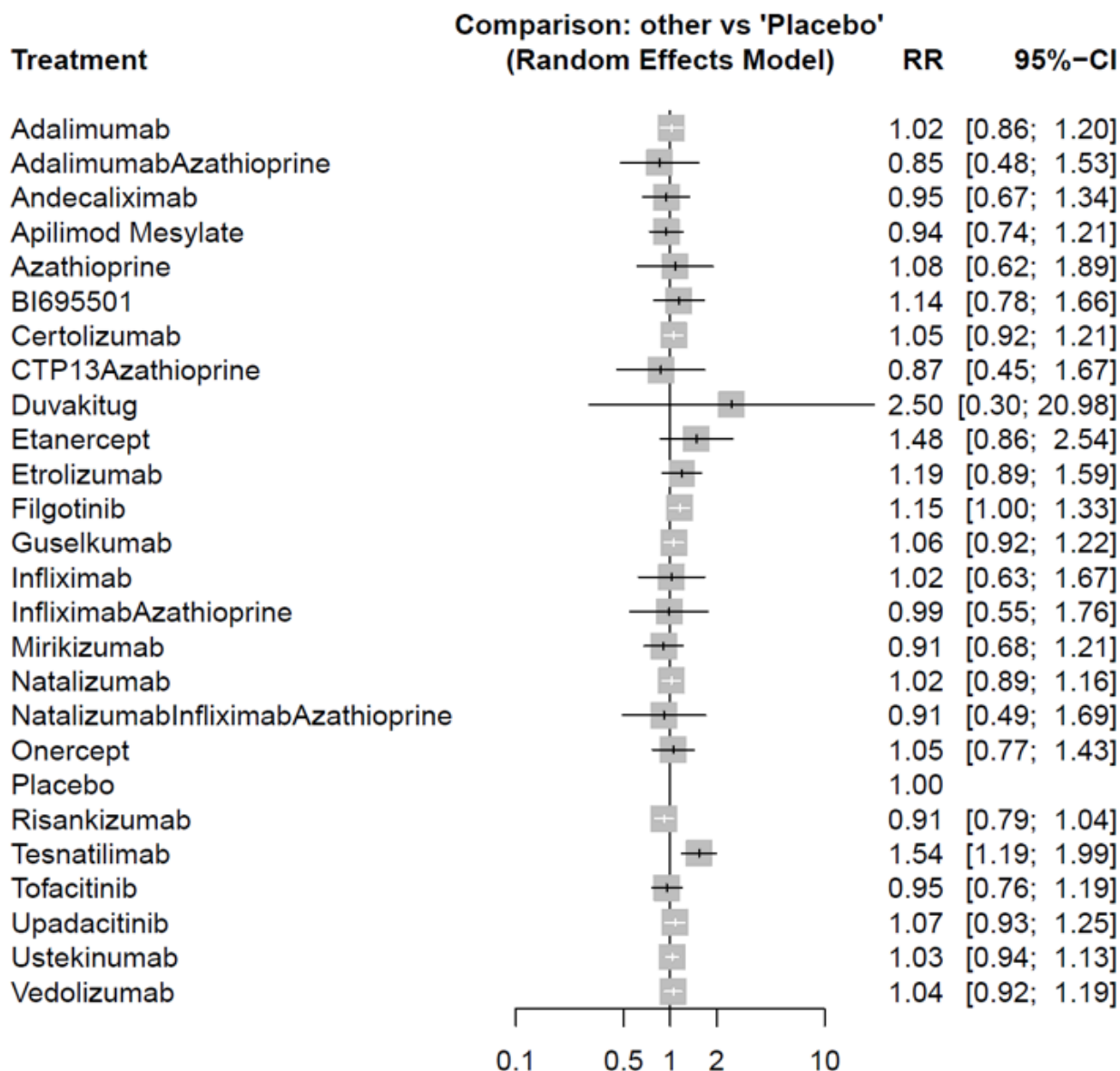


Figure 44. SUCRA rankings for total adverse events during induction

	SUCRA (common)	SUCRA (random)
Risankizumab	0.8240	0.7636
AdalimumabAzathioprine	0.6608	0.7520
Mirikizumab	0.7744	0.7332
Tofacitinib	0.7128	0.6844
NatalizumabInfliximabAzathioprine	0.6264	0.6576
Andecaliximab	0.6648	0.6536
Apilimod Mesylate	0.7128	0.6484
CTP13Azathioprine	0.6876	0.6344
Placebo	0.6104	0.5956
Infliximab	0.4852	0.5728
InfliximabAzathioprine	0.4692	0.5524
Natalizumab	0.5160	0.5496
Ustekinumab	0.5484	0.5328
Adalimumab	0.6060	0.5248
Onercept	0.4896	0.4836
Vedolizumab	0.5068	0.4820
Guselkumab	0.4840	0.4672
Certolizumab	0.4972	0.4560
Upadacitinib	0.4676	0.4308
Azathioprine	0.3068	0.4304
BI695501	0.4188	0.3900
Filgotinib	0.2272	0.3152
Etrolizumab	0.2668	0.2876
Duvakitug	0.1948	0.1836
Etanercept	0.1772	0.1464
Tesnatilimab	0.0644	0.0720

- based on 100 simulations

A total of 26 treatments were included in this network from a total of 53 studies (n = 17,897 participants), and 61 pairwise comparisons were carried out.

Direct comparisons to placebo included one study on infliximab (n = 94), one on mirikizumab (n = 191), three on guselkumab (n = 1052), three on risankizumab (n = 1551), two on adalimumab (n = 388), eight on ustekinumab (n = 2754), three on filgotinib (n = 1071), three on upadacitinib (n = 1241), five on vedolizumab (n = 1341), three on natalizumab (n = 1662), one on etrolizumab (n = 385), four on certolizumab (n = 1485), two on tofacitinib (n = 419), two on tesnatilimab (n = 341), one on onercept (n = 207), one on andecaliximab (n = 187), one on etanercept (n = 43), one on apilimod mesylate (n = 220) and one on duvakitug (n = 92).

There is high-certainty evidence that ustekinumab leads to a similar number of total adverse events during induction compared to placebo (RR 1.03, 95% CI 0.94 to 1.13).

There is moderate-certainty evidence that adalimumab (RR 1.02, 95% CI 0.86 to 1.2), vedolizumab (RR 1.04, 95% CI 0.92 to 1.19) and

upadacitinib (RR 1.07, 95% CI 0.93 to 1.25) are probably no different to placebo for total adverse events.

There is low-certainty evidence that mirikizumab (RR 0.91, 95% CI 0.68 to 1.21), apilimod mesylate (RR 0.94, 95% CI 0.74 to 1.21), tofacitinib (RR 0.95, 95% CI 0.76 to 1.19), natalizumab (RR 1.02, 95% CI 0.89 to 1.16), certolizumab (RR 1.05, 95% CI 0.92 to 1.21) and etrolizumab (RR 1.19, 95% CI 0.89 to 1.59) may lead to a similar number of total adverse events during induction compared to placebo.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The tau² for this outcome was 0.01 and the I² = 60.9% (total Q = 81.76, P value < 0.0001). The Chi² test results and P values for assessing inconsistency within study designs (Q = 53.54, P value = 0.0005) showed the designs within filgotinib and placebo (Q = 19.64, P value < 0.0001), guselkumab, placebo and ustekinumab (Q = 13.1, P value = 0.0014), and placebo and risankizumab (9.92, P

value = 0.007) to be statistically significant. Between study designs ($Q = 28.22$, P value = 0.0004), all the designs were found to have statistically significant results (Q range 13.19 to 28.22, all P values < 0.05).

Biologic drugs for maintenance of remission in Crohn's disease

The extraction data for the maintenance studies can be found in Table 6. Outcome measurements ranged from 26 weeks to 60 weeks.

Critical outcome: Clinical relapse

The results for this outcome are presented in Summary of findings 3, with placebo as the reference treatment. A GORDON plot of the network estimates ordered by GRADE certainty and SUCRA ranking for the treatments not crossing the line of no effect with high, moderate and low certainty can be seen in Figure 45. A comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all treatments compared to placebo, and SUCRA rankings for this outcome can be seen in Figure 46, Figure 47, Figure 48, Figure 49 and Figure 50. The comparison-adjusted funnel plot did not reveal large imbalances.

Figure 45. Gordon plot for clinical relapse for biologics for maintenance of remission in Crohn's Disease, with placebo as reference treatment.

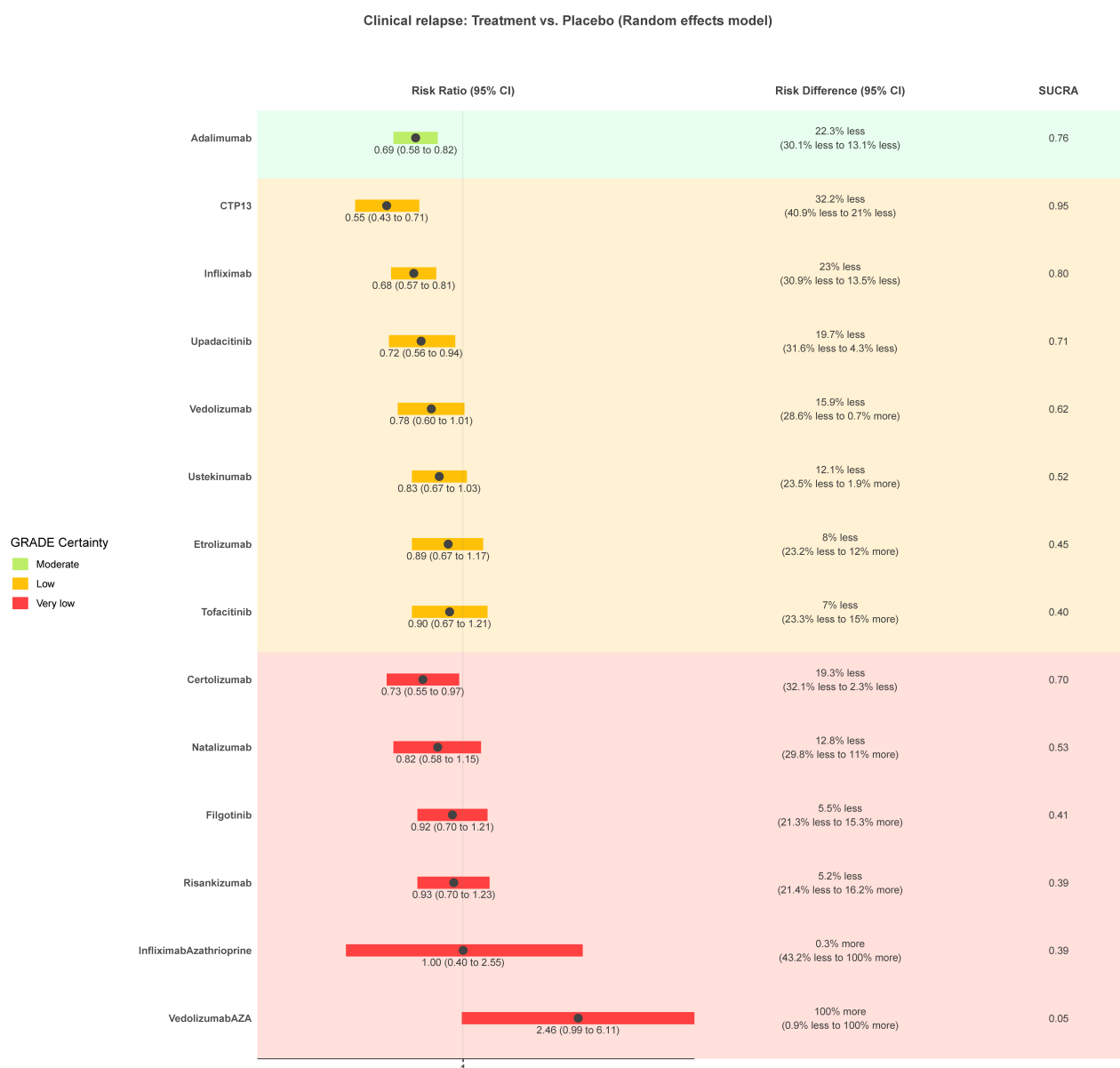


Figure 46. Comparison-adjusted funnel plot for clinical relapses during maintenance of clinical remission

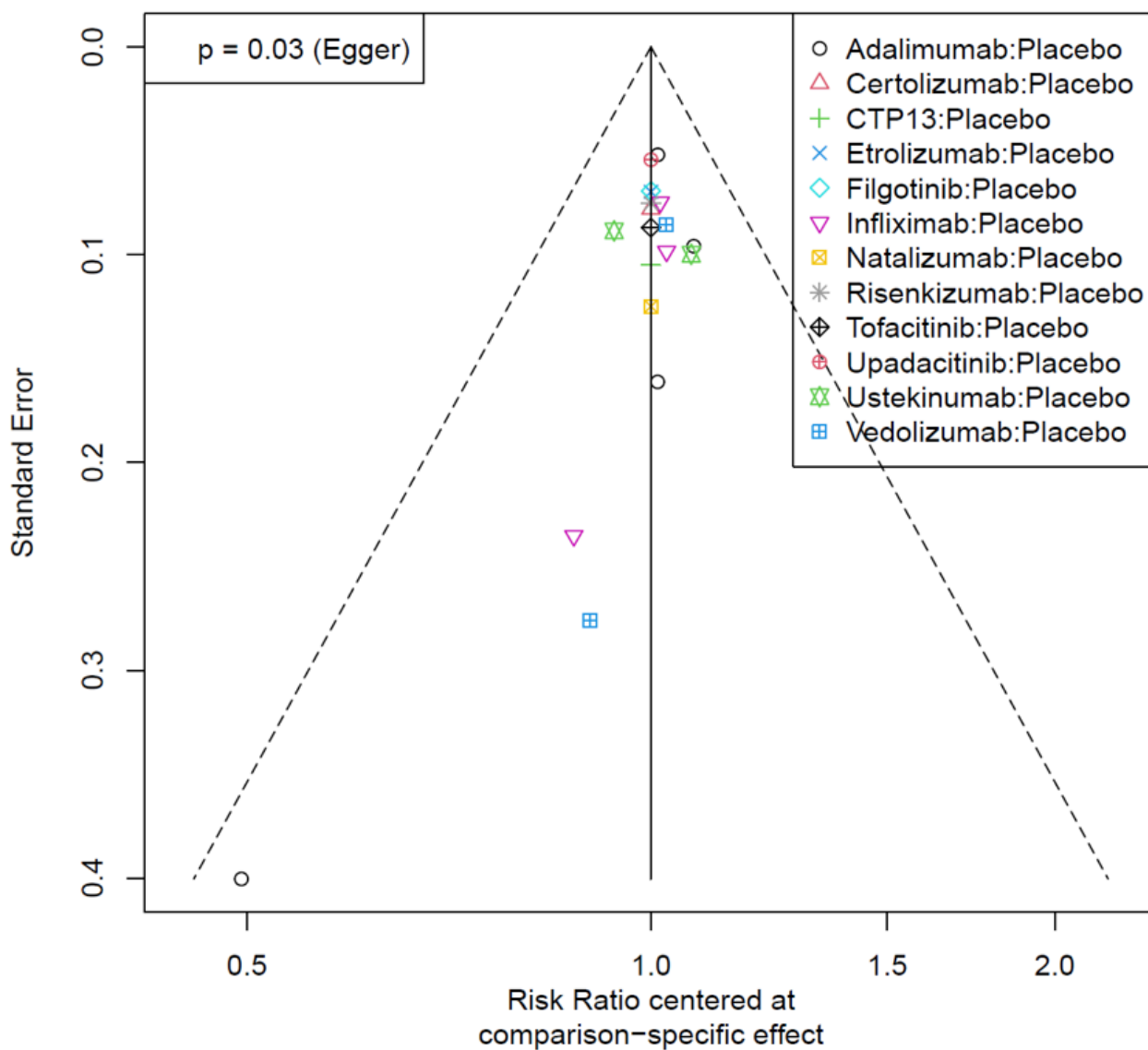


Figure 47. Network plot for maintenance of clinical remission (shaded regions denote denotes multi-arm studies)

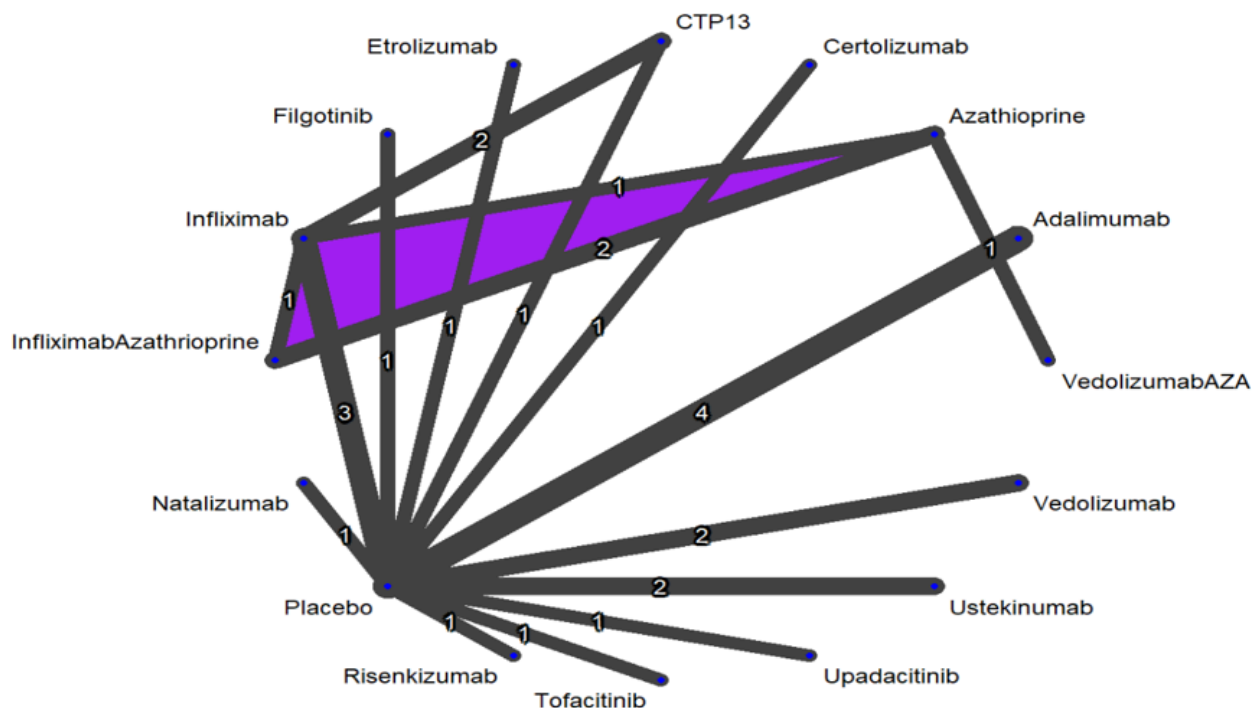


Figure 48. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for maintenance of clinical remission.

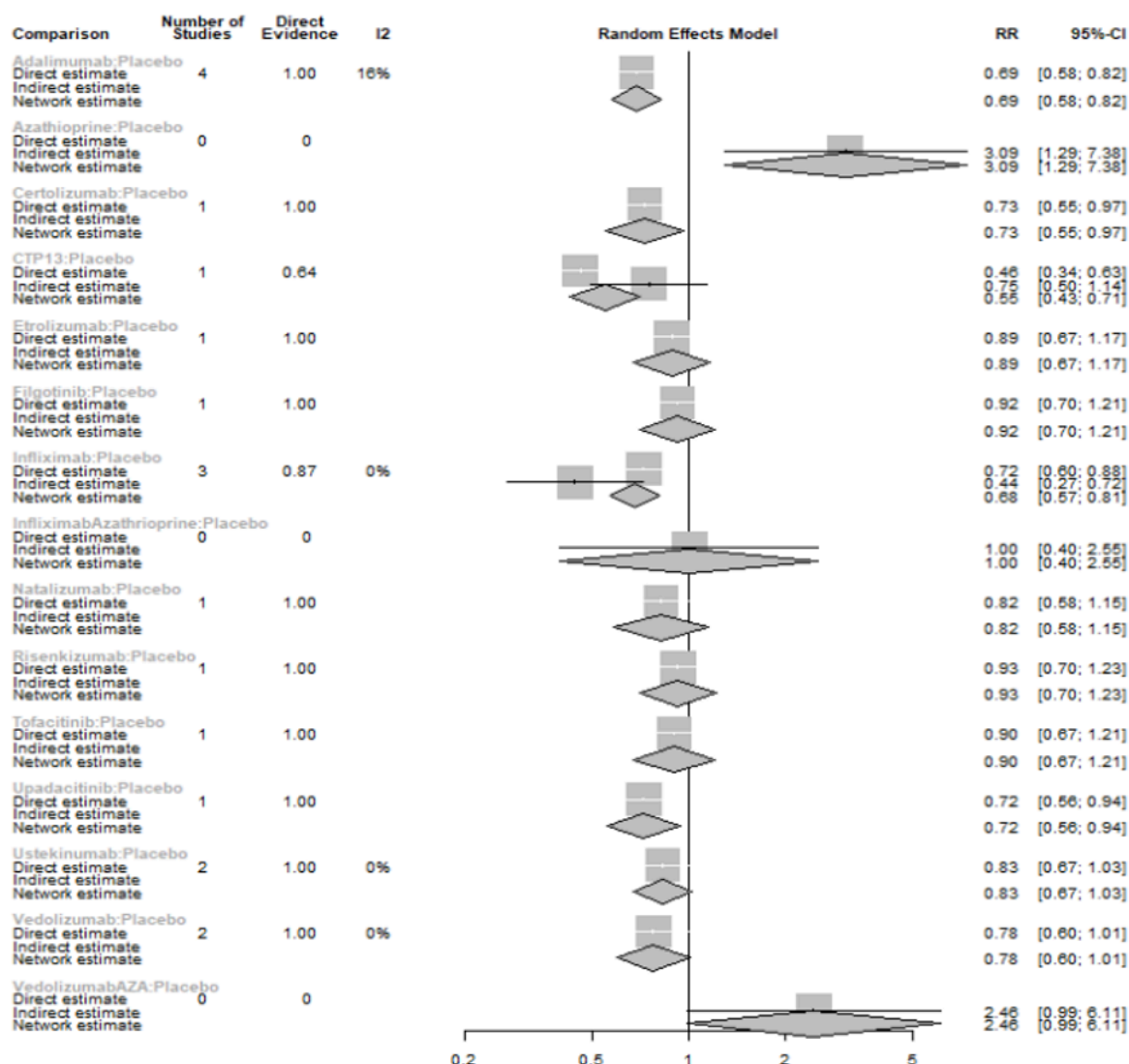


Figure 49. Forest plot for clinical relapse during maintenance of clinical remission

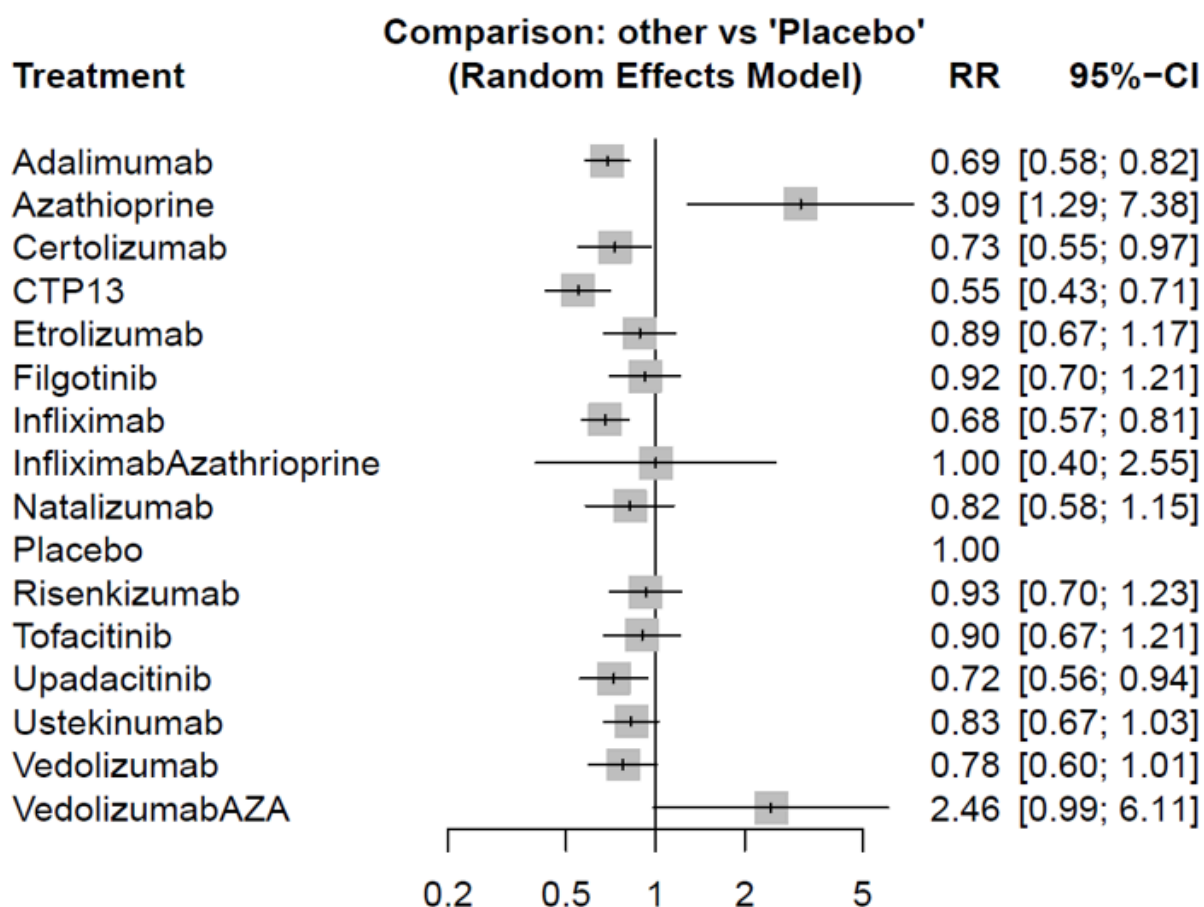


Figure 50. SUCRA rankings for maintenance of clinical remission

	SUCRA (common)	SUCRA (random)
CTP13	0.9960	0.9600
Infliximab	0.7907	0.8020
Adalimumab	0.8100	0.7820
Certolizumab	0.7580	0.7193
Upadacitinib	0.7553	0.6687
Vedolizumab	0.6040	0.6107
Natalizumab	0.5253	0.5680
Ustekinumab	0.5140	0.5000
Etrolizumab	0.4507	0.4540
Tofacitinib	0.4207	0.4433
Risenkizumab	0.3673	0.3893
Filgotinib	0.3793	0.3813
InfliximabAzathrioprine	0.3160	0.3713
Placebo	0.2407	0.2780
VedolizumabAZA	0.0480	0.0493
Azathioprine	0.0240	0.0227

- based on 100 simulations

All studies included in the analysis defined remission as a CDAI score of less than 150, except Jorgensen 2017 (NORSWITCH), which defined remission as a reduction in the HBI (Harvey-Bradshaw Index) scale of ≤ 4 points and partial Mayo Score ≤ 2 points, and Sands 2004 (ACCENT II) Maintenance (Responders) as a reduction of at least 50% from baseline in the number of draining fistulas at consecutive visits four or more weeks apart.

A total of 16 treatments were included in this network from a total of 24 studies ($n = 6543$ participants), and 26 pairwise comparisons were carried out.

Direct comparisons to placebo included one study on CT-P13 ($n = 325$), three on infliximab ($n = 603$), four on adalimumab ($n = 731$), one on certolizumab ($n = 428$), one on upadacitinib ($n = 502$), two on vedolizumab ($n = 434$), two on ustekinumab ($n = 542$), one on natalizumab ($n = 428$), one on tofacitinib ($n = 180$), one on etrolizumab ($n = 434$), one on risankizumab ($n = 542$) and one on filgotinib ($n = 329$).

We did not judge the evidence for any treatments at high certainty using GRADE for this outcome.

There is moderate-certainty evidence that adalimumab (RR 0.69, 95% CI 0.58 to 0.82, SUCRA = 0.76, NNT = 2, moderate effect size) is probably more effective than placebo in preventing clinical relapse in Crohn's disease.

There is low-certainty evidence that CTP13 (RR 0.55, 95% CI 0.43 to 0.71, SUCRA = 0.95, large effect size), infliximab (RR 0.68, 95% CI 0.57 to 0.81, SUCRA = 0.8, moderate effect size) and upadacitinib (RR 0.72, 95% CI 0.56 to 0.94, SUCRA = 0.71, small effect size) may be more effective than placebo in preventing clinical relapse in Crohn's disease.

There is low-certainty evidence that there may be no difference between vedolizumab (RR 0.78, 95% CI 0.6 to 1.01), ustekinumab (RR 0.83, 95% CI 0.67 to 1.03), etrolizumab (RR 0.89, 95% CI 0.67 to 1.17), tofacitinib (RR 0.9, 95% CI 0.67 to 1.21) and placebo in preventing clinical relapse in Crohn's disease.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The τ^2 for this outcome was 0.01 and the $I^2 = 47\%$ (total $Q = 18.88$, P value = 0.0418). The χ^2 test results and P values for assessing inconsistency within study designs ($Q = 9.02$, P value = 0.3402) showed no statistically significant differences. Between study designs ($Q = 9.86$, P value = 0.0072), the designs between CTP13, infliximab and placebo ($Q = 4.46$, P value = 0.0346), and purine analogues and infliximab with purine analogues ($Q = 5.39$, P value = 0.0202) were statistically significant.

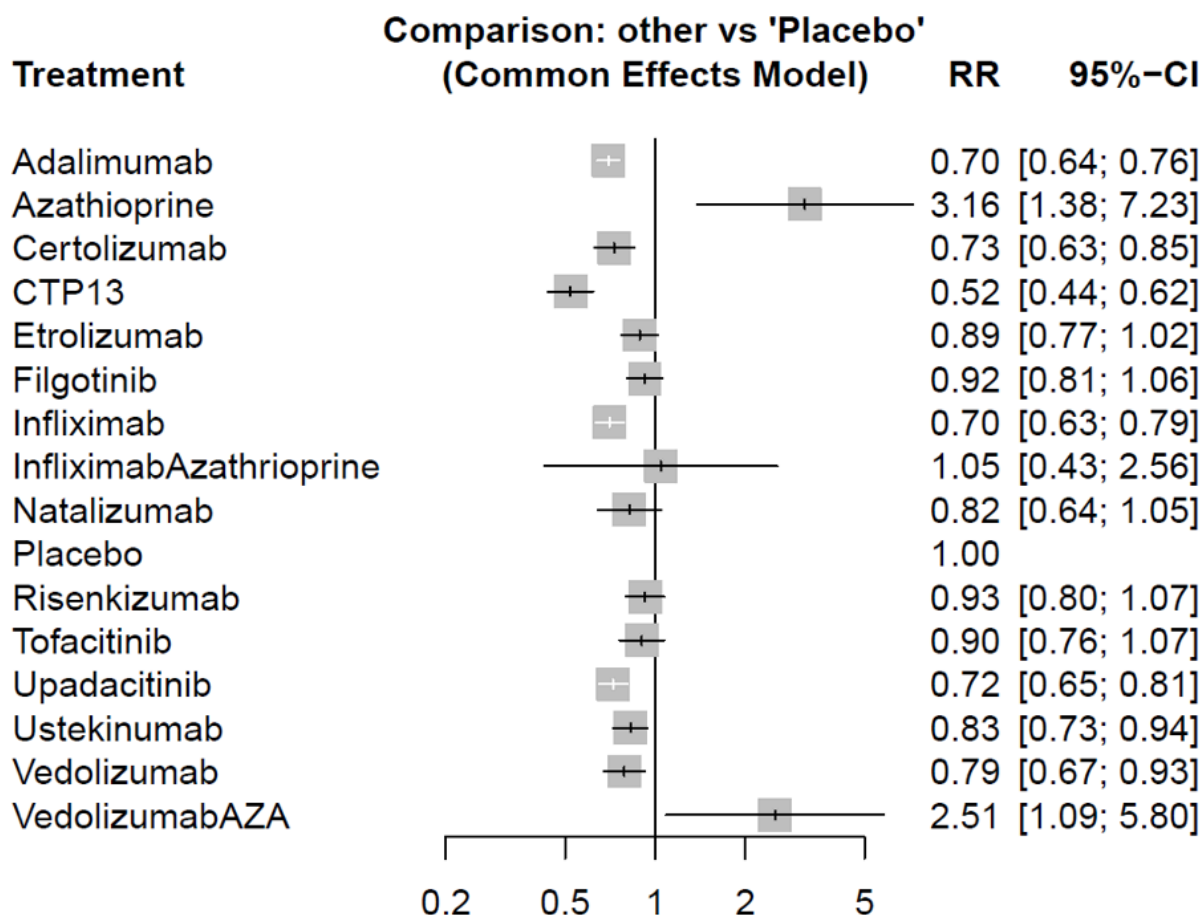
Subgroup analyses

The subgroup analyses on biologic naivety could not be conducted as there were no maintenance RCTs on biologically naive patients.

Sensitivity analyses

The sensitivity analysis using a fixed-effect model revealed no differences to the random-effects model analysis (Figure 51).

Figure 51. Common (fixed) effect sensitivity analysis for maintenance of clinical remission



The rest of the sensitivity analyses could not be conducted as there were not enough studies to form networks for comparison based on the sensitivity analysis plans. We judged six studies to be at higher risk of bias (Ferrante 2022 - FORTIFY; Louis 2023 - SPARE; Röder 2021 [648]; Schreiber 2007 (PRECISE 2); Vermeire 2025 - DIVERSITY (maintenance); Watanabe 2012 - Maintenance). Three studies had less than 20% of participants using concomitant immunosuppressants (Loftus 2023 (U-ENDURE); Panes 2017 Maintenance; Röder 2021). Two studies had all participants in clinical remission at baseline (Buhl 2022 (STOP-IT); Sandborn 2007b (CLASSIC II)). One study was on a fistulating population (Sands 2004 (ACCENT II) Maintenance (Responders)).

One study was not fully published as a peer reviewed publication (Röder 2021).

Important outcomes**Clinical loss of response**

A GORDON plot, comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all treatments compared to placebo, and SUCRA rankings can be seen in Figure 52, Figure 53, Figure 54, Figure 55, Figure 56 and Figure 57. The comparison-adjusted funnel plot did not reveal large imbalances.

Figure 52. Gordon plot for loss of clinical response for biologics for maintenance of remission in Crohn's disease, with placebo as reference treatment.

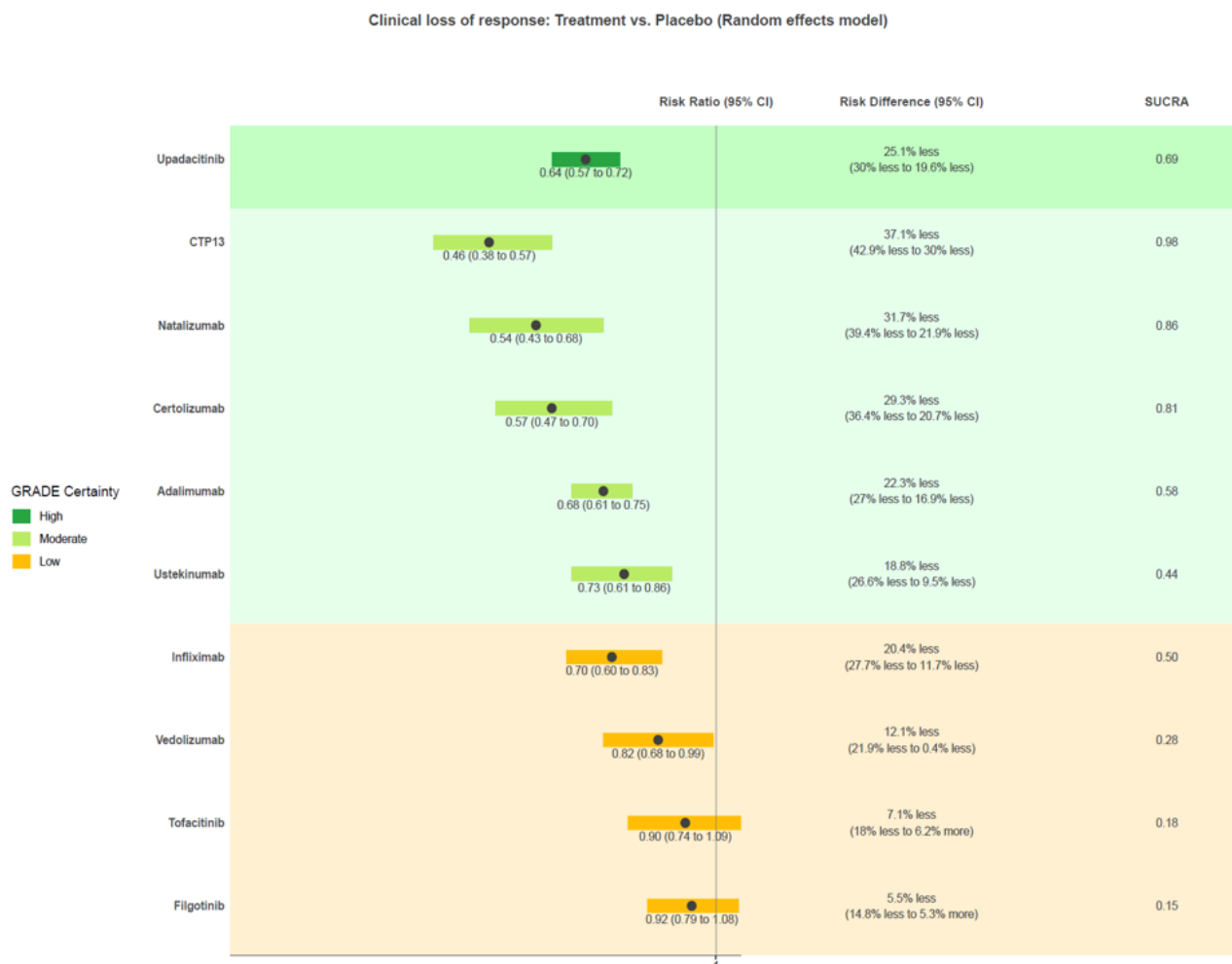


Figure 53. Comparison-adjusted funnel plot for loss of clinical response during maintenance

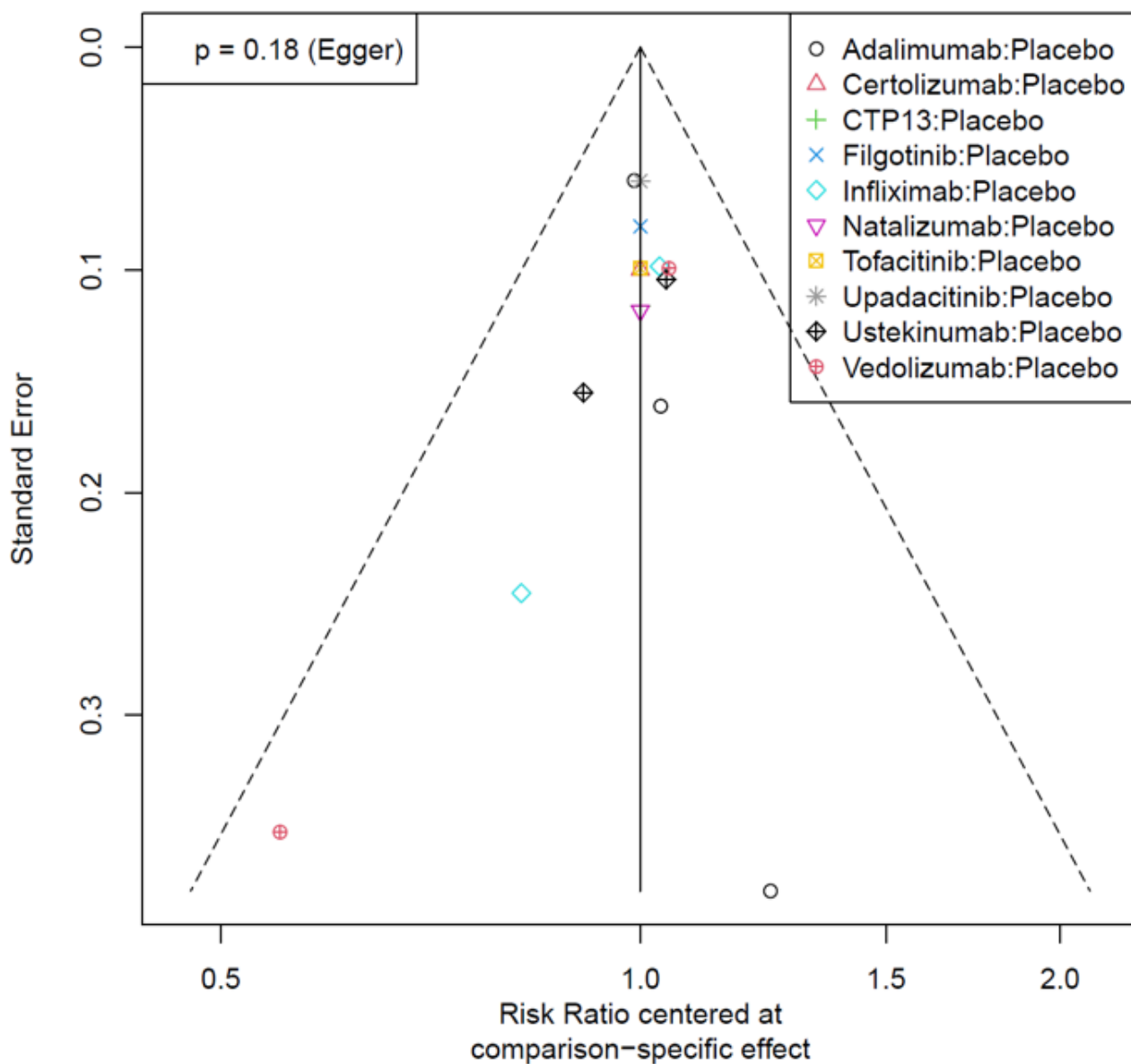


Figure 54. Network plot for maintenance of clinical response

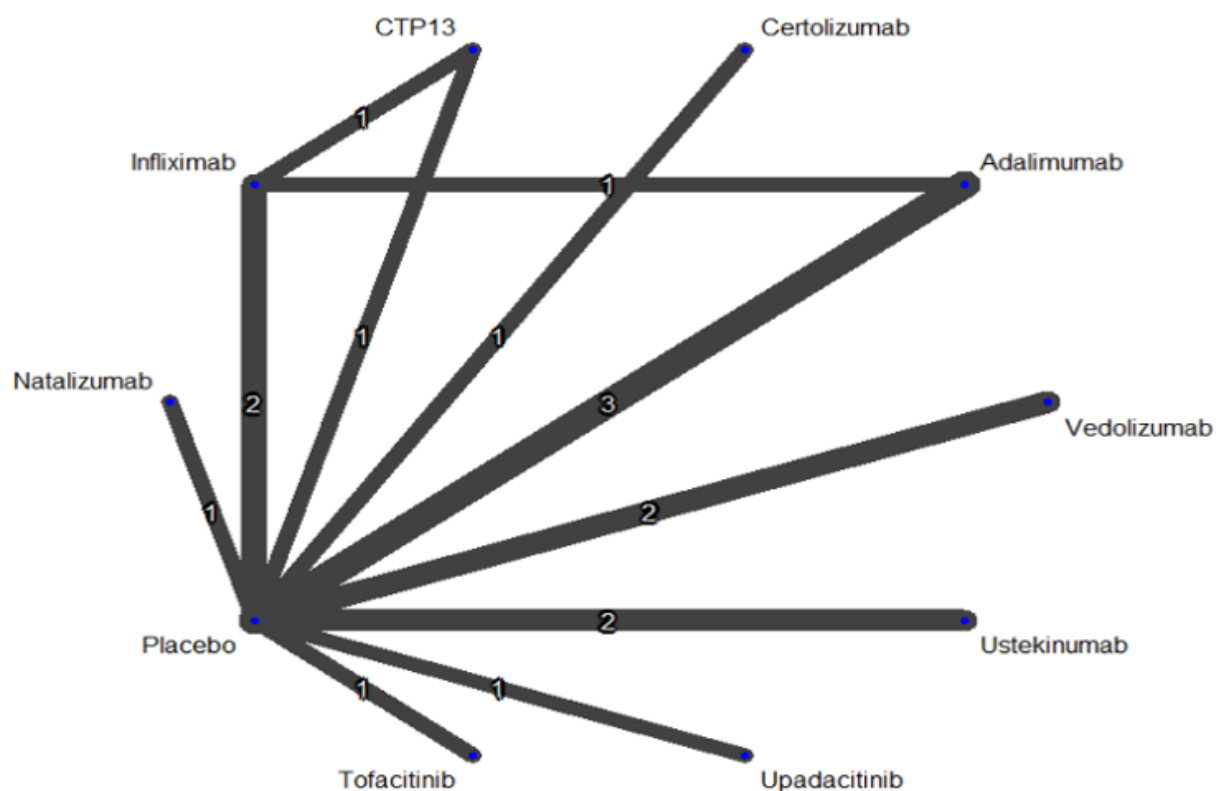


Figure 55. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for maintenance of clinical response.

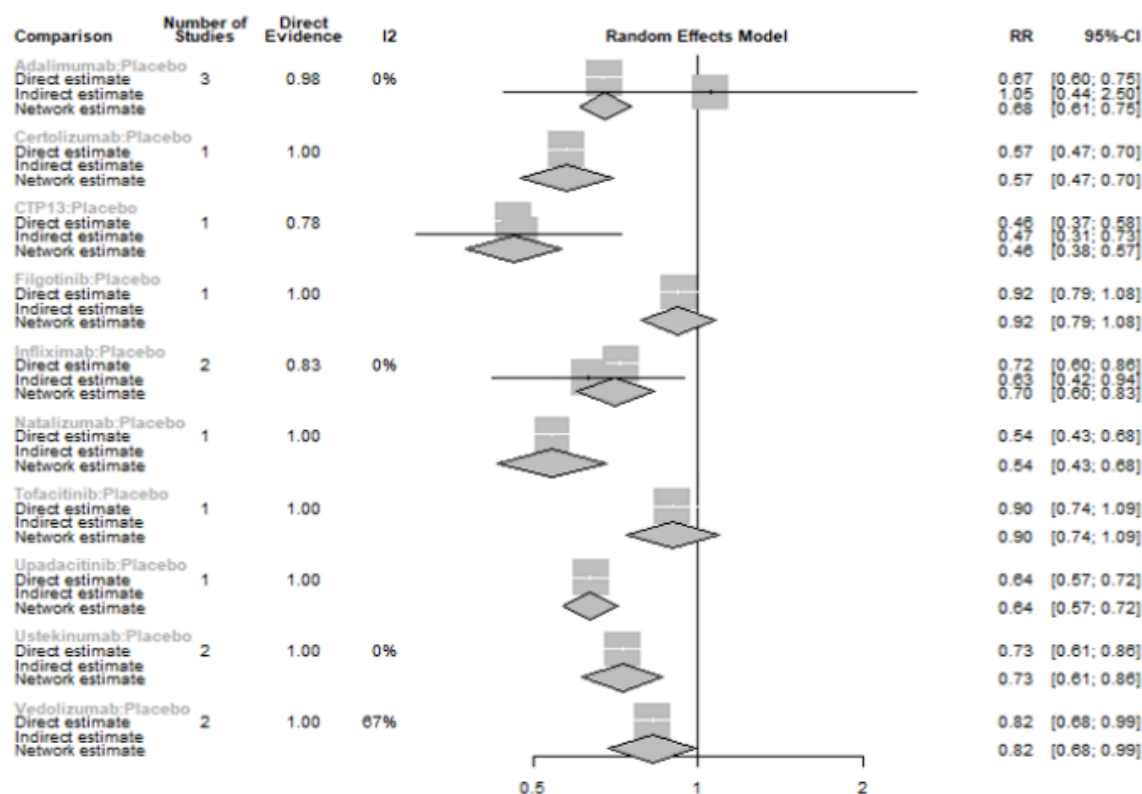


Figure 56. Forest plot for loss of clinical response during maintenance

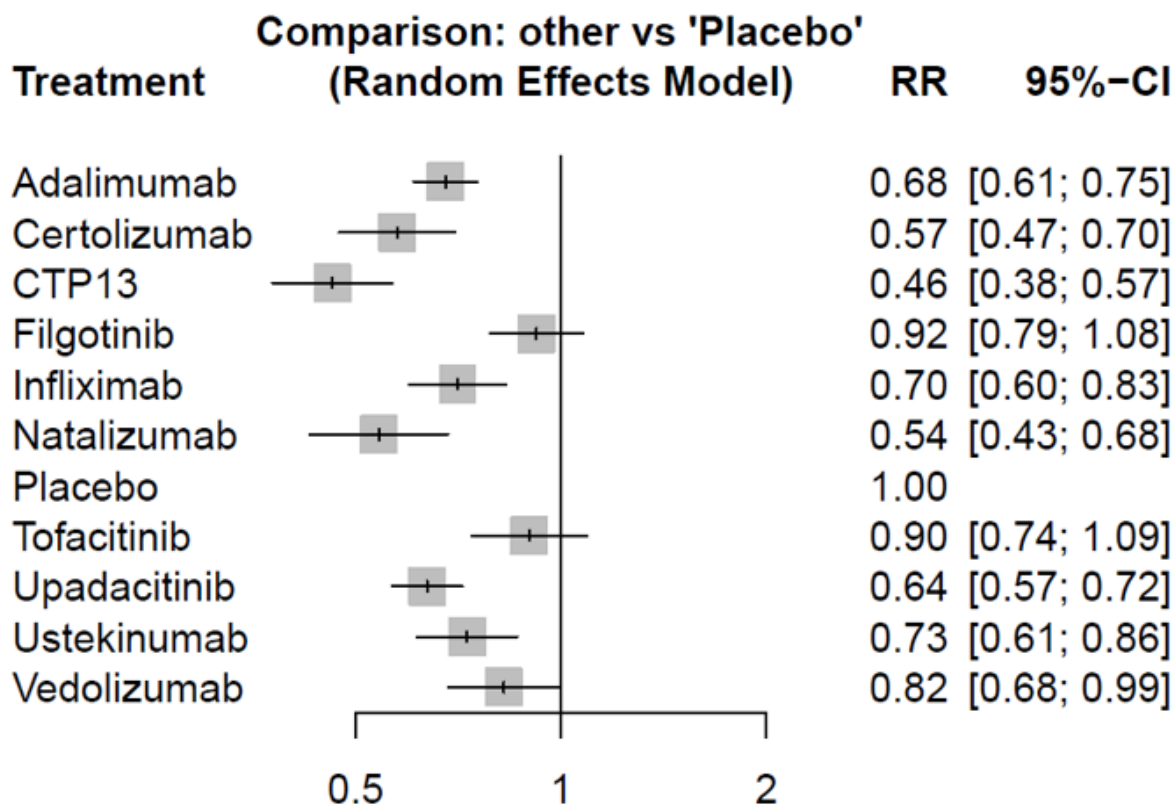


Figure 57. SUCRA rankings for maintenance of clinical response

	SUCRA (common)	SUCRA (random)
CTP13	0.981	0.972
Natalizumab	0.860	0.867
Certolizumab	0.793	0.809
Upadacitinib	0.666	0.664
Adalimumab	0.582	0.562
Infliximab	0.503	0.535
Ustekinumab	0.477	0.457
Vedolizumab	0.280	0.286
Tofacitinib	0.185	0.182
Filgotinib	0.138	0.138
Placebo	0.035	0.028

- based on 100 simulations

All studies included in the analysis defined response as a reduction of 70 or 100 points on the CDAI scale, except Jorgensen 2017 (NORSWITCH), which defined disease worsening as a reduction on the HBI scale, and Sands 2004 (ACCENT II) Maintenance (Responders), which defined loss of response as the recrudescence of draining fistulas, the need for a change in medication for Crohn's disease or the need for additional therapy for persistent or worsening luminal disease activity, the need for a surgical procedure for Crohn's disease, or the discontinuation of the study medication owing to a perceived lack of efficacy.

A total of 11 treatments were included in this network from a total of 15 studies (n = 4268 participants), and 17 pairwise comparisons were carried out.

Direct comparisons to placebo included one study on CT-P13 (n = 325), two on infliximab (n = 268), three on adalimumab (n = 627), one on certolizumab (n = 428), one on upadacitinib (n = 502), two on vedolizumab (n = 434), two on ustekinumab (n = 542), one on natalizumab (n = 428), one on tofacitinib (n = 180) and one on filgotinib (n = 329).

There is high-certainty evidence that upadacitinib is effective in preventing loss of response compared to placebo (RR 0.64, 95% CI 0.57 to 0.72, SUCRA = 0.69, moderate effect size).

There is moderate-certainty evidence that CTP13 (RR 0.46, 95% CI 0.38 to 0.57, SUCRA = 0.98, large effect size), natalizumab (RR 0.54, 95% CI 0.43 to 0.68, SUCRA = 0.86, moderate effect size), certolizumab (RR 0.57, 95% CI 0.47 to 0.7, SUCRA = 0.81, moderate effect), adalimumab (RR 0.68, 95% CI 0.61 to 0.75, SUCRA = 0.58, small effect) and ustekinumab (RR 0.73, 95% CI 0.61 to 0.86, SUCRA

= 0.44, small effect) are probably more effective than placebo in preventing loss of clinical response in Crohn's disease.

There is low-certainty evidence that infliximab (RR 0.70, 95% CI 0.60 to 0.83, SUCRA = 0.5, small effect) and vedolizumab (RR 0.82, 95% CI 0.68 to 0.99, SUCRA = 0.28, trivial effect) may be more effective than placebo in preventing loss of clinical response in Crohn's disease.

There is low-certainty evidence that there may be no difference between tofacitinib (RR 0.9, 95% CI 0.74 to 1.09), filgotinib (RR 0.92, 95% CI 0.79 to 1.08) and placebo in preventing loss of clinical response in Crohn's disease.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The tau² for this outcome was 0 and the I² = 0% (total Q = 5.79, P value = 0.56). The Chi² test results and P values for assessing inconsistency within study designs (Q = 4.75, P value = 0.44) showed no statistically significant differences. Between study designs (Q = 1.04, P value = 0.59) there were no statistically significant differences.

Endoscopic relapse

There was not enough data to conduct a network meta-analysis for this outcome as only six studies reported data (Buhl 2022 (STOP-IT); Colombel 2023 - LIBERTY-CD [649, 650, 651]; Ferrante 2022 - FORTIFY; Loftus 2023 (U-ENDURE); Rutgeerts 2012 (EXTEND); Sandborn 2023 - BERGAMOT Maintenance). The extracted data can be found in Table 6.

Withdrawals due to adverse events

A GORDON plot, comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all

treatments compared to placebo, and SUCRA rankings can be seen in [Figure 58](#), [Figure 59](#), [Figure 60](#), [Figure 61](#), [Figure 62](#) and [Figure 63](#). The comparison-adjusted funnel plot did not reveal large imbalances.

Figure 58. Gordon plot for withdrawals due to adverse events for biologics for maintenance of remission in Crohn's disease, with placebo as reference treatment

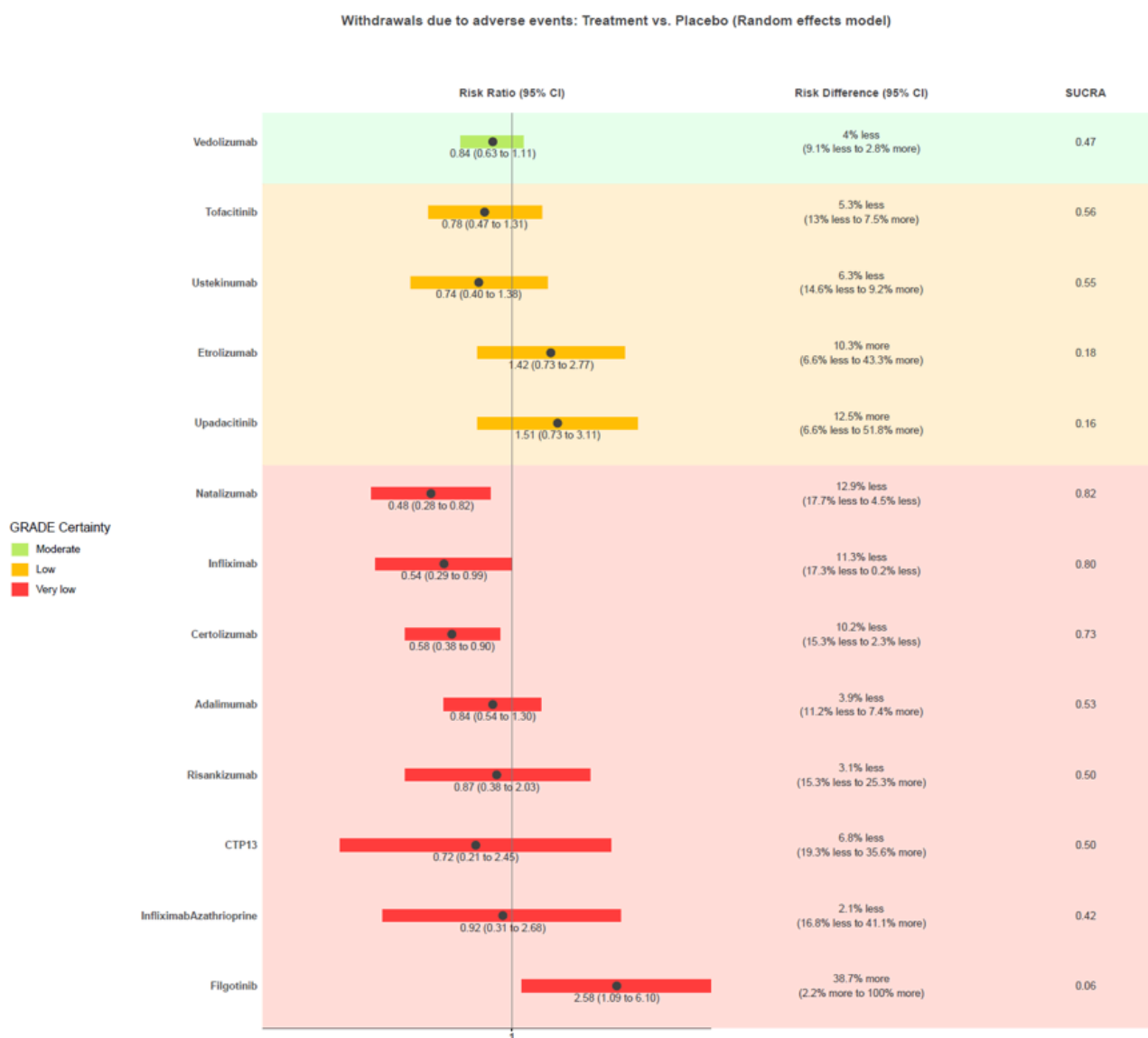


Figure 59. Comparison-adjusted funnel plot for withdrawals due to adverse events during maintenance

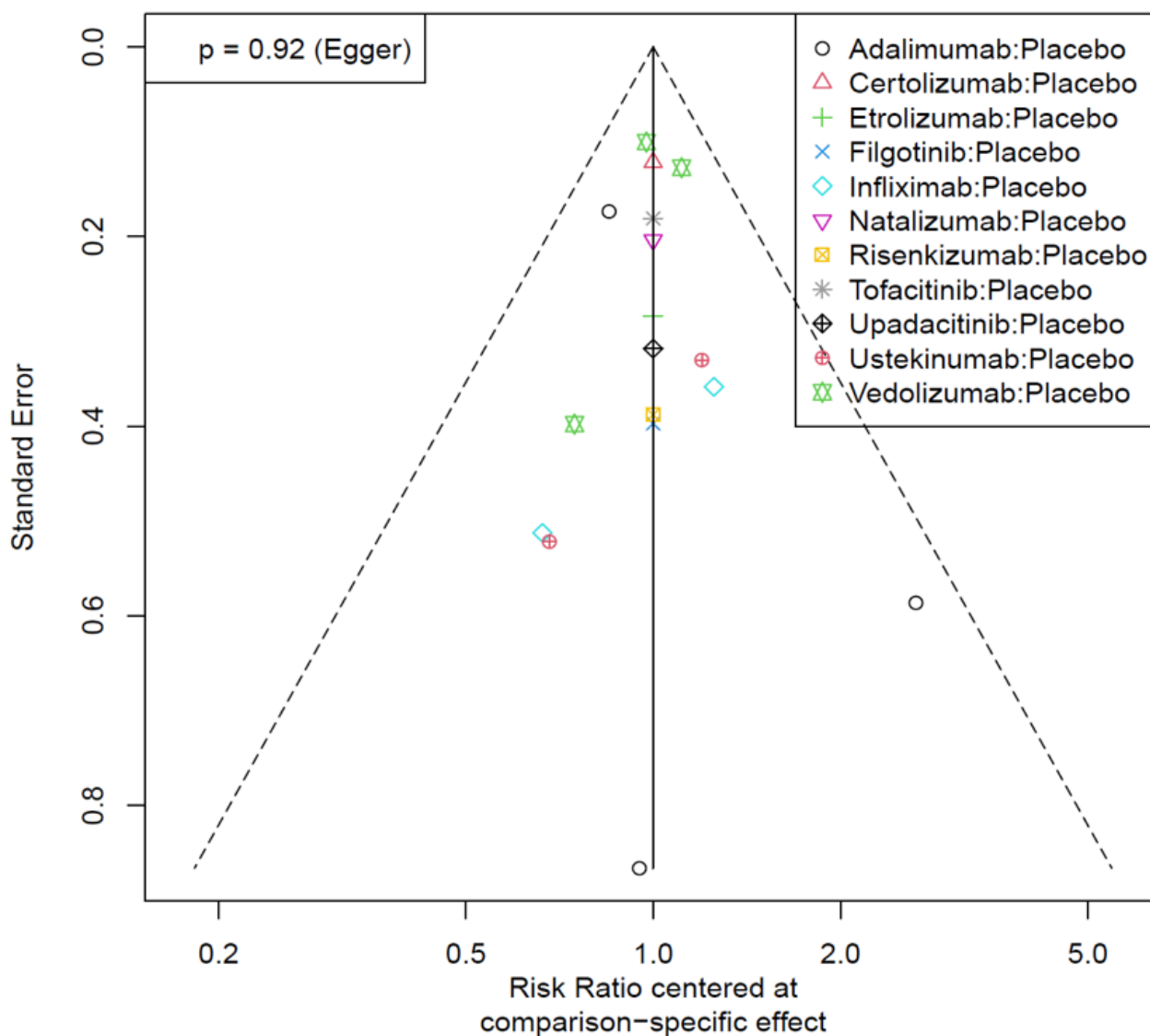


Figure 60. Network plot for withdrawals due to adverse events during maintenance (shaded region denotes multi-arm studies)

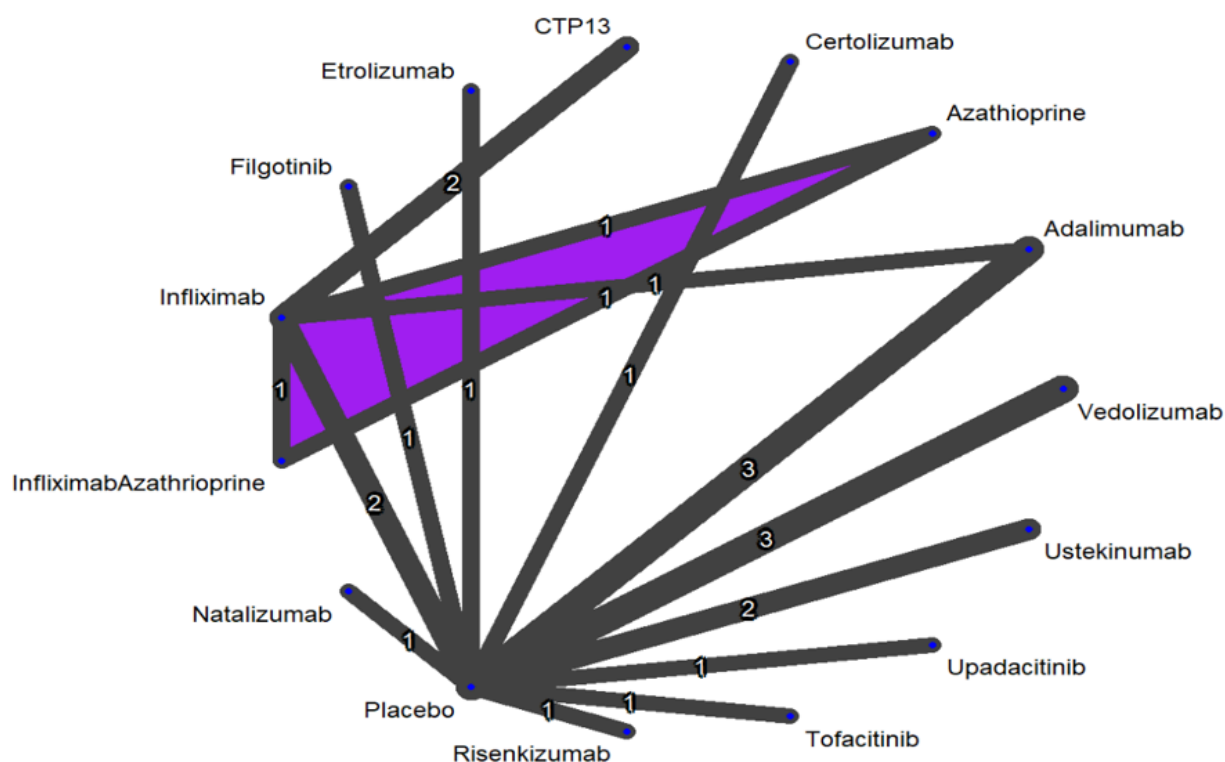


Figure 61. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for withdrawals due to adverse events during maintenance

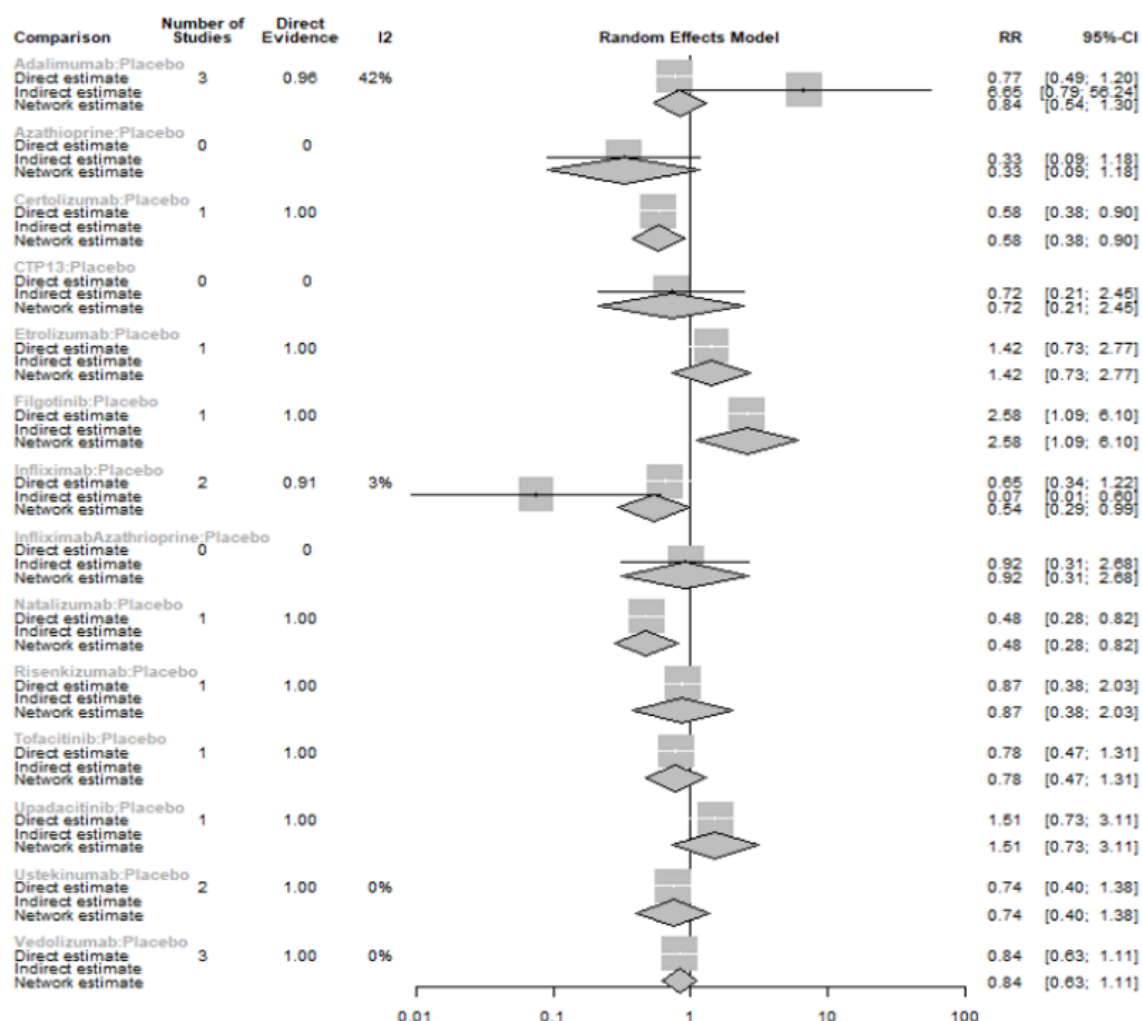


Figure 62. Forest plot for withdrawals due to adverse events during maintenance

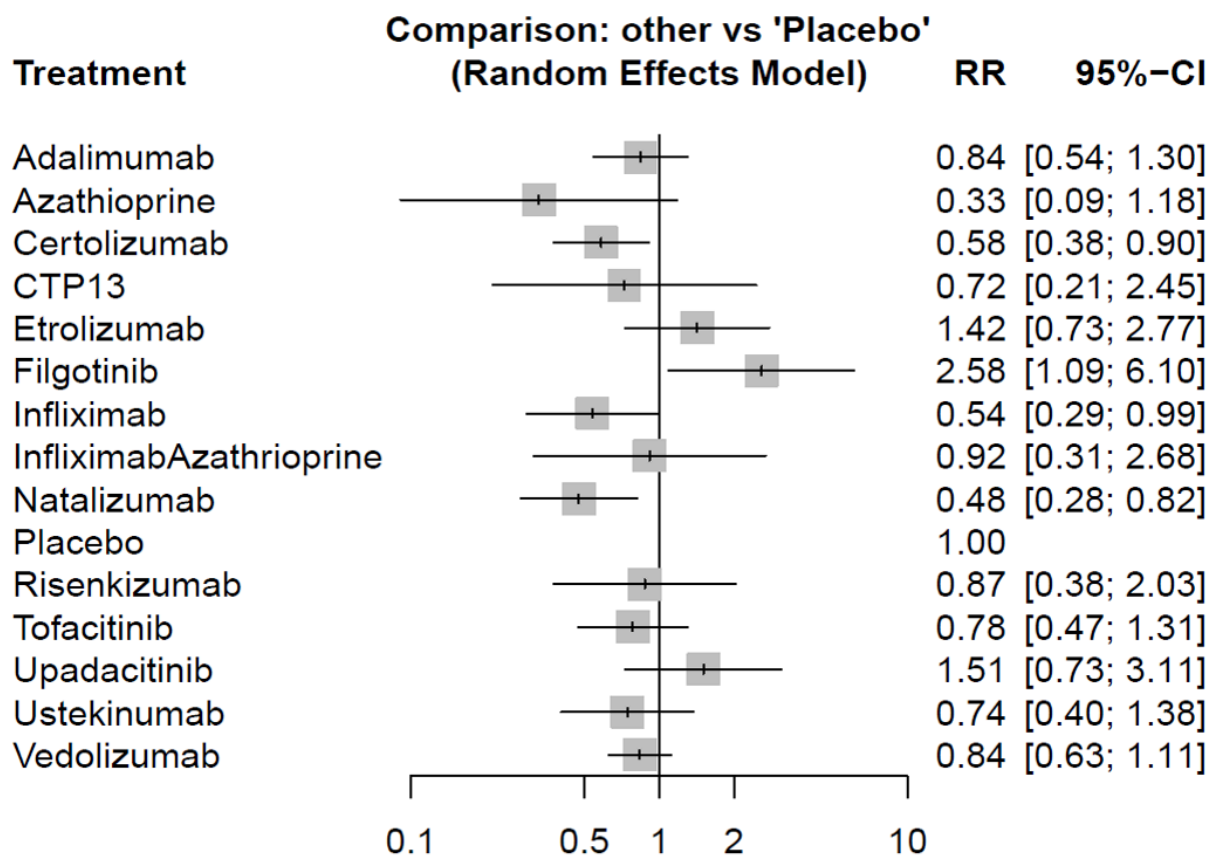


Figure 63. SUCRA rankings for withdrawals due to adverse events during maintenance

	SUCRA (common)	SUCRA (random)
Azathioprine	0.9064	0.8707
Natalizumab	0.8586	0.8507
Infliximab	0.7850	0.7671
Certolizumab	0.7643	0.7329
CTP13	0.5607	0.5964
Ustekinumab	0.5621	0.5600
Tofacitinib	0.5057	0.5500
Adalimumab	0.5479	0.4829
Vedolizumab	0.4671	0.4743
InfliximabAzathrioprine	0.4336	0.4650
Risenkizumab	0.4536	0.4650
Placebo	0.3079	0.3314
Etrolizumab	0.1821	0.1657
Upadacitinib	0.1336	0.1579
Filgotinib	0.0314	0.0300

- based on 100 simulations

A total of 15 treatments were included in this network from a total of 21 studies (n = 6149 participants), and 23 pairwise comparisons were carried out.

Direct comparisons to placebo included one study on CT-P13 (n = 325), two on infliximab (n = 268), three on adalimumab (n = 234), one on certolizumab (n = 428), three on vedolizumab (n = 895), two on ustekinumab (n = 542), one on natalizumab (n = 428), one on tofacitinib (n = 180), one on etrolizumab (n = 434) and one on filgotinib (n = 329).

We did not judge the evidence for any treatments at high certainty using GRADE for this outcome.

There is moderate-certainty evidence that vedolizumab (RR 0.84, 95% CI 0.63 to 1.11) is probably no different to placebo in withdrawals due to adverse events during maintenance.

There is low-certainty evidence that tofacitinib (RR 0.78, 95% CI 0.47 to 1.31), ustekinumab (RR 0.74, 95% CI 0.4 to 1.38), etrolizumab (RR 1.42, 95% CI 0.73 to 2.77) and upadacitinib (RR 1.51, 95% CI 0.73 to

3.11) may be no different to placebo in withdrawals due to adverse events during maintenance.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The tau² for this outcome was 0.351 and the I² = 31.8% (total Q = 11.74, P value = 0.1633). The Chi² test results and P values for assessing inconsistency within study designs (Q = 7.38, P value = 0.39) showed no statistically significant differences. Between study designs (Q = 4.36, P value = 0.0368) there were no local designs that could be assessed.

Serious adverse events

A GORDON plot, comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all treatments compared to placebo, and SUCRA rankings can be seen in [Figure 64](#), [Figure 65](#), [Figure 66](#), [Figure 67](#), [Figure 68](#) and [Figure 69](#). The comparison-adjusted funnel plot revealed potential imbalances.

Figure 64. Gordon plot for serious adverse events for biologics for maintenance of remission in Crohn's disease, with placebo as reference treatment

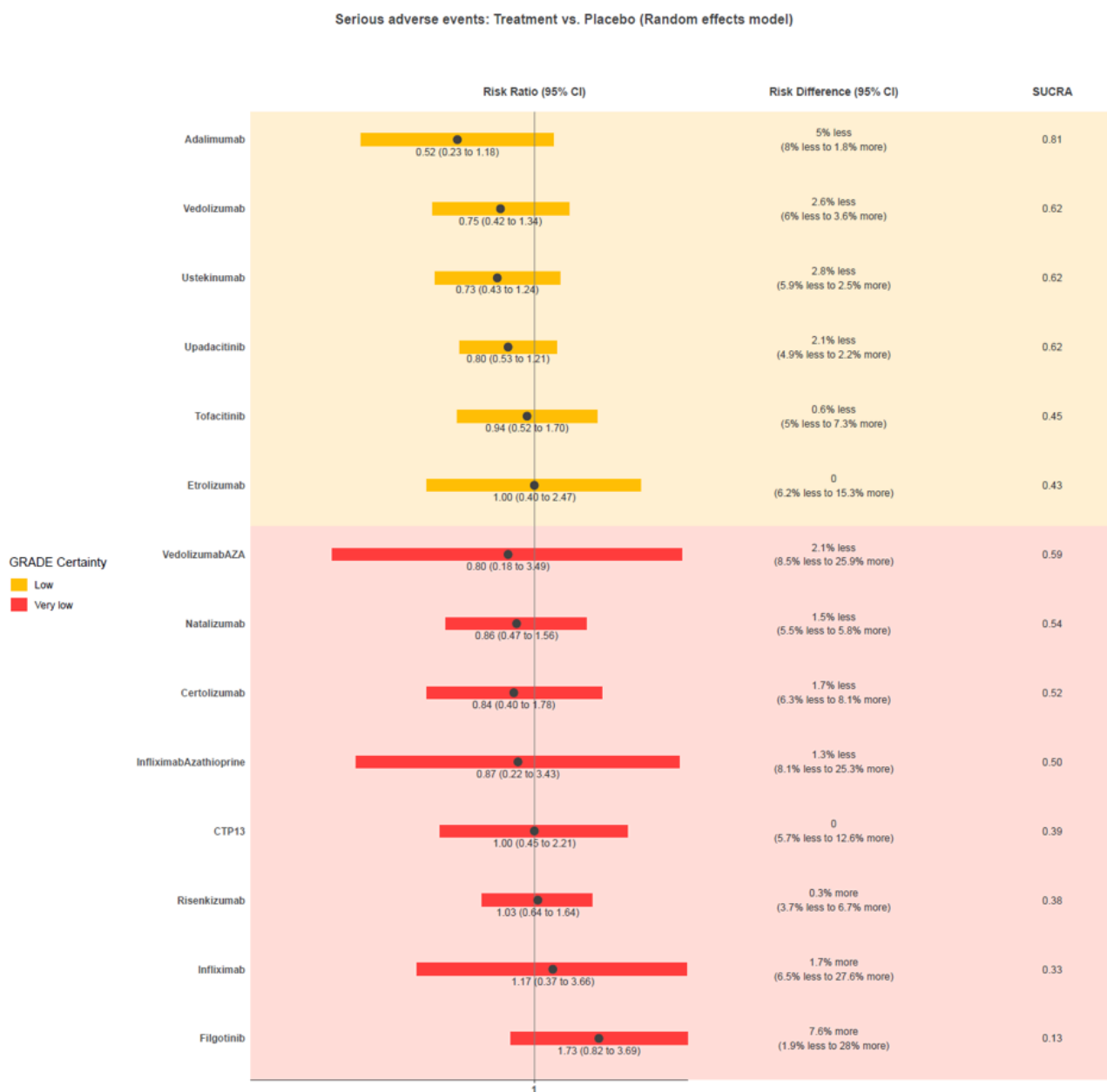


Figure 65. Comparison-adjusted funnel plot for serious adverse events during maintenance

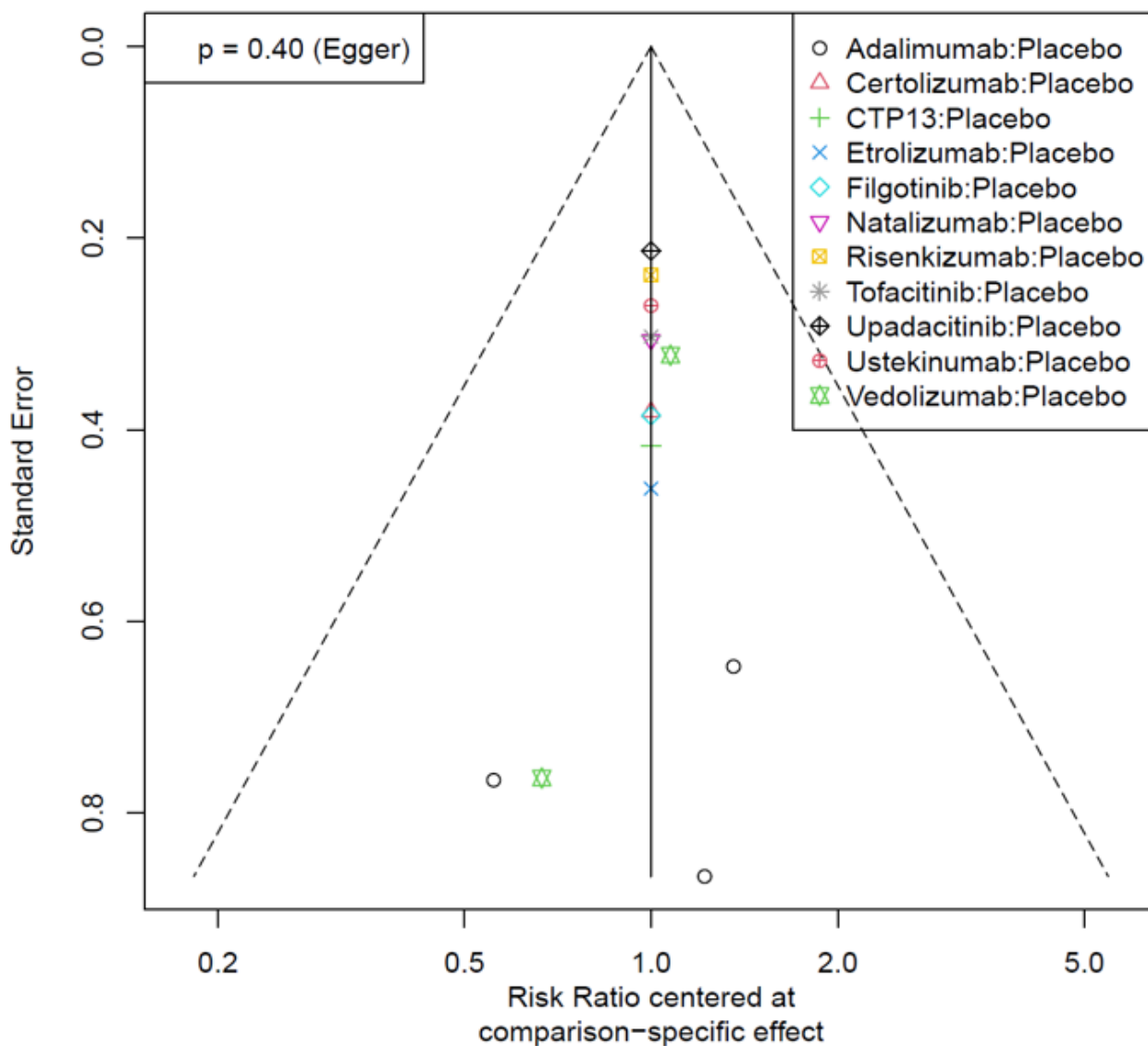


Figure 66. Network plot for serious adverse events during maintenance (shaded region denotes multi-arm studies)

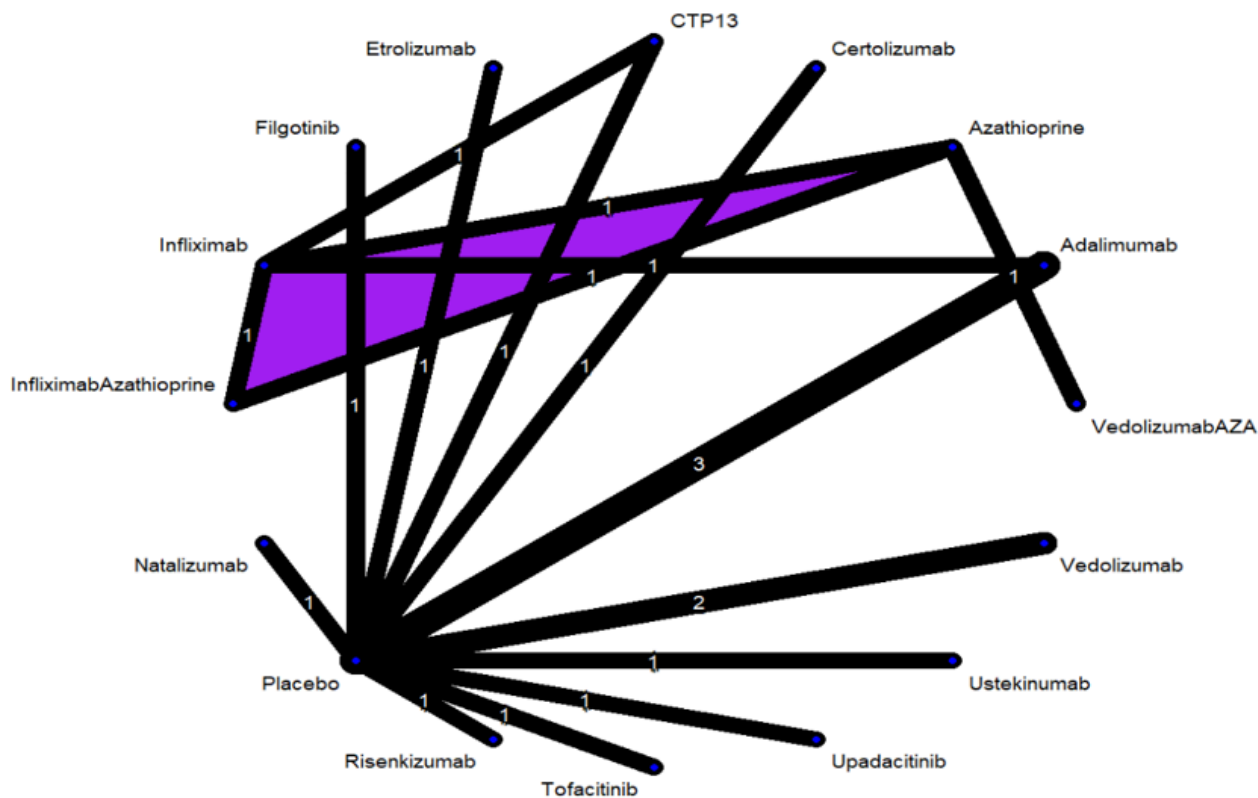


Figure 67. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for serious adverse events during maintenance

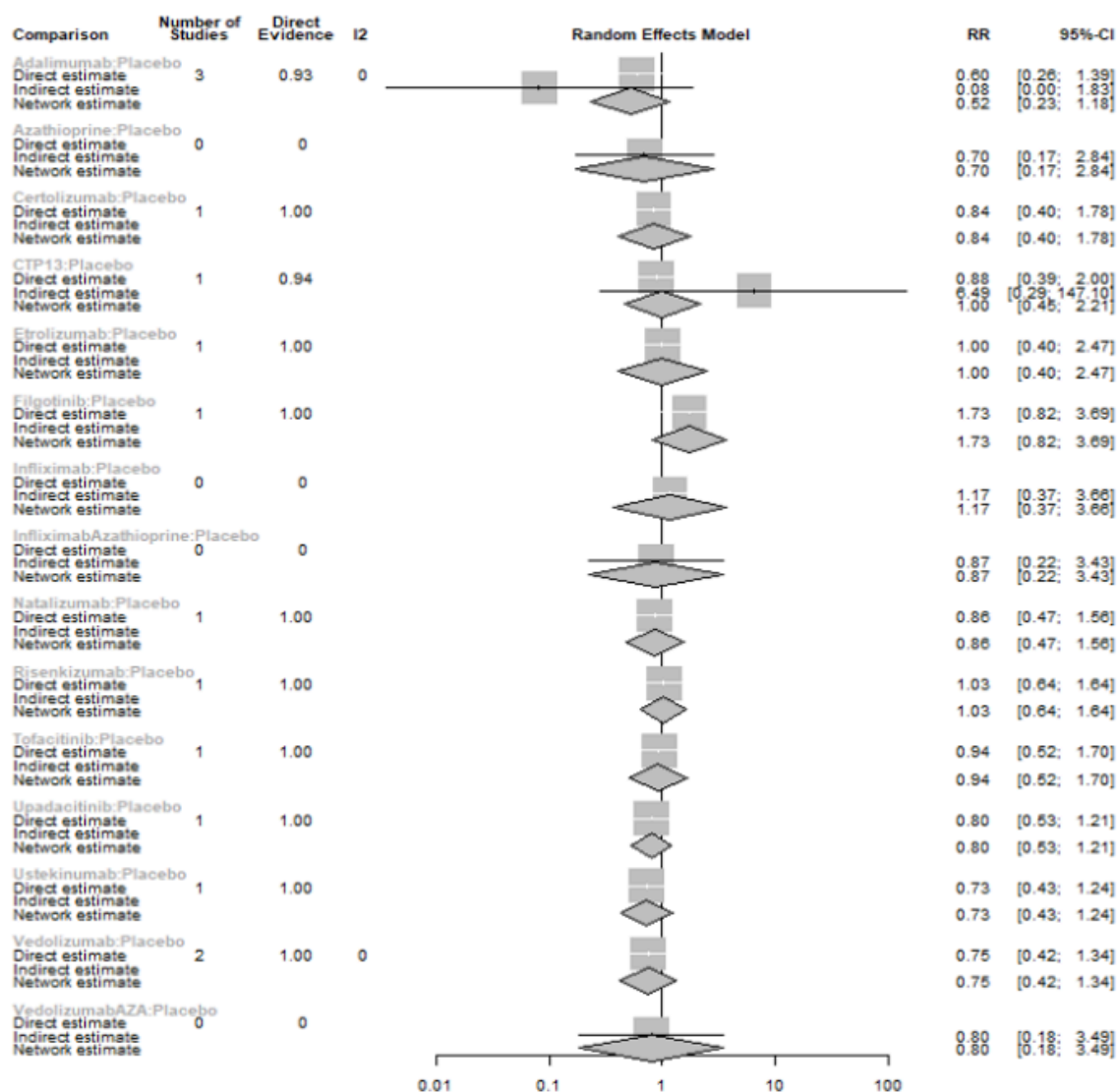


Figure 68. Forest plot for serious adverse events during maintenance

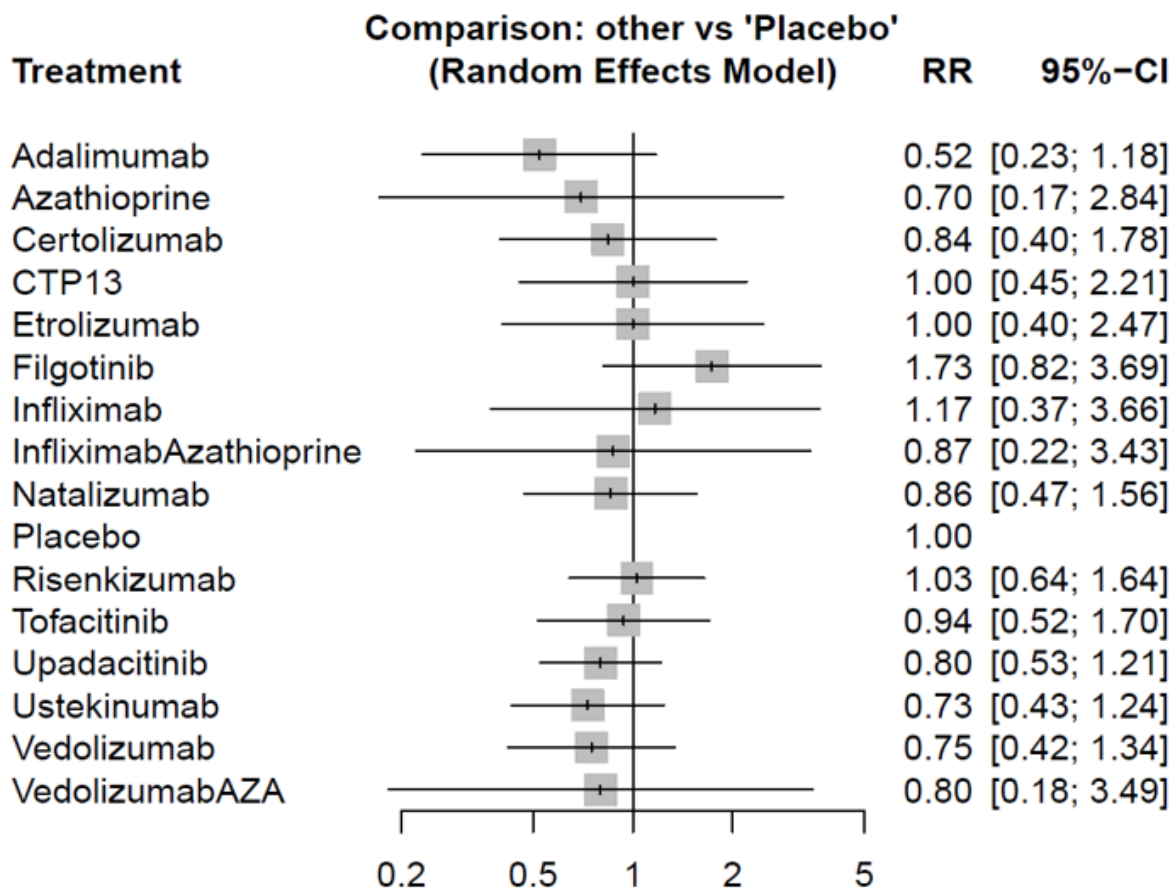


Figure 69. SUCRA rankings for serious adverse events during maintenance

	SUCRA (common)	SUCRA (random)
Adalimumab	0.8380	0.8320
Ustekinumab	0.6487	0.6553
Vedolizumab	0.6460	0.6227
Upadacitinib	0.6047	0.6087
Azathioprine	0.6020	0.6007
InfliximabAzathioprine	0.5567	0.5633
VedolizumabAZA	0.5180	0.5373
Certolizumab	0.5253	0.5187
Natalizumab	0.4933	0.4893
Tofacitinib	0.4667	0.4673
Etrolizumab	0.4367	0.4500
CTP13	0.4053	0.4087
Placebo	0.3793	0.3800
Risenkizumab	0.4247	0.3713
Infliximab	0.3347	0.3327
Filgotinib	0.1200	0.1620

A total of 16 treatments were included in this network from a total of 18 studies (n = 6032 participants), and 20 pairwise comparisons were carried out.

Direct comparisons to placebo included one study on CT-P13 (n = 325), three on adalimumab (n = 234), one on certolizumab (n = 428), two on vedolizumab (n = 434), one on ustekinumab (n = 397), one on natalizumab (n = 428), one on tofacitinib (n = 180), one on etrolizumab (n = 434), one on risankizumab (n = 542), one upadacitinib (n = 673) and one on filgotinib (n = 329).

We did not judge the evidence for any treatments at high or moderate certainty using GRADE for this outcome.

There is low-certainty evidence that adalimumab (RR 0.52, 95% CI 0.23 to 1.18), vedolizumab (RR 0.75, 95% CI 0.42 to 1.34), ustekinumab (RR 0.73, 95% CI 0.43 to 1.24), upadacitinib (RR 0.8, 95% CI 0.53 to 1.21), tofacitinib (RR 0.94, 95% CI 0.52 to 1.7) and etrolizumab (RR 1, 95% CI 0.4 to 2.47) may be no different to placebo for serious adverse events during maintenance.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The tau² for this outcome was 0 and the I² = 0% (total Q = 2.66, P value = 0.6162). The Chi² test results and P values for assessing inconsistency within study designs (Q = 1.19, P value = 0.755) showed no statistically significant differences. Between study designs (Q = 1.47, P value = 0.2256) there were no local designs that could be assessed.

Total adverse events

A GORDON plot, comparison-adjusted funnel plot, network plot, direct, indirect and network estimates plot, forest plot of all treatments compared to placebo, and SUCRA rankings can be seen in [Figure 70](#), [Figure 71](#), [Figure 72](#), [Figure 73](#), [Figure 74](#) and [Figure 75](#). The comparison-adjusted funnel plot did not reveal large imbalances.

Figure 70. Gordon plot for total adverse events for biologics for maintenance of remission in Crohn's disease, with placebo as reference treatment

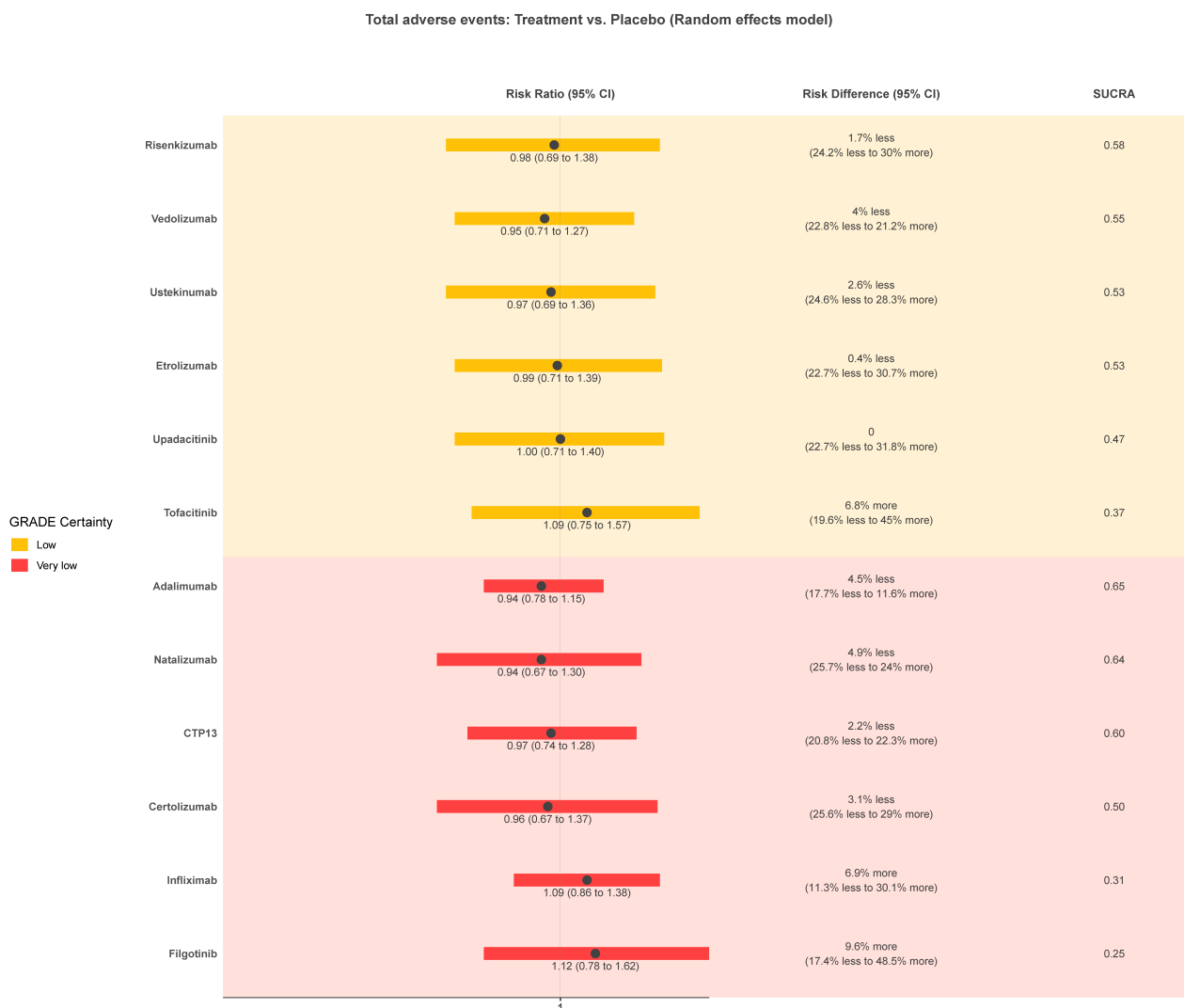


Figure 71. Comparison-adjusted funnel plot for total adverse events during maintenance

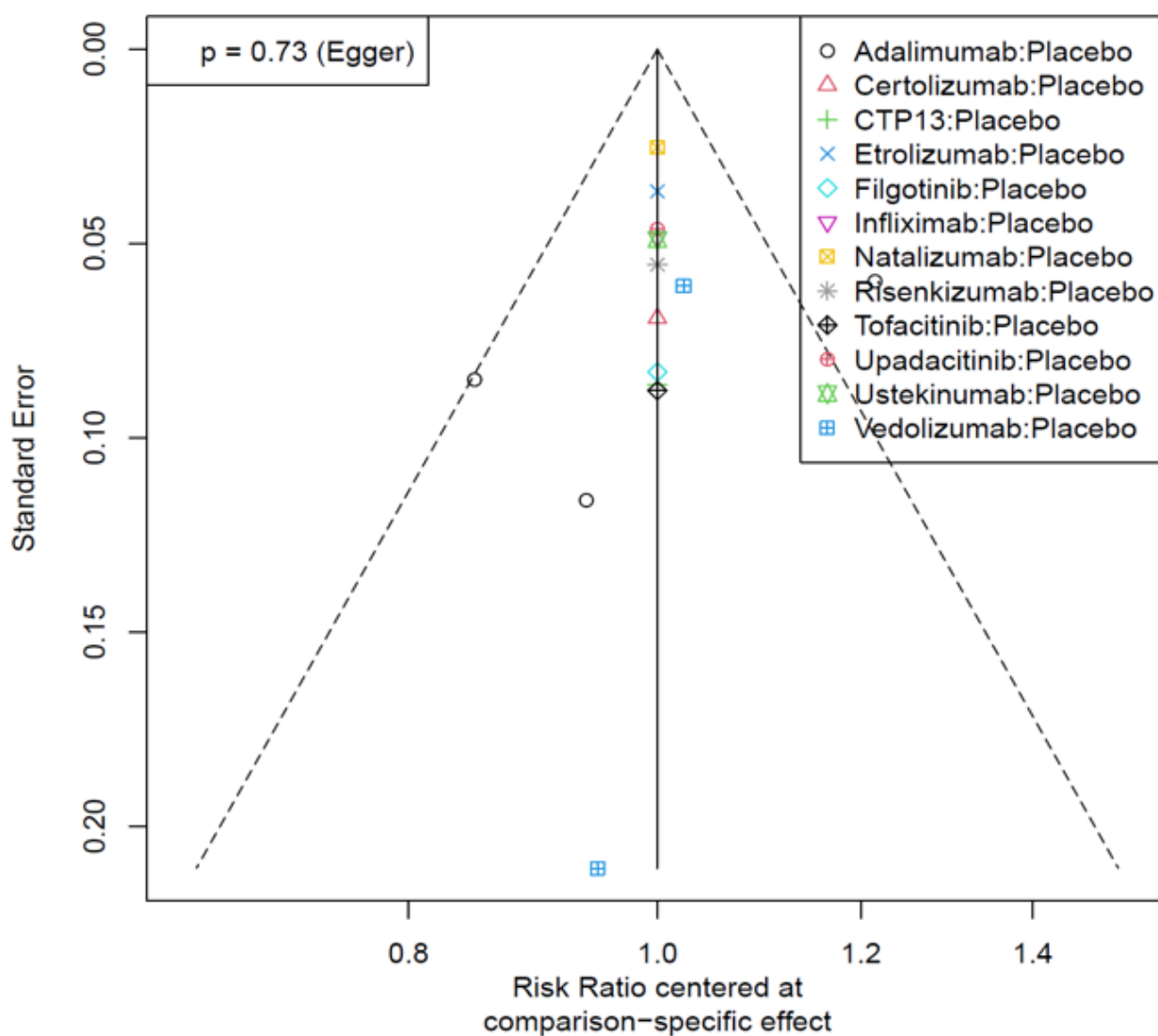


Figure 72. Network plot for total adverse events during maintenance

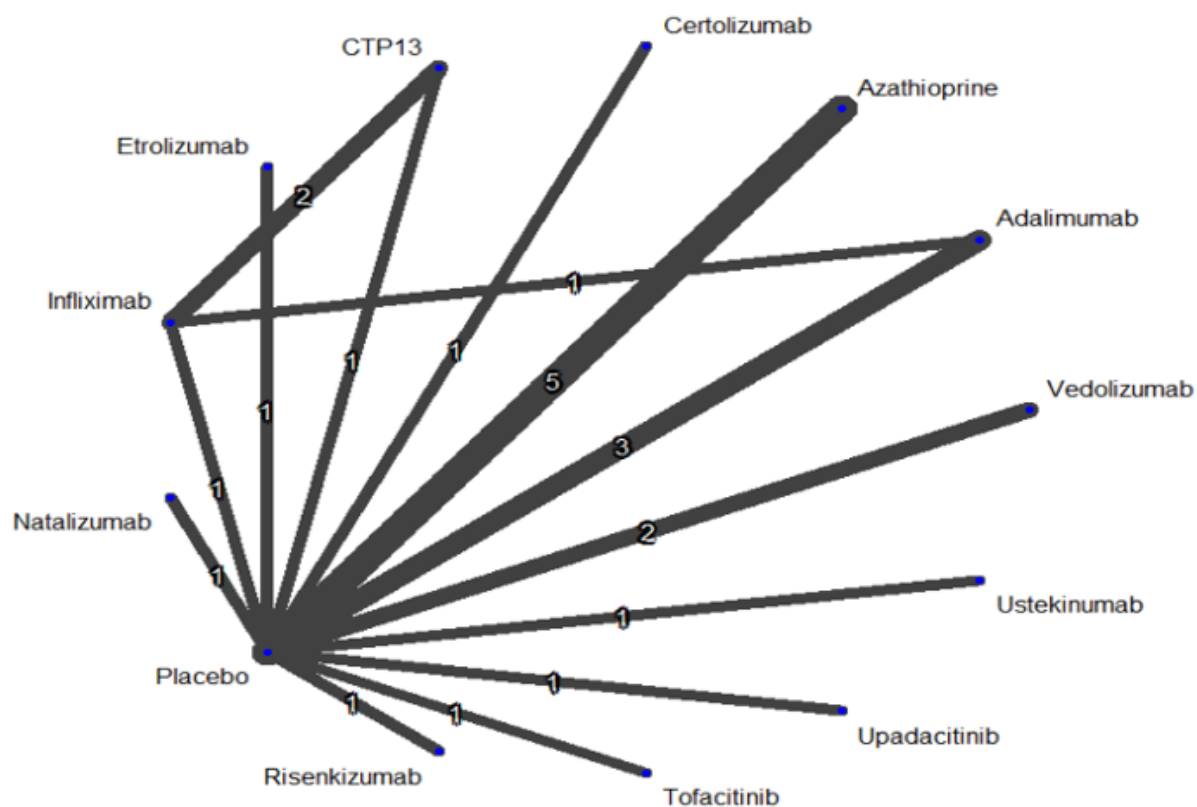


Figure 73. Direct, indirect, and network estimates, number of direct studies vs placebo, contribution of direct evidence to the network estimate, and I^2 for pairwise comparisons for total adverse events during maintenance

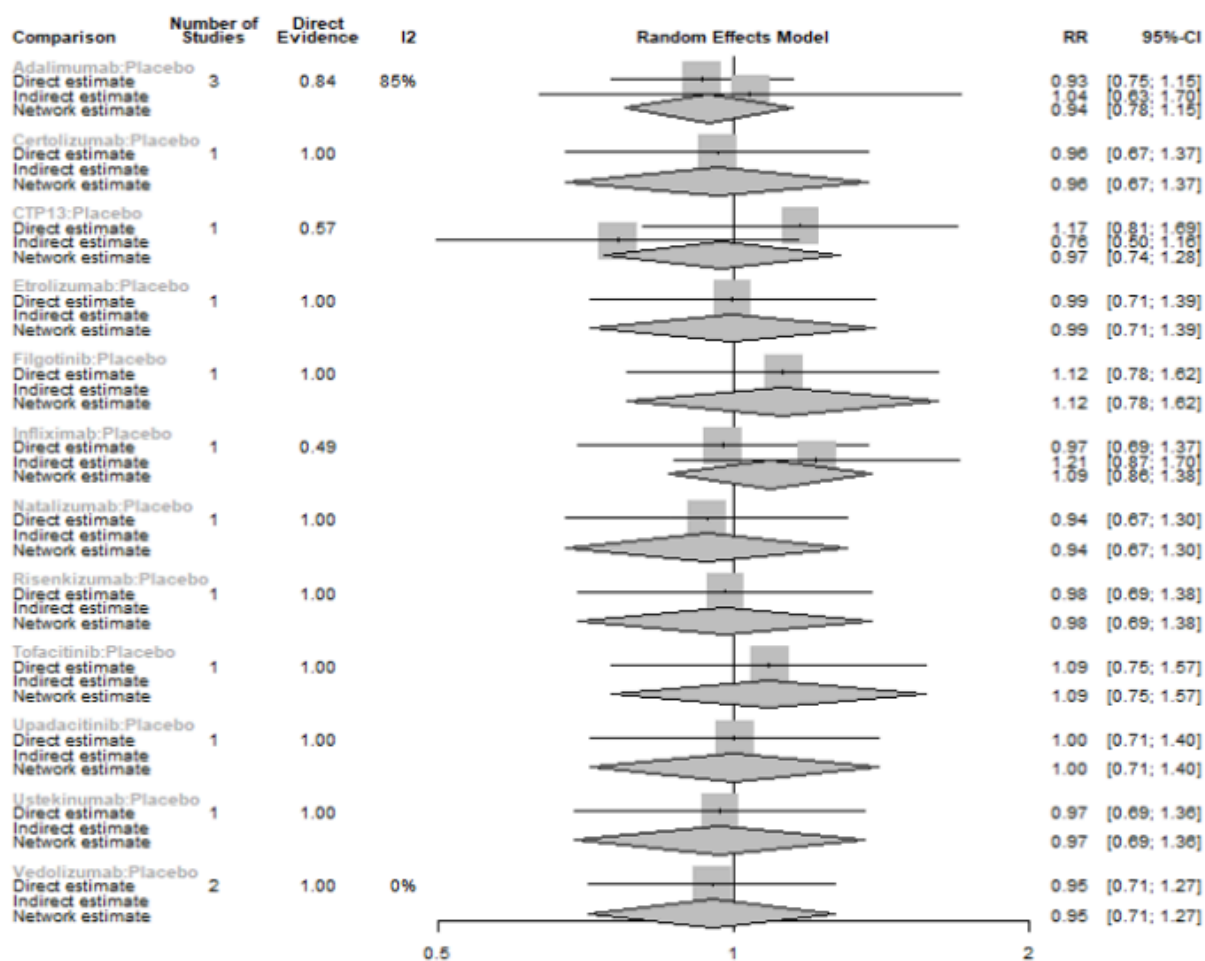


Figure 74. Forest plot for total adverse events during maintenance

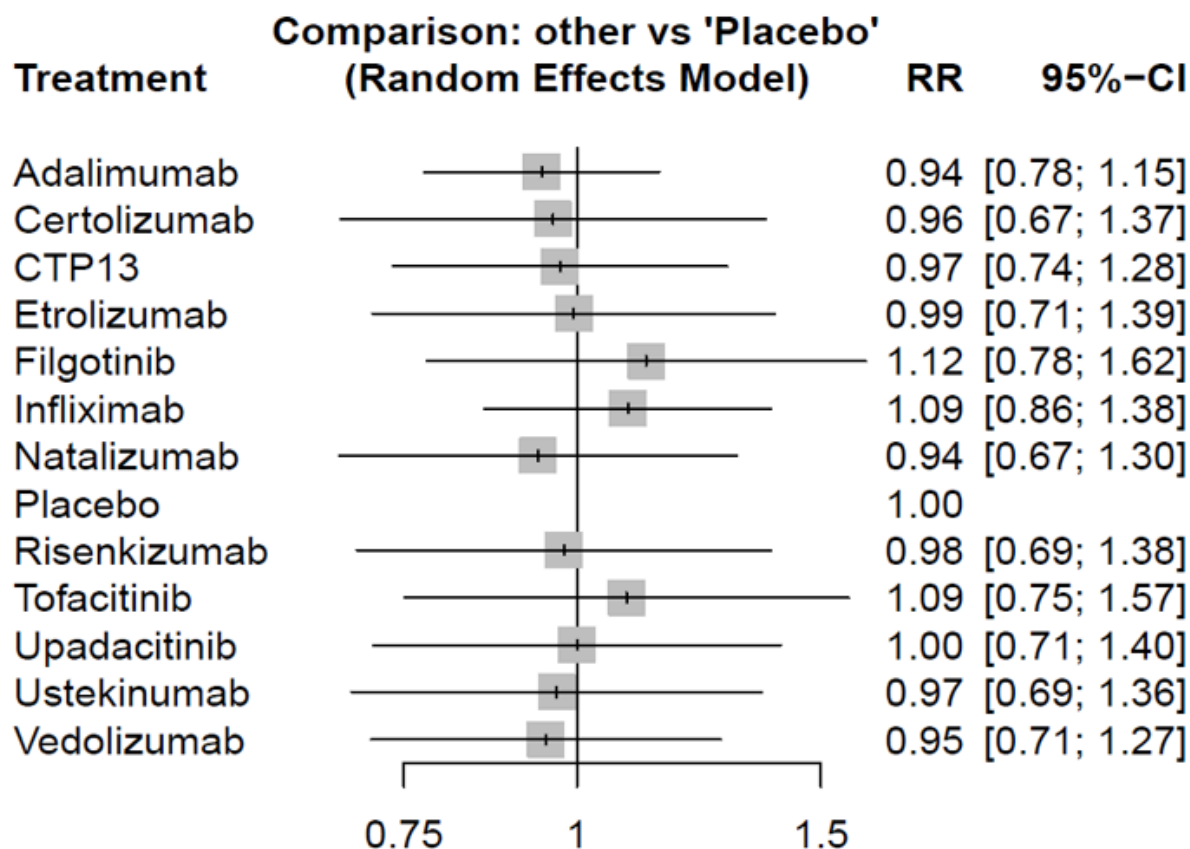


Figure 75. SUCRA rankings for total adverse events during maintenance

	SUCRA (common)	SUCRA (random)
Adalimumab	0.6000	0.6617
Natalizumab	0.8492	0.6342
Certolizumab	0.6683	0.5875
CTP13	0.4108	0.5642
Vedolizumab	0.6250	0.5450
Ustekinumab	0.6083	0.5192
Placebo	0.4458	0.5133
Risankizumab	0.6125	0.5092
Upadacitinib	0.5683	0.5025
Etrolizumab	0.4808	0.4683
Filgotinib	0.0942	0.3458
Tofacitinib	0.1908	0.3417
Infliximab	0.3458	0.3075

- based on 100 simulations

A total of 13 treatments were included in this network from a total of 18 studies (n = 5151 participants), and 18 pairwise comparisons were carried out.

Direct comparisons to placebo included one study on CT-P13 (n = 325), one on infliximab (n = 73), three on adalimumab (n = 234), one on certolizumab (n = 428), two on vedolizumab (n = 434), one on ustekinumab (n = 397), one on natalizumab (n = 428), one on tofacitinib (n = 180), one on etrolizumab (n = 434), one on risankizumab (n = 542), one upadacitinib (n = 673) and one on filgotinib (n = 329).

We did not judge the evidence for any treatments at high or moderate certainty using GRADE for this outcome.

There is low-certainty evidence that risankizumab (RR 0.98, 95% CI 0.69 to 1.38), vedolizumab (RR 0.95, 95% CI 0.71 to 1.27), ustekinumab (RR 0.97, 95% CI 0.69 to 1.36), etrolizumab (RR 0.99, 95% CI 0.71 to 1.39), upadacitinib (RR 1, 95% CI 0.71 to 1.4) and tofacitinib (RR 1.09, 95% CI 0.75 to 1.57) may be no different to placebo for total adverse events during maintenance.

The evidence for all other treatments was of very low certainty and no conclusions can be drawn.

The tau² for this outcome was 0.0278 and the I² = 73.6% (total Q = 22.73, P value = 0.0009). The Chi² test results and P values for assessing inconsistency within study designs (Q = 15.27, P value

= 0.0042) showed the design within adalimumab and placebo (Q = 13.18, P value = 0.0014) to be statistically significant. Between study designs (Q = 7.47, P value = 0.0239), the designs between adalimumab and infliximab (Q = 7.39, P value = 0.0066) and adalimumab and placebo (Q = 7.39, P value = 0.0066) were statistically significant.

Equity assessment

No equity assessment was planned for this review.

Reporting biases

Assessing publication bias with funnel plots was not possible as no pairwise comparisons included at least 10 studies.

DISCUSSION

Summary of main results

Network meta-analysis showed that for our critical outcome for induction therapy, 'clinical remission', ustekinumab (number needed to benefit, NNTB = 6) had high-certainty evidence for efficacy, while adalimumab with purine analogues (NNTB = 3), guselkumab (NNTB = 4), adalimumab (NNTB = 4) upadacitinib (NNT = 7), vedolizumab (NNT = 10) and natalizumab (NNT = 15) had moderate-certainty evidence.

Risankizumab, BI695501 and the combination of infliximab with purine analogues were high on the rankings with low-certainty evidence.

It is of note that subgroup analyses for studies' different participant baseline characteristics regarding previous biologic exposure found no major differences in the magnitude or ranking of effects.

Upadacitinib (NNTB = 3) and risankizumab (NNTB = 6) showed efficacy for endoscopic remission with moderate-certainty evidence.

Upadacitinib, ustekinumab and vedolizumab were found to be similar to placebo with moderate and high-certainty evidence across all safety outcomes. For other treatments, no clear conclusions can be drawn due to low or very low-certainty evidence.

Network meta-analysis showed that for our critical outcome for maintenance therapy, 'clinical relapse', adalimumab (NNT = 2) had moderate-certainty evidence for efficacy, and CTP13, infliximab and upadacitinib had low-certainty evidence. Vedolizumab, ustekinumab, tofacitinib and etrolizumab may be no different to placebo.

With regard to our safety outcomes, vedolizumab showed moderate-certainty evidence for no difference to placebo in withdrawals due to adverse events and low-certainty evidence of no difference for serious and total events. All other treatments had low or very low-certainty evidence for no difference to placebo.

Results were limited by a sparse network.

Limitations of the evidence included in the review

We thoroughly reviewed the studies for quality and assessed bias. Overall, the risk of bias of studies was predominately low.

There were results for outcomes on GRADE analysis with high or moderate certainty in our network analysis, but imprecision was the main reason for downgrading, while remaining lack of clarity for risk of bias also played a role in downgrading.

Reporting of adverse events was also sparse during maintenance and long term and so this was reflected in the GRADE analysis.

The evidence is complete and applicable in a number of ways. Evidence on the full range of biological therapies has been included and this has allowed effect, magnitude and certainty to be considered. The results of sensitivity analysis enhance the reliability of the findings and subgroup analysis, accounting for the key issue of previous biologic exposure at induction, adds to the utility of the findings. The transitivity assumption that underpins the use of our methods appears to be met and there was limited incoherence in our outcomes.

There are a number of ways in which the evidence is lacking. The majority of studies included in this review used the CDAI to recruit patients to studies and assess clinical remission and response at different endpoints, which is considered the standard tool in Crohn's disease research. However, other measures of disease state and in turn endpoints for studies, such as endoscopic measures, were limited in reporting and networks could not be produced.

Moreover, heterogeneity of participants at entry is a factor that we have attempted to transparently present and account for in subgroup and sensitivity analysis, but was still a key area in which the primary evidence and, in turn, the findings of the review are limited when compared to other contexts of study [652, 653, 654].

The role of concomitant purine analogues is a key issue that must be commented on. Whilst most studies did not explicitly seek to investigate concomitant purine use or compare to purines, many studies included more than 50% of people on such therapy and, as such, assumptions were made to manage this issue. This may limit the utility of our results.

A limited number of head-to-head trials have been conducted thus far (Allez 2023 (TRIDENT- Part 2); Peyrin-Biroulet 2024 - SEQUENCE; Sandborn 2022 - GALAXI 1; Sands 2022 (SEAVUE)), which have produced outcome results that have been used in our networks. However, our networks remain relatively sparse, with the clear majority of the evidence comparing single agents to placebo.

The reporting of adverse events is another area of concern. It is not uncommon to experience difficulties in reporting related to the heterogeneity of thresholds for defining serious or severe events and, as such, withdrawals are often the most available measure. For common effects, randomised trial data can be sufficient to consider safety [655], but this is not the case for rarer and possibly more devastating outcomes where long-term safety data are not addressed by this synthesis.

For our analyses, we have used all dosages compared in the original RCTs, including non-licensed and dosages not used in clinical practice, and data from phase 2b trials that may not have progressed to phase 3. Whilst we do not think this might have affected our efficacy data, it might have potentially affected our safety outcomes and must be considered by those utilising our review.

In this review we have included biologic and advanced treatments irrespective of whether they have received approval by regulatory authorities. This was to capture the complete evidence base in our networks and because new treatments may share similar medication class properties with their same-class approved treatments.

We have excluded non-randomised, treat-through maintenance designs according to our review inclusion criteria as they are not RCTs, ensuring only trials with properly randomised designs have been included.

Studies solely comparing purine analogue or steroid monotherapy to placebo have not been included in this review. They appear in some of our plots because some studies included them in comparison to advanced treatments, and we have included all studies on advanced treatment irrespective of what they were compared to. As purine analogue and steroid monotherapies were not in the scope of the review, no assessments have been made of these treatments, and they are not discussed in this review.

Finally, the sample size of trials has resulted in issues with precision and limited analyses for maintenance therapy. This is a pervasive issue within the field [656, 657].

Limitations of the review processes

Gaps in information to judge risk of bias were initially pervasive. We chose to contact the study authors for clarification or additional information, with many authors responding. The use of such unpublished data can also be seen as a source of bias.

We are aware of the possibility of industry funding influencing the validity of the results. Funding from manufacturing companies or any conflicts of interests from both primary studies and the review team have been reported, but are almost universal within the field and so must be considered as a constant and overarching potential source of bias.

The methodological approaches and assumptions in the review have been transparently reported, discussed and the impact of such assumptions tested in sensitivity analysis. However, of particular note is the handling of concomitant purine analogues. This was an unexpectedly pervasive issue within the evidence. The approach used represents the consensus of the team, editorial and methods support from Cochrane, and clinical advice from experts in the field. However, this must be considered a key potential source of bias.

Agreements and disagreements with other studies or reviews

The results broadly support current international guidelines in the UK [658], Europe [659] and North America [660] on the role of established therapies. For novel therapies, these are not discussed in any of these guidelines and, as such, the evidence in this review may support them in updates. None of these guidelines employed network meta-analysis data and so this is not reflected in their guidance. Most of these guidelines propose a role for a combination of purine and biologic. However, within the network, purine combinations had low or very low-certainty evidence for efficacy and safety. The newest British Society of Gastroenterology guidelines on IBD have used network meta-analyses for their formulation and have arrived at their recommendations using the GRADE evidence to decision framework based on evidence similar to the results of this review [661]. It is worth noting that the results of any network meta-analysis should not be interpreted as a clinical guideline on their own. Any evidence needs the clinical interpretation of a guideline development group, which needs to carefully consider all aspects that need to be considered when making recommendations.

Our results agree with the most recent Cochrane pairwise meta-analyses on the individual agents for which they have been conducted in terms of the similarity in effect estimations and GRADE certainty assessments [20, 21, 22, 23, 24, 25, 26].

Broadly, our results agree with recent NMAs on the subject, even though one key difference is that these tend to rely on statistical rankings and omit assessments of the certainty of the evidence. Our review relies primarily on GRADE assessments, which, in combination with our pre-defined thresholds for magnitude size, has allowed us to make more transparent and nuanced conclusions [662, 663].

AUTHORS' CONCLUSIONS

Implications for practice

For people with active Crohn's disease, for induction of clinical remission, we found the following results from our network meta-analysis:

- The most effective interventions when compared to placebo were ustekinumab (number needed to treat for an additional beneficial outcome, NNTB = 6), adalimumab with purine analogues (NNTB = 3), adalimumab (NNTB = 4), guselkumab (NNTB = 4), upadacitinib (NNTB = 7), vedolizumab (NNTB = 10) and natalizumab (NNTB = 15), which were all effective with moderate and high-certainty evidence.
- Risankizumab, infliximab with purine analogues, infliximab, filgotinib and BI695501 (adalimumab biosimilar) had low-certainty evidence for efficacy.
- Tofacitinib and apilimod mesylate may be no different to placebo.
- Upadacitinib, ustekinumab and vedolizumab were found to be similar to placebo with moderate and high-certainty evidence across all safety outcomes.

For people with Crohn's disease in remission, for maintenance therapy, we found the following results for prevention of clinical relapse from our network meta-analysis:

- Adalimumab (NNTB = 2) had moderate-certainty evidence for efficacy. CTP13, infliximab and upadacitinib had low-certainty evidence for efficacy.
- Vedolizumab, ustekinumab, tofacitinib and etrolizumab may be no different to placebo.
- All treatments had mostly low or very low-certainty evidence regarding safety outcomes.

Results were limited by a sparse network.

Implications for research

This network meta-analysis has supported evidence generation, but head-to-head comparative trials for key interventional subclasses or specific agents may be needed, as guided by key stakeholders. One example would be the use of biosimilar versions of medications out of patent.

The role of concomitant purine analogues has been a key confounding factor and future studies may not only want to investigate specifically the role of combinations, but also consider stratifying populations or purposefully excluding participants based on this key class of therapy to avoid such heterogeneity.

Given that the majority of evidence focusses on the outcomes of clinical remission and relapse, future studies may want to further consider other outcomes of clear interest to clinicians and patients, particularly endoscopic remission.

Finally, safety data are limited, especially during maintenance and in the long term. Whilst future studies with longer follow-ups could provide increased data, it may be that study designs outside of randomised trials are employed for such outcomes.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD012751.pub2](https://doi.org/10.1002/14651858.CD012751.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Appendix 1: Approved dose and route of administration

ADDITIONAL INFORMATION

Acknowledgements

Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Grigorios Leontiadis, McMaster University; Toby Lasserson, Cochrane Evidence Production and Methods Directorate;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Samuel Hinsley and Hannah Payne, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Cynthia Stafford, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Jenny Bellorini, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Joana Torres, Hospital da Luz, Lisbon, Portugal (clinical/content review); Siddharth Singh, Division of Gastroenterology, University of California San Diego (clinical/content review); Brian Duncan (consumer review); Jennifer Hilgart, Cochrane (methods review); Sofia Tsokani, Methods Support Unit, Cochrane (statistical reviewer); Jo Platt, Central Editorial Information Specialist (search review).

Contributions of authors

Morris Gordon: secured funding, designed and developed, screened, extracted, resolved conflicts, assessed certainty, contributed to writing and editing, advised on, approved the final version prior to submission and is a guarantor of the review.

Vassiliki Sinopoulou: screened, extracted, assessed certainty, contributed to writing and editing, did administrative work, advised on, approved the final version prior to submission.

Muhammad Ibrahim Khan, Masitara Masitara, Alireza Sherafat, Rose Cherayath: screened, extracted, did administrative work,

contributed to writing and editing, approved the final version prior to submission.

Assem Elleithy, Fareed Alqusous, Sanjana Phlananthachai: screened, did administrative work, approved the final version prior to submission.

Anthony K Akobeng: contributed to validation, reviewed and approved the final version prior to submission.

Gordon William Moran: contributed to supervision, analysis, validation, reviewed and approved the final version prior to submission.

Suzanna C Freeman: provided statistical network meta-analysis guidance.

The authors of the protocol of this review (published in 2017) are no longer included on the author byline: Pieter Hindryckx, GY Zou, Brian G Feagan, Sushil K Garg, Jasvinder A Singh, Triana Lobaton, Siddharth Singh, Claire E Parker, Tran M Nguyen, Orli M Silverberg, Reena Khanna and Vipul Jairath. Some of the content retained in this review reflects their contributions.

Declarations of interest

Vassiliki Sinopoulou has nothing to disclose.

Morris Gordon is Co-Editor of Cochrane Gut. He was not involved in the editorial process for this review.

Gordon Moran was a consultant for Abbvie for a regulatory submission. Gordon discloses a collaborator grant from both AstraZeneca AB and Janssen Biotech. He also discloses a contract/grant with both Bristol Myers Squibb Company and Pfizer.

Anthony Akobeng is Co-ordinating Editor, Cochrane Gut. He was not involved in the editorial process for this review. Anthony has published opinions on the treatment of Crohn's disease in a number of peer-reviewed medical journals and is a Consultant Clinical Adviser for the National Institute for Care and Health Excellence, UK.

Fareed Al Qusous has nothing to disclose.

Rose Cherayath has nothing to disclose.

Assem Elleithy has nothing to disclose.

Suzanne Freeman: Suzanne Freeman is a member of the NIHR Evidence Synthesis Group @CRSU, a member of the NIHR Complex Reviews Support Unit and holds an NIHR Advanced Fellowship. All of these grants/funds are paid directly to the University of Leicester, who pay Suzanne's salary. Suzanne is also an academic member of the Biostatistics Research Group at the University of Leicester and is supported by the NIHR Applied Research Collaboration (ARC) East Midlands and Leicester NIHR Biomedical Research Centre (BRC). These funds are all paid to the University of Leicester. The University of Leicester pay Suzanne's salary with funding for some of her time from the National Institute for Health Research.

Muhammad Ibrahim Khan has nothing to disclose.

Masitara Masitara has nothing to disclose.

Sanjana Phlananthachai has nothing to disclose.

Alireza Sherafat has nothing to disclose.

Sources of support

Internal sources

- None, Other

None

External sources

- NIHR grant, UK

This study/project is funded by an NIHR Evidence Synthesis Programme Grant (NIHR132748). The views expressed are those of the author(s) and not necessarily those of the NIHR or the Department of Health and Social Care.

Registration and protocol

A protocol for this review was published in 2017: doi.org/10.1002/14651858.CD012751 [29].

History

Protocol first published: Issue 8, 2017

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (see [Supplementary material 1](#); [Supplementary material 2](#); [Supplementary material 3](#); [Supplementary material 4](#); [Supplementary material 5](#); [Supplementary material 6](#); [Supplementary material 7](#)): full search strategies for each database; full citations of each unique report for all studies (included, ongoing or awaiting classification, or excluded at the full text screen) in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates and individual data rows. Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. The following scripts and artefacts were used to generate analyses outside of RevMan: R statistical package, gordonplots.shinyapps.io/shiny_gordon_plots/. Template data extraction forms from (Covidence, Excel, etc.) are available from the authors on reasonable request.

REFERENCES

1. Boyapati R, Satsangi J, Ho G-T. Pathogenesis of Crohn's disease. *F1000Prime Reports* 2015;**7**:44.
2. Molodecky NA, Soon IS, Rabi DM, Ghali WA, Ferris M, Chernoff G, et al. Increasing incidence and prevalence of the inflammatory bowel diseases with time, based on systematic review. *Gastroenterology* 2012;**142**:46-54.
3. Windsor JW, Kaplan GG. Evolving epidemiology of IBD. *Current Gastroenterology Reports* 2019;**21**:1-9.
4. D'Haens G, Baert F, van Assche G, Caenepeel P, Vergauwe P, Tuynman H, et al. Early combined immunosuppression or conventional management in patients with newly diagnosed Crohn's disease: an open randomised trial. *Lancet* 2008;**371**:660-7.
5. Nguyen NH, Solitano V, Vuyyuru SK, MacDonald JK, Syversen SW, Jørgensen KK, et al. Proactive therapeutic drug monitoring versus conventional management for inflammatory bowel diseases: a systematic review and meta-analysis. *Gastroenterology* 2022;**163**(4):937-49.
6. Dulai PS, Sandborn WJ. Next-generation therapeutics for inflammatory bowel disease. *Current Gastroenterology Reports* 2016;**18**:51.
7. Wetwittayakhleng P, Bessissow T, Lakatos PL. Novel and emerging drugs for the treatment of Crohn's disease: a review of phase II and III trials. *Expert Opinion on Emerging Drugs* 2024;**29**(1):19-34.
8. Levin AD, Wildenberg ME, van den Brink GR. Mechanism of action of anti-TNF therapy in inflammatory bowel disease. *Journal of Crohn's & Colitis* 2016;**10**:989-97.
9. Lima MSR, de Lima VCO, Piuvezam G, de Azevedo KPM, Maciel BLL, Morais AHA. Mechanisms of action of anti-inflammatory proteins and peptides with anti-TNF-alpha activity and their effects on the intestinal barrier: A systematic review. *PLOS One* 2022;**17**(8):e0270749.
10. Sandborn WJ, Hanauer SB, Katz S, Safdi M, Wolf M, Baerg RD. Etanercept for active Crohn's disease: a randomized, double-blind, placebo-controlled trial. *Gastroenterology* 2001;**121**(5):1088-94.
11. Rutgeerts P, Sandborn WJ, Fedorak RN, Rachmilewitz D, Tarabar D, Gibson P. Onercept for moderate-to-severe Crohn's disease: a randomized, double-blind, placebo-controlled trial. *Clinical Gastroenterology and Hepatology* 2006;**4**(7):888-93.
12. Hebuterne X, Lemann M, Bouhnik Y, Dewit O, Dupas JL, Mross M, et al. Endoscopic improvement of mucosal lesions in patients with moderate to severe ileocolonic Crohn's disease following treatment with certolizumab pegol. *Gut* 2013;**62**(2):201-8.
13. Feagan BG, Rutgeerts P, Sands BE, Hanauer S, Colombel JF, Sandborn WJ, et al. Vedolizumab as induction and maintenance therapy for ulcerative colitis. *New England Journal of Medicine* 2013;**369**:699-710.
14. French J, Rosario M, Dirks NL, Milton A, Fox I, Gastonguay MR. Vedolizumab exposure-response relationship during induction therapy in adults with Crohn's disease. *Gastroenterology* 2014;**146**(5 Suppl 1):S592-3.
15. Rosario M, Dirks NL, Milch C, Parikh A, Bargfrede M, Wyant T, et al. A review of the clinical pharmacokinetics, pharmacodynamics, and immunogenicity of vedolizumab. *Clinical Pharmacokinetics* 2017;**56**(11):1287-301. [DOI: [10.1007/s40262-017-0546-0](https://doi.org/10.1007/s40262-017-0546-0)]
16. Feagan BG, Sandborn WJ, Gasink C, Jacobstein D, Lang Y, Friedman JR, et al. Ustekinumab as induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2016;**375**(20):1946-60.
17. Hindryckx P, Baert F, Hart A, Armuzzi A, Panès J, Peyrin-Biroulet L. Clinical trials in luminal Crohn's disease: a historical perspective. *Journal of Crohns and Colitis* 2014;**8**:1339-50.
18. Clark JD, Flanagan ME, Telliez JB. Discovery and development of Janus kinase (JAK) inhibitors for inflammatory diseases. *Journal of Medicinal Chemistry* 2014;**57**:523-38.
19. Kitsou K, Kokkotis G, Rivera-Nieves J, Bamias G. Targeting the sphingosine-1-phosphate pathway: new opportunities in inflammatory bowel disease management. *Drugs* 2024;**84**(10):1179-97.
20. Abbass M, Cepek J, Parker CE, Nguyen TM, MacDonald JK, Feagan BG, et al. Adalimumab for induction of remission in Crohn's disease. *Cochrane Database of Systematic Reviews* 2019, Issue 11. Art. No: CD012878. [DOI: [10.1002/14651858.CD012878.pub2](https://doi.org/10.1002/14651858.CD012878.pub2)]
21. Gordon M, Radford SJ, Eldragini ME, Darie AM, Sinopoulou V, Akobeng AK, et al. Infliximab for medical induction of remission in Crohn's disease. *Cochrane Database of Systematic Reviews* 2023, Issue 11. Art. No: CD012623. [DOI: [10.1002/14651858.CD012623.pub2](https://doi.org/10.1002/14651858.CD012623.pub2)]
22. Gordon M, Sinopoulou V, Akobeng AK, Sarian A, Moran GW. Infliximab for maintenance of medically-induced remission in Crohn's disease. *Cochrane Database of Systematic Reviews* 2024, Issue 2. Art. No: CD012609. [DOI: [10.1002/14651858.CD012609.pub2](https://doi.org/10.1002/14651858.CD012609.pub2)]
23. Hui S, Sinopoulou V, Gordon M, Aali G, Krishna A, Ding NS, et al. Vedolizumab for induction and maintenance of remission in Crohn's disease. *Cochrane Database of Systematic Reviews* 2023, Issue 7. Art. No: CD013611. [DOI: [10.1002/14651858.CD013611.pub2](https://doi.org/10.1002/14651858.CD013611.pub2)]
24. Okabayashi S, Yamazaki H, Yamamoto R, Anan K, Matsuoka K, Kobayashi T, et al. Certolizumab pegol for maintenance of medically induced remission in Crohn's disease. *Cochrane Database of Systematic Reviews* 2022, Issue 6. Art. No: CD013747. [DOI: [10.1002/14651858.CD013747.pub2](https://doi.org/10.1002/14651858.CD013747.pub2)]
25. Townsend CM, Nguyen TM, Cepek J, Abbass M, Parker CE, MacDonald JK, et al. Adalimumab for maintenance of remission in Crohn's disease. *Cochrane Database of*

Systematic Reviews 2020, Issue 5. Art. No: CD012877. [DOI: [10.1002/14651858.CD012877.pub2](https://doi.org/10.1002/14651858.CD012877.pub2)]

26. Yamazaki H, So R, Matsuoka K, Kobayashi T, Shinzaki S, Matsuura M, et al. Certolizumab pegol for induction of remission in Crohn's disease. *Cochrane Database of Systematic Reviews* 2019, Issue 8. Art. No: CD012893. [DOI: [10.1002/14651858.CD012893.pub2](https://doi.org/10.1002/14651858.CD012893.pub2)]

27. Sandborn WJ, Feagan BG, Rutgeerts P, Hanauer S, Colombel JF, Sands BE, et al. Vedolizumab as induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2013;**369**:711-21.

28. Sands BE, Feagan BG, Rutgeerts P, Colombel JF, Sandborn WJ, et al. Effects of vedolizumab induction therapy for patients with Crohn's disease in whom tumor necrosis factor antagonist treatment failed. *Gastroenterology* 2014;**147**:618-27.

29. Hindryckx P, Zou GY, Feagan BG, Garg SK, Singh, et al. Biologic drugs for induction and maintenance of remission in Crohn's disease: a network meta-analysis. *Cochrane Database of Systematic Reviews* 2017;**8**:CD012751. [DOI: [10.1002/14651858.CD012751](https://doi.org/10.1002/14651858.CD012751)]

30. Lefebvre C, Glanville J, Briscoe S, Littlewood A, Marshall C, Metzendorf MI, et al. Searching for and selecting studies. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews Interventions*. 2nd edition. Chichester (UK): John Wiley & Sons, 2019:67-107.

31. Glanville J, Foxlee R, Wisniewski S, Noel-Storr A, Edwards M, Dooley G. Translating the Cochrane EMBASE RCT filter from the Ovid interface to Embase.com: a case study. *Health Information and Libraries Journal* 2019;**36**(3):264-77. [PMID: 31328866]

32. Aali G, Shokraneh F. No limitations to language, date, publication type, and publication status in search step of systematic reviews. *Journal of Clinical Epidemiology* 2021;**133**:165-7. [DOI: [10.1016/j.jclinepi.2021.02.002](https://doi.org/10.1016/j.jclinepi.2021.02.002)] [PMID: 33571633]

33. Review Manager Web (RevMan Web). Version 1.22.0. The Cochrane Collaboration, 2020. Available at <https://revman.cochrane.org>.

34. Higgins JP, Altman DG, Sterne JAC (editors). Chapter 8: Assessing risk of bias in included studies. In: Higgins JPT, Green S (editors). *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1.0 [updated March 2011]. The Cochrane Collaboration, 2011. <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v5.1.0>.

35. Higgins JP, Jackson D, Barrett JK, Lu G, Ades AE, White IR. Consistency and inconsistency in network meta-analysis: concepts and models for multi-arm studies. *Research Synthesis Methods* 2012;**3**(2):98-110.

36. Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Cochrane, 2022. Available from <https://www.cochrane.org/handbook>.

37. Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;**315**(7109):629-34.

38. Chaimani A, Salanti G. Visualizing assumptions and results in network meta-analysis: the network graphs package. *Stata Journal* 2015;**15**(4):905-50.

39. Sterne JA, Sutton AJ, Ioannidis JP, Terrin N, Jones DR, Lau J, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ* 2011;**343**:d4002.

40. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Controlled Clinical Trials* 1986;**7**(3):177-88.

41. R Development Core Team. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing 2009.

42. Rucker G, Schwarzer G, Krahn U, König, J. Package "netmeta": Network meta-analysis using frequentist methods (Version 0.7-0). <http://cran.r-project.org/web/packages/netmeta/netmeta.pdf> 2015.

43. Salanti G, Ades AE, Ioannidis JP. Graphical methods and numerical summaries for presenting results from multiple-treatment meta-analysis: an overview and tutorial. *Journal of Clinical Epidemiology* 2011;**64**(2):163-71.

44. Chaimani A, Caldwell DM, Li T, Higgins JP, Salanti G. Chapter 11: Undertaking network meta-analyses. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from <https://www.cochrane.org/handbook>.

45. Balduzzi S, Rücker G, Nikolakopoulou A, Papakonstantinou T, Salanti G, Efthimiou O, et al. Netmeta: an R package for network meta-analysis using frequentist methods. *Journal of Statistical Software* 2023;**23**(106):1-40.

46. Schünemann H, Brozek J, Guyatt G, Oxman A, editors. Handbook for Grading the Quality of Evidence and the Strength of Recommendations Using the GRADE Approach (updated October 2013). GRADE Working Group, 2013.

47. GRADEpro GDT. Version accessed 6 August 2023. Hamilton (ON): McMaster University (developed by Evidence Prime), 2022.

48. Brignardello-Petersen R, Bonner A, Alexander PE, Siemieniuk RA, Furukawa TA, Rochwerf B, et al. Corrigendum to "Advances in the GRADE approach to rate the certainty in estimates from a network meta-analysis" [J Clin Epidemiol 2018; 93: 36-44]. *Journal of Clinical Epidemiology* 2018;**98**:162.

49. Puhan MA, Schünemann HJ, Murad MH, Li T, Brignardello-Petersen R, Singh JA, et al. A GRADE Working Group approach for rating the quality of treatment effect estimates from network meta-analysis. *BMJ* 2014;**349**:g5630.

50. Schünemann HJ, Higgins JP, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins JPT, Thomas

J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.2 (updated February 2021). Cochrane, 2021. Available from <https://www.cochrane.org/handbook>.

51. Brignardello-Petersen R, Mustafa RA, Siemieniuk RA, Murad MH, Agoritsas T, Izcovich A et al. GRADE approach to rate the certainty from a network meta-analysis: addressing incoherence. *Journal of Clinical Epidemiology* 2019;**1**(108):77-85.

52. Gordon M, Shaban N, Sinopoulou V, Vuyyuru S, Radford S, Magro F, et al. Developing IBD outcome effect size thresholds to inform research, guidelines, and clinical decisions. *Inflammatory Bowel Diseases* 2025;**31**(10):2798-2804. [DOI: [10.1093/ibd/izaf085](https://doi.org/10.1093/ibd/izaf085)]

53. Allez M, Sands BE, Feagan BG, D'Haens G, De Hertogh G, Randall CW, et al. A phase 2b, randomised, double-blind, placebo-controlled, parallel-arm, multicenter study evaluating the safety and efficacy of tesnatilimab in patients with moderately to severely active Crohn's disease. *Journal of Crohn's & Colitis* 2023;**no volume**:jjad047.

54. EUCTR2016-000634-21. A phase 2b, randomized, double-blind, placebo-controlled, parallel-group, multicenter protocol to evaluate the safety and efficacy of JNJ-64304500 in subjects with moderately to severely active Crohn's disease. <https://clinicaltrials.gov/ct2/show/study?term=EUCTR2016-000634-21&rank=1> (first received 24 August 2016).

55. NCT02877134. Safety and efficacy study of JNJ-64304500 in participants with moderately to severely active Crohn's disease. <https://clinicaltrials.gov/ct2/show/study?term=NCT02877134&rank=1> 2016.

56. Allez M, Sands BE, Feagan BG, D'Haens G, De Hertogh G, Randall CW, et al. A phase 2b, randomised, double-blind, placebo-controlled, parallel-arm, multicenter study evaluating the safety and efficacy of tesnatilimab in patients with moderately to severely active Crohn's disease. *Journal of Crohn's & Colitis* 2023;**no volume**:jjad047.

57. Chen B, Gao X, Zhong J, Ren J, Zhu X, Liu Z, et al. Efficacy and safety of adalimumab in Chinese patients with moderately to severely active Crohn's disease: results from a randomized trial. *Therapeutic Advances in Gastroenterology* 2020;**13**:1-13.

58. Chen B, Chen M, Gao X, Zhong J, Ren J, Zhu X, et al. Efficacy and safety of adalimumab in Chinese patients with moderately to severely active Crohn's disease. *United European Gastroenterology Journal* 2018;**6**:A240.

59. NCT02499783. Study to evaluate the safety and efficacy of adalimumab in Chinese subjects with moderate to severe Crohn's disease. <https://clinicaltrials.gov/ct2/show/study?term=NCT02499783&rank=1> (first received 16 July 2015).

60. Colombel JF, Sandborn WJ, Rutgeerts P, Enns R, Hanauer SB, Panaccione R, et al. Adalimumab for maintenance of clinical response and remission in patients with Crohn's disease: the CHARM trial. *Gastroenterology* 2007;**132**(1):52-65.

61. Panaccione R, Sandborn W, Colombel J, Pollack P, Chen N, Chao J, et al. Impact of adalimumab therapy after 12 weeks in

patients with Crohn's disease who were nonresponders at week 4. *Inflammatory Bowel Diseases* 2008;**14**(Suppl 3):S28.

62. NCT00077779. Adalimumab for the induction and maintenance of clinical remission in subjects with Crohn's disease. <https://clinicaltrials.gov/show/study?term=NCT00077779&rank=1> (first received 6 February 2004).

63. Colombel JF, Sandborn WJ, Reinisch W, Mantzaris GJ, Kornbluth A, Rachmilewitz D, et al. Infliximab, azathioprine, or combination therapy for Crohn's disease. *New England Journal of Medicine* 2010;**362**(15):1383-95.

64. Cottone M, Papi C, Orlando A. Infliximab, azathioprine or combination therapy in the treatment of active Crohn's disease. *Expert Review of Gastroenterology & Hepatology* 2010;**4**(6):709-12.

65. Peyrin-Biroulet L, Reinisch W, Colombel JF, Mantzaris GJ, Kornbluth A, Diamond R, et al. Relationships between clinical remission, c-reactive protein normalization and mucosal healing in Crohn's disease: analyses from the sonic trial. *Journal of Crohn's & Colitis* 2013;**7**:S8.

66. Peyrin-Biroulet L, Reinisch W, Colombel JF, Mantzaris GJ, Kornbluth A, Diamond R, et al. Relationships between clinical remission, C-reactive protein normalization and mucosal healing in Crohn's disease: analyses from the sonic trial. *Gastroenterology* 2013;**144**(5 Suppl 1):S228.

67. Peyrin-Biroulet L, Reinisch W, Colombel JF, Mantzaris GJ, Kornbluth A, Diamond R, et al. Clinical disease activity, c-reactive protein normalisation and mucosal healing in Crohn's disease in the sonic trial. *Gut* 2014;**63**(1):88-95.

68. EUCTR2004-002815-10-GB. Multicenter, randomized, double-blind, active-controlled trial comparing REMICADE® (infliximab) and REMICADE® plus azathioprine to azathioprine in the treatment of patients with crohn's disease naive to both immunomodulators and biologic therapy (study of biologic and immunomodulator naive patients in crohn's disease) SONIC - SONIC. <https://clinicaltrials.gov/ct2/show/study?term=EUCTR2004-002815-10-GB&rank=1> (first received 26 May 2005).

69. Colombel JF, Reinisch W, Mantzaris GJ, Kornbluth A, Rutgeerts P, Tang KL, et al. Randomised clinical trial: deep remission in biologic and immunomodulator naive patients with Crohn's disease - a sonic post hoc analysis. *Alimentary Pharmacology & Therapeutics* 2015;**41**(8):734-46.

70. Colombel JF, Rutgeerts P, Yan S, Bala M, Bao W, Olson A, et al. Infliximab maintenance treatment results in lower hospitalization rate in Crohn's disease patients. *Gastroenterology* 2002;**122**(4 Suppl 1):A613.

71. Sandborn WJ, Rutgeerts PJ, Reinisch W, Mantzaris GJ, Kornbluth A, Rachmilewitz D, et al. One year data from the sonic study: a randomized, double-blind trial comparing infliximab and infliximab plus azathioprine to azathioprine in patients with Crohn's disease naive to immunomodulators and biologic therapy. *Gastroenterology* 2009;**136**(5 Suppl 1):751f.

72. Sandborn W, Rutgeerts P, Reinisch W, Mantzaris G, Kornbluth A, Rachmilewitz D, et al. Sonic study: a randomized,

double-blind trial comparing infliximab and infliximab plus azathioprine to azathioprine in patients with Crohn's disease naive to immunomodulators and biologic therapy. *Inflammatory Bowel Diseases* 2009;**15**(Suppl 2):S13.

73. Colombel JF, Rutgeerts P, Reinisch W, Mantzaris GJ, Kornbluth A, Rachmilewitz D, et al. Sonic: a randomized, double-blind, controlled trial comparing infliximab and infliximab plus azathioprine to azathioprine in patients with Crohn's disease naive to immunomodulators and biologic therapy. *UEGW* 2008;**no volume**:no pagination.

74. D'Haens G, Van Deventer S, Van Hogezaand R, Chalmers D, Kothe C, Baert F, et al. Endoscopic and histological healing with infliximab anti-tumor necrosis factor antibodies in Crohn's disease: a European multicenter trial. *Gastroenterology* 1999;**116**:1029-34.

75. D'Haens GR, Van Deventer SJ, Van Hogezaand R, Chalmers DM, Braakman TA, Bijl H, et al. Endoscopic healing with cA2 anti-TNF antibodies in Crohn's disease: a European multicenter trial. *Endoscopy* 1996;**28**:S31.

76. D'Haens G, Rutgeerts P, Braakman T. Anti-tumor necrosis factor-alpha (cA2) induces remission and endoscopic healing in active Crohn's disease: a double-blind, placebo-controlled study. *Acta Gastroenterologica Belgica* 1997;**60**:no pagination.

77. D'Haens G, Baert F, van Assche G, Vermeire S, Rutgeerts P, Caenepeel P, et al. Early combined immunosuppression or conventional management in patients with newly diagnosed Crohn's disease: an open randomised trial. *Lancet* 2008;**371**:660-7.

78. D'Haens G, Panaccione R, Baert F, Bossuyt P, Colombel JF, Danese S, et al. Risankizumab as induction therapy for Crohn's disease: results from the phase 3 ADVANCE and MOTIVATE induction trials. *Lancet* 2022;**399**(10340):2015-30.

79. D'Haens GR, Colombel JF, Bossuyt P, Danese S, Lim A, Lindsay J, et al. 775a risankizumab induction therapy in patients with moderate-to-severe Crohn's disease with intolerance or inadequate response to conventional and/or biologic therapy: results from the phase 3 ADVANCE study. *Gastroenterology* 2021;**161**(2):e28.

80. NCT03105128. A study of the efficacy and safety of risankizumab in participants with moderately to severely active Crohn's disease. <https://clinicaltrials.gov/ct2/show/NCT03105128> (first received 7 April 2017).

81. D'Haens GR, Sandborn WJ, Loftus Jr EV, Hanauer SB, Schreiber S, Peyrin-Biroulet L, et al. Higher vs standard adalimumab induction dosing regimens and two maintenance strategies: randomized SERENE CD trial results. *Gastroenterology* 2022;**162**(7):1876-90.

82. EUCTR2013-001746-33-SK. A blinded study to evaluate higher versus standard adalimumab doses for induction and maintenance therapy in subjects with moderately to severely active Crohn's disease and evidence of mucosal ulceration. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2013-001746-33-SK> 2014.

83. NCT02065570. Study to evaluate efficacy and safety of two drug regimens in subjects with moderate to severe Crohn's disease. <https://ClinicalTrials.gov/show/NCT02065570> 2014.

84. Duan Z, Luo J, Li W. Efficacy of infliximab combined with azathioprine for moderate to severe Crohn's disease. *Chinese Journal of Gastroenterology* 2013;**18**(4):229-32.

85. Feagan BG, Vandervoort MK, Greenberg GR, Wild G, Bitton A, Cohen A, et al. Treatment of active Crohn's disease with MLN0002, a humanized antibody to the alpha4beta7 integrin. *Clinical Gastroenterology and Hepatology* 2008;**6**:1370-7.

86. NCT00655135. Phase 2 study of the safety and efficacy of LDP-02 in mildly to moderately active Crohn's patients. <https://ClinicalTrials.gov/show/NCT00655135> 2000.

87. Feagan B, Gasink C, Lang Y, Friedman J, Blank M, Johanns J. Ustekinumab induction therapy for Crohn's disease refractory to conventional or tumor necrosis factor antagonist therapy. *New England Journal of Medicine* 2015;**375**(20):1946-60.

88. Adedokun O, Xu Z, Gasink C, Friedman J, Szapary P, Lang Y, et al. Pharmacokinetics and exposure-response relationships of intravenously administered ustekinumab during induction treatment in patients with Crohn's disease: results from the uniti-1 and uniti-2 studies. *Journal of Crohn's & Colitis* 2016;**10**(Suppl 1):S23-4.

89. Cohen SA, Griffiths AM, Hyams JS, Rosh JR, Turner D, Strauss R, et al. Clinical characteristics, clinical response, and clinical remission status in adults treated with ustekinumab in the phase 3 UNITI studies at week 6 and week 44 by age at Crohn's disease onset. *Gastroenterology* 2021;**160**(6 Suppl):S-44.

90. Feagan B, Gasink C, Lang Y, Friedman J, Johanns J, Gao L, et al. A multicenter, double-blind, placebo (pbo)-controlled ph3 study of ustekinumab (ust), a human mab to IL-12/23P40, in pts with moderately-severely active Crohn's disease (cd) who are naive or not refractory to anti-tnf: UNITI-2. *Gut* 2016;**65**(Suppl 1):A6-7.

91. Rutgeerts P, Gasink C, Lang Y, Johanns J, Gao L, Blank M, et al. A multicenter, double-blind, placebo (pbo)-controlled ph3 study of ustekinumab (ust), a human IL-12/23P40 mab, in moderate-severe Crohn's disease (CD) refractory to anti-TNFA: UNITI-1. *Gut* 2016;**65**(Suppl 1):A3-4.

92. Rutgeerts P, Gasink C, Chan D, Lang Y, Pollack P, Jacobstein D, et al. Efficacy of ustekinumab for induction and maintenance of endoscopic healing in patients with Crohn's disease. *United European Gastroenterology Journal* 2016;**4**(5 Suppl 1):A44.

93. Feagan B, Gasink C, Pollack P, Jacobstein D, Gao L, Johanns J, et al. Effect of maintenance ustekinumab on corticosteroid-free clinical outcomes in patients with Crohn's disease. *Gastroenterology* 2017;**152**(5 Suppl 1):S595.

94. Sandborn W, Gasink C, Lang Y, Johanns J, Gao L, Blank M, et al. A multicenter, double-blind, placebo-controlled phase 3 study of ustekinumab, a human IL-1/-3p40 mab, in moderate-

service Crohn's disease refractory to anti-TFNalpha: UNITI-1. *Inflammatory Bowel Diseases* 2016;**22**(Suppl 1):S1.

95. Sandborn W, Yeager B, Gasink C, Hoops T, Gao L, Wang Y, et al. Ustekinumab IV induction results in Crohn's disease symptom improvement within the first week in anti-TNF refractory patients. *United European Gastroenterology Journal* 2017;**5**(8):1141-2.

96. Sandborn W, Yeager B, Gasink C, Long Gao L, Hoops T, Wang Y, et al. Ustekinumab IV induction results in Crohn's disease symptom improvement within the first week in anti-TNF refractory patients. *American Journal of Gastroenterology* 2017;**112**(Suppl 1):S403.

97. Sandborn W, Yeager B, Hoops T, Gasink C, Gao L, Wang Y, et al. Ustekinumab IV induction results in Crohn's disease symptom improvement within the first week in anti-TNF refractory patients. *American Journal of Gastroenterology* 2018;**113**(Suppl 1):S4.

98. Gasink C, Lang Y, Jacobstein D, Johannis J, Rutgeerts P, De Villiers W, et al. Assessment of serum c-reactive protein, faecal lactoferrin, and faecal calprotectin in patients with Crohn's disease: results from the uniti-1 and uniti-2 ustekinumab induction studies. *Journal of Crohn's & Colitis* 2016;**10**(Suppl 1):S390-1.

99. Rutgeerts P, Gasink C, Lang Y, Johannis J, Gao L, Blank M, et al. A multicentre, double-blind, placebo-controlled phase 3 study of ustekinumab, a human interleukins-12/23p40 mab, in moderate-severe Crohn's disease refractory to anti-tumour necrosis factor alpha: UNITI-1. *Journal of Crohn's & Colitis* 2016;**10**(Suppl 1):S11-2.

100. Narula N, Wong EC, Dulai PS, Marshall JK, Colombel JF, Reinisch W. Week 6 calprotectin best predicts likelihood of long-term endoscopic healing in Crohn's disease: a post-hoc analysis of the UNITI/IM-UNITI trials. *Journal of Crohn's & Colitis* 2021;**15**(3):462-70.

101. EUCTR2010-022758-18-BE. Study to evaluate the safety and efficacy of ustekinumab in patients with moderately to severely active Crohn's disease who have failed or are intolerant to tumor necrosis factor (TNF) antagonist therapy. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2010-022758-18-BE> 2011.

102. Feagan B, Gasink C, Lang Y, Friedman J, Johannis J, Gao L. A multicenter, double-blind, placebo-controlled ph3 study of ustekinumab, a human monoclonal antibody to IL-12/23p40, in patients with moderately-severely active Crohn's disease who are not naive or not refractory to anti-TNFa: uniti-2. *United European Gastroenterology Journal* 2015;**3**(6):563-4.

103. Feagan B, Gasink C, Lang Y, Friedman J, Johannis J, Gao L, et al. A multicenter, double-blind, placebo-controlled ph3 study of ustekinumab, a human monoclonal antibody to IL-12/23p40, in patients with moderately-severely active Crohn's disease who are naive or not refractory to anti-TNFa: Uniti-2. *Canadian Journal of Gastroenterology and Hepatology* 2016;**78**(2):383-94.

104. Feagan B, Ghosh S, Dieleman L, Sandborn W, Targan S, Gasink C, et al. Ustekinumab as induction and maintenance

therapy for Crohn's disease. *New England Journal of Medicine* 2016;**375**(20):1946-60.

105. Rutgeerts P, Gasink C, Chan D, Lang Y, Pollack P, Jacobstein D, et al. Efficacy of ustekinumab for inducing endoscopic healing in patients with Crohn's disease. *Gastroenterology* 2018;**155**(4):1045-58.

106. NCT01369329. A study to evaluate the safety and efficacy of ustekinumab in patients with moderately to severely active Crohn's disease who have failed or are intolerant to tumor necrosis factor (TNF) antagonist therapy (UNITI-1). <https://ClinicalTrials.gov/show/NCT01369329> (first received 8 June 2011).

107. Obando C, Ding Z, Gao L, Pires A, Muser E, Gasink C. Reduced healthcare costs of Crohn's disease-related hospitalizations and surgeries in patients initiated with ustekinumab in the IM-UNITILTE study. *Journal of Managed Care and Specialty Pharmacy* 2018;**24**:S73.

108. Ghosh S, Gasink C, Gao L, Obando C. Number needed to treat to achieve clinical response at week 8 along with response or remission at week 52 with ustekinumab treatment vs. placebo from the phase 3 UNITI Crohn's disease studies, by population. *Journal of Crohn's & Colitis* 2018;**12**(Suppl 1):S450-1.

109. Sandborn W, Gasink C, Chan D, Lang Y, Pollack P, Jacobstein D, et al. Endoscopic healing in induction and maintenance with ustekinumab in the phase 3 UNITI Crohn's disease program. *American Journal of Gastroenterology* 2016;**111**(Supplement 1):S278-9.

110. DRKS00005759. A phase 3, randomized, double-blind, placebo-controlled, parallel-group, multicenter study to evaluate the safety and efficacy of ustekinumab induction therapy in subjects with moderately to severely active Crohn's disease who have failed or are intolerant to TNF antagonist therapy (UNITI-1). https://www.drks.de/drks_web/navigate.do?navigationId=trial.HTML&TRIAL_ID=DRKS00005759 (first received 10 May 2017).

111. Feagan BG, Sandborn WJ, Gasink C, Jacobstein D, Lang Y, Friedman JR, et al. Ustekinumab as induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2016;**375**(20):1946-60.

112. Cohen SA, Griffiths AM, Hyams JS, Rosh JR, Turner D, Strauss R, et al. Clinical characteristics, clinical response, and clinical remission status in adults treated with ustekinumab in the phase 3 UNITI studies at week 6 and week 44 by age at Crohn's disease onset. *Gastroenterology* 2021;**160**(6 Suppl):S-44.

113. Adedokun OJ, Xu Z, Gasink C, Friedman J, Szapary P, Lang Y, et al. Pharmacokinetics and exposure-response relationships of intravenously administered ustekinumab during induction treatment in patients with Crohn's disease: results from the UNITI-1 and UNITI-2 studies. *Journal of Crohn's & Colitis* 2016;**10**:S23-4.

114. Adedokun OJ, Xu Z, Gasink C, Szapary P, Johannis J, Gao LL, et al. Pharmacokinetics and exposure-response relationships of ustekinumab during IV induction and SC

maintenance treatment of patients with Crohn's disease with ustekinumab: results from the UNITI-1, UNITI-2, and IM-UNITI studies. *Gastroenterology* 2016;**1**:S408.

115. Li K, Hayden K, Wadman E, Bhagat S, Emrich S, Friedman JR, et al. Molecular response to ustekinumab in moderate-to-severe Crohn's disease by serum protein analysis: results from the UNITI-1 and UNITI-2 phase 3 induction studies. *Journal of Crohn's & Colitis* 2016;**10**:S26-7.

116. Gasink C, Lang Y, Jacobstein D, Johanns J, Rutgeerts P, De Villiers W, et al. Assessment of serum c-reactive protein, faecal lactoferrin, and faecal calprotectin in patients with Crohn's disease: results from the UNITI-1 and UNITI-2 ustekinumab induction studies. *Journal of Crohn's & Colitis* 2016;**10**(Suppl 1):S-390-1.

117. EUCTR2010-022759-42-NL. A study to evaluate the safety and efficacy of ustekinumab induction therapy in patients with moderately to severely active Crohn's disease. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2010-022759-42-NL> 2011.

118. Narula N, Wong EC, Dulai PS, Marshall JK, Colombel JF, Reinisch W. Week 6 calprotectin best predicts likelihood of long-term endoscopic healing in Crohn's disease: a post-hoc analysis of the UNITI/IM-UNITI trials. *Journal of Crohn's & Colitis* 2021;**15**(3):462-70.

119. Feagan B, Gasink C, Lang Y, Friedman JR, Johanns J, Gao LL, et al. A multicenter, double-blind, placebo-controlled p3 study of ustekinumab, a human monoclonal antibody to IL-12/23p40, in patients with moderately-severely active Crohn's disease who are naive or not refractory to anti-TNFa: uniti-2. *United European Gastroenterology Journal* 2015;**3**(6):563-4.

120. Gasink C, Chan D, Gao LL, Schenkel B, Han C. Assessment of sleep impairment in patients with Crohn's disease: results from the ustekinumab CERTIFI study. *Gastroenterology* 2013;**1**:S231.

121. Rutgeerts P, Gasink C, Blank M, Lang Y, Johanns J, Gao LL, et al. A multicentre, double-blind, placebo-controlled phase 3 study of ustekinumab, a human interleukins-12/23p40 mab, in moderate-severe Crohn's disease refractory to anti-tumour necrosis factor alpha: uniti-1. *Journal of Crohn's & Colitis* 2016;**10**:S11-2.

122. Sandborn W, Gasink C, Jacobstein D, Gao LL, Johanns J, Targan S, et al. Assessment of serum c-reactive protein, fecal lactoferrin, and fecal calprotectin in patients with moderate-severely active Crohn's disease: results from the im UNITI maintenance study. *Gastroenterology* 2016;**1**:S982.

123. Sands BE, Han C, Gasink C, Szapary P, Gao LL, Lang Y, et al. Ustekinumab improves general health status and disease-specific health related quality of life of patients with moderate to severe Crohn's disease: results from the UNITI and IMUNITI phase 3 clinical trials. *Gastroenterology* 2016;**1**:S1004.

124. Li K, Hayden K, Wadman E, Bhagat S, Emrich S, Friedman J, et al. Molecular response to ustekinumab in moderate-to-severe Crohn's disease by serum protein analysis:

results from UNITI-1 induction, UNITI-2 induction, and IMUNITI maintenance studies. *Gastroenterology* 2016;**1**:S377.

125. Sandborn W, Gasink C, Blank M, Lang Y, Johanns J, Gao LL, et al. A multicenter, double-blind, placebo-controlled phase3 study of ustekinumab, a human IL-13p40 mab, in moderate-service Crohn's disease refractory to anti-TFNalpha: UNITI-1. *Inflammatory Bowel Diseases* 2016;**22**:S1.

126. Feagan B, Gasink C, Lang Y, Friedman J, Johanns J, Gao L, et al. A multicenter, double-blind, placebo-controlled ph3 study of ustekinumab, a human monoclonal antibody to IL-12/23p40, in patients with moderately-severely active Crohn's disease who are naive or not refractory to anti-TNFA: uniti-2. *Canadian Journal of Gastroenterology and Hepatology* 2016;**no volume**:no pagination.

127. Colombel J, Sloan S, Gasink C, Gao L, Jacobstein D, Lee S, et al. Response and remission after 16 weeks of ustekinumab: an all patients analysis from the UNITI Crohn's studies. *American Journal of Gastroenterology* 2017;**112**(Suppl 1):S397.

128. Colombel J, Sloan S, Gasink C, Gao L, Jacobstein D, Lee S, et al. Response and remission after 16 weeks of ustekinumab-an all patients analysis from the UNITI Crohn's studies. *United European Gastroenterology Journal* 2017;**5**(5 Suppl 1):A308.

129. NCT01369342. A study to evaluate the safety and efficacy of ustekinumab induction therapy in patients with moderately to severely active Crohn's disease (UNITI-2). <https://ClinicalTrials.gov/show/NCT01369342> (first received 8 June 2011).

130. Ghosh S, Gasink C, Gao L, Obando C. Number needed to treat to achieve clinical response at week 8 along with response or remission at week 52 with ustekinumab treatment vs. placebo from the phase 3 UNITI Crohn's disease studies, by population. *Gastroenterology* 2018;**154**(6 Suppl 1):S-391.

131. Feagan B, Sattin B, Tornatore V, Gasink C, Gao L, Sloan S, et al. Outcomes of maintenance ustekinumab therapy for Crohn's disease based on inflammatory burden: a post-hoc analysis of the UNITI trials. *Gastroenterology* 2018;**154**(6 Suppl 1):S-586.

132. Feagan B, Gasink C, Lang Y, Friedman JR, Johanns J, Gao LL, et al. A multicenter, double-blind, placebo (pbo)-controlled ph3 study of ustekinumab (ust), a human mab to IL-12/23P40, in pts with moderately-severely active Crohn's disease (cd) who are naive or not refractory to antitnfa: uniti-2. *Gut* 2016;**65**(Suppl 1):A6-7.

133. Sands BE, Kramer BC, Gasink C, Jacobstein D, Gao LL, Ma T, et al. Association of ustekinumab serum concentrations and perianal fistula resolution in the Crohn's disease UNITI program. *Gastroenterology* 2019;**156**(6 Suppl 1):S-1100.

134. Sands BE, Kramer B, Gasink C, Gao LL, Ma T, Jacobstein D, et al. Association of ustekinumab serum concentrations and perianal fistula resolution in the Crohn's disease UNITI program. *United European Gastroenterology Journal* 2019;**7**(8 Suppl):913.

135. Feagan BG, Sandborn WJ, D'Haens G, Panés J, Kaser A, Ferrante M, et al. Induction therapy with the selective

interleukin-23 inhibitor risankizumab in patients with moderate-to-severe Crohn's disease: a randomised, double-blind, placebo-controlled phase 2 study. *Lancet* 2017;**389**(10080):1699-709.

136. Panaccione R, Schreiber S, Peyrin-Biroulet L, Loftus EV, Dubinsky M, Ferrante M, et al. Risankizumab as induction therapy in patients with moderately to severely active Crohn's disease who failed 1 vs >1 prior biologic treatment: results from the motivate study. *American Journal of Gastroenterology* 2021;**116**(Suppl):S348.

137. NCT02031276. Efficacy, safety and pharmacokinetics of BI 655066/ABBV-066 (Risankizumab) in patients with active, moderate-to-severe Crohn's disease. <https://clinicaltrials.gov/ct2/show/NCT02031276> (first received 9 January 2014).

138. Ghosh S, Goldin E, Gordon FH, Malchow HA, Rask-Madsen J, Rutgeerts P, et al. Natalizumab for active Crohn's disease. *New England Journal of Medicine* 2003;**348**(1):24-32.

139. Ghosh S. Alpha-integrin blockade in inflammatory bowel disease. *Annals of the Rheumatic Diseases* 2003;**62**(Suppl 2):70-2.

140. Rutgeerts P, Donoghue S, Palmer T. Quality of life improvements in a phase 2 study of natalizumab for active Crohn's disease. *Gastroenterology* 2003;**124**(4):A378.

141. Antegren Publication Committee. A randomised, double-blind, placebo-controlled, pan-European study of a recombinant humanised antibody to $\alpha 4$ integrin (AntegrenTM) in moderate to severely active Crohn's disease. *Gastroenterology* 2003;**120**(5):A127-8.

142. Gordon FH, Lai CW, Hamilton MI, Allison MC, Srivastava ED, Fouweather MG, et al. A randomized placebo-controlled trial of a humanized monoclonal antibody to $\alpha 4$ integrin in active Crohn's disease. *Gastroenterology* 2001;**121**(2):268-74.

143. Antegren. A randomised, double-blind, placebo-controlled, pan-European study of a recombinant humanised antibody to $\alpha 4$ integrin (AntegrenTM) in moderate to severely active Crohn's disease. *Gastroenterology* 2001;**120**:A127-8.

144. Gordon FH, Hamilton MI, Donoghue S, Amlot PL, Greenlees C, Rowley-Jones D, et al. Treatment of active ulcerative colitis with a recombinant humanised antibody to $\alpha 4$ integrin (Antegren). *Gastroenterology* 1999;**116**:A726.

145. Gordon FH, Lai CW, Hamilton MI, Allison MC, Fouweather M, Donoghue S. Randomised doubleblind placebo controlled trial of recombinant humanised antibody to $\alpha 4$ integrin (AntegrenTM) in active Crohn's disease. *Gastroenterology* 1999;**116**:A726.

146. Hanauer SB, Feagan BG, Lichtenstein GR, Mayer LF, Schreiber S, Colombel JF, et al. Maintenance infliximab for Crohn's disease: the ACCENT I randomised trial. *Lancet* 2002;**359**(9317):1541-9.

147. Feagan B, Yan S, Bala M, Bao W, Olson A, Hanauer S. Infliximab maintenance therapy improves health related

quality of life (HRQL) in Crohn's disease over 54 weeks. *Gastroenterology* 2002;**122**(4 Suppl 1):A615.

148. Hanauer SB, Wagner CL, Bala M, Mayer L, Travers S, Diamond RH, et al. Incidence and importance of antibody responses to infliximab after maintenance or episodic treatment in Crohn's disease. *Clinical Gastroenterology and Hepatology* 2004;**2**(7):542-53.

149. Reinisch W, Wang Y, Oddens BJ, Link R. C-reactive protein, an indicator for maintained response or remission to infliximab in patients with Crohn's disease: a post-hoc analysis from ACCENT I. *Alimentary Pharmacology & Therapeutics* 2012;**35**(5):568-76.

150. Fasanmade A, Olson A, Bao W, Pendley C, Davis H, Mayer L. Relationship between infliximab pharmacokinetics and improvement in Crohn's disease. *Gastroenterology* 2002;**122**(4 Suppl 1):A617-8.

151. Feagan BG, Yan S, Bala M, Bao W, Lichtenstein GR. The effects of infliximab maintenance therapy on health-related quality of life. *American Journal of Gastroenterology* 2003;**98**(10):2232-8.

152. Lichtenstein GR, Bao W, Keenan G, Olson A, Hanauer S, Colombel JF. Infliximab treatment allows Crohn's disease to reduce or discontinue concomitant corticosteroid use: ACCENT 1 year results. *Gastroenterology* 2002;**122**(4 Suppl 1):A615.

153. Hanauer SB, Lichtenstein GR, Colombel J, Mayer L, Schreiber S, Rachmilewicz D, et al. Maintenance infliximab (remicade) is safe, effective and steroid-sparing in Crohn's disease: preliminary results from the ACCENT I trial. *Gastroenterology* 2001;**120**(5 Suppl 1):A-21.

154. Piche T. Infliximab in maintenance treatment of Crohn's disease: results of the ACCENT 1 study. *Hepato-Gastro* 2002;**9**(5):391.

155. Rutgeerts P, Diamond RH, Bala M, Olson A, Lichtenstein GR, Bao W, et al. Scheduled maintenance treatment with infliximab is superior to episodic treatment for the healing of mucosal ulceration associated with Crohn's disease. *Gastrointestinal Endoscopy* 2006;**63**(3):433-42.

156. Rutgeerts P, Van Assche G, Van Deventer S, Bao W, Keenan G, Olson A, et al. Infliximab maintenance treatment strategy results in mucosal healing in patients with Crohn's disease. *Gastroenterology* 2002;**122**(4 Suppl 1):A618.

157. Rutgeerts P, Feagan BG, Lichtenstein GR, Mayer LF, Schreiber S, Colombel JF, et al. Comparison of scheduled and episodic treatment strategies of infliximab in Crohn's disease. *Gastroenterology* 2004;**126**(2):402-13.

158. Sands B, D'Haens PG, Rutgeerts P, Bala M, Patel K, Keenan G, et al. Change in c-reactive protein correlates with change in endoscopic measures of disease activity in Crohn's disease patients treated with infliximab. *Gastroenterology* 2002;**122**(4 Suppl 1):P284.

159. Cornillie F, Hanauer SB, Diamond RH, Wang J, Tang KL, Xu Z, et al. Postinduction serum infliximab trough level and

decrease of c-reactive protein level are associated with durable sustained response to infliximab: a retrospective analysis of the ACCENT I trial. *Gut* 2014;**63**(11):1721-7.

160. Geboes K, Rutgeerts P, Opdenakker G, Olson A, Patel K, Wagner CL, et al. Endoscopic and histologic evidence of persistent mucosal healing and correlation with clinical improvement following sustained infliximab treatment for Crohn's disease. *Current Medical Research and Opinion* 2005;**21**(11):1741-54.

161. Geboes K, Rutgeerts P, Wagner C, Olson A, Cornillie F, Bao W, et al. Infliximab treatment results in the sustained reduction of inflammatory markers in the mucosa of patients with Crohn's disease. *Gastroenterology* 2002;**122**(4 Suppl 1):A150.

162. Lichtenstein G, Yan S, Bala M, Hanauer S. Remission in patients with Crohn's disease is associated with improvement in employment and quality of life and a decrease in hospitalizations and surgeries. *American Journal of Gastroenterology* 2004;**99**(1):91-6.

163. NCT00207662. A safety and efficacy study of infliximab (remicade) in patients with moderate to severe active Crohn's disease. <https://ClinicalTrials.gov/show/NCT00207662> (first received July 2000).

164. Hanauer SB, Sandborn WJ, Rutgeerts P, Fedorak RN, Lukas M, MacIntosh D, et al. Human anti-tumor necrosis factor monoclonal antibody (adalimumab) in Crohn's disease: the CLASSIC-I trial. *Gastroenterology* 2006;**130**(2):323-33.

165. Sandborn WJ, Colombel JF, Castillo MM, Zhou Q, Thakkar RB. Induction of remission with adalimumab in patients with moderate Crohn's disease: subanalysis of CLASSIC I. *Gut* 2012;**61**:A228-9.

166. Sandborn W, Colombel JF, Castillo M, Zhou Q, Thakkar R. Induction of remission with adalimumab in patients with moderate Crohn's disease: subanalysis of CLASSIC I. *American Journal of Gastroenterology* 2011;**106**:S436-7.

167. Hanauer S, Liedert B, Balser S, Brockstedt E, Moschetti V, Schreiber S. Safety and efficacy of BI 695501 versus adalimumab reference product in patients with advanced Crohn's disease (VOLTAIRE-CD): a multicentre, randomised, double-blind, phase 3 trial. *Lancet Gastroenterology & Hepatology* 2021;**6**:P816-25.

168. Hanauer SB, Schreiber S, Balser SE, Brockstedt E, Moschetti V, Liedert BJ. BI 695501 demonstrates similar efficacy and comparable safety to adalimumab reference product in patients with active Crohn's disease: final analysis of the phase III VOLTAIRE-CD study. *Gastroenterology* 2020;**158**:S-1193.

169. Hanauer S, Schreiber S, Balser S, Liedert B. Biosimilar BI 695501 demonstrates non-inferior efficacy and comparable safety to adalimumab reference product in patients with active Crohn's disease. *American Journal of Gastroenterology* 2019;**114**:S13-4.

170. Hanauer SB, Schreiber S, Balser S, Brockstedt E, Moschetti V, McCabe D, et al. BI 695501 demonstrates

similar efficacy and comparable safety to adalimumab reference product in patients with active Crohn's disease: final analysis of the phase III VOLTAIRE-CD study. *United European Gastroenterology Journal* 2020;**8**:439-40.

171. NCT02871635. BI 695501 versus humira in patients With active Crohn's disease: a trial comparing efficacy, endoscopic improvement, safety, and immunogenicity. <https://ClinicalTrials.gov/show/NCT02871635> (first received 18 August 2016).

172. EUCTR2016-000612-14-CZ. BI 695501 versus Humira in patients with active Crohn's disease: a trial comparing efficacy, endoscopic improvement, safety, and immunogenicity. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2016-000612-14-CZ> 2016.

173. Lemann M, Mary JY, Duclos B, Veyrac M, Dupas JL, Delchier JC, et al. Infliximab plus azathioprine for steroid-dependent Crohn's disease patients: a randomized placebo-controlled trial. *Gastroenterology* 2006;**130**(4):1054-61.

174. Loftus Jr EV, Panés J, Lacerda AP, Peyrin-Biroulet L, D'Haens G, Panaccione R, et al. Upadacitinib induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2023;**388**(21):1966-80.

175. D'Haens G, Panaccione R, Baert F, Bossuyt P, Colombel JF, Danese S, et al. Risankizumab as induction therapy for Crohn's disease: results from the phase 3 ADVANCE and MOTIVATE induction trials. *Lancet* 2022;**399**(10340):2015-30.

176. NCT03104413. A study to assess the efficacy and safety of risankizumab in participants with moderately to severely active Crohn's disease who failed prior biologic treatment. <https://clinicaltrials.gov/ct2/show/NCT03104413> (first received 7 April 2017).

177. Panaccione R, Schreiber S, Peyrin-Biroulet L, Loftus EV, Dubinsky M, Ferrante M, et al. S754 risankizumab as induction therapy in patients with moderately to severely active Crohn's disease who failed 1 vs> 1 prior biologic treatment: results from the MOTIVATE study. *Official Journal of the American College of Gastroenterology* 2021;**116**:S348.

178. Panés J, Sandborn WJ, Schreiber S, Sands BE, Vermeire S, D'Haens G, et al. Tofacitinib for induction and maintenance therapy of Crohn's disease: results of two phase IIb randomised placebo-controlled trials. *Gut* 2017;**66**(6):1049-59.

179. EUCTR2011-001733-16-HR. A randomized, double-blind, placebo-controlled, parallel group, multi-centre study to investigate the safety and efficacy of CP-690,550 for induction therapy in subjects with moderate to severe Crohn's disease. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2011-001733-16-HR> 2014.

180. Present DH, Rutgeerts P, Targan S, Hanauer SB, Mayer L, van Hogezaand RA, et al. Infliximab for the treatment of fistulas in patients with Crohn's disease. *New England Journal of Medicine* 1999;**340**(18):1398-405.

181. van Deventer SJ, Rutgeerts P, Targan SR, Hanauer SB, Mayer L, Present D, et al. Anti-tumor necrosis factor antibody

cA2 treatment induces clinical remissions in patients with active Crohn's disease. *Gut* 1996;**39**(Suppl 3):A188.

182. NCT00004941. Phase III randomized study of anti-tumor necrosis factor chimeric monoclonal antibody (cA2) for patients with enterocutaneous fistulae as a complication of Crohn's disease. <https://clinicaltrials.gov/ct2/show/NCT00004941> (first received 25 February 2000).

183. Sandborn WJ, Feagan BG, Hanauer SB, Present DH, Sutherland LR, Kamm MA, et al. An engineered human antibody to TNF (CDP571) for active Crohn's disease: a randomized double-blind placebo-controlled trial. *Gastroenterology* 2001;**120**(6):1330-8.

184. Sandborn WJ, Feagan BG, Hanauer SB, Present DH, Sutherland LR, Kamm MA, et al. Etanercept for active Crohn's disease: a randomized, double-blind, placebo-controlled trial. *Gastroenterology* 2001;**121**(5):1088-94.

185. Sandborn WJ, Colombel JF, Enns R, Feagan BG, Hanauer SB, Lawrance IC, et al. Natalizumab induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2005;**353**(18):1912-25.

186. Lawrance I, Colombel J, Enns R, Feagan B, Hanauer S, Panaccione R. Pharmacodynamic results from a phase 3 study evaluating natalizumab therapy of active Crohn's disease (ENACT-1). *Gastroenterology* 2004;**126**:A211.

187. Sandborn J, Hass S, White J, Niecko T, Feagan B. Health-related quality of life during natalizumab maintenance therapy for Crohn's disease. *American Journal of Gastroenterology* 2007;**102**:2737-46.

188. Enns R, Colombel J, Feagan B, Hanauer S, Lawrance I, Panaccione R. Safety, tolerability and immunogenicity of natalizumab in a phase III study of active Crohn's disease therapy (ENACT-1). *Gastroenterology* 2004;**126**:A462.

189. Plevy S, Dryden G, Lu Y, Hogge G, De V, Willem J. The efficacy of natalizumab in Crohn's disease patients who smoke. *Gastroenterology* 2009;**136**:A523.

190. Rutgeerts P, Allison M, Cortot A, D'Haens G, Dahlerup J, Hebuterne X. Effects of natalizumab on gut healing in a phase 3 study of active Crohn's disease therapy (ENACT-1). *Gastroenterology* 2004;**126**:A208.

191. NCT00032786. Safety and efficacy of natalizumab in the treatment of Crohn's disease. <https://ClinicalTrials.gov/show/NCT00032786> (first received 4 April 2002).

192. NCT00032799. Safety and efficacy of natalizumab in the treatment of Crohn's disease. <https://ClinicalTrials.gov/show/NCT00032799> (first received 4 April 2002).

193. Sandborn WJ, Colombel JF, Enns RA, Feagan BG, Hanauer SB, Lawrance IC, et al. Efficacy assessment of natalizumab in patients with Crohn's disease and prior history of anti-TNF therapy: results from ENACT-1. *Gastroenterology* 2004;**126**:A-76 Abstract.

194. Sandborn WJ, Rutgeerts P, Enns R, Hanauer SB, Colombel JF, Panaccione R, et al. Adalimumab induction therapy for Crohn disease previously treated with infliximab: a randomized trial. [Summary for patients in *Ann Intern Med*. 2007 Jun 19;146(12):I20; PMID: 17470825]. *Annals of Internal Medicine* 2007;**146**(12):829-38.

195. EUCTR2004-002934-20. A multi-center, randomized, double-blind, placebo-controlled study of the human anti-TNF monoclonal antibody adalimumab for the induction of clinical remission in subjects with moderate to severe Crohn's disease who have lost response or are intolerant to infliximab. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2004-002934-20/BE> (first received 8 July 2005).

196. NCT00105300. Study of the human anti-TNF monoclonal antibody adalimumab for the induction of clinical remission in subjects with Crohn's disease. <https://ClinicalTrials.gov/show/NCT00105300> (first received 8 March 2004).

197. Sandborn WJ, Feagan BG, Stoinov S, Honiball PJ, Honiball P, Rutgeerts P, et al. Certolizumab pegol for the treatment of Crohn's disease. *New England Journal of Medicine* 2007;**357**(3):228-38.

198. Sandborn W, Lee S, Randall C, Gutierrez A, Schwartz D, Ambarkhane S, et al. Long-term safety and efficacy of certolizumab pegol in the treatment of Crohn's disease: 7-year results from the PRECISE 3 study. *Alimentary Pharmacology & Therapeutics* 2014;**40**(8):903-16.

199. NCT00552058. Study to evaluate efficacy and safety of certolizumab pegol for induction of remission in patients with Crohn's disease. <https://ClinicalTrials.gov/show/NCT00552058> (first received 1 November 2007).

200. Rubin D, Randall C, Coarse J, Kosutic G, Spearman M, Ambarkhane S, et al. Crohn's disease behavior as a risk factor for loss of maintenance of remission in patients treated with certolizumab pegol: results from the PRECISE 3 study. *American Journal of Gastroenterology* 2014;**102**(Suppl 2):S508-9.

201. Sandborn W, Feagan B, Stoinov S, Honiball P, Rutgeerts P, McColm J. Certolizumab pegol administered subcutaneously is effective and well tolerated in patients with active Crohn's disease: results from a 26-week, placebo controlled phase III study (PRECISE 1). *Gastroenterology* 2006;**130**(4 Suppl 2):A107.

202. Schreiber S, Feagan B, Hanauer S, Rutgeerts P, McColm J, Sandborn W. Safety and tolerability of subcutaneous (sc) certolizumab pegol in active Crohn's disease (CD): results from two phase III studies (PRECISE program). *Gastroenterology* 2006;**106**(4 Suppl 2):A479.

203. Schreiber S, Hanauer S, Feagan B, Bloomfield R, Rutgeerts P, Sandborn W. Long-term treatment with certolizumab pegol for up to 18 months in patients with active Crohn's disease: PRECISE 3 efficacy results. *Gastroenterology* 2007;**132**(4 Suppl 2):A503.

204. Sandborn W, Ullman T, Pierre-Louis B, Binion D. Fecal calprotectin concentration and clinical remission in patients with active Crohn's disease treated with certolizumab pegol:

results from PRECISE 1. *American Journal of Gastroenterology* 2011;**106**(Suppl 2):S441.

205. Feagan B, Coteur G, Keininger D, Tan S, Schreiber S. Clinically meaningful improvement in health-related quality of life in a randomized controlled trial of certolizumab pegol maintenance therapy for Crohn's disease. *American Journal of Gastroenterology* 2009;**104**(8):1976-83.

206. Sandborn W, Schreiber S, Feagan B, Rutgeerts P, Younes Z, Bloomfield R, et al. Induction therapy with certolizumab pegol in patients with moderate to severe Crohn's disease: a placebo-controlled trial. *American Journal of Gastroenterology* 2010;**105**(Suppl 1):S419.

207. NCT00152490. A study to test the effect of cdp870 in the treatment of Crohn's disease over 26 weeks, comparing CDP870 to a dummy drug (placebo). <https://ClinicalTrials.gov/show/NCT00152490> (first received 9 September 2005).

208. Sandborn WJ, Feagan BG, Fedorak RN, Scherl E, Fleisher MR, Katz S, et al. A randomized trial of ustekinumab, a human interleukin-12/23 monoclonal antibody, in patients with moderate-to-severe Crohn's disease. *Gastroenterology* 2008;**135**(4):1130-41.

209. Toedter GP, Blank M, Lang Y, Chen D, Sandborn WJ, De Villiers WJ. Relationship of C-reactive protein with clinical response after therapy with ustekinumab in Crohn's disease. *American Journal of Gastroenterology* 2009;**104**(11):2768-73.

210. Sandborn W, Feagan B, Fedorak R, Scherl E, Fleisher M, Katz S. A multicenter, randomized, phase 2A study of human monoclonal antibody to IL-12/23P40 (CNTO 1275) in patients with moderately to severely active Crohn's disease. *Inflammatory Bowel Diseases* 2008;**14**(Suppl 1):S10.

211. Sandborn WJ, Feagan BG, Fedorak R, Scherl E, Fleisher M, Katz S. A multicenter, randomized, phase 2a study of human monoclonal antibody to IL-12/23p40 (CNTO 1275) in patients with moderately to severely active Crohn's disease. *Gastroenterology* 2008;**132**(4 Suppl 1):A51.

212. NCT00265122. A study of the safety and efficacy of ustekinumab (CNTO 1275) in participants with Crohn's disease. <https://clinicaltrials.gov/ct2/show/NCT00265122> (first received 14 December 2005).

213. Sandborn WJ, Feagan BG, Fedorak RN, Scherl E, Fleisher MR, Katz S, et al. A randomized trial of ustekinumab, a human interleukin-12/23 monoclonal antibody, in patients with moderate-to-severe Crohn's disease. *Gastroenterology* 2008;**135**(4):1130-41.

214. Sandborn WJ, Schreiber S, Feagan BG, Rutgeerts P, Younes ZH, Bloomfield R, et al. Certolizumab pegol for active Crohn's disease: a placebo-controlled, randomized trial. *Clinical Gastroenterology and Hepatology* 2011;**9**(8):670-8.e3.

215. Sandborn W, Binion D, Rubin D, Ullman T, Pierre-Louis B, Loftus E. Antibodies against certolizumab pegol (CZP), plasma concentrations of CZP and efficacy in patients with Crohn's disease receiving continuous CZP therapy with or without

concomitant immunosuppressants. *American Journal of Gastroenterology* 2011;**106**:S439-40.

216. Sandborn W, Schreiber S, Feagan B, Rutgeerts P, Younes Z, Bloomfield R, et al. Induction therapy with certolizumab pegol in patients with moderate to severe Crohn's disease: a placebo-controlled trial. *American Journal of Gastroenterology* 2010;**105**:S419.

217. Sandborn WJ, Younes ZH, Pierre-Louis B, D'Haens GR. Efficacy of certolizumab pegol induction and maintenance therapy in patients with Crohn's disease not in remission following active or placebo induction. *Gastroenterology* 2012;**1**:S565.

218. Sandborn WJ, Gasink C, Gao LL, Blank MA, Johanns J, Guzzo C, et al. Ustekinumab induction and maintenance therapy in refractory Crohn's disease. *New England Journal of Medicine* 2012;**367**(16):1519-28.

219. Sands BE, Gasink C, Gao LL, Blank MA, Johanns J, Guzzo C, et al. Health related quality of life results through week 22 from the CERTIFI study, a multicenter, randomized, double-blind, placebo-controlled phase2b study of ustekinumab in patients with moderately to severely active Crohn's disease. *Inflammatory Bowel Diseases* 2011;**17**:S24.

220. Feagan B, Gasink C, Gao LL, Blank M, Johanns J, Guzzo C, et al. Health related quality of life results through week 22 from the CERTIFI study, a multicenter, randomized, double-blind, placebo-controlled phase2b study of ustekinumab in patients with moderately to severely active Crohn's disease. *Journal of Crohn's & Colitis* 2012;**6**:S129-30.

221. NCT00771667. A study of safety and effectiveness of ustekinumab in patients with moderate to severe active Crohn's disease who have been previously treated with anti-TNF therapy. <https://ClinicalTrials.gov/show/NCT00771667>.

222. Feagan B, Gasink C, Gao L, Blank M, Johanns J, Guzzo C, et al. A multicenter, randomized, double-blind, placebo-controlled phase 2b study of ustekinumab, a human monoclonal antibody to IL-12/23P40, in patients with moderately to severely active Crohn's disease: results through week 36 from the CERTIFI trial 2011 ACG presidential poster. *American Journal of Gastroenterology* 2011;**106**:S463.

223. EUCTR2008-000649-77. A phase 2b, multicenter, randomized, double-blind, placebo-controlled, parallel-group study to evaluate the efficacy and safety of ustekinumab therapy in subjects with moderately to severely active Crohn's disease previously treated with TNF antagonist therapy. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2008-000649-77/results> (first received 12 September 2008).

224. Feagan B, Gasink C, Gao L, Blank M, Johanns J, Guzzo C, et al. A multicenter, randomized, double-blind, placebo-controlled phase 2b study of ustekinumab, a human monoclonal antibody to IL-12/23P40, in patients with moderately to severely active Crohn's disease: results through week 36 from the CERTIFI trial 2011 ACG presidential poster. *American Journal of Gastroenterology* 2011;**106**:S463.

- 225.** Gasink C, Friedman J, Gao L, Chan D, Sandborn W, Feagan B. Evaluation of an interim Crohn's disease outcome measure (PRO-2) based on 2 patient-reported components (stool frequency, abdominal pain) of the Crohn's disease activity index (CDAI) in the ustekinumab CERTIFI study. *American Journal of Gastroenterology* 2014;**109**:S497.
- 226.** Toedter G, Wu X, Gao LL, Gasink C. Reductions in fecal calprotectin and lactoferrin following ustekinumab induction therapy in patients with moderate to severe Crohn's disease who have previously failed or been intolerant of TNF antagonist therapies. *Gastroenterology* 2011;**140**:S264.
- 227.** Ding T, Telesco S, Monast C, Brodmerkel C, Yatsunenkov T, Das A, et al. The gut microbiome differentiates clinical phenotypes in moderate to severe Crohn's disease: results from the CERTIFI study. *Canadian Journal of Gastroenterology and Hepatology. Conference* 2016;**148**(4):S713.
- 228.** Sandborn WJ, Gasink C, Gao LL, Blank M, Johanns J, Guzzo C, et al. A multicenter, randomized, double-blind, placebo-controlled phase 2b study of ustekinumab, a human monoclonal antibody to IL-12/23p40, in patients with moderately to severely active Crohn's disease: results through week 22 from the CERTIFI trial. *Gastroenterology* 2011;**140**(5):S109.
- 229.** Sandborn WJ, Gasink C, Gao LL, Blank MA, Johanns J, Guzzo C, et al. Ustekinumab induction and maintenance therapy in refractory Crohn's disease. *New England Journal of Medicine* 2012;**367**(16):1519-28.
- 230.** Sandborn WJ, Ghosh S, Panes J, Vranic I, Wang W, Niezychowski W et al. Study A3921043 Investigators A phase 2 study of tofacitinib, an oral Janus kinase inhibitor, in patients with Crohn's disease. *Clinical Gastroenterology and Hepatology* 2017;**12**(9):1485-93.
- 231.** Sandborn WJ, Panés J, Danese S, Sharafali Z, Hassanali A, Jacob-Moffatt R, et al. Etrolizumab as induction and maintenance therapy in patients with moderately to severely active Crohn's disease (BERGAMOT): a randomised, placebo-controlled, double-blind, phase 3 trial. *Lancet Gastroenterology & Hepatology* 2023;**8**(1):43-55.
- 232.** William S, Julian P, Jennifer J, Azra H, Rhian J, Zaineb S, et al. Etrolizumab as induction therapy in moderate to severe Crohn's disease: results from BERGAMOT cohort 1. *Official Journal of the American College of Gastroenterology* 2018;**113**:S3.
- 233.** NCT02394028. A study to assess whether etrolizumab is a safe and efficacious treatment for participants with moderately to severely active Crohn's disease (BERGAMOT). <https://clinicaltrials.gov/ct2/show/NCT02394028> (first received 20 March 2015).
- 234.** Sandborn WJ, Vermeire S, Tyrrell H, Lacey S, Hassanali A, Tole S, et al. Etrolizumab for the treatment of ulcerative colitis and Crohn's disease: an overview of the phase 3 clinical program. *Advances in Therapy* 2020;**37**:3417-31.
- 235.** Sandborn WJ, Panés J, Danese S, Sharafali Z, Hassanali A, Jacob-Moffatt R, et al. Etrolizumab as induction and maintenance therapy in patients with moderately to severely active Crohn's disease (BERGAMOT): a randomised, placebo-controlled, double-blind, phase 3 trial. *Lancet Gastroenterology & Hepatology* 2023;**8**(1):43-55.
- 236.** Sandborn WJ, Feagan BG, Loftus EV, Peyrin-Biroulet L, Van Assche G, D'Haens G, et al. Efficacy and safety of upadacitinib in a randomized trial of patients with Crohn's disease. *Gastroenterology* 2020;**158**(8):2123-38.
- 237.** EUCTR2014-003240-12-HU. A multicenter, randomized, double-blind, placebo-controlled study of ABT-494 for the induction of symptomatic and endoscopic remission in subjects with moderately to severely active Crohn's disease who have inadequately responded to or are intolerant to I. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2014-003240-12-HU> 2015.
- 238.** Sandborn WJ, D'Haens GR, Reinisch W, Panes J, Chan D, Gonzalez S, et al. Guselkumab for the treatment of Crohn's disease: induction results from the phase 2 GALAXI-1 study. *Gastroenterology* 2022;**162**(6):1650-64.e8.
- 239.** Leong RW, D'Haens G, Rubin DT, Gonzalez S, Chan D, Johanns J, et al. Effect of guselkumab induction therapy on endoscopic outcome measures in patients with moderately to severely active Crohn's disease: week 12 results from the phase 2 GALAXI 1 study. *Journal of Gastroenterology and Hepatology* 2021;**36**(Suppl 3):169-70.
- 240.** Sands BE, Danese S, Andrews JM, Weisel K, Chan D, Johanns J, et al. The effect of guselkumab induction therapy on inflammatory biomarkers in patients with moderately to severely active Crohn's disease: week 12 results from the phase 2 GALAXI 1 study. *Gastroenterology* 2021;**160**(6 Suppl):S-351.
- 241.** D'Haens G, Rubin DT, Panes J, Gonzalez S, Chan D, Johanns J, et al. The effect of guselkumab induction therapy on endoscopic outcome measures in patients with moderately to severely active Crohn's disease: week 12 results from the phase 2 GALAXI 1 study. *Gastroenterology* 2021;**160**(6 Suppl):S-91.
- 242.** Andrews J, Danese S, Sands B, Weisel K, Chan D, Johanns J, et al. Effect of guselkumab induction therapy on inflammatory biomarkers in patients with moderately to severely active Crohn's disease: week 12 results from the phase 2 GALAXI 1 study. *Journal of Gastroenterology and Hepatology* 2021;**36**(Suppl 3):108-9.
- 243.** Reinisch W, Sandborn W, Hisamatsu T, Weisel K, Gonzalez S, Frustaci ME, et al. Early pro-2 symptom remission following guselkumab induction treatment: results through week 12 of the phase 2 GALAXI 1 study. *United European Gastroenterology Journal* 2021;**9**(Suppl 8):495-7.
- 244.** Panes J, Panaccione R, D'Haens G, Han C, Gonzalez S, Weisel K, et al. Patient-reported outcomes of response and remission following guselkumab induction treatment as measured by the inflammatory bowel disease questionnaire: results through week 12 of the phase 2 GALAXI 1 study. *United European Gastroenterology Journal* 2021;**9**(Suppl 8):497-8.
- 245.** NCT03466411. A study of the efficacy and safety of guselkumab in participants with moderately to severely active

Crohn's disease (GALAXI). <https://clinicaltrials.gov/ct2/show/NCT03466411> (first received 15 March 2018).

246. Panes J, Panaccione R, D'Haens G, Han C, Gonzalez S, Weisel K, et al. Patient-reported outcomes of response and remission following guselkumab induction treatment as measured by the inflammatory bowel disease questionnaire: results through week 12 of the phase 2 GALAXI 1 study. *United European Gastroenterology Journal* 2021;**9**(Suppl 8):497-8.

247. Afzali A, Sandborn WJ, Hisamatsu T, Weisel K, Gonzalez S, Frustaci ME, et al. Early pro-2 symptom remission following guselkumab induction treatment: results through week 12 of the phase 2 GALAXI 1. *American Journal of Gastroenterology* 2021;**116**(Suppl):S344-5.

248. D'Haens G, Rubin DT, Panes J, Gonzalez S, Chan D, Johanns J, et al. The effect of guselkumab induction therapy on endoscopic outcome measures in patients with moderately to severely active Crohn's disease: week 12 results from the phase 2 GALAXI-1 study. *Journal of Gastroenterology and Hepatology* 2021;**36**(Suppl 2):115.

249. Sands BE, Danese S, Andrews JM, Weisel K, Chan D, Johanns J, et al. The effect of guselkumab induction therapy on inflammatory biomarkers in patients with moderately-to-severely active Crohn's disease: week 12 results from the phase 2 GALAXI-1 study. *Journal of Gastroenterology and Hepatology* 2021;**36**(Suppl 2):124-5.

250. Rubin DT, Panaccione R, D'Haens G, Han C, Gonzalez S, Weisel K, et al. Patient-reported outcomes of response and remission following guselkumab induction treatment as measured by the inflammatory bowel disease questionnaire: results through week 12 of the phase 2 GALAXI 1 study. *American Journal of Gastroenterology* 2021;**116**(Suppl):S400.

251. Reinisch W, Sandborn W, Hisamatsu T, Weisel K, Gonzalez S, Frustaci ME, et al. Early pro-2 symptom remission following guselkumab induction treatment: results through week 12 of the phase 2 GALAXI 1. *United European Gastroenterology Journal* 2021;**9**(Suppl 8):495-7.

252. Andrews J, Danese S, Sands B, Weisel K, Chan D, Johanns J, et al. Effect of guselkumab induction therapy on inflammatory biomarkers in patients with moderately to severely active Crohn's disease: week 12 results from the phase 2 GALAXI 1 study. *Journal of Gastroenterology and Hepatology* 2021;**36**(Suppl 3):108-9.

253. Sandborn WJ, Chan D, Johanns J, Lang G, Adedokun OJ, Afzali A, et al. The efficacy and safety of guselkumab induction therapy in patients with moderately to severely active Crohn's disease: week 12 interim analyses from the phase 2 GALAXI 1 study. *United European Gastroenterology Journal* 2020;**8**(8 Suppl):64.

254. Early pro-2 symptom remission following guselkumab induction treatment: results through week 12 of the phase 2 GALAXI 1 study. Afzali A, Sandborn WJ, Hisamatsu T, Weisel K, Gonzalez S, Frustaci ME, et al. *American Journal of Gastroenterology* 2021;**116** (Suppl):S344-5.

255. Sands BE, Anderson FH, Bernstein CN, Chey WY, Feagan BG, Fedorak RN, et al. Infliximab maintenance therapy for fistulizing Crohn's disease. *New England Journal of Medicine* 2004;**350**(9):876-85.

256. Sands BE, Blank MA, Diamond RH, Barrett JP, Van Deventer SJ. Maintenance infliximab does not result in increased abscess development in fistulizing Crohn's disease: results from the ACCENT II study. *Alimentary Pharmacology & Therapeutics* 2006;**23**:1127-36.

257. Wagner C, Hanauer SB, Bala M, Patel K, Travers S, Olson A, et al. Relationship of antibodies to infliximab to duration of response and infusion reactions for Crohn's disease patients receiving episodic infliximab retreatment in ACCENT I. *Gastroenterology* 2003;**124**(4 Suppl 1):A378.

258. Sands BE, Blank MA, Diamond RH, Barrett JP, Van Deventer SJ. Maintenance infliximab does not result in increased abscess development in fistulizing Crohn's disease: results from the ACCENT II study. *Alimentary Pharmacology & Therapeutics* 2006;**23**(8):1127-36.

259. Lichtenstein GR, Yan S, Bala M, Blank M, Sands BE. Infliximab maintenance treatment reduces hospitalizations, surgeries, and procedures in fistulizing Crohn's disease. *Gastroenterology* 2005;**128**(4):862-9.

260. Sands B, Blank M, Marster P, van Deventer S. Long-term treatment of rectovaginal fistulas in Crohn's disease: response to infliximab in the ACCENT II trial. *Gastroenterology* 2003;**124**(4 Suppl 1):A380.

261. Sands BE, Blank MA, Patel K, Van Deventer SJ. Long-term treatment of rectovaginal fistulas in Crohn's disease: response to infliximab in the ACCENT II study. *Clinical Gastroenterology and Hepatology* 2004;**2**(10):912-20.

262. Sands BE, Yan S, Bala M, Blank MA, van Deventer SJ. Maintenance treatment with infliximab reduces the likelihood of hospitalization in patients with fistulizing Crohn's disease. *Gastroenterology* 2004;**126**(4 Suppl 2):A466.

263. Fasanmade AA, Marsters P, Munsanje E, Graham MA, Davis HM, van Deventer S. Infliximab pharmacokinetics and improvement in fistulizing Crohn's disease. *Gastroenterology* 2003;**124**(4 Suppl 1):A61.

264. van Deventer SJ, Blank M, Waldschmidt S, Sands B. Infliximab does not result in increased abscess development in fistulizing Crohn's disease: results from ACCENT II. *Gastroenterology* 2003;**124**(4 Suppl 1):A379.

265. NCT00207766. A study of the safety and efficacy of infliximab (Remicade) in subjects with fistulizing Crohn's disease. <https://clinicaltrials.gov/ct2/show/NCT00207766> (first received 27 April 2010).

266. Sands B, Kozarek R, Spainhour J, Barish CF, Becker S, Goldberg L, et al. Safety and tolerability of concurrent natalizumab treatment for patients with Crohn's disease not in remission while receiving infliximab. *Inflammatory Bowel Diseases* 2007;**13**(1):2-11.

- 267.** NCT00055536. Safety and efficacy of natalizumab in combination with remicade in the treatment of Crohn's disease. <https://ClinicalTrials.gov/show/NCT00055536> (first received 6 March 2003).
- 268.** Sands BE, Jacobson EW, Sylwestrowicz T, Younes Z, Dryden G, Fedorak R, et al. Randomized, double-blind, placebo-controlled trial of the oral interleukin-12/23 inhibitor apilimod mesylate for treatment of active Crohn's disease. *Inflammatory Bowel Diseases* 2010;**16**(7):1209-18.
- 269.** Sands B, Feagan B, Rutgeerts P, Colombel JF, Sandborn W, Sy R, et al. Effects of vedolizumab induction therapy for patients with Crohn's disease in whom tumor necrosis factor antagonist treatment failed. *Gastroenterology* 2014;**147**(3):618-27.
- 270.** Sands B, Feagan B, Rutgeerts P, Colombel JF, Sandborn W, Sy R, et al. Vedolizumab induction therapy for patients with Crohn's disease and prior anti-tumour necrosis factor antagonist failure: a randomised, placebo-controlled, double-blind, multicentre trial. *Journal of Crohn's & Colitis* 2013;**7**(Suppl 1):S5-6.
- 271.** Sandborn WJ, Rutgeerts P, Xu J, Abhyankar B, Fox I. Efficacy of induction treatment with vedolizumab for patients with Crohn's disease who have experienced tumour necrosis factor antagonist failure or are tumour necrosis factor antagonist naive. *Journal of Crohn's & Colitis* 2014;**8**:S50.
- 272.** Sands BE, Van Assche G, Tudor D, Tan T. Vedolizumab in combination with steroids for induction therapy in Crohn's disease: an exploratory analysis of the GEMINI 2 and GEMINI 3 studies. *Journal of Crohn's & Colitis* 2018;**12**(Suppl 1):S68.
- 273.** Sands B, Van Assche G, Tudor D, Akhundova-Unadkat G, Tan T, Curtis R, et al. Vedolizumab in combination with corticosteroids for induction therapy in Crohn's disease: a post hoc analysis of GEMINI 2 and 3. *Inflammatory Bowel Diseases* 2019;**25**(8):1375-82.
- 274.** Dulai P, Peyrin-Biroulet L, Demuth D, Lasch K, Hahn K, Lindner D, et al. Early intervention with vedolizumab on longer term surgery rates in Crohn's disease: post hoc analysis of the GEMINI Phase 3 and long-term safety programs. *Journal of Crohn's & Colitis* 2020;**15**(2):195-202.
- 275.** Sands BE, Sandborn W, Rutgeerts PJ, Xu J, Abhyankar B, Fox I. Efficacy of vedolizumab induction therapy in patients with Crohn's disease who have experienced tumor necrosis factor antagonist failure or are tumor necrosis factor antagonist naive. *Gastroenterology* 2014;**146**(5 Suppl 1):S-149.
- 276.** Sands B, Sandborn W, Van Assche G, Lukas M, Xu J, James A, et al. Vedolizumab as induction and maintenance therapy for Crohn's disease in patients naive to or who have failed tumor necrosis factor antagonist therapy. *Inflammatory Bowel Diseases* 2017;**23**(1):97-106.
- 277.** Steinwurz F, Carvalho NS, Flaquer FS. Evaluation of the use of vedolizumab in 46 patients with inflammatory bowel disease in Brazil. *American Journal of Gastroenterology* 2017;**112**(Suppl 1):S428-9.
- 278.** NCT01224171. Study of vedolizumab in patients with moderate to severe Crohn's disease. <https://ClinicalTrials.gov/show/NCT01224171> (first received 19 October 2010).
- 279.** Sands B, Feagan B, Rutgeerts P, Colombel JF, Sandborn W, Sy R, et al. Vedolizumab induction therapy for patients with crohn's disease and prior anti-tnf antagonist failure: a randomized, placebo-controlled, double-blind, multicenter trial. *Inflammatory Bowel Diseases* 2012;**18**:S24-5.
- 280.** EUCTR2009-016488-12-HU. A phase 3, randomized, placebo-controlled, blinded, multicenter study of the induction of clinical response and remission by vedolizumab in patients with moderate to severe Crohn's disease. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2009-016488-12-HU> (first received 2010).
- 281.** Sands BE, Irving PM, Hoops T, Izanec JL, Gao LL, Gasink C, et al. Ustekinumab versus adalimumab for induction and maintenance therapy in biologic-naive patients with moderately to severely active Crohn's disease: a multicentre, randomised, double-blind, parallel-group, phase 3b trial. *Lancet* 2022;**399**(10342):2200-11.
- 282.** Loftus EV, Danese S, Panaccione R, Hoops T, Izanec JL, Ding Z, et al. Health-related quality of life with ustekinumab vs adalimumab for induction and maintenance therapy in biologic-naive patients with moderate-to-severe Crohn's disease: LBDQ in the SEAVUE study. *American Journal of Gastroenterology* 2021;**116**(Suppl):S355-7.
- 283.** Lewis JD, Allez M, Irving PM, Hoops T, Izanec JL, Ding Z, et al. Health-related quality of life with ustekinumab vs adalimumab for induction and maintenance therapy in biologic-naive patients with moderate-to-severe Crohn's disease: Promis-29 in the SEAVUE study. *American Journal of Gastroenterology* 2021;**116**(Suppl):S395.
- 284.** Danese S, Sands BE, Irving P, Hoops T, Izanec JL, Ma T, et al. The pharmacokinetics and immunogenicity of ustekinumab and adalimumab in patients with moderate-to-severe Crohn's disease: results from the SEAVUE study. *United European Gastroenterology Journal* 2021;**9**(10):1215-7.
- 285.** NCT03464136. Safety and efficacy of adalimumab versus ustekinumab for one year. <https://ClinicalTrials.gov/show/NCT03464136> (first received 13 March 2018).
- 286.** Sands BE, Irving PM, Hoops T, Izanec JL, Gao LL, Gasink C, et al. 775d ustekinumab versus adalimumab for induction and maintenance therapy in moderate-to-severe Crohn's disease: the SEAVUE study. *Gastroenterology* 2021;**161**(2):e30-1.
- 287.** EUCTR2017-004209-41. A study to compare the efficacy and safety of ustekinumab with adalimumab in the treatment of subjects with moderately-to-severely active Crohn's disease, not undergone treatment with biologic. <https://trialsearch.who.int/Trial2.aspx?TrialID=EUCTR2017-004209-41-NL> (first received 21 June 2018).
- 288.** Sands BE, Peyrin-Biroulet L, Kierkus J, Higgins PD, Fischer M, Jairath V, et al. Efficacy and safety of mirikizumab in a randomized phase 2 study of patients with Crohn's disease. *Gastroenterology* 2022;**162**(2):495-508.

- 289.** Chen M, Gao X, Cao Q, Rossiter G, Kitagawa T, Sun Y, et al. Efficacy and safety of intravenous vedolizumab treatment in Chinese patients with moderate-to-severe Crohn's disease. *Clinics and Research in Hepatology and Gastroenterology* 2025;**49**(6):102591.
- 290.** NCT03234907. Vedolizumab intravenous (IV) compared to placebo in chinese participants with Crohn's disease. <https://ClinicalTrials.gov/show/NCT03234907> (first received 31 July 2017).
- 291.** NCT03926130. A study of mirikizumab (LY3074828) in participants with Crohn's disease. <https://ClinicalTrials.gov/show/NCT03926130> (first received 24 April 2019).
- 292.** Ferrante M, D'Haens G, Jairath V, Danese S, Chen M, Ghosh S et al. Efficacy and safety of mirikizumab in patients with moderately-to-severely active Crohn's disease: a phase 3, multicentre, randomised, double-blind, placebo-controlled and active-controlled, treat-through study. *Lancet* 2024;**404**(10470):2423-36.
- 293.** NCT05197049. A study of guselkumab subcutaneous therapy in participants with moderately to severely active Crohn's disease (GRAVITI). <https://clinicaltrials.gov/study/NCT05197049> (first received January 2022).
- 294.** Hart A, Panaccione R, Steinwurz F, Danese S, Hisamatsu T, Cao Q, et al. Efficacy and safety of guselkumab subcutaneous induction and maintenance in participants with moderately to severely active Crohn's disease: results from the phase 3 GRAVITI study. *Gastroenterology* 2025;**no volume**:no page.
- 295.** Jairath V, Kierkus J, Duvall GA, Danese S, Sands BE, Ratiu-Duma B, et al. OP40 Duvakitug (TEV-48574), an anti-TL1a monoclonal antibody, demonstrates efficacy and favourable safety as an induction treatment in adults with moderately to severely active Crohn's disease: results from a phase 2b, randomised, double-blind, placebo-controlled dose-ranging, basket trial (RELIEVE UCCD). *Journal of Crohn's & Colitis* 2019;**19**(Suppl 1):i77-8.
- 296.** Loftus Jr EV, Panés J, Lacerda AP, Peyrin-Biroulet L, D'Haens G, Panaccione R, et al. Upadacitinib induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2023;**388**(21):1966-80.
- 297.** Abbvie. Second phase 3 induction study confirms upadacitinib (RINVOQ®) improved clinical and endoscopic outcomes in patients with Crohn's disease. <https://news.abbvie.com/news/press-releases/news-type/rd-news/second-phase-3-induction-study-confirms-upadacitinib-rinvoq-improved-clinical-and-endoscopic-outcomes-in-patients-with-crohns-disease.htm>.
- 298.** NCT03345849. A study of the efficacy and safety of upadacitinib (ABT-494) in participants with moderately to severely active Crohn's disease who have inadequately responded to or are intolerant to conventional and/or biologic therapies. <https://clinicaltrials.gov/ct2/show/NCT03345849> (first received 17 November 2017).
- 299.** Matsumoto T, Motoya S, Watanabe K, Hisamatsu T, Nakase H, Yoshimura N, et al. Adalimumab monotherapy and a combination with azathioprine for Crohn's disease: a prospective, randomized trial. *Journal of Crohn's & Colitis* 2016;**10**(11):1259-66.
- 300.** UMIN000005146. Study of combination therapy of adalimumab and an immunomodulator for Crohn's disease(DIAMOND study). https://center6.umin.ac.jp/cgi-open-bin/ctr_e/ctr_view.cgi?recptno=R000006111 (first registered February 2011).
- 301.** Panaccione R, Feagan BG, Afzali A, Rubin DT, Reinisch W, Panés J, et al. Efficacy and safety of intravenous induction and subcutaneous maintenance therapy with guselkumab for patients with Crohn's disease (GALAXI-2 and GALAXI-3): 48-week results from two phase 3, randomised, placebo and active comparator-controlled, double-blind, triple-dummy trials. *Lancet* 2025;**no volume**:no page.
- 302.** NCT03466411. A study of the efficacy and safety of guselkumab in participants with moderately to severely active Crohn's disease (GALAXI). <https://clinicaltrials.gov/study/NCT03466411> (first posted April 2018).
- 303.** Panaccione R, Feagan BG, Afzali A, Rubin DT, Reinisch W, Panés J, et al. Efficacy and safety of intravenous induction and subcutaneous maintenance therapy with guselkumab for patients with Crohn's disease (GALAXI-2 and GALAXI-3): 48-week results from two phase 3, randomised, placebo and active comparator-controlled, double-blind, triple-dummy trials. *Lancet* 2025;**no volume**:no page.
- 304.** NCT03466411. A study of the efficacy and safety of guselkumab in participants with moderately to severely active Crohn's disease (GALAXI). <https://clinicaltrials.gov/study/NCT03466411> (first published in April 2018).
- 305.** Peyrin-Biroulet L, Chapman JC, Colombel JF, Caprioli F, D'Haens G, Ferrante M, et al. Risankizumab versus ustekinumab for moderate-to-severe Crohn's disease. *New England Journal of Medicine* 2024;**18**(391(3)):213-23.
- 306.** Peyrin-Biroulet L. SEQUENCE 3b study risankizumab versus ustekinumab for patients with moderate to severe Crohn's disease. In: European Gastroenterology Week (UEGW 2023). 2023.
- 307.** NCT04524611. Study comparing intravenous (IV)/subcutaneous (SC) risankizumab to IV/SC ustekinumab to assess change in Crohn's Disease Activity Index (CDAI) in adult participants with moderate to severe Crohn's disease (CD). <https://ClinicalTrials.gov/show/NCT04524611> (first received 24 August 2020).
- 308.** EUCTR2020-002674-26. A study to compare the efficacy of risankizumab and ustekinumab in subjects with moderately to severe active Crohns disease. <https://trialsearch.who.int/Trial2.aspx?TrialID=EUCTR2020-002674-26-NL> (first received 18 November 2020).
- 309.** Schreiber S, Rutgeerts P, Fedorak RN, Khaliq-Kareemi M, Kamm MA, Boivin M, et al. A randomized, placebo-controlled trial of certolizumab pegol (CDP870) for treatment of Crohn's disease. [Erratum appears in *Gastroenterology*.

2005 Nov;**129**(5):1808 Note: dosage error in article text]. *Gastroenterology* 2005;**129**(3):807-18.

310. Schreiber S, Rutgeerts P, Fedorak RN, Khaliq-Kareemi M, Kamm MA, Boivin M, et al. Erratum: a randomized, placebo-controlled trial of certolizumab pegol (CDP870) for treatment of Crohn's disease (*Gastroenterology* (2005) 129 (807-818)). *Gastroenterology* 2005;**129**:1808.

311. Schreiber S, Feagan B, Keininger DL, O'Neil L, Fedorak RN, Rutgeerts P. Certolizumab pegol, a monthly subcutaneously administered FC-free anti-TNFalpha, improves health-related quality of life in patients with moderate to severe Crohn's disease. *International Journal of Colorectal Disease* 2008;**23**:289-96.

312. NCT00291668. Clinical study of CDP870/certolizumab pegol in patients with active Crohn's disease. <https://ClinicalTrials.gov/show/NCT00291668> 2006.

313. Schreiber S, Siegel CA, Friedenberg KA, Younes ZH, Seidler U, Bhandari BR et al. A phase 2, randomized, placebo-controlled study evaluating matrix metalloproteinase-9 inhibitor, andecaliximab, in patients with moderately to severely active Crohn's disease. *Journal of Crohn's & Colitis* 2018;**12**(9):1014-20.

314. Schreiber S, Siegel C, Friedenberg K, Seidler U, Bhandari BR, Younes Z, et al. Andecaliximab (anti-MMP9) induction therapy for Crohn's disease: a double-blind, randomized, placebo-controlled, phase 2 study. *United European Gastroenterology Journal* 2017;**5**(5 Suppl 1):A302-3.

315. NCT02630966. Vedolizumab IV 300 mg in the treatment of fistulizing Crohn's disease (ENTERPRISE). <https://clinicaltrials.gov/ct2/show/NCT02630966> (first received 15 December 2015).

316. EUCTR2015-000852-12. A randomized double-blind phase 4 study to evaluate the safety and proportion of subjects with fistula healing in 2 dose regimens of entyvio (vedolizumab iv) in the treatment of fistulizing Crohn's disease. <https://trialsearch.who.int/Trial2.aspx?TrialID=EUCTR2015-000852-12-ES> (first received 27 July 2016).

317. Schwartz DA, Peyrin-Biroulet L, Lasch K, Adsul S, Danese S. Efficacy and safety of 2 vedolizumab intravenous regimens for perianal fistulizing Crohn's disease: ENTERPRISE study. *Clinical Gastroenterology and Hepatology* 2021;**no volume**:no pagination.

318. Targan SR, Hanauer SB, Van Deventer SJ, Mayer L, Present DH, Braakman T, et al. A short-term study of chimeric monoclonal antibody cA2 to tumor necrosis factor alpha for Crohn's disease. *New England Journal of Medicine* 1997;**337**(15):1029-35.

319. D'Haens G, Van Deventer S, Van Hogezaand R, Chalmers D, Kothe C, Baert F, et al. Endoscopic and histological healing with infliximab anti-tumor necrosis factor antibodies in Crohn's disease: a European multicenter trial. *Gastroenterology* 1999;**116**(5):1029-34.

320. Lichtenstein GR, Bala M, Han C, DeWoody K, Schaible T. Infliximab improves quality of life in patients with Crohn's disease. *Inflammatory Bowel Diseases* 2002;**8**(4):237-43.

321. NCT00269854. An efficacy and safety study of anti-TNF monoclonal antibody in patients with Crohn's disease. <https://ClinicalTrials.gov/show/NCT00269854> 1995.

322. Targan SR, Feagan BG, Fedorak RN, Lashner BA, Panaccione R, Present DH, et al. Natalizumab for the treatment of active Crohn's disease: results of the encore trial. *Gastroenterology* 2007;**132**(5):1672-83.

323. Panaccione R, Colombel J, Enns R, Feagan B, Hanauer S, Lawrance I. Natalizumab induces and maintains continuous response and remission in patients with active Crohn's disease despite prior anti-tumor necrosis factor-alpha treatment. *Inflammatory Bowel Diseases* 2008;**14**:S10.

324. Panaccione R, Feagan BG, Hass S, Panjabi S. The impact of natalizumab on the quality of life of patients exposed to or failing anti-TNFalpha therapy in the ENCORE trial. *Gastroenterology* 2009;**136**:A180.

325. Present D, Feagan B, Fedorak R, Lashner B, Panaccione R, Rutgeerts P. Natalizumab induces sustained response and remission in Crohn's patients after previous infliximab failure: results from the ENCORE trial. *Gastroenterology* 2007;**132**:507.

326. Targan SR, Feagan B, Fedorak R, Lashner B, Panaccione R, Present D. Natalizumab induces sustained response and remission with active Crohn's disease: results from the ENCORE trial. *Gastroenterology* 2006;**130**:A108.

327. Panaccione R, Feagan B, Fedorak R, Lashner B, Present D, Rutgeerts P. Natalizumab induces sustained response and remission in patients with Crohn's disease activity index scores 330: results from the ENCORE trial. *UEGW Abstract Database* 2007;**no volume**:no pagination.

328. Rutgeerts P, Feagan B, Fedorak R, Lashner B, Panaccione R, Present D. Natalizumab induces sustained response and remission in Crohn's disease patients after previous anti-TNF alpha therapy failure: results from the ENCORE trial. *UEGW Abstract Database* 2007;**no volume**:no pagination.

329. Spehlmann M, Feagan B, Fedorak R, Lashner B, Panaccione R, Present D. Natalizumab induces sustained response and remission in patients with active Crohn's disease: ENCORE trial results. *UEGW Abstract Database* 2006;**no volume**:no pagination.

330. NCT00078611. A clinical trial of natalizumab in individuals with moderately to severely active Crohn's disease. <https://ClinicalTrials.gov/show/NCT00078611> (first received 8 March 2004).

331. Vermeire S, Schreiber S, Petryka R, Kuehbachner T, Hebuterne X, Roblin X, et al. Clinical remission in patients with moderate-to-severe Crohn's disease treated with filgotinib (the FITZROY study): results from a phase 2, double-blind, randomised, placebo-controlled trial. *Lancet* 2017;**389**:266-75.

- 332.** Vermeire S, Schreiber S, Petryka R, Kuehbach T, Hebuterne X, Roblin X, et al. Clinical remission in patients with moderate-to-severe Crohn's disease treated with filgotinib (the FITZROY study): results from a phase 2, double-blind, randomised, placebo-controlled trial. *Lancet* 2017;**389**:266-75.
- 333.** Vermeire S, Schreiber S, Rubin DT, D'Haens G, Reinisch W, Watanabe M, et al. Efficacy and safety of filgotinib as induction and maintenance therapy for Crohn's disease (DIVERSITY): a phase 3, double-blind, randomised, placebo-controlled trial. *Lancet Gastroenterology & Hepatology* 2025;**1**(10):138-53.
- 334.** NCT02914561. Filgotinib in the induction and maintenance of remission in adults with moderately to severely active Crohn's disease (DIVERSITY1). <https://clinicaltrials.gov/study/NCT02914561> (first published October 2016).
- 335.** Vermeire S, Schreiber S, Rubin DT, D'Haens G, Reinisch W, Watanabe M, et al. Efficacy and safety of filgotinib as induction and maintenance therapy for Crohn's disease (DIVERSITY): a phase 3, double-blind, randomised, placebo-controlled trial. *Lancet Gastroenterology & Hepatology* 2025;**1**(10):138-53.
- 336.** NCT02914561. Filgotinib in the induction and maintenance of remission in adults with moderately to severely active Crohn's disease (DIVERSITY1). <https://clinicaltrials.gov/study/NCT02914561> (first published October 2016).
- 337.** Watanabe M, Hibi T, Lomax KG, Paulson SK, Chao J, Alam MS, et al. Adalimumab for the induction and maintenance of clinical remission in Japanese patients with Crohn's disease. *Journal of Crohn's & Colitis* 2012;**6**(2):160-73.
- 338.** Hibi T, Watanabe M, Camez AA, Alam S, Chao J, Mulani P, et al. Japanese patients with moderately to severely active Crohn's disease experience improved quality of life with adalimumab treatment. *Gastroenterology* 2010;**138**(5 Suppl 1):S321.
- 339.** Watanabe K, Motoya S, Ogata H, Kanai T, Matsui T, Suzuki Y, et al. Effects of vedolizumab in Japanese patients with Crohn's disease: a prospective, multicenter, randomized, placebo-controlled phase 3 trial with exploratory analyses. *Journal of Gastroenterology* 2020;**55**(3):291-306.
- 340.** NCT02038920. Phase III study of MLN0002 (300 mg) in treatment of Crohn's disease. <https://ClinicalTrials.gov/show/NCT02038920> 2014.
- 341.** Ogata H, Motoya S, Watanabe K, Kanai T, Matsui T, Suzuki Y, et al. A phase 3 study of vedolizumab for induction and maintenance therapy in Japanese patients with moderate to severe Crohn's disease. *Gastroenterology* 2019;**156**(6 Suppl 1):S-1109.
- 342.** Winter TA, Wright J, Ghosh S, Jahnsen J, Innes A, Round P. Intravenous CDP870, a pegylated fab' fragment of a humanized antitumour necrosis factor antibody, in patients with moderate-to-severe Crohn's disease: an exploratory study. *Alimentary Pharmacology & Therapeutics* 2004;**20**:1337-46.
- 343.** Ye BD, Pesegova M, Alexeeva O, Osipenko M, Lahat A, Dorofeyev A, et al. Efficacy and safety of biosimilar ct-p13 compared with originator infliximab in patients with active Crohn's disease: an international, randomised, double-blind, phase 3 non-inferiority study. *Lancet* 2019;**393**:1699-707.
- 344.** NCT02096861. Demonstrate noninferiority in efficacy and to assess safety of CT-P13 in patients with active Crohn's disease. <https://ClinicalTrials.gov/show/NCT02096861> 2014.
- 345.** Ye BD, Leszczyszyn J, Dudkowiak R, Lahat A, Ben-Horin S, Gawdis-Wojnarska B, et al. Exposure-response relationship of subcutaneous infliximab (ct-p13 sc) in patients with active Crohn's disease and ulcerative colitis: analysis from a multicenter, randomized controlled pivotal trial. *United European Gastroenterology Journal* 2020;**8**:385-6.
- 346.** Kim YH, Ye BD, Pesegova M, Alexeeva O, Osipenko M, Lahat A, et al. Phase III randomised, double-blind, controlled trial to compare biosimilar infliximab (ct-p13) with innovator infliximab in patients with active Crohn's disease: early efficacy and safety results. *Journal of Crohn's & Colitis* 2017;**11**:S62.
- 347.** Kim Y, Ye BD, Pesegova M, Alexeeva O, Osipenko M, Lahat A, et al. Phase III randomized controlled trial to compare biosimilar infliximab (ct-p13) with innovator infliximab in patients with active Crohn's disease: 1-year maintenance and switching results. *United European Gastroenterology Journal* 2017;**5**:1139-40.
- 348.** Kim YH, Ye BD, Pesegova M, Alexeeva O, Osipenko M, Lahat A, et al. Phase III randomized, double-blind, controlled trial to compare biosimilar infliximab (ct-p13) with innovator infliximab (inx) in patients with active Crohn's disease: early efficacy and safety results. *Gastroenterology* 2017;**152**:S65.
- 349.** Ye BD, Kim YH, Pesegova M, Alexeeva O, Osipenko M, Lahat A, et al. Phase III randomized controlled trial to compare biosimilar infliximab (ct-p13) with innovator infliximab in patients with active Crohn's disease: 1-year maintenance and switching results. *Gastroenterology* 2018;**154**:S-168.
- 350.** Buhl S, Steenholdt C, Brynskov J, Christensen KR, Dorn-Rasmussen M, Thomsen OØ, et al. Discontinuation of infliximab therapy in patients with Crohn's disease. *NEJM Evidence* 2022;**1**(8):EVIDoa2200061.
- 351.** Buhl SS, Steenholdt C, Brynskov J, Thomsen OO, Ainsworth MA, Bendtzen K. Discontinuation of infliximab therapy in patients with Crohn's disease in sustained complete remission (the STOP IT study): protocol for a double-blind, randomised, placebo-controlled, multicentre trial. *BMJ Open* 2014;**4**(12):e005887.
- 352.** NCT01817426. Discontinuation of infliximab therapy in patients with Crohn's disease during sustained complete remission (STOP IT). <https://clinicaltrials.gov/ct2/show/NCT01817426> (first received 25 March 2013).
- 353.** Buhl S, Steenholdt C, Brynskov J, Christensen KR, Dorn-Rasmussen M, Thomsen OO, et al. Discontinuation of infliximab therapy in patients with Crohn's disease in sustained, complete clinical-biochemical-endoscopic remission: a double-blinded, placebo-controlled, randomized clinical trial. *Gastroenterology* 2021;**160**(6 Suppl):S-152.

- 354.** EUCTR2012-002702-51-FI. Discontinuation of biologic therapy (infliximab) in patients with Crohn's disease during sustained complete absence of disease activity: a Nordic multi-center, double blinded, randomized, placebo controlled study. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2012-002702-51/DK> 2015.
- 355.** Colombel JF, Sandborn WJ, Rutgeerts P, Enns R, Hanauer SB, Panaccione R, et al. Adalimumab for maintenance of clinical response and remission in patients with Crohn's disease: the CHARM trial. *Gastroenterology* 2007;**132**(1):52-65.
- 356.** Rubin DT, Bensimon AG, Yu AP, Wu E, Chao J, Mulani P. Increased risk of clinical relapse is associated with longer disease duration in Crohn's disease. *Gastroenterology* 2009;**136**(5 Suppl 1):A197.
- 357.** NCT00077779. Adalimumab for the induction and maintenance of clinical remission in subjects with Crohn's disease. <https://ClinicalTrials.gov/show/NCT00077779> (first received 6 February 2004).
- 358.** Vermeire S, Colombel J, Robinson AM, Huang B, Thakkar RB, Pollack PF. Effect of concomitant immunomodulator use on the efficacy of adalimumab in Crohn's disease patients stratified by prior anti-TNF use. *Journal of Crohn's & Colitis* 2011;**5**(1):S68-9.
- 359.** Panaccione R, Colombel JF, Sandborn WJ, D'Haens G, Zhou Q, Pollack PF, et al. Adalimumab maintains remission of Crohn's disease after up to 4 years of treatment: data from CHARM and ADHERE. *Alimentary Pharmacology & Therapeutics* 2013;**38**(10):1236-47.
- 360.** Schreiber S, Reinisch W, Colombel JF, Sandborn W, Robinson A, Huang B, et al. Subgroup analysis of the placebo-controlled CHARM trial: increased remission rates for adalimumab-treated patients with early Crohn's disease. *Journal of Crohn's & Colitis* 2012;**6**:S106.
- 361.** Schreiber S, Reinisch W, Colombel JF, Sandborn WJ, Hommes DW, Robinson AM, et al. Subgroup analysis of the placebo-controlled CHARM trial: Increased remission rates through 3 years for adalimumab-treated patients with early Crohn's disease. *Journal of Crohn's & Colitis* 2013;**7**(3):213-21.
- 362.** Panaccione R, Colombel JF, Sandborn WJ, Robinson A, Chao J, Mulani P, et al. Long-term maintenance of clinical remission with reduced dosing frequency of adalimumab in patients with moderate to severe Crohn's disease. *Gastroenterology* 2010;**138**(5 Suppl 1):S468-9.
- 363.** Colombel JF, Schwartz DA, Sandborn WJ, Kamm MA, D'Haens G, Rutgeerts P, et al. Adalimumab for the treatment of fistulas in patients with Crohn's disease. *Gut* 2009;**58**(7):940-8.
- 364.** Kamm MA, Hanauer SB, Panaccione R, Colombel JF, Sandborn WJ, Pollack PF, et al. Adalimumab sustains steroid-free remission after 3 years of therapy for Crohn's disease. *Alimentary Pharmacology & Therapeutics* 2011;**34**(3):306-17.
- 365.** Sandborn WJ, Rutgeerts PJ, Colombel JF, Reinisch W, Loftus EV, Tang J, et al. Adalimumab 160-/80-mg induction regimen is associated with better outcomes compared with the 80-/40-mg regimen during maintenance therapy in patients with Crohn's disease. *Gastroenterology* 2011;**140**(5 Suppl 1):S263.
- 366.** Panaccione R, Colombel JF, Sandborn WJ, Rutgeerts P, D'Haens GR, Robinson AM, et al. Adalimumab sustains clinical remission and overall clinical benefit after 2 years of therapy for Crohn's disease. *Alimentary Pharmacology & Therapeutics* 2010;**31**(12):1296-309.
- 367.** Colombel JF, Sandborn W, Castillo M, Zhou Q, Thakkar R. Efficacy and safety of adalimumab in moderate compared with severe Crohn's disease: pooled data from the CHARM and EXTEND trials. *American Journal of Gastroenterology* 2011;**106**:S434-5.
- 368.** Colombel JF, Sandborn WJ, Rutgeerts P, Kamm MA, Yu AP, Wu EQ, et al. Comparison of two adalimumab treatment schedule strategies for moderate-to-severe Crohn's disease: results from the CHARM trial. *American Journal of Gastroenterology* 2009;**104**(5):1170-9.
- 369.** Colombel JF. The CHARM trial of adalimumab in Crohn's disease. *Gastroenterology and Hepatology (Quebec)* 2006;**2**(7):486-8.
- 370.** Colombel JF, Sandborn WJ, Castillo MM, Zhou Q, Thakkar RB. Efficacy and safety of adalimumab in moderate compared with severe Crohn's disease: pooled data from the CHARM and EXTEND trials. *Gut* 2012;**61**:A229.
- 371.** Feagan BG, Panaccione R, Sandborn WJ, D'Haens GR, Schreiber S, Rutgeerts PJ, et al. Effects of adalimumab therapy on incidence of hospitalization and surgery in Crohn's disease: results from the CHARM study. *Gastroenterology* 2008;**135**(5):1493-9.
- 372.** Loftus EV, Feagan BG, Colombel JF, Rubin DT, Wu EQ, Yu AP, et al. Effects of adalimumab maintenance therapy on health-related quality of life of patients with Crohn's disease: patient-reported outcomes of the CHARM trial. *American Journal of Gastroenterology* 2008;**103**(12):3132-41.
- 373.** Loftus E, Yang M, Mulani P, Chao J. Adalimumab maintenance therapy improves patient-reported outcomes in patients with moderate crohn's disease: subanalysis of CHARM. *American Journal of Gastroenterology* 2011;**106**:S449-50.
- 374.** Rubin DT, Mulani P, Chao J, Pollack PF, Bensimon AG, Yu AP, et al. Effect of adalimumab on clinical laboratory parameters in patients with Crohn's disease: results from the CHARM trial. *Inflammatory Bowel Diseases* 2012;**18**(5):818-25.
- 375.** Plevy S, Schreiber S, Colombel J, Pollack P, Chao J, Mulani P. Resolution of flare or nonresponse with no dosage increase in most adalimumab-treated patients with Crohn's disease. *Canadian Journal of Gastroenterology* 2009;**23**(Suppl SA):no pagination.
- 376.** Colombel E, Colombel J, Pollack P, Majethia S, Chen N. Adalimumab treatment associated with rapid and significant improvements in health-related quality of life in patients with Crohn's disease. *Canadian Journal of Gastroenterology* 2009;**23**(Suppl. SA):no pagination.

- 377.** Wasvary HJ. Adalimumab for maintenance of clinical response and remission in patients with Crohn's disease: the CHARM trial. *Diseases of the Colon and Rectum* 2007;**50**(8):1291.
- 378.** Loftus EV, Colombel JF, Rutgeerts P, Rubin DT, Chen N, Mulani P, et al. Quality-of-life improvements in patients with Crohn's disease treated for 3 years with adalimumab in an open-label extension of CHARM. *Gastroenterology* 2009;**136**(5 Suppl 1):A179.
- 379.** Loftus EV, Rubin DT, Chen N, Mulani P, Chao J. Remission and quality-of-life improvements sustained with adalimumab in anti-TNF-naïve patients with Crohn's disease: 3-year data from CHARM. *Gastroenterology* 2009;**136**(5 Suppl 1):A180.
- 380.** Panaccione R, Sandborn W, Colombel J, Pollack P, Chen N, Chao J, et al. Response with continued adalimumab therapy for up to 12 weeks in patients with Crohn's disease who were nonresponders at week 4. *Canadian Journal of Gastroenterology* 2009;**23**(Suppl SA):no pagination.
- 381.** Panaccione R, Colombel J, Sandborn W, Rutgeerts P, D'Haens G, Lomax K, et al. Adalimumab maintains long-term remission in moderately to severely active Crohn's disease through 2 years. *Canadian Journal of Gastroenterology* 2009;**23**(Suppl SA):no pagination.
- 382.** Panaccione R, Colombel J, Sandborn W, Rutgeerts P, D'Haens G, Lomax K, et al. Adalimumab therapy maintains remission for up to 3 years in patients with moderately to severely active Crohn's disease: results from an open-label extension of CHARM. *Inflammatory Bowel Diseases* 2009;**15**(Suppl 2):S43.
- 383.** Rubin D, Rutgeerts P, Colombel J, Wu E, Yu A, Chao J, et al. Adalimumab maintenance therapy and health-related quality of life in TNF-antagonist-naïve patients with Crohn's disease. *Canadian Journal of Gastroenterology* 2009;**23**(Suppl SA):no pagination.
- 384.** Rutgeerts P, Colombel JF, Sandborn WJ, Rutgeerts PJ, D'Haens GR, Lomax KG, et al. Maintenance of long-term remission in patients with moderately to severely active Crohn's disease treated for 3 years with adalimumab: results from the open-label ADHERE study. *Gastroenterology* 2009;**136**(5 Suppl 1):A652.
- 385.** Ghosh S, Wu E, Bensimon A, Mulani P, Yang M, Chao J. C-reactive protein levels in moderate to severe Crohn's disease patients decreased rapidly and significantly during adalimumab therapy. *American Journal of Gastroenterology* 2011;**106**(Suppl 2):S450.
- 386.** Colombel JF, Panaccione R, Louis E, Skup M, Yang M, Pollack PF, et al. Quality of life for patients with deep remission vs. clinical remission and deep remission vs. absence of mucosal ulceration: 3-year data from CHARM/ADHERE. *Gastroenterology* 2013;**144**(5 Suppl 1):S629.
- 387.** D'Haens GR, Sandborn WJ, Loftus Jr EV, Hanauer SB, Schreiber S, Peyrin-Biroulet L, et al. Higher vs standard adalimumab induction dosing regimens and two maintenance strategies: randomized SERENE CD trial results. *Gastroenterology* 2022;**162**(7):1876-90.
- 388.** NCT02065570. Study to evaluate efficacy and safety of two drug regimens in subjects with moderate to severe crohn's disease. <https://ClinicalTrials.gov/show/NCT02065570> (first received 19 February 2014).
- 389.** Feagan B, Gasink C, Lang Y, Friedman J, Blank M, Johannis J. Ustekinumab induction therapy for Crohn's disease refractory to conventional or tumor necrosis factor antagonist therapy. *New England Journal of Medicine* 2015;**375**(20):1946-60.
- 390.** Li K, Friedman JR, Chan D, Pollack P, Yang F, Jacobstein D, et al. Effects of ustekinumab on histologic disease activity in patients with Crohn's disease. *Gastroenterology* 2019;**157**:1019-31.
- 391.** Sandborn WJ, Sands BE, Colombel JF, Gasink C, Patel R, Jacobstein D, et al. P034 efficacy of ustekinumab in Crohn's disease at maintenance week 56: IM-UNITI STUDY...Crohn's & Colitis Congress, February 7-9, 2019, Las Vegas, Nevada. *Inflammatory Bowel Diseases* 2019;**25**:S16.
- 392.** Sands BE, Han C, Gasink C, Szapary P, Gao L, Lang Y, et al. Ustekinumab improves general health status and disease-specific health related quality of life of patients with moderate to severe Crohn's disease: results from the UNITI and IMUNITI phase 3 clinical trials. *Gastroenterology* 2016;**150**(4 Suppl 1):S1004.
- 393.** NCT01369355. A study to evaluate the safety and efficacy of ustekinumab maintenance therapy in patients with moderately to severely active Crohn's disease (IM-UNITI). <https://ClinicalTrials.gov/show/NCT01369355> (first received 8 June 2011).
- 394.** Li K, Chan D, Pollack P, Jacobstein D, Brodmerkel C, Gasink C, et al. Efficacy of ustekinumab for induction and maintenance of histological healing in patient's with Crohn's disease. *Gastroenterology* 2017;**152**:S595.
- 395.** Li K, Pollack P, Jacobstein D, Brodmerkel C, Chan D, Feagan B, et al. Efficacy of ustekinumab for induction and maintenance of histological healing in patient's with Crohn's disease. *Gut* 2017;**66**:A123.
- 396.** Sands B, Colombel JF, Gasink C, Jacobstein D, Gao LL, Johannis J, et al. Efficacy and safety of dose adjustment and delayed response to ustekinumab in moderate-severe Crohn's disease patients: results from the IM-UNITI maintenance study. *American Journal of Gastroenterology* 2016;**111**(Supplement 1):S302-3.
- 397.** Cohen SA, Griffiths AM, Hyams JS, Rosh JR, Turner D, Strauss R, et al. Clinical characteristics, clinical response, and clinical remission status in adults treated with ustekinumab in the phase 3 UNITI studies at week 6 and week 44 by age at Crohn's disease onset. *Gastroenterology* 2021;**160**(6 Suppl):S-44.
- 398.** Narula N, Wong EC, Dulai PS, Marshall JK, Colombel JF, Reinisch W. Week 6 calprotectin best predicts likelihood of long-term endoscopic healing in Crohn's disease: a post-hoc analysis of the UNITI/IM-UNITI trials. *Journal of Crohn's & Colitis* 2021;**15**(3):462-70.

- 399.** Sandborn WJ, Ghosh S, Han C, Gasink C, Zou B, Feagan BG. Long-term (5-YEAR) maintenance of clinically meaningful improvement in health-related quality of life in patients with moderate to severe Crohn's disease treated with ustekinumab in the IM-UNITI long-term extension study. *Gastroenterology* 2021;**160**(6 Suppl):S-559.
- 400.** EUCTR2010-022760-12-NL. A study to evaluate the safety and efficacy of ustekinumab maintenance therapy in patients with moderately to severely active Crohn's disease. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2010-022760-12-NL> 2011.
- 401.** Sands BE, Han C, Gasink C, Jacobstein D, Szapary P, Gao LL, et al. The effects of ustekinumab on health-related quality of life in patients with moderate to severe Crohn's disease. *Journal of Crohn's & Colitis* 2018;**12**(8):883-95.
- 402.** Sandborn WJ, Rebuck R, Wang Y, Zou B, Adedokun OJ, Gasink C, et al. Five-year efficacy and safety of ustekinumab treatment in Crohn's disease: the IM-UNITI trial. *Clinical Gastroenterology and Hepatology* 2021;**20**:578-590.e4. [DOI: [10.1016/j.cgh.2021.02.025](https://doi.org/10.1016/j.cgh.2021.02.025)]
- 403.** Sandborn WJ, Sands BE, Colombel JF, Gasink C, Patel R, Jacobstein D, et al. Efficacy of ustekinumab in Crohn's disease at maintenance week 56: IM-UNITI study. *Inflammatory Bowel Diseases* 2019;**25**(Suppl 1):S16-7.
- 404.** Sandborn WJ, Sands B, Colombel JF, Gasink C, Patel R, Jacobstein D, et al. Efficacy of ustekinumab in Crohn's disease at maintenance week 56: IM-UNITI study. *Journal of Crohn's & Colitis* 2019;**13**(Suppl 1):S274.
- 405.** Sandborn WJ, Rutgeerts P, Gasink C, Jacobstein D, Gao L, Johanns J, et al. Long term efficacy and safety of ustekinumab for Crohn's disease: results from IM-UNITI long-term extension through 2 years. *Gut* 2017;**66**(Suppl 2):A22.
- 406.** Sandborn WJ, Rutgeerts P, Gasink C, Jacobstein D, Gao L, Johanns J, et al. Long-term efficacy and safety of ustekinumab for Crohn's disease: results from IM-UNITI long-term extension through two years. *Gastroenterology* 2017;**152**(5 Suppl 1):S585.
- 407.** Sandborn WJ, Rutgeerts P, Gasink C, Jacobstein D, Zou B, Johanns J, et al. Long-term efficacy and safety of ustekinumab for Crohn's disease through the second year of therapy. *Alimentary Pharmacology & Therapeutics* 2018;**48**(1):65-77.
- 408.** Sandborn WJ, Rutgeerts P, Lynch J, Rebuck R, Wang Y, Zou B, et al. Efficacy and safety of ustekinumab for Crohn's disease through 5 years: final results from the IM-UNITI long-term extension. *American Journal of Gastroenterology* 2020;**115**(Suppl):S323-4.
- 409.** Sandborn WJ, Feagan BG, Gasink C, Jacobstein D, Gao LL, Johanns J, et al. A ph3 randomised, multicenter, double-blind, placebo (PBO)-controlled study of ustekinumab (UST) maintenance therapy in moderate-severe Crohn's disease (CD) pts: results from IM-Uniti. *Gut* 2016;**66**(Suppl 1):A32-4.
- 410.** Sandborn WJ, Rutgeerts P, Jacobstein D, Zou B, Johanns J, Gasink C, et al. Efficacy and safety of ustekinumab for Crohn's disease: results from IM-UNITI long-term extension through 3 years. *United European Gastroenterology Journal* 2018;**6**(8 Suppl):A121-2.
- 411.** Sands B, Colombel J, Gasink C, Jacobstein D, Gao L, Johanns J, et al. Efficacy & safety of dose adjustment & delayed response to ustekinumab in moderate-severe Crohn's disease patients: results from IM-UNITI maintenance study. *Gut* 2017;**66**(Suppl 2):A23.
- 412.** Sands B, Gasink C, Lang Y, Jacobstein D, Ott E, Ramachandran P, et al. Safety of ustekinumab with and without concomitant corticosteroids or immunosuppressants in patients with moderately-to-severely active Crohn's disease. *American Journal of Gastroenterology* 2018;**113**(Suppl 1):S24.
- 413.** Colombel J, Sands B, Gasink C, Sloan S, Oortwijn A, Van Kruchten R, et al. Combined clinical and biological response concomitant CRP and faecal calprotectin reductions in induction and maintenance from the phase 3 ustekinumab Crohn's disease studies. *Journal of Crohn's & Colitis* 2018;**12**(Suppl 1):S243.
- 414.** Loftus EV, Sattin B, Jacobstein D, Gasink C, Sloan S, Feagan BG, et al. Tolerability of ustekinumab in induction and maintenance for the treatment of Crohn's disease. *American Journal of Gastroenterology* 2017;**112**(Suppl 1):S364.
- 415.** Lofus E, Sattin B, Jacobstein D, Gasink C, Sloan S, Feagan B, et al. Tolerability of ustekinumab in induction and maintenance for the treatment of Crohn's disease. *American Journal of Gastroenterology* 2018;**113**(Supplement 1):S23-4.
- 416.** Adedokun OJ, Xu Z, Gasink C, Jacobstein D, Szapary P, Johanns J, et al. Exposure-response to SC ustekinumab in moderate-severe Crohn's disease: results from the IM-UNITI maintenance study. *United European Gastroenterology Journal* 2017;**5**(5 Suppl 1):A3.
- 417.** Sandborn W, Feagan BG, Gasink C, Jacobstein D, Gao LL, Johanns J, et al. A phase 3 randomized, multicenter, double-blind, placebo-controlled study of ustekinumab maintenance therapy in moderate-severe Crohn's disease patients: results from IM-UNITI. *Gastroenterology* 2016;**150**(4 Suppl 1):S157-8.
- 418.** Li K, Chan D, Pollack P, Jacobstein D, Brodmerkel C, Gasink C, et al. Efficacy of ustekinumab for induction and maintenance of histological healing in patients with Crohn's disease. *Journal of Crohn's & Colitis* 2017;**11**:S314-5.
- 419.** Sandborn WJ, Ghosh S, Han C, Gasink C, Zou B, Feagan BG. Long-term (5-year) maintenance of clinically meaningful improvement in health-related quality of life in patients with moderate to severe Crohn's disease treated with ustekinumab in the IM-UNITI long-term extension study. *Journal of Gastroenterology and Hepatology* 2021;**36**(Suppl 2):108-9.
- 420.** Ferrante M, Panaccione R, Baert F, Bossuyt P, Colombel JF, Danese S, et al. Risankizumab as maintenance therapy for moderately to severely active Crohn's disease: results from the multicentre, randomised, double-blind, placebo-controlled, withdrawal phase 3 FORTIFY maintenance trial. *Lancet* 2022;**399**(10340):2031-46.

- 421.** NCT03105102. A study of the efficacy and safety of risankizumab in participants with Crohn's disease. <https://clinicaltrials.gov/ct2/show/NCT03105102> (first received 7 April 2017).
- 422.** Ferrante M, Panaccione R, Feagan B, Sandborn W, Panés J, Peyrin-Biroulet L, et al. OP008 - Efficacy and safety of open-label maintenance therapy with subcutaneous risankizumab in patients with moderate-to-severe Crohn's disease. In: UEG Week Virtual 2021, October 3-5. 2021.
- 423.** Hanauer SB, Feagan BG, Lichtenstein GR, Mayer LF, Schreiber S, Colombel JF, et al. Maintenance infliximab for Crohn's disease: the ACCENT I randomised trial. *Lancet* 2002;**359**(9317):1541-9.
- 424.** Jørgensen KK, Olsen IC, Goll GL, Lorentzen M, Bolstad N, Haavardsholm EA, et al. Switching from originator infliximab to biosimilar CT-P13 compared with maintained treatment with originator infliximab (NOR-SWITCH): a 52-week, randomised, double-blind, non-inferiority trial. *Lancet* 2017;**389**(10086):2304-16.
- 425.** NCT02148640. The NOR-SWITCH study. <https://ClinicalTrials.gov/show/NCT02148640> (first received 28 May 2014).
- 426.** Jørgensen K, Olsen I, Goll G, Lorentzen M, Bolstad N, Berset I. Biosimilar infliximab (ct-p13) is not inferior to originator infliximab: explorative IBD subgroup-analyses in Crohn's disease and ulcerative colitis from the NOR-SWITCH trial. *Journal of Crohn's & Colitis* 2017;**11**:S62-3.
- 427.** Goll G, Olsen I, Kvien T, Jørgensen K, Jahnsen J, Lorentzen M. Biosimilar infliximab (ct-p13) is not inferior to originator infliximab: results from a 52-week randomized switch trial in Norway. *Arthritis & Rheumatology* 2016;**68**:4389-92.
- 428.** Goll GL, Sexton J, Jørgensen KK, Olsen IC, Bolstad N, Warren D. Long-term efficacy and safety of biosimilar infliximab (ct-p13) after switching from originator infliximab: open-label extension of the NOR-SWITCH trial. *Journal of Internal Medicine* 2019;**285**:653-69.
- 429.** Louis E, Resche-Rigon M, Laharie D, Satsangi J, Ding N, Siegmund B, et al. Withdrawal of infliximab or concomitant immunosuppressant therapy in patients with Crohn's disease on combination therapy (SPARE): a multicentre, open-label, randomised controlled trial. *Lancet Gastroenterology & Hepatology* 2023;**8**(3):215-27.
- 430.** NCT02177071. A prospective randomized controlled trial comparing infliximab-antimetabolites combination therapy to anti-metabolites monotherapy and infliximab monotherapy in Crohn's disease patients in sustained steroid-free remission on combination therapy. <https://ClinicalTrials.gov/show/NCT02177071> 2015.
- 431.** Louis EJ, Resche-Rigon M, Laharie D, Satsangi J, Ding N, Preiss J, et al. Withdrawal of infliximab or anti-metabolite therapy in Crohn's Disease patients in sustained remission on combination therapy: a randomized unblinded controlled trial (SPARE). *Journal of Crohn's & Colitis* 2022;**16**(Suppl 1):i001.
- 432.** Louis E, Resche-Rigon M, Laharie D, Ding N, Irving P, Lamb C, et al. Treatment de-escalation in Crohn's disease patients in remission under infliximab and immunosuppressant therapy: the SPARE trial. *Gut* 2022;**71**:A27.
- 433.** Panés J, Sandborn WJ, Schreiber S, Sands BE, Vermeire S, D'Haens G, et al. Tofacitinib for induction and maintenance therapy of Crohn's disease: results of two phase IIb randomised placebo-controlled trials. *Gut* 2017;**66**(6):1049-59.
- 434.** Rutgeerts P, D'Haens G, Targan S, Vasiliauskas E, Hanauer SB, Present DH, et al. Efficacy and safety of retreatment with anti-tumor necrosis factor antibody (infliximab) to maintain remission in Crohn's disease. *Gastroenterology* 1999;**117**(4):761-9.
- 435.** Rutgeerts P, Van Assche G, Sandborn WJ, Wolf DC, Geboes K, Colombel JF, et al. Adalimumab induces and maintains mucosal healing in patients with Crohn's disease: data from the EXTEND trial. *Gastroenterology* 2012;**142**(5):1102-11.
- 436.** Colombel JF, Schreiber S, Rutgeerts P, Sandborn W, Yang M, Lomax K, et al. Effect of disease duration on "deep remission": results from the EXTEND trial. *American Journal of Gastroenterology* 2010;**105**(Suppl 1):S428.
- 437.** Colombel J, Sandborn WJ, Louis E, Panaccione R, Yang M, Chao J, et al. Economic impact of deep remission in adalimumab-treated patients with Crohn's disease: results from EXTEND. *Journal of Crohn's & Colitis* 2011;**5**(1):S41-2.
- 438.** Reinisch W, Colombel JF, D'Haens G, Sandborn W, Rutgeerts P, Petersson J, et al. Characterization of mucosal healing with adalimumab treatment in patients with moderately to severely active Crohn's disease from extend. *American Journal of Gastroenterology* 2014;**109**:S475.
- 439.** Rutgeerts P, Reinisch W, Sandborn WJ, D'Haens G, Colombel JF, Petersson J, et al. Evaluation of site versus central assessment of endoscopies in Crohn's disease: comparison using data from EXTEND. *Journal of Crohn's & Colitis* 2014;**8**:S30-1.
- 440.** Sandborn WJ, Wolf D, Colombel JF, Panes J, Eichner S, lezzi A, et al. Adalimumab achieves efficacy in mucosal healing regardless of baseline disease severity in patients with Crohn's disease: data from EXTEND. *Journal of Crohn's & Colitis* 2014;**8**:S306-7.
- 441.** Sandborn W, Colombel JF, Panaccione R, Louis E, Yang M, Pollack P, et al. Importance of deep remission compared with mucosal healing only: results from EXTEND. *Journal of Crohn's & Colitis* 2012;**6**:S112.
- 442.** Colombel JF, Rutgeerts P, Sandborn W, Yang M, Lomax K, Pollack P, et al. Achievement of early deep remission predicts better long-term outcomes for adalimumab-treated patients with Crohn's disease: data from extend. *American Journal of Gastroenterology* 2010;**105**:S434-5.
- 443.** Colombel J, Schreiber S, Rutgeerts P, Sandborn W, Yang M, Lomax K, et al. Effect of disease duration on 'deep remission':

results from the EXTEND trial. *Inflammatory Bowel Diseases* 2011;**17**:S44.

444. Rutgeerts PJ, Reinisch W, Sandborn W, D'Haens GR, Colombel JF, Petersson J, et al. Evaluation of site versus central assessment of endoscopies in Crohn's disease: comparison using data from EXTEND. *Gastroenterology* 2014;**1**:S-585.

445. Sandborn WJ, Colombel JF, Lomax KG, Pollack PF, Thakkar R, Camez A, et al. Achievement of early deep remission is associated with lower rates of weekly dosing for adalimumab-treated patients with Crohn's disease: data from EXTEND. *Gut* 2011;**60**:A136-7.

446. Sandborn WJ, Wolf DC, Colombel JF, Panes J, Eichner S, lezzi A, et al. Adalimumab achieves efficacy in mucosal healing regardless of baseline disease severity in patients with Crohn's disease: data from EXTEND. *Journal of Gastroenterology and Hepatology (Australia)* 2014;**29**:118.

447. Sandborn W, Wolf DC, Colombel JF, Panes J, Eichner S, lezzi A, et al. Adalimumab achieves efficacy in mucosal healing regardless of baseline disease severity in patients with Crohn's disease: data from EXTEND. *Gastroenterology* 2014;**146**(5 Suppl 1):S-589.

448. Sandborn W, Colombel JF, Lomax K, Pollack P, Thakkar R, Camez A, et al. Achievement of early deep remission is associated with lower rates of weekly dosing for adalimumab-treated patients with Crohn's disease: data from EXTEND. *American Journal of Gastroenterology* 2010;**105**:S442-3.

449. Reinisch W, Colombel JF, D'Haens G, Sandborn WJ, Rutgeerts P, Petersson J, et al. Characterization of mucosal healing with adalimumab treatment in patients with moderately to severely active Crohn's disease from EXTEND. *United European Gastroenterology Journal* 2014;**1**:A537.

450. Balzola F, Cullen G, Ho GT, Russel RK, Wehkamp J. Adalimumab induces and maintains mucosal healing in patients with Crohn's disease: data from the EXTEND trial. *Inflammatory Bowel Disease Monitor* 2012;**12**(4):151-2.

451. Colombel JF, Rutgeerts PJ, Sandborn WJ, Camez AA, Pollack PF, Chen N, et al. Adalimumab treatment results in deep remission for patients with moderate to severe ileocolonic Crohn's disease: results from EXTEND. *Gastroenterology* 2010;**138**(5 Suppl 1):S518.

452. Colombel JF, Rutgeerts P, Van Assche G, Sandborn W, Wolf D, Reinisch W, et al. Clinical benefit of adalimumab dose adjustment for patients with moderately to severely active Crohn's disease in EXTEND. *American Journal of Gastroenterology* 2014;**109**(Suppl 2):S474.

453. Colombel JF, Rutgeerts PJ, Sandborn WJ, Yang M, Pollack PF, Thakkar RB, et al. Adalimumab induces deep remission in patients with Crohn's disease. *Clinical Gastroenterology and Hepatology* 2014;**12**(3):414.

454. Sandborn WJ, Panaccione R, Thakkar R, Lomax KG, Chen N, Chao J, et al. Crohn's disease mucosal healing in adalimumab-treated patients is affected by disease duration:

results from EXTEND. *Gastroenterology* 2010;**138**(5 Suppl 1):S164.

455. Colombel J, Sandborn WJ, Louis E, Panaccione R, Yang M, Chao J, et al. Identification of subgroups of adalimumab-treated patients with Crohn's disease who experience high rates of deep remission. *Journal of Crohn's & Colitis* 2011;**5**(1):S41.

456. Rutgeerts P, D'Haens GR, Van Assche GA, Sandborn WJ, Wolf DC, Colombel JF, et al. Adalimumab induces and maintains mucosal healing in patients with moderate to severe ileocolonic Crohn's disease - first results of the EXTEND trial. *Gastroenterology* 2009;**136**(5 Suppl 1):A116.

457. Rutgeerts P, Van Assche G, D'Haens G, Sandborn W, Wolf D, Colombel J, et al. Sustained mucosal healing in adalimumab-treated patients with moderate to severe ileocolonic Crohn's disease: results of the EXTEND trial. *Inflammatory Bowel Diseases* 2009;**15**(Suppl 2):S13.

458. Rutgeerts PJ, Reinisch W, Thakkar R, Wu EQ, Kaltenboeck A, Yang M, et al. Early mucosal healing status predicts long-term clinical benefits for adalimumab-treated patients with moderate to severe Crohn's disease. *Gastroenterology* 2010;**138**(5 Suppl 1):S85.

459. Sandborn W, Colombel J, Rutgeerts P, Yang M, Lomax K, Pollack P, et al. Achievement of early deep remission predicts better long-term outcomes for adalimumab-treated patients with Crohn's disease: data from EXTEND. *Inflammatory Bowel Diseases* 2011;**17**(Suppl 1):S60.

460. Sandborn W, Colombel J, Lomax K, Pollack P, Thakkar R, Chen N, et al. Achievement of early deep remission is associated with lower rates of weekly dosing for adalimumab-treated patients with Crohn's disease: data from extend. *Inflammatory Bowel Diseases* 2011;**17**(Suppl 1):S61.

461. NCT00348283. Effects of adalimumab on mucosal healing in subjects with Crohn's disease involving the colon. <https://clinicaltrials.gov/ct2/show/NCT00348283> (first received 11 February 2010).

462. Sandborn WJ, Colombel JF, Enns R, Feagan BG, Hanauer SB, Lawrence IC, et al. Natalizumab induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2005;**353**(18):1912-25.

463. Sandborn W, Colombel J, Enns R, Feagan B, Hanauer S, Lawrence I. Efficacy of natalizumab in maintaining clinical response and remission in Crohn's disease: comparison of sustained response and remission rates through 12 months vs point-in-time response and remission rates at month 12. *Gastroenterology* 2005;**128**(4 Suppl 2):A586.

464. Feagan B, Panaccione R, Hass S, Panjabi S. The impact of natalizumab on the quality of life of patients exposed to or failing anti-TNFalpha therapy in the ENACT-2 trial. *Gastroenterology* 2009;**136**:A179-80.

465. Targan S, Colombel J, Enns R, Feagan B, Hanauer S, Lawrence I. Maintenance therapy with natalizumab (ENACT-2) for patients with active Crohn's disease despite immunosuppressant use. *Gastroenterology* 2005;**128**:A587.

- 466.** Panaccione R, Colombel J, Enns R, Feagan B, Hanauer S, Lawrance I. Efficacy of natalizumab in patients with Crohn's disease and prior history of infliximab therapy: 12-month results from ENACT-2. *Gastroenterology* 2005;**128**:A584.
- 467.** Dudley-Brown S, Nag A, Cullinan C, Ayers M, Hass S, Panjabi S. Health-related quality-of-life evaluation of Crohn disease patients after receiving natalizumab therapy. *Gastroenterology Nursing* 2009;**32**(5):327-39.
- 468.** Sandborn W, Colombel J, Enns R, Feagan B, Hanauer S, Lawrance I, et al. Maintenance therapy with natalizumab does not require use of concomitant immunosuppressants for sustained efficacy in patients with active Crohn's disease: results from the ENACT-2 study. *Gastroenterology* 2006;**130**:A-482 Abstract 1137.
- 469.** Sandborn WJ, Hanauer SB, Rutgeerts P, Fedorak RN, Lulcas M, MacIntosh DG, et al. Adalimumab for maintenance treatment of Crohn's disease: results of the CLASSIC II trial. *Gut* 2007;**56**(9):1232-9.
- 470.** Colombel J, Rutgeerts P, Sandborn WJ, Reinisch W, Loftus Jr EV, Tang J, et al. Impact of induction dosing on maintenance outcome with adalimumab in Crohn's disease. *Journal of Crohn's & Colitis* 2011;**5**(1):S88.
- 471.** NCT00055497. Remission in subjects with Crohn's disease, 1 year phase. <https://ClinicalTrials.gov/show/NCT00055497> (first received 5 March 2002).
- 472.** NCT01070303. Remission in subjects with Crohn's disease, open label extension. <https://ClinicalTrials.gov/show/NCT01070303> 2002.
- 473.** Sandborn WJ, Abreu MT, D'Haens G, Colombel J, Vermeire S, Mitchev K, et al. Certolizumab pegol in patients with moderate to severe Crohn's disease and secondary failure to infliximab. *Clinical Gastroenterology and Hepatology* 2010;**8**:688-95.
- 474.** Abreu MT, Vermeire S, Rutgeerts PJ, Ullman TA, Wolf DC, Sandborn WJ. Certolizumab pegol effectiveness in Crohn's disease patients with secondary failure to infliximab is independent of infliximab dosing: results from the welcome study. *Gastroenterology* 2009;**136**:A181.
- 475.** NCT00308581. Certolizumab in Crohn's disease patients with loss of response or intolerance to infliximab. <https://ClinicalTrials.gov/show/NCT00308581> (first received 29 March 2006).
- 476.** Sandborn WJ, Vermeire S, D'Haens GR, Colombe J, Fedorak RN, Spehlmann ME, et al. Welcome: A randomized, double-blind, controlled trial comparing certolizumab pegol 400 mg every 2 weeks with every 4 weeks for maintenance of response and remission in patients with moderate to severe Crohn's disease with secondary failure to infliximab. *Gastroenterology* 2009;**136**:A27.
- 477.** Feagan B, Sandborn W, Pierre-Louis B, Fourment C, Abreu M. Erratum: P-145 welcome 54-week open-label extension: efficacy of certolizumab pegol in Crohn's disease patients with secondary failure and anti-infliximab antibodies (Inflammatory Bowel Diseases (2013) 19 (S82)). *Inflammatory Bowel Diseases* 2014;**20**:E7.
- 478.** Feagan BG, Sandborn WJ, Wolf DC, Mitchev K, Jamoul C, Rutgeert PJ. Certolizumab pegol improves health-related quality of life in Crohn's disease patients for whom infliximab treatment was not successful: results from the welcome study. *Gastroenterology* 2009;**136**:A179.
- 479.** Sandborn W, Vermeire S, Rutgeerts P, D'Haens G, Colombel J, Fedorak R, et al. Efficacy of certolizumab pegol is not affected by baseline anti-infliximab antibody status in patients with Crohn's disease with secondary infliximab failure. *Inflammatory Bowel Diseases* 2009;**15**:S13-4.
- 480.** Sandborn WJ, Vermeire S, Ernault E, Lichtenstein GR. Crohn's disease patients with persistently elevated crp plasma concentration have higher rates of remission with certolizumab pegol 400 mg every 2 weeks vs every 4 weeks. *Inflammatory Bowel Diseases* 2011;**17**:S21.
- 481.** Sandborn WJ, Vermeire S, Ernault E, Lichtenstein GR. Crohn's disease patients with persistently elevated crp plasma concentration have higher rates of remission with certolizumab pegol 400 mg every 2 weeks vs. every 4 weeks. *Gastroenterology* 2011;**140**:S263.
- 482.** Feagan BG, Sandborn WJ, Wolf DC, Coteur G, Purcaru O, Brabant Y, et al. Randomised clinical trial: Improvement in health outcomes with certolizumab pegol in patients with active Crohn's disease with prior loss of response to infliximab. *Alimentary Pharmacology & Therapeutics* 2011;**33**:541-50.
- 483.** Fedorak R, Sandborn W, Abreu M, D'Haens G, Colombel J, Mitchev K, et al. Certolizumab pegol is efficacious in crohn's disease patients who have failed infliximab regardless of concomitant therapy or reason for failure. *Canadian Journal of Gastroenterology* 2009;**23**:S430-1.
- 484.** Sandborn WJ, Gasink C, Gao LL, Blank MA, Johanns J, Guzzo C, et al. Ustekinumab induction and maintenance therapy in refractory Crohn's disease. *New England Journal of Medicine* 2012;**367**(16):1519-28. [DOI: [10.1056/Nejm0a1203572](https://doi.org/10.1056/Nejm0a1203572)]
- 485.** Sandborn WJ, Feagan BG, Rutgeerts P, Hanauer S, Colombel JF, Sands BE, et al. Vedolizumab as induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2013;**369**(8):711-21. [DOI: [10.1056/Nejm0a1215739](https://doi.org/10.1056/Nejm0a1215739)]
- 486.** Dubinsky M, D'Haens GR, Candela N, Peterson T, Uddin SM, Lasch K, et al. Symptomatic improvement with vedolizumab therapy for Crohn's disease stratified by disease activity and prior tumor necrosis factor antagonist failure: post hoc analyses from the GEMINI 2 trial. *Gastroenterology* 2021;**160**(6 Suppl):S-414.
- 487.** Sandborn WJ, Feagan BG, Loftus EV, Peyrin-Biroulet L, Van Assche G, D'Haens G, et al. Efficacy and safety of upadacitinib in a randomized trial of patients With Crohn's disease.. *Gastroenterology* 2020;**158**(8):2123.
- 488.** Sands BE, Anderson FH, Bernstein CN, Chey WY, Feagan BG, Fedorak RN, et al. Infliximab maintenance therapy

for fistulizing Crohn's disease. *New England Journal of Medicine* 2004;**350**(9):876-85.

489. Sands BE, Blank MA, Diamond RH, Barrett JP, Van Deventer SJ. Maintenance infliximab does not result in increased abscess development in fistulizing Crohn's disease: results from the ACCENT II study. *Alimentary Pharmacology & Therapeutics* 2006;**23**:1127-36.

490. Wagner C, Hanauer SB, Bala M, Patel K, Travers S, Olson A, et al. Relationship of antibodies to infliximab to duration of response and infusion reactions for Crohn's disease patients receiving episodic infliximab retreatment in ACCENT I. *Gastroenterology* 2003;**124**(4 Suppl 1):A378.

491. NCT00207766. A study of the safety and efficacy of infliximab (Remicade) in subjects with fistulizing Crohn's disease. <https://clinicaltrials.gov/ct2/show/NCT00207766> (first received 27 April 2010).

492. Fasanmade AA, Marsters P, Munsanje E, Graham MA, Davis HM, van Deventer S. Infliximab pharmacokinetics and improvement in fistulizing Crohn's disease. *Gastroenterology* 2003;**124**(4 Suppl 1):A61.

493. Lichtenstein GR, Yan S, Bala M, Blank M, Sands BE. Infliximab maintenance treatment reduces hospitalizations, surgeries, and procedures in fistulizing Crohn's disease. *Gastroenterology* 2005;**128**(4):862-9.

494. van Deventer SJ, Blank M, Waldschmidt S, Sands B. Infliximab does not result in increased abscess development in fistulizing Crohn's disease: results from ACCENT II. *Gastroenterology* 2003;**124**(4 Suppl 1):A379.

495. Sands BE, Yan S, Bala M, Blank MA, van Deventer SJ. Maintenance treatment with infliximab reduces the likelihood of hospitalization in patients with fistulizing Crohn's disease. *Gastroenterology* 2004;**126**(4 Suppl 2):A466.

496. Sands BE, Blank MA, Patel K, Van Deventer SJ. Long-term treatment of rectovaginal fistulas in Crohn's disease: response to infliximab in the ACCENT II study. *Clinical Gastroenterology and Hepatology* 2004;**2**(10):912-20.

497. Sands B, Blank M, Marster P, van Deventer S. Long-term treatment of rectovaginal fistulas in Crohn's disease: response to infliximab in the ACCENT II trial. *Gastroenterology* 2003;**124**(4 Suppl 1):A380.

498. Sands BE, Blank MA, Diamond RH, Barrett JP, Van Deventer SJ. Maintenance infliximab does not result in increased abscess development in fistulizing Crohn's disease: results from the ACCENT II study. *Alimentary Pharmacology & Therapeutics* 2006;**23**(8):1127-36.

499. NCT03466411. A study of the efficacy and safety of guselkumab in participants with moderately to severely active Crohn's disease. <https://ClinicalTrials.gov/show/NCT03466411> (first received 15 March 2018).

500. Schreiber S, Khaliq-Kareemi M, Lawrance IC, Thomsen OO, Hanauer SB, McColm J, et al. Maintenance therapy with certolizumab pegol for Crohn's disease. [Erratum appears in N

Engl J Med. 2007 Sep 27;357(13):1357]. *New England Journal of Medicine* 2007;**357**(3):239-50.

501. Khaliq-Kareemi M, Schreiber S, Lawrance I, Thomsen O, Hanauer S, Bloomfield R, et al. Certolizumab pegol is effective in the treatment of Crohn's disease patients with open fistulas. *Canadian Journal of Gastroenterology* 2009;**23**(Suppl SA):no pagination.

502. Schreiber S, Lawrance IC, Thomsen OO, Hanauer SB, Bloomfield R, Sandborn WJ. Randomised clinical trial: certolizumab pegol for fistulas in Crohn's disease - subgroup results from a placebo-controlled study. *Alimentary Pharmacology & Therapeutics* 2011;**33**(2):185-93.

503. Hanauer SB, Panes J, Colombel JF, Bloomfield R, Schreiber S, Sandborn WJ. Clinical trial: impact of prior infliximab therapy on the clinical response to certolizumab pegol maintenance therapy for Crohn's disease. *Alimentary Pharmacology & Therapeutics* 2010;**32**(3):384-93.

504. Sandborn WJ, Pierre-Louis B, Ullman TA, Binion DG. Fecal calprotectin concentration and clinical response to certolizumab pegol inpatients with active Crohn's disease: results from PRECISE 2. *Inflammatory Bowel Diseases* 2011;**17**:S24-5.

505. Sandborn W, Ullman T, Pierre-Louis B, Binion D. Inflammatory biomarkers and clinical remission in patients with active Crohn's disease: results from PRECISE 2. *American Journal of Gastroenterology* 2011;**106**:S441-2.

506. Sandborn W, Feagan B, Lee S, Hanauer S. Baseline corticosteroid use and corticosteroid-free clinical remission in Crohn's disease patients treated with certolizumab pegol in the PRECISE 2 trial. *Inflammatory Bowel Diseases* 2012;**18**:S23-4.

507. Scherl E, Pierre-Louis B, Lee S. Formation of autoantibodies in patients with moderate to severe Crohn's disease, receiving continuous and interrupted certolizumab pegol (CZP) therapy with or without concomitant immunosuppressants. *American Journal of Gastroenterology* 2011;**106**:S440.

508. Schreiber S, Colombel JF, Bloomfield R, Nikolaus S, Scholmerich J, Panes J, et al. Increased response and remission rates in short-duration crohn's disease with subcutaneous certolizumab pegol: an analysis of PRECISE 2 randomized maintenance trial data. *American Journal of Gastroenterology* 2010;**105**(7):1574-82.

509. Schreiber S, Lawrance I, Thomsen O, Hanauer S, Bloomfield R, Sandborn W. Certolizumab pegol is effective at maintaining response and remission in patients with fistulising Crohn's disease: 3-year results from the PRECISE 3 study. *Gastroenterology* 2010;**138**(5 Suppl 1):S165.

510. Feagan B, Lichtenstein G, Schreiber S, Sandborn W. The long-term, 30 months, efficacy and tolerability of certolizumab pegol therapy for crohn's disease. *Canadian Journal of Gastroenterology* 2009;**23**(Suppl SA):s29.

511. Feagan B, Hanauer S, Coteur G, Schreiber S. Evaluation of a daily practice composite score for the assessment of Crohn's

disease: the treatment impact of certolizumab pegol. *Alimentary Pharmacology & Therapeutics* 2011;**33**(10):1143-51.

512. Lichtenstein G, Thomsen O, Schreiber S, Lawrance I, Hanauer S, Bloomfield R, et al. Continuous therapy with certolizumab pegol maintains remission of patients with Crohn's disease for up to 18 months. *Clinical Gastroenterology and Hepatology* 2010;**8**(7):600-9.

513. Schreiber S, Khaliq-Kareemi M, Lawrance I, Thomsen O, Hanauer S, Bloomfield R. Certolizumab pegol demonstrates efficacy in maintaining response and remission in patients with active Crohn's disease regardless of their immunosuppressant treatment status at entry to the PRECISE 2 study. *Gastroenterology* 2010;**138**(5 Suppl 1):S165.

514. Sandborn W, Schreiber S, Hanauer S, Colombel J, Bloomfield R, Lichtenstein G. Long-term maintenance of remission with no dose escalation after re-induction with certolizumab pegol in patients with Crohn's disease exacerbation: 3-year results from the PRECISE 4 study. *Inflammatory Bowel Diseases* 2009;**15**(Suppl 2):S14.

515. Sandborn W, Schreiber S, Hanauer S, Colombel J, Bloomfield R, Lichtenstein G. Long-term maintenance of remission with no dose escalation after re-induction with certolizumab pegol in patients with Crohn's disease exacerbation who were naive to anti-TNF treatment: 4-year results from the precise 4 study. *American Journal of Gastroenterology* 2010;**105**(Suppl 1):S418.

516. Sandborn W, Schreiber S, Hanauer S, Colombel J, Bloomfield R, Lichtenstein G. Sustained remission with no dose escalation after re-induction with certolizumab pegol in patients with Crohn's disease exacerbation who were naive to anti-TNF treatment: 4-year results from PRECISE. *Inflammatory Bowel Diseases* 2011;**17**(Suppl 1):S60.

517. Sandborn W, Rubin D, Sen D, Feagan B. Sustained remission with no dose escalation after re-induction with certolizumab pegol in patients with Crohn's disease exacerbation: 1-year results from PRECISE 4. *American Journal of Gastroenterology* 2011;**106**(Suppl 2):S439.

518. NCT00152425. Study to test the effect of CDP870 in the treatment of Crohn's disease over 26 weeks, comparing CDP870 to a dummy drug (placebo), following 3 doses of active drug (CDP870). <https://ClinicalTrials.gov/show/NCT00152425> (first received 9 September 2005).

519. Schreiber S, Ben-Horin S, Leszczyszyn J, Dudkowiak R, Lahat A, Gawdis-Wojnarska B, et al. Randomized controlled trial: subcutaneous versus intravenous infliximab CT-P13 maintenance in inflammatory bowel disease. *Gastroenterology* 2021;**160**:2340-53.

520. Schreiber S, D'Haens GR, Cummings F, Irving PM, Ye BD, An Y, et al. Switching from intravenous to subcutaneous infliximab in patients with active inflammatory bowel disease: post-hoc analysis of pre/post switch outcomes from a multicentre, randomised controlled pivotal. *United European Gastroenterology Journal* 2021;**9**(SUPPL 8):553-4.

521. D'Haens GR, Reinisch W, Schreiber S, Cummings F, Irving PM, Ye B, et al. Comparison of combination subcutaneous infliximab and an immunomodulator versus subcutaneous infliximab monotherapy: Post-hoc analysis of a randomised clinical trial. *United European Gastroenterology Journal* 2021;**9**(SUPPL 8):549-50.

522. NCT02883452. A phase I study to evaluate pharmacokinetics, efficacy and safety of CT-P13 subcutaneous in patients with active Crohn's disease and ulcerative colitis. <https://ClinicalTrials.gov/show/NCT02883452> (first received 30 August 2016).

523. Van Assche G, Vermeire S, Ballet V, Gabriels F, Noman M, D'Haens G, et al. Switch to adalimumab in patients with Crohn's disease controlled by maintenance infliximab: prospective randomised switch trial. *Gut* 2012;**61**:229-34.

524. Van Assche GA, Vermeire S, Ballet V, Noman M, D'Haens GR, Humblet E, et al. Switch to adalimumab in patients with Crohn's disease controlled by maintenance infliximab. The prospective randomized SWITCH study. *Gastroenterology* 2010;**138**:S85.

525. Balzola F, Cullen G, Ho G, Russell RK, Wehkamp J. Switch to adalimumab in patients with Crohn's disease controlled by maintenance infliximab: prospective randomised switch trial. *Inflammatory Bowel Disease Monitor* 2012;**12**:150.

526. Vermeire S, Schreiber S, Petryka R, Kuehbach T, Hebuterne X, Roblin X, et al. Clinical remission in patients with moderate-to-severe Crohn's disease treated with filgotinib (the FITZROY study): results from a phase 2, double-blind, randomised, placebo-controlled trial. *Lancet* 2017;**389**:266-75.

527. Vermeire S, D'Haens G, Baert F, Danese S, Kobayashi T, Loftus EV, et al. Efficacy and safety of subcutaneous vedolizumab in patients with moderately to severely active Crohn's disease: results from the visible 2 randomised trial. *Journal of Crohn's & Colitis* 2021;**1**:12.

528. D'Haens G, Baert FJ, Danese S, Kobayashi T, Loftus EV, Sandborn WJ, et al. Early onset of response with intravenous vedolizumab induction therapy in patients with moderately to severely active Crohn's disease treated in the visible 2 study. *Gastroenterology* 2021;**160**(6 Suppl):S-352.

529. D'Haens GR, Sandborn WJ, Zhang W, Kisfalvi K, Wang S, Vermeire S. Effects of subcutaneous vedolizumab on health-related quality of life and work productivity in patients with Crohn's disease: results from the phase 3 visible 2 study. *Gastroenterology* 2020;**158**(6 Suppl 1):S-437.

530. D'Haens GR, Sandborn W, Zhang W, Kisfalvi K, Wang S, Vermeire S. Effects of subcutaneous vedolizumab on health-related quality of life and work productivity in patients with Crohn's disease: results from the phase 3 visible 2 trial. *United European Gastroenterology Journal* 2020;**8**(8 Suppl):409-10.

531. Loftus EV, Sandborn WJ, Wolf D, Danese S, Chen J, Yao X, et al. Efficacy and safety of 2 or 3 vedolizumab intravenous infusions as induction therapy for ulcerative colitis and Crohn's disease: results from VISIBLE 1 and 2. *Journal of Crohn's & Colitis* 2019;**13**(Suppl 1):S361-2.

- 532.** Loftus EV, Sandborn WJ, Wolf DC, Danese S, Chen J, Yao X, et al. Efficacy and safety of 2 or 3 vedolizumab intravenous infusions as induction therapy for ulcerative colitis and Crohn's disease: results from VISIBLE 1 and 2. *Gastroenterology* 2019;**156**(6 Suppl 1):S-1095.
- 533.** Vermeire S, Sandborn W, Danese S, Baert F, Kobayashi T, Loftus E, et al. Efficacy and safety of vedolizumab subcutaneous formulation in patients with moderately to severely active Crohn's disease: results of the VISIBLE 2 study. *Journal of Gastroenterology and Hepatology (Australia)* 2020;**35**(Suppl 1):115.
- 534.** Vermeire S, D'Haens GR, Baert FJ, Danese S, Kobayashi T, Loftus EV, et al. 950 the visible 2 phase 3 study of efficacy and safety of vedolizumab SC for moderate-to-severe Crohn's disease. *Gastroenterology* 2020;**158**(6 Suppl 1):S-194.
- 535.** NCT02611817. Efficacy and safety of vedolizumab subcutaneous (SC) as maintenance therapy in Crohn's disease (CD). <https://ClinicalTrials.gov/show/NCT02611817> (first received 23 November 2015).
- 536.** Vermeire S, Schreiber S, Rubin DT, D'Haens G, Reinisch W, Watanabe M, et al. Efficacy and safety of filgotinib as induction and maintenance therapy for Crohn's disease (DIVERSITY): a phase 3, double-blind, randomised, placebo-controlled trial. *Lancet Gastroenterology & Hepatology* 2025;**1**(10):138-53.
- 537.** NCT02914561. Filgotinib in the induction and maintenance of remission in adults with moderately to severely active Crohn's disease (DIVERSITY1). <https://clinicaltrials.gov/study/NCT02914561> (first published October 2016).
- 538.** Hanzel J, Jansen JM, Ter Steege RW, Gecse KB, D'Haens GR. Multiple switches from the originator infliximab to biosimilars is effective and safe in inflammatory bowel disease: a prospective multicenter cohort study. *Inflammatory Bowel Diseases* 2022;**28**(4):495-501.
- 539.** Volkers AG, Jansen JM. Similar trial-efficacy of infliximab-biosimilar compared to infliximab-biological in patients with inflammatory bowel disease in remission-a randomized, controlled, double blind, phase 4 noninferiority trial. *United European Gastroenterology Journal* 2017;**5**(5 Suppl 1):A307.
- 540.** NCT02452151. Efficacy and safety of infliximab-biosimilar (inflectra) compared to infliximab-innovator (remicade) in patients with inflammatory bowel disease in remission: the SIMILAR trial. <https://ClinicalTrials.gov/show/NCT02452151> 2015.
- 541.** Watanabe M, Hibi T, Lomax KG, Paulson SK, Chao J, Alam MS, et al. Adalimumab for the induction and maintenance of clinical remission in Japanese patients with Crohn's disease. *Journal of Crohn's & Colitis* 2012;**6**(2):160-73.
- 542.** NCT00445432. A study of adalimumab for the maintenance of clinical remission in Japanese subjects with Crohn's disease. <https://ClinicalTrials.gov/show/NCT00445432> 2007.
- 543.** Watanabe K, Motoya S, Ogata H, Kanai T, Matsui T, Suzuki Y, et al. Effects of vedolizumab in Japanese patients with Crohn's disease: a prospective, multicenter, randomized, placebo-controlled phase 3 trial with exploratory analyses. *Journal of Gastroenterology* 2020;**55**(3):291-306.
- 544.** Young D, Harris C, Rahmany S, Iria I, Gonçalves J, Addison J, et al. A randomised, crossover trial exploring the patient perspective and effectiveness of biosimilar adalimumab transition: IBD reference and biosimilar adalimumab cross over study (iBaSS). *International Journal of Clinical Pharmacy* 2024;**46**(5):1091-101.
- 545.** EudraCT number 2018-004967-30. Inflammatory bowel disease (IBD) reference product (humira-adalimumab) and biosimilar product (imraldi-adalimumab) cross over study. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2018-004967-30> (first registered in January 2019).
- 546.** Baima J, Lima T, Moreira A, De Barros J, Dorna M, Da Silva R, et al. Randomized clinical trial for clinical response evaluation in Crohn's disease: infliximab versus adalimumab. *Inflammatory Bowel Diseases* 2016;**22**:S30.
- 547.** EUCTR2005-004104-37. A phase IIIb multicenter, open-label induction and double-blind comparison of two maintenance schedules evaluating clinical benefit and tolerability of certolizumab pegol, a pegylated Fab' fragment of humanized antibody to tumor necrosis factor (TNF) over 26 weeks in subjects suffering from Crohn's disease (CD) with prior loss of response or intolerance to infliximab. <https://www.clinicaltrialsregister.eu/ctr-search/search?query=2005-004104-37> (first received 23 March 2006).
- 548.** EUCTR2007-001913-41. A phase IIIb, multinational, randomized, double-blind, placebo-controlled trial to assess the efficacy and safety of certolizumab pegol, a pegylated Fab' fragment of a humanized anti-TNF-alpha monoclonal antibody, administered subcutaneously at weeks 0, 2 and 4 in subjects with moderately to severely active Crohn's disease. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2007-001913-41/results> (first received 16 October 2007).
- 549.** EUCTR2007-003239-21. A phase IIIb, open label, randomized, exploratory clinical study exploring efficacy, safety and patients' satisfaction with Certolizumab pegol treatment in moderate to severe Crohn's disease when treatment is administered in two different settings : either at hospital or at home. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2007-003239-21-FR> (first received 15 January 2008).
- 550.** EUCTR2008-001131-35. Azathioprine maintenance treatment versus Infliximab maintenance treatment in Crohn's disease patients in remission: a randomized multicenter trial - Azorix-trial. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2008-001131-35-NL> (first received 29 May 2008).
- 551.** NTR1404. Azathioprine maintenance treatment versus infliximab maintenance treatment in Crohn's disease patients in remission (Azorix trial). <https://trialsearch.who.int/?TrialID=NTR1404> (first received 1 August 2008).
- 552.** EUCTR2008-004926-18-DE. Comparison of the human TNF-alpha antibody adalimumab with infliximab in induction and maintenance of steroid-free remission

in patients with moderate to severe Crohn's disease - CHAMP. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2008-004926-18-DE> (first received 30 October 2008).

553. EUCTR2012-001723-12-ES. Adalimumab intralesional in intestinal strictures of crohn's disease patients. <https://www.clinicaltrialsregister.eu/ctr-search/search?query=2012-001723-12> (first received 14 August 2013).

554. EUCTR2020-002701-26. A clinical study to evaluate the safety and effectiveness of JNJ-64304500 as an add-on therapy to standard of care biologic therapy with anti-tumor necrosis factor alpha or anti-interleukin 12/23 in Responder Not Remitter participants with active Crohn's disease [Uno studio clinico di Fase 2a, randomizzato, in doppio cieco, controllato con placebo, a gruppi paralleli, multicentrico, di fattibilità (Proof-of-Concept), volto a valutare la sicurezza e l'efficacia di JNJ-64304500 come terapia aggiuntiva alla terapia biologica standard di cura con anti-fattore di necrosi tumorale alfa o anti-interleuchina 12/23 in partecipanti rispondenti non remittenti con malattia di Crohn attiva]. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2020-002701-26-IT> (first received 7 April 2021).

555. Huang, P, Yu FL, Li L. The advantages of early application of infliximab to Crohn's disease treatment. *Chinese Pharmaceutical Journal* 2012;**47**(8):642-4.

556. JPRN-UMIN000005689. GMA (granulocyte and monocyte adsorption) early combined with azathioprine vs Infliximab plus azathioprine for induction of remission in active Crohn's disease : an open randomized trial. https://center6.umin.ac.jp/cgi-open-bin/ctr_e/ctr_view.cgi?recptno=R000006728 (first received 31 May 2011).

557. JPRN-UMIN000010524. Optimization of adalimumab switch therapy for loss of response with infliximab in Crohn's disease -OASISstudy. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=JPRN-UMIN000010524> (first received 17 April 2013).

558. NCT00349752. Corticosteroid sparing effect of certolizumab in crohn's disease. <https://ClinicalTrials.gov/show/NCT00349752> (first received 10 July 2006).

559. EUCTR2006-003871-11. An open-label, multi-center trial to examine the long term safety, efficacy, and corticosteroid-sparing effect of certolizumab pegol in patients with moderate to severe Crohn's disease who have failed tapering of corticosteroids, needed corticosteroids after completion of tapering, or have successfully completed the trial C87059 (COSPAR I). <https://trialsearch.who.int/?TrialID=EUCTR2006-003871-11-DE> (first received 16 October 2006).

560. EUCTR2006-003870-88. A phase IIIb, multi-center, double-blind, placebo-controlled, randomized trial to examine the corticosteroid-sparing effect of certolizumab pegol in patients with moderate to severe Crohn's disease (COSPAR 1). <https://www.clinicaltrialsregister.eu/ctr-search/trial/2006-003870-88/DE#D> (first received 29 September 2006).

561. NCT00796250. Efficacy of infliximab in the treatment of patients affected by corticoiddependent crohn's disease (P02732). <https://ClinicalTrials.gov/show/NCT00796250> (first received 24 November 2008).

562. NCT01674413. Calprotectin-directed humira® maintenance therapy (CADHUM). <https://ClinicalTrials.gov/show/NCT01674413> (first received 28 August 2012).

563. NCT01696942. Cimzia versus mesalamine for crohn's recurrence. <https://ClinicalTrials.gov/show/NCT01696942> (first received 2 October 2012).

564. NCT02771457. Managing infliximab reinduction after temporary discontinuation of drug. <https://ClinicalTrials.gov/show/NCT02771457> (first received 13 May 2016).

565. NCT03221166. Thalidomide versus infliximab in new onset crohn's disease with poor prognostic factors. <https://ClinicalTrials.gov/show/NCT03221166> (first received 18 July 2017).

566. NCT03370601. Evaluation of the clinical and immunological impact of two therapeutic strategies in chronic inflammatory diseases. <https://ClinicalTrials.gov/show/NCT03370601> (first received 12 December 2017).

567. NCT03393247. The therapy effect of azathioprine initial or 14 weeks after combined with infliximab on CD. <https://ClinicalTrials.gov/show/NCT03393247> (first received 8 January 2018).

568. NCT04131322. Loss of response of adalimumab biosimilar vs adalimumab original, in inflammatory bowel disease. <https://ClinicalTrials.gov/show/NCT04131322> (first received 18 October 2019).

569. NCT04655807. A study of JNJ-64304500 as add-on therapy in participants with active Crohn's disease. <https://ClinicalTrials.gov/show/NCT04655807> (first received 7 December 2021).

570. NCT05057273. Safety and efficacy of oral BT-11 in crohn's disease patients with moderate to severe disease. <https://clinicaltrials.gov/ct2/show/NCT05057273> (first received 27 September 2021).

571. Rogler G, Stallmach A, Fichtner-Feigl S, Schreiber S, Sturm A, Ramsden E, et al. QAX576, an anti-interleukin (IL)-13 monoclonal antibody, for the treatment of patients with fistulising Crohn's disease (CD): results of a proof-of concept study. *United European Gastroenterology Journal* 2014;**2**:A225-6.

572. Wu KC, Ran ZH, Gao X, Chen M, Zhong J, Sheng JQ et al. Adalimumab induction and maintenance therapy achieve clinical remission and response in Chinese patients with Crohn's disease. *Intestinal Research* 2016;**14**(2):152.

573. NCT02015793. Study to Evaluate the Pharmacokinetics, Safety and Efficacy of Two Treatment Modules in Chinese Subjects With Moderate to Severe Crohn's Disease. <https://ClinicalTrials.gov/show/NCT02015793> 2013.

- 593.** Loftus Jr EV, Panés J, Lacerda AP, Peyrin-Biroulet L, D'Haens G, Panaccione R, et al. Upadacitinib induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2023;**388**(21):1966-80.
- 594.** Sandborn WJ, Feagan BG, Rutgeerts P, Hanauer S, Colombel JF, Sands BE, et al. Vedolizumab as induction and maintenance therapy for Crohn's disease. *New England Journal of Medicine* 2013;**369**(8):711-21.
- 595.** Feagan B, Sandborn W, Colombel JF, O'Byrne S, Khalid J, Brayshaw N, et al. Effect of vedolizumab treatment on extraintestinal manifestations in patients with Crohn's disease: a GEMINI 2 post hoc analysis. *Gastroenterology* 2017;**152**(5 Suppl 1):S597.
- 596.** D'Haens G, Colombel JF, Dubinsky M, Abhyankar B, James A, Lasch K. The efficacy of vedolizumab by disease activity and prior tumour necrosis factor-alpha antagonist failure in patients with ulcerative colitis or Crohn's disease: post-hoc analyses from the GEMINI 1 and GEMINI 2 studies. *Journal of Crohn's & Colitis* 2016;**10**:S58.
- 597.** Feagan B, Schwartz D, Danese S, Rubin DT, Abhyankar B, Smyth M, et al. Vedolizumab for the treatment of fistulising Crohn's disease: an exploratory analysis of data from GEMINI 2. *Journal of Crohn's & Colitis* 2015;**9**:S333-4.
- 598.** French J, Rosario M, Dirks NL, Milton A, Fox I, Gastonguay MR. Vedolizumab exposure-response relationship during induction therapy in adults with Crohn's disease. *Gastroenterology* 2014;**1**:S592-3.
- 599.** Sands BE, Shafran I, Farraye FA, Cheifetz AS, Abhyankar B, Sankoh S, et al. Efficacy and safety of retreatment with vedolizumab in patients with Crohn's disease. *Journal of Crohn's & Colitis* 2015;**9**:S377-8.
- 600.** Sands BE, Cohen R, Isaacs K, Fedorak RN, Abhyankar B, Sankoh S, et al. Infusion-related reactions with vedolizumab treatment in patients with uc or cd during the GEMINI 1 and GEMINI 2 clinical trials. *Journal of Crohn's & Colitis* 2015;**9**:S392-3.
- 601.** Sandborn WJ, Vermeire S, Sands BE, Feagan BG, Shetzline M, Xu J, et al. Evaluation of patient-reported outcomes from the Crohn's disease activity index to define remission in the GEMINI 2 study. *United European Gastroenterology Journal* 2015;**1**:A604-5.
- 602.** Colombel JF, Loftus Jr EV, Siegel CA, Lewis JD, Smyth M, Xu J, et al. Efficacy of vedolizumab with concomitant corticosteroid or immunomodulator use in patients with Crohn's disease in GEMINI 2. *Journal of Crohn's & Colitis* 2015;**9**:S344-5.
- 603.** Loftus Jr EV, Siegel CA, Panaccione R, Sandborn WJ, Smyth MD, Green A, et al. Corticosteroid dose reduction with vedolizumab treatment of Crohn's disease during the GEMINI 2 trial. *American Journal of Gastroenterology* 2015;**110**:S789.
- 604.** Feagan B, Lissos T, Lasch K, Cao C, James A, Khalid JM, et al. Vedolizumab demonstrates early symptomatic improvement in Crohn's disease (CD): a GEMINI 2 post hoc analysis. *American Journal of Gastroenterology* 2017;**112**(Suppl 1):S370-1.
- 605.** Feagan B, Sandborn WJ, Colombel JF, O'Byrne S, Geransar P, Khalid JM, et al. Effect of vedolizumab treatment on extraintestinal manifestations in patients with Crohn's disease: a GEMINI 2 post hoc analysis. *Gut* 2017;**66**(Suppl 2):A237-8.
- 606.** Sands B, Dubinsky M, Vermeire S, Sankoh S, Rosario M, Milch C, et al. Effects of increased vedolizumab dosing frequency on disease activity in ulcerative colitis and Crohn's disease. *American Journal of Gastroenterology* 2014;**109**(Suppl 2):S478-9.
- 607.** Feagan B, Schwartz D, Danese S, Rubin D, Lissos T, Lasch K, et al. Efficacy of vedolizumab in fistulising Crohn's disease: exploratory analyses of data from GEMINI 2. *Journal of Crohn's & Colitis* 2018;**12**(5):621-6.
- 608.** NCT00783692. Study of vedolizumab (MLN0002) in patients with moderate to severe Crohn's disease. (first received 2 November 2008).
- 609.** Hardi R, Bokemeyer B, Sandborn W, Reinisch W, Shafran I, Stallmach A, et al. Sustained clinical remission with vedolizumab in moderate-to-severe Crohn's disease: a GEMINI 2 post hoc analysis of week 14 remitters. *Journal of Crohn's & Colitis* 2018;**12**(Suppl 1):S485.
- 610.** Banerjee R, Chuah SW, Hilmi ID, Wu DC, Yang SK, Demuth D, et al. Efficacy and safety of vedolizumab in Crohn's disease in patients from Asian countries in the GEMINI 2 study. *Intestinal Research* 2021;**19**(1):83-94.
- 611.** Dubinsky M, D'Haens GR, Candela N, Peterson T, Uddin SM, Lasch K, et al. Symptomatic improvement with vedolizumab therapy for Crohn's disease stratified by disease activity and prior tumor necrosis factor antagonist failure: post hoc analyses from the GEMINI 2 trial. *Gastroenterology* 2021;**160**(6 Suppl):S-414.
- 612.** Loftus Jr EV, Siegel CA, Panaccione R, Sandborn WJ, Smyth M, Green A, et al. Corticosteroid dose reduction with vedolizumab treatment of Crohn's disease. *United European Gastroenterology Journal* 2015;**65**:A254-5.
- 613.** Sandborn W, Reinisch W, Loftus EV, Panaccione R, Doll HA, Patel H, et al. Effect of vedolizumab on health-related quality of life in patients with Crohn's disease. *Gastroenterology* 2015;**148**:S842.
- 614.** Sandborn W, Sands B, Colombel JF, Feagan B, Hanauer S, Rutgeerts P, et al. Vedolizumab induction therapy for Crohn's disease: results of GEMINI II, a randomized, placebo-controlled, double-blind, multicenter phase 3 trial. *American Journal of Gastroenterology* 2012;**107**:S624-5.
- 615.** Sands B, Hanauer S, Colombel JF, Danese S, Abreu MT, Ahuja V, et al. Reductions in corticosteroid use in patients with ulcerative colitis or Crohn's disease treated with vedolizumab. *United European Gastroenterology Journal* 2013;**108**:A370.
- 616.** Hanauer SB, Feagan BG, MacIntosh DG, Xu J, Milch C, Fox I, et al. Efficacy of vedolizumab in Crohn's disease by

prior treatment failure in GEMINI II, a randomized, placebo-controlled, double-blind, multicenter study. *Gastroenterology* 2013;**144**:S772.

617. Feagan BG, Schwartz DA, Danese S, Rubin DT, Abhyankar B, Smyth MD, et al. Vedolizumab for the treatment of fistulizing Crohn's disease: an exploratory analysis of data from GEMINI 2. *Gastroenterology* 2015;**148**:S274.

618. Feagan BG, Siegel CA, Melmed GY, Isaacs K, Lasch K, Rosario M, et al. Efficacy of vedolizumab with and without continued immunosuppressant use in GEMINI 1 and GEMINI 2. *United European Gastroenterology Journal* 2015;**3**:A17-8.

619. D'Haens GR, Colombel JF, Dubinsky M, Abhyankar B, James A, Lasch K. The efficacy of vedolizumab by disease activity and prior tumor necrosis factor a antagonist failure in patients with ulcerative colitis or Crohn's disease: post hoc analyses from the GEMINI 1 and GEMINI 2 studies. *Gastroenterology* 2016;**150**:S804-5.

620. Feagan B, Siegel CA, Melmed G, Isaacs K, Lasch K, Rosario M, et al. Efficacy of vedolizumab maintenance therapy with and without continued immunosuppressant use in GEMINI 1 and GEMINI 2. *American Journal of Gastroenterology* 2015;**110**:S791.

621. Sandborn WJ, Feagan BG, Reinisch W, Smyth MD, Xu J, Parikh A, et al. Effects of continued vedolizumab therapy for Crohn's disease in week 6 induction therapy nonresponders. *Gastroenterology* 2014;**146**:S588.

622. Sandborn W, Danese S, Abhyankar B, Reinisch W, Xu J, Lasch K, et al. Vedolizumab results in durable clinical remission in Crohn's disease. *American Journal of Gastroenterology* 2014;**109**:S476-7.

623. Loftus E, Colombel JF, Siegel C, Lewis J, Abhyankar B, Sankoh S, et al. Safety of vedolizumab alone or with concomitant corticosteroids and/or immunosuppressants in patients with ulcerative colitis or Crohn's disease. *American Journal of Gastroenterology* 2014;**109**:S478.

624. Rosario M, Fox I, Milch C, Parikh A, Feagan B, Sandborn W, et al. Pharmacokinetic/pharmacodynamic relationship and immunogenicity of vedolizumab in adults with inflammatory bowel disease: additional results from GEMINI 1 and 2. *Inflammatory Bowel Diseases* 2013;**19**:S80.

625. Rosario M, Wyant T, Milch C, Parikh A, Feagan B, Sandborn WJ, et al. Pharmacokinetic and pharmacodynamic relationship and immunogenicity of vedolizumab in adults with inflammatory bowel disease: additional results from the GEMINI 1 and 2 studies. *Journal of Crohn's & Colitis* 2014;**8**:S42-3.

626. Sandborn W, Danese S, Abhyankar B, Reinisch W, Xu J, Lasch K, et al. Durability of clinical remission with vedolizumab in Crohn's disease. *United European Gastroenterology Journal* 2014;**1**:A537-8.

627. Sandborn WJ, Vermeire S, Sands BE, Feagan B, Shetzline M, Xu J, et al. Defining remission using patient-reported outcome components of the Crohn's disease activity

index: post-hoc analyses of vedolizumab data from the GEMINI 2 study. *American Journal of Gastroenterology* 2015;**110**:S792-3.

628. Rutgeerts PJ, Sandborn W, Reinisch W, Xu J, Milch C, Fox I, et al. Sustained therapeutic benefit of vedolizumab throughout 1 year in Crohn's disease in GEMINI ii a randomized placebo-controlled double-blind, multicenter study. *Gastroenterology* 2013;**144**:S21.

629. Sands BE, Dubinsky MC, Vermeire S, Sankoh S, Rosario M, Milch C. Effects of increased vedolizumab dosing frequency on disease activity in ulcerative colitis and Crohn's disease. *United European Gastroenterology Journal* 2014;**2**:A67.

630. Sands BE, Shafran I, Farraye FA, Cheifetz AS, Abhyankar B, Sankoh S, et al. Efficacy and safety of retreatment with vedolizumab in patients with Crohn's disease. *Gastroenterology* 2015;**148**:S278.

631. Sands B, Hanauer S, Colombel JF, Danese S, Abreu M, Ahuja V, et al. Reductions in corticosteroid use in patients with ulcerative colitis or Crohn's disease treated with vedolizumab. *American Journal of Gastroenterology* 2013;**108**:S503.

632. Yajnik V, Khan N, Dubinsky M, Axler J, Green A, Abhyankar B, et al. Efficacy and safety of vedolizumab with advancing age in patients with Crohn's disease: results from the GEMINI 2 study. *Journal of Crohn's & Colitis* 2015;**9**:S244-5.

633. Yajnik V, Khan N, Dubinsky M, Axler J, Green A, Abhyankar B, et al. Efficacy and safety of vedolizumab with advancing age in patients with Crohn's disease: results from the GEMINI 2 study. *American Journal of Gastroenterology* 2015;**110**:S790.

634. Rubin D, Feagan B, Dryden G, Melmed G, Lasch K, Abhyankar B, et al. The effect of vedolizumab on extraintestinal manifestations in patients with Crohn's disease in GEMINI 2. *Inflammatory Bowel Diseases* 2016;**22**:S42-3.

635. Colombel JF, Loftus EV, Siegel CA, Lewis J, Smyth MD, Xu J, et al. Efficacy of vedolizumab with concomitant corticosteroid or immunomodulator use in patients with Crohn's disease from GEMINI 2. *Gastroenterology* 2015;**148**:S277.

636. Hanauer S, Colombel JF, Feagan B, Rutgeerts P, Sandborn W, Sands B, et al. Vedolizumab maintenance therapy for Crohn's disease: results of GEMINI II, a randomized, placebo-controlled, double-blind, multicenter phase 3 trial. *American Journal of Gastroenterology* 2012;**107**:S620-1.

637. Colombel JF, Siegel CA, Abhyankar B, Loftus Jr EV, Lewis JD, Sankoh S, et al. Safety of vedolizumab alone or with concomitant corticosteroids and/or immunosuppressants in patients with ulcerative colitis or Crohn's disease. *United European Gastroenterology Journal* 2014;**2**:A82.

638. Sandborn W, Reinisch W, Loftus Jr EV, Panaccione R, Doll H, Patel H, et al. Effect of vedolizumab on health-related quality of life (HRQOL) in patients with Crohn's disease. *Journal of Crohn's & Colitis* 2015;**9**:S227.

639. Colombel JF, Sands BE, Feagan BG, Loftus EV, Sankoh S, Fox I, et al. Integrated safety analysis of vedolizumab for

the treatment of ulcerative colitis or Crohn's disease. *Gastroenterology* 2013;**5**(144):S113.

640. Sands BE, Cohen RD, Isaacs KL, Fedorak RN, Abhyankar B, Sankoh S, et al. Infusion-related reactions with vedolizumab treatment in patients with UC or CD during the GEMINI 1 and GEMINI 2 clinical trials. *Gastroenterology* 2015;**148**(4):S232-3.

641. Sandborn WJ, Feagan B, Reinisch W, Smyth M, Xu J, Parikh A, et al. Efficacy of continued vedolizumab therapy in patients with Crohn's disease who did not respond to vedolizumab induction therapy at week 6. *Journal of Crohn's & Colitis* 2014;**8**:S274-5.

642. Dulai P, Peyrin-Biroulet L, Hahn K, Khalife N, Lindner D, Lasch K, et al. The impact of early disease control with vedolizumab on surgery rates among patients with Crohn's disease: a post hoc analysis of the GEMINI trials. *Gastroenterology* 2019;**156**(6):S1100.

643. Rutgeerts P, Sandborn W, Reinisch W, Xu J, Milch C, Fox I, et al. Sustained therapeutic benefit of vedolizumab throughout 1 year in Crohn's disease in GEMINI ii a randomized placebo-controlled double-blind, multicenter study. *Gastroenterology* 2013;**144**(5 Suppl 1):S21.

644. Vermeire S, Feagan B, Mody R, Previtali A, Abhyankar B. Response and remission rates with up to 3 years of vedolizumab treatment in patients with Crohn's disease. *Gastroenterology* 2016;**150**(4 Suppl 1):S803-4.

645. Loftus E, Siegel C, Panaccione R, Sandborn W, Smyth M, James A, et al. Corticosteroid dose reduction with vedolizumab treatment of Crohn's disease. *Gut* 2016;**65**(Suppl 1):A147.

646. EUCTR2008-002783-33. A phase 3, randomized, placebo-controlled, blinded, multicenter study of the induction and maintenance of clinical response and remission by vedolizumab (MLN0002) in patients with moderate to severe Crohn's disease. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=EUCTR2008-002783-33-EE> (first received 27 October 2008).

647. Sandborn WJ, Panés J, Danese S, Sharafali Z, Hassanali A, Jacob-Moffatt R, et al. Etrolizumab as induction and maintenance therapy in patients with moderately to severely active Crohn's disease (BERGAMOT): a randomised, placebo-controlled, double-blind, phase 3 trial. *Lancet Gastroenterology & Hepatology* 2023;**8**(1):43-55.

648. Röder H. Kontrollierte Einführung des Infliximab-Biosimilars CT-P13 Bei Patienten Mit Morbus Crohn und Colitis Ulcerosa in Einer Großen Deutschen Zentrumskohorte (Doctoral Dissertation). LMU, 2021.

649. EUCTR2019-001087-30. Evaluating efficacy and safety of subcutaneous CT-P13 (CT-P13 SC) as maintenance therapy in patients with Crohn's disease. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2019-001087-30/LV> (first received 4 June 2019).

650. Colombel JF, Hanauer SB, Sandborn W, Sands BE, Schreiber S, Danese S, et al. Subcutaneous infliximab (CT-P13 SC) as maintenance therapy for Crohn's disease: a phase 3,

randomised, placebo-controlled study (LIBERTY-CD). *Journal of Crohn's & Colitis* 2023;**17**:i161-2.

651. NCT03945019. CT-P13 (infliximab) subcutaneous administration in patients with moderately to severely active Crohn's disease. <https://ClinicalTrials.gov/show/NCT03945019> (first received 10 May 2019).

652. Gjuladin-Hellon T, Iheozor-Ejiofor Z, Gordon M, Akobeng AK. Azathioprine and 6-mercaptopurine for maintenance of surgically-induced remission in Crohn's disease. *Cochrane Database of Systematic Reviews* 2019, Issue 8. Art. No: CD010233. [DOI: [10.1002/14651858.CD010233.pub3](https://doi.org/10.1002/14651858.CD010233.pub3)]

653. Gjuladin-Hellon T, Gordon M, Iheozor-Ejiofor Z, Akobeng AK. Oral 5-aminosalicylic acid for maintenance of surgically-induced remission in Crohn's disease. *Cochrane Database of Systematic Reviews* 2019, Issue 6. Art. No: CD008414. [DOI: [10.1002/14651858.CD008414.pub3](https://doi.org/10.1002/14651858.CD008414.pub3)]

654. Iheozor-Ejiofor Z, Gordon M, Clegg A, Freeman SC, Gjuladin-Hellon T, MacDonald JK, Akobeng AK. Interventions for maintenance of surgically induced remission in Crohn's disease: a network meta-analysis. *Cochrane Database of Systematic Reviews* 2019, Issue 9. Art. No: CD013210. [DOI: [10.1002/14651858.CD013210.pub2](https://doi.org/10.1002/14651858.CD013210.pub2)]

655. Gordon M, Grafton-Clarke C, Akobeng A, Macdonald J, Chande N, Hanauer S, et al. Pancreatitis associated with azathioprine and 6-mercaptopurine use in Crohn's disease: a systematic review. *Frontline Gastroenterology* 2021;**12**:423-36.

656. Gordon M, Lakunina S, Sinopoulou V, Akobeng A. Minimum sample size estimates for trials in inflammatory bowel disease: a systematic review of a support resource. *World Journal of Gastroenterology* 2021;**27**:7572.

657. Iheozor-Ejiofor Z, Lakunina S, Gordon M, Akintelure D, Sinopoulou V, Akobeng A. Sample-size estimation is not reported in 24% of randomised controlled trials of inflammatory bowel disease: a systematic review. *UEG Journal* 2021;**9**:47-53.

658. Lamb CA, Kennedy NA, Raine T, Hendy PA, Smith PJ, Limdi JK, et al. British Society of Gastroenterology consensus guidelines on the management of inflammatory bowel disease in adults. *Gut* 2019;**68**(3):s1-106.

659. Torres J, Bonovas S, Doherty G, Kucharzik T, Gisbert JP, Raine T, et al. ECCO guidelines on therapeutics in Crohn's disease: medical treatment. *Journal of Crohn's & Colitis* 2020;**14**:4-22.

660. Feuerstein J, Ho E, Schmidt E et al. AGA clinical practice guidelines on the medical management of moderate to severe luminal and perianal fistulizing Crohn's disease. *Gastroenterology* 2021;**160**:2496-508.

661. Moran GW, Gordon M, Sinopoulou V, Radford SJ, Darie AM, Vuyyuru SK, et al. British Society of Gastroenterology guidelines on inflammatory bowel disease in adults: 2025. *Gut* 2025;**74**(Suppl 2):s1-s101.

662. Shehab M, Alrashed F, Alrashidi A, Hassan A, Ma C, Narula N, et al. Network meta-analysis: comparative efficacy of biologics and small molecules in the induction and maintenance of remission in Crohn's disease. *Alimentary Pharmacology & Therapeutics* 2025;**62**(5):472-82.

663. Attauabi M, Steenholdt C, Poulsen A, Gubatan J, Burisch J, Nielsen OH, et al. Network meta-analysis: Comparative onset of early effect of biologics and small molecules in moderately to severely active luminal Crohn's disease. *Alimentary Pharmacology & Therapeutics* 2024;**60**(2):124-43.

ADDITIONAL TABLES

Table 1. Induction studies: characteristics

Study ID	Numbers randomised	Purine analogues use	Biologic naive or not
Chen 2020	IG (ada): 102	n (%)	Naive
	CG (placebo): 103	Azathioprine: CG: 59 (57%) IG: 60 (59%) Mercaptopurine CG: 2 (2%) IG: 0 > 50%	
Colombel 2007 (CHARM) - Induction (non-responders)	CG (placebo): 91	n (%)	Mixed naive and not 152 (54.5%) out of 279 participants were naive at recruitment
	IG1 (Ada eow): 88	AZA: 88/279	
	IG2 (Ada weekly): 100	6-MP: 27/279 < 50%	
Colombel 2010 (SONIC)	CG (AZA only): 170	Part of the intervention	Naive
	IG1 (IFX only): 169		
	IG2 (combination therapy): 169		
D'Haens 1999 (refractory CD)	CG (placebo): 8	Concomitant azathioprine, n	Naive
	IG (ifx 5 mg): 7	CG (placebo): 3/8	
	IG (ifx 10 mg): 7	IG (ifx 5 mg): 3/7	
	IG (ifx 20 mg): 8	IG (ifx 10 mg): 1/7	
		IG (ifx 20 mg): 5/8 < 50%	
D'Haens 2008	CG (corticosteroids): 66	Part of the intervention	Naive
	IG (ifx + aza): 67		
D'Haens 2022 (ADVANCE)	CG (placebo): 186	Unclear (abstract only)	Not
	IG1 (ris 600 mg): 373		
	IG2 (ris 1200 mg): 372		

Table 1. Induction studies: characteristics (Continued)

D'Haens 2022 (SERENE - Induc- tion)	IG Ada (high): 308 CG Ada (standard): 206	Immunosuppressant use (n, %) IG: 78 (25.3%) CG: 61 (29.6%) < 50%	Not The study allowed enrol- ment of participants with secondary loss of response or intolerance to infliximab (up to 25% of the total study pop- ulation).
Duan 2013	IG (ifx): 8 IG (aza): 8 IG (comb): 8	Part of the intervention	Unclear
Feagan 2008c	CG (placebo): 58 IG1 (Ved 0.5 mg): 62 IG2 (Ved 2 mg): 65	Participants on immunosuppressives were excluded < 50%	Naive
Feagan 2015a (UNITI-1)	CG (placebo): 247 IG (ust 130 mg): 245 IG2 (ust 6 mg/kg): 249	Immunosuppressants including azathio- prine, mercaptopurine and methotrexate, n (%) CG (placebo): 81 (32.8%) IG (ust 8 weeks): 74 (30.2%) IG (ust 12 weeks): 78 (31.3%) < 50%	Not History of disease refractory to treatment with TNF antag- onist, n (%) CG (placebo): 246 (99.6%) IG (ust 160 mg): 243 (99.2%) IG2 (ust 6 mg/kg): 246 (98.8%)
Feagan 2015b(UNITI-2)	CG (placebo): 210 IG (ust 130 mg): 209 IG2 (ust 6 mg/kg): 209	Immunosuppressants including azathio- prine, mercaptopurine and methotrexate, n (%) CG (placebo): 73 (34.8%) IG (ust 8 weeks): 74 (35.4%) IG (ust 12 weeks): 72 (34.4%) < 50%	Mixed naive and not No history of TNF antagonist treatment, n (%) CG (placebo): 131 (62.4%) IG (ust 160 mg): 152 (72.7%) IG2 (ust 6 mg/kg): 144 (68.9%)
Feagan 2017	CG (placebo): 39 IG (Ris 200 mg): 41 IG (Ris 600 mg): 41	Immunosuppressants and corticos- teroids, n CG (placebo): 5 (13%) IG (Ris 200 mg): 3 (7%) IG (Ris 600 mg): 3 (7%) Immunosuppressants, n (%) CG (placebo): 8 (21%) IG (Ris 200 mg): 7 (17%) IG (Ris 600 mg): 5 (12%)	Not (more than 90% were not naive)

Table 1. Induction studies: characteristics (Continued)

< 50%			
Ghosh 2003	CG (placebo): 63	Azathioprine or mercaptopurine, (%)	Naive
	IG (nat 1 x 3 mg): 68	CG (placebo): 22 (35%)	
	IG (nat 2 x 3 mg): 66	IG (nat 1 x 3 mg): 26 (38%)	
	IG (nat 2 x 6 mg): 51	IG (nat 2 x 3 mg): 17 (26%)	
< 50%			
Gordon 2001	CG (placebo): 12	Azathioprine, n (%)	Naive
	IG (nat): 18	CG: 2 (17%)	
		IG: 6 (33%)	
< 50%			
Hanauer 2002 (AC-CENT I) - Induction (non-response)	CG (placebo): 78	6-mercaptopurine and azathioprine, n (%)	Naive
	IG1 (ifx 5 mg): 79	63 (27%) of all randomised non-responders	
	IG2 (ifx 5 + 10 mg): 81		
< 50%			
Hanauer 2006 (CLASSIC I)	CG (placebo): 74	Azathioprine, n (%)	Naive
	IG1 (Ada 40/20 mg): 74	CG: 13 (18%)	
	IG2 (Ada 80/40 mg): 75	IG1: 13 (18%)	
	IG3 (Ada 160/80 mg): 76	IG2: 9 (12%)	
		IG3: 11 (14%)	
		6-mercaptopurine, n (%)	
		CG: 8 (11%)	
		IG1: 6 (8%)	
Hanauer 2021 (VOLTAIRE-CD)	CG (ada): 75	IG2: 10 (13%)	
	IG (BI 695501): 72	IG3: 10 (13%)	
		< 50%	
Lemann 2006	CG (aza): 58	Unclear	Naive, except for previous exposure to infliximab
	IG (aza + ifx): 57		CG: 7 (10%)
			IG: 5 (7%)
Allez 2023 (TRI-DENT- Part 1)	CG (placebo): 72	Unclear	Naive
			Not naive

Table 1. Induction studies: characteristics (Continued)

IG (tesnatilimab) 73			
Allez 2023 (TRI-DENT- Part 2)	Tesnatilimab (400 mg or 150 mg or 50 mg subcutaneous at week 0, then half dosage at weeks 2, 4, 8, 12, 16 and 20) (n = 148) Ustekinumab (6 mg/kg intravenous injection at week 0, 90 mg subcutaneous at week 8 and 16) (n = 47)	Unclear	Not naive
Panaccione 2022 (MOTIVATE)	CG (placebo): 187 IG1 (ris 600 mg): 191 IG2 (ris 1200 mg): 191	Unclear	Mixed naive and not Failed one or more biologics, n CG: 37 IG1: 81 IG2: 77
Panes 2017 Induction	CG (placebo): 92 IG (tof 5 mg): 86 IG (tof 10 mg): 86 IG (tof 15 mg): 16 (the intervention was discontinued)	Participants receiving azathioprine, 6-mercaptopurine or methotrexate were required to stop these agents at least 2 weeks before randomisation < 50%	Mixed naive and not Prior use of TNFi, n (%) CG (placebo): 70 (76.9%) IG (tof 5 mg): 68 (79.1%) IG (tof 10 mg): 66 (76.7%) IG (tof1 5 mg): 11 (68.8%)
Present 1999	CG (placebo): 31 IG1 (ifx 5 mg): 31 IG2 (ifx 10 mg): 32	Mercaptopurine or azathioprine, n (%) CG: 9 (29%) IG1: 12 (39%) IG2: 17 (53%) < 50%	Naive
Sandborn 2001a	CG (placebo 8-weekly): 27 CG (placebo 12 weekly): 31 IG1 (CDP571 10 mg/kg 8-weekly): 29 IG1 (CDP571 10 mg/kg 12-weekly): 25 IG2 (CDP571 20 mg/kg 8-weekly): 28 IG2 (CDP571 20 mg/kg 12-weekly): 29	Azathioprine or 6-mercaptopurine, n CG (placebo 8-weekly): 9 CG (placebo 12-weekly): 18 IG1 (CDP571 10 mg/kg 8-weekly): 18 IG1 (CDP571 10 mg/kg 12-weekly): 13 IG2 (CDP571 20 mg/kg 8-weekly): 14 IG2 (CDP571 20 mg/kg 12-weekly): 17 < 50%	Naive
Sandborn 2001b	CG (placebo): 20 IG (etanercept): 23	Mercaptopurine or azathioprine, n CG: 7	Mixed naive and not

Table 1. Induction studies: characteristics (Continued)

		IG: 10 < 50%	Prior treatment with inflix- imab CG: 8 IG: 12
Sandborn 2005a (ENACT 1)	CG (placebo): 181 IG (nat): 724	Immunosuppressants, n (%) CG: 52 (29) IG: 247(34) < 50%	Mixed naive and not Previous receipt of therapy against TNF- α , n (%) CG: 69 (38%) IG: 291 (40%)
Sandborn 2007a	CG (placebo): 166 IG (ada): 159	Immunosuppressants, n (%) CG: 85 (51%) IG: 73 (46%) > 50%	Not
Sandborn 2007c (PRECISE 1)	CG (placebo): 329 IG (cert): 333	Immunosuppressants, n (%) CG: 66 (20%) IG: 69 (21%) < 50%	Mixed naive and not Previous ifx therapy, n (%) CG: 85 (26%) IG: 100 (30%)
Sandborn 2008 (Population 1)	CG1 (placebo SC): 26 CG2 (placebo IV): 27 IG1 (ust SC): 25 IG2 (ust IV): 26	Immunosuppressants, n (%) CG1: 10 (39%) CG2: 19 (37%) IG1: 2 (20%) IG2: 10 (39%) < 50%	Mixed naive and not Participants with prior inflix- imab use CG1: 16 (32.65%) CG2: 11 (22.45%) IG1: 14 (28.57%) IG2: 8 (16.33%)
Sandborn 2011	CG (placebo): 216 IG (cert): 223	Immunosuppressants, n (%) CG: 67 (31.2%) IG: 77 (34.5%) < 50%	Naive
Sandborn 2012 (CERTIFI) - Induc- tion	CG (placebo): 132 IG1 (ust 1 mg): 131 IG2 (ust 3 mg): 132 IG3 (ust 6 mg): 131	Immunomodulatory drug CG: 30 (22.7%) IG1: 33 (25.2%) IG2: 28 (21.2%) IG3: 35 (26.7%) < 50%	Not

Table 1. Induction studies: characteristics (Continued)

Sandborn 2012 (CERTIFI) - Induction (non-response)	CG (placebo): 110 IG (ust): 109	< 50% (assumption based on the above, not provided separately for this population)	Not (as above)
Sandborn 2013 (GEMINI II) - Induction	CG (placebo): 148 IG1 (ved): 220	Immunosuppressive agents, n (%) CG: 25 (16.9%) IG: 37 (16.8%) < 50%	Mixed naive and not Prior TNF antagonist therapy, n (%) CG: 72 (48.6%) IG: 111 (50.5%)
Sandborn 2014	CG (placebo): 34 IG1 (Tof 1 mg): 36 IG2 (Tof 5 mg): 34 IG3 (Tof 15 mg): 35	None except for 2 participants in IG3 < 50%	Mixed naive and not Anti-TNFα use within the previous 12 months, n CG: 1 IG1: 4 IG2: 1 IG3: 4
Sandborn 2023 - BERGAMOT (cohort 1)	CG (placebo): 59 IG1 (etro 105 mg): 120 IG2 (etro 210 mg): 121	Participants were refractory to immunosuppressives, but concurrent medications are not reported	73% of participants were TNF-experienced
Sandborn 2020d (CELEST) - Induction	CG (placebo): 37 IG (Upa 3 mg twice daily): 39 IG (Upa 6 mg twice daily): 37 IG (Upa 12 mg twice daily): 36 IG (Upa 24 mg twice daily): 36 IG (Upa 24 mg once daily): 35	None, participants had to discontinue these drugs < 50%	Not (< 5% were naive)
Sandborn 2022 - GALAXI 1	CG (placebo): 70 IG (Gus 200 mg): 73 IG (Gus 600 mg): 73 IG (Gus 1200 mg): 73 IG (UST): 71	Immunomodulatory therapy, n (%) CG: 26 (42.6%) IG G200: 15 (24.6%) IG G600: 18 (28.6%) IG G1200: 25 (41.0%) IG UST: 26 (41.3%) < 50%	Mixed naive and not Biologic naive, n (%) CG: 19 (31.1%) IG G200: 25 (41.0%) IG G600: 24 (38.1%) IG G1200: 25 (41.0%) IG UST: 19 (30.2%)
Sands 2004 (AC-CENT II) Induction (Non-responders)	CG (placebo): 44 IG (ifx): 43	Mercaptopurine or azathioprine, n (%) 53/87 (61) > 50%	Naive

Table 1. Induction studies: characteristics (Continued)

Sands 2007	CG (ifx): 27 IG1 (ifx + nat): 52	Azathioprine, mercaptopurine or methotrexate, n (%) CG: 15 (56%) IG1: 15 (56%) > 50%	Not
Sands 2010	CG (placebo): 73 IG1 (AM 50 mg): 73 IG2 (AM 100 mg): 74	Azathioprine or mercaptopurine, n (%) CG: 1 (1%) IG1: 2 (3%) IG2: 4 (5%) < 50%	Mixed naive and not Previous biologic use, n (%) CG: 38 (52%) IG1: 47 (64%) IG2: 45 (61%)
Sands 2014 (GEMINI III)	CG (placebo): 207 IG (ved): 209	Immunosuppressive use, n (%) CG: 69 (33%) IG: 69 (33%) < 50%	Not All participants had experienced an inadequate response, loss of response, or intolerance to TNF antagonists, immunosuppressives or corticosteroids within the past 5 years. Results presented separately for naive and not.
Sands 2022 (SEAVUE)	CG (ada): 195 IG (ust): 191	Unclear (abstract only)	Naive
Schreiber 2005	CG (placebo): 73 IG1 (cert 100 mg): 74 IG2 (cert 200 mg): 72 IG3 (cert 400 mg): 73	Immunomodulators, n (%) CG: 26 (35.6%) IG1: 26 (35.1%) IG2: 29 (40.3%) IG3: 27 (37.5%) < 50%	Mixed naive and not CG: 16 (21.9%) IG1: 18 (24.3%) IG2: 17 (23.6%) IG3: 12 (16.7%)
Schreiber 2018a	CG (placebo): 28 IG1 (Ande 150 mg Q2W): 53 IG2 (Ande 150 mg QW): 53 IG3 (Ande 300 mg QW): 53	Immunomodulators, n (%) CG: 19 (35.8%) IG1: 17 (32.1%) IG2: 8 (15.1%) IG3: 8 (28.6%) < 50%	Not (vast majority) CG: 46 (86.8) IG1: 45 (84.9) IG2: 43 (81.1) IG3: 24 (85.7)
Schwartz 2021 - ENTERPRISE	IG1 (ved): 16 IG2 (ved+): 18	Immunosuppressants, n (%) IG1: 4 (28.6%) IG2: 3 (21.4%)	Mixed naive and not "In each regimen, 11 patients (78.6%) had prior anti-TNF therapy at baseline"

Table 1. Induction studies: characteristics (Continued)

Table 2. Induction studies: characteristics (continued)			
	The study compared vedolizumab regimens	< 50%	
Targan 1997	CG (placebo): 25	Mercaptopurine, n (%)	Naive
	IG1 (ifx 5 mg): 27	CG: 4 (16%)	
	IG2 (ifx 10 mg): 28	IG1: 4 (15%)	
	IG3(ifx 20 mg): 28	IG2: 4 (14%)	
		IG3: 4 (14%)	
	Azathioprine, n (%)		
	CG: 7 (28%)		
	IG1: 5 (19%)		
	IG2: 4 (14%)		
	IG3: 8 (29%)		
< 50%			
Targan 2007 (EN-CORE)	CG (placebo): 250	Immunosuppressants, n (%)	Mixed naive and not
	IG (nat): 259	CG: 96 (38%)	Prior use of anti-tumour necrosis factor agents, n (%)
		IG: 97 (37%)	
		< 50%	CG: 112 (45%) IG: 130 (50%)
Vermeire 2017 FITZROY - induction	CG (placebo): 44	Immunomodulators (e.g. thiopurines and methotrexate) must have been discontinued at least 25 days before the first dose of the study drug	Mixed naive and not
	IG (fil): 130		Previous anti-TNF exposure, n (%)
			CG: 28 (64%)
			IG: 73 (56%)
Vermeire 2017 FITZROY - induction non-responders	CG (placebo): 9	< 50% (as above)	Mixed naive and not
	IG2 (fil 200 mg): 25		(as above, not provided separately)
Watanabe 2012 - Induction	CG (placebo): 23	Immunosuppressants, n (%)	Mixed naive and not
	IG1 (ada 80/40 mg): 34	CG: 8 (34.8%)	Previous use of infliximab or biologic at baseline
		IG1: 11 (32.4%)	
	IG2 (ada 160/80 mg): 33	IG2: 10 (30.3%)	CG: 13 (56.5%)
		< 50%	IG1: 20 (58.8%)
			IG2: 19 (57.6%)
			Previous exposure to anti-TNF agents other than adalimumab was allowed; primary non-responders to prior anti-TNF therapy were excluded

Table 1. Induction studies: characteristics (Continued)

Watanabe 2020 - Induction	CG (placebo): 78	Corticosteroids and immunomodulators, n (%)	Mixed naive and not
	IG (ved): 79	CG: 11 (14.1%)	Prior anti-TNFa treatment, n (%)
		IG: 9 (11.4%)	CG: 62 (79.5%)
		< 50%	IG: 61 (77.2%)
Winter 2004	CG (placebo): 25	Participants receiving immunosuppressive therapy, n (%)	Mixed naive and not
	IG1 (cert 1.25 mg): 2	CG: 11 (44)	Prior anti-TNFa treatment, n (%)
	IG2 (cert 5 mg): 25	IG1: 0 (0)	CG: 3 (13%)
	IG3 (cert 10 mg): 17	IG2: 11 (44)	IG1: 1 (50%)
	IG4 (cert 20 mg): 23	IG3: 9 (53)	IG2: 6 (23%)
		IG4: 10 (44)	IG3: 5 (29%)
		< 50%	IG4: 7 (30%)
Ye 2019	CG (ifx): 109	Treatment with immunomodulators, n (%)	Naive
	IG (CT-P13): 111	CG: 80(73%)	
		IG: 84 (76%)	
		> 50%	
Loftus 2023 (U-EX-CEL)	CG (placebo): 176	Unclear (press release only)	Not
	IG (upa): 350		

6-mp: 6-mercaptopurine

Ada/ada: adalimumab

Ande: andecaliximab

AM: apilimod mesylate

Aza/aza: azathioprine

Cert: certolizumab

CG: control group

Eow: every other week

Etro: etrolizumab

Fil: filgotinib

Gus: guselkumab

IFX/ifx: infliximab

IG: intervention group

IV: intravenous

Nat: natalizumab

QW/Q2W: once every week/once every two weeks

Ris: risankizumab

SC: subcutaneous

TNF: tumour necrosis factor

TNFi: TNF inhibitor

Tof: tofacitinib

Upa/upa: upadacitinib

Ust/UST: ustekinumab

Ved/ved: vedolizumab

Table 2. Induction studies: intervention details

Study ID	Intervention medications IG	Intervention medications CG	Previous experience of biologic therapy	Medications up to study start	Medications that had to be discontinued prior to starting study	Mandatory medications per study protocol	Concomitant medications during study
Allez 2023 (TRIDENT-Part 1)	Tesnatilimab (400 mg subcutaneous at Week 0, 200 mg every 2 weeks)	Placebo (4 subcutaneous every 2 weeks)	Participants were previously exposed to biologic therapies	Bio-NF patients were either currently receiving corticosteroids and/ or immunomodulators	Not reported	Not reported	Unclear
Allez 2023 (TRIDENT-Part 2)	Tesnatilimab (400 or 150 or 50 mg subcutaneous at week 0, then half dosage at weeks 2, 4, 8, 12, 16, and 20) Ustekinumab (6 mg/kg intravenous injection at week 0, 90 mg subcutaneous at week 8 and 16)	Placebo (subcutaneous every 2 weeks from Week 0–4, Week 8, every 2 weeks from Week 12–16, and Week 20)	Participants were previously exposed to biologic therapies	Bio-NF patients were either currently receiving corticosteroids and/ or immunomodulators	Not reported	Not reported	Unclear
Chen 2020	Participants were randomised to double-blind adalimumab 160/80 mg at weeks 0/2 and 40 mg at weeks 4/6	Placebo at weeks 0/2 followed by blinded adalimumab 160/80 mg at weeks 4/6	Participants were naïve to anti-TNF therapy	Corticosteroid use: CG: 32 (31%) IG: 31 (30%) Immunosuppressant use: CG: 65 (63%) IG: 61 (60%) Azathioprine: CG: 59 (57%) IG: 60 (59%)	NR	NR	Doses of other CD-related medications (e.g. immunosuppressants, aminosalicylates, and CD-related antibiotics) were to remain stable during the study except if treatment-related toxicities were considered moderate to severe in the opinion of the investigator.

Table 2. Induction studies: intervention details (Continued)

Mercaptopurine: CG: 2 (2%) IG: 0 Methotrexate: CG: 4 (4%) IG: 1 (1%)							
Colombel 2007 (CHARM) - Induction (non-responders)	IG1 (eow): adalimumab 40 mg once every other week, subcutaneous IG2 (weekly): adalimumab 40 mg weekly, subcutaneous	Placebo (no further details)	Previous TNF-antagonist exposure: n (%) 152 (54.5) out of 279 participants	Concurrent therapies for CD, including stable dosages (for at least 4 weeks before screening) of azathioprine, 6-mercaptopurine, methotrexate, 5-aminosalicylates, sulfasalazine, oral mesalamine, and CD-related antibiotics, were permitted, as were stable dosages (for at least 2 weeks before screening) of prednisone (≤ 30 mg/day or equivalent) or budesonide (≤ 9 mg/day) (participants could not be on both prednisone and budesonide). Participants who had received infliximab or any TNF antagonist other than adalimumab more than 12 weeks before screening could be enrolled provided that they did not exhibit initial nonresponse to the agent (i.e. no clinical response to first injection as judged by the investigator). Female participants of childbearing potential were required to use an effective form of birth control.	Participants could not be enrolled if: currently, on parenteral nutrition; had received investigational chemical agents within 30 days or investigational biologic therapy within 3 months before screening; had received antibiotic treatment for non-CD-related infections within 3 weeks before screening; had received treatment with adalimumab or participated in an adalimumab clinical study; had received enema therapy within 2 weeks before screening; had received cyclosporine, mycophenolate mofetil, or tacrolimus with-	All participants entered a 2-week screening period before their baseline assessments. At the baseline visit (week 0), all eligible participants received open-label adalimumab 80 mg subcutaneously, followed by a 40-mg dose at week 2.	From all three groups n (%) out of 279 participants Any corticosteroid: 129; prednisone: 86; budesonide: 39 Any immunosuppressive: 126; azathioprine: 88; 6-mercaptopurine: 27; methotrexate: 29 5-aminosalicylates: 100

Table 2. Induction studies: intervention details (Continued)

					in 8 weeks of screening		
Colombel 2010 (SONIC)	IG1: infliximab (IFX) only group - intravenous infusions of infliximab at 5 mg/kg of body weight at wk 0, 2, 6, 14, 22 plus daily oral placebo through week 30 IG2: combination therapy - intravenous infusion of infliximab at 5 mg/kg of body weight at wk 0, 2, 6, 14, 22 plus daily oral azathioprine at 2.5 mg/kg of body weight through week 30	CG: azathioprine only group - daily oral azathioprine at 2.5 mg/kg of body weight through week 30 plus intravenous placebo infusions at week 0, 2, 6, 14, 22	It was ensured that there was previous use of anti-TNF alpha drugs but no mention of other biologics.	Participants who received systemic corticosteroids during main study: Baseline IG1: IFX only - 40 participants at mean dose of 24.8 ± 16.2 mg/day Baseline IG2: 24.9 ± 12.8 mg/day Baseline CG: 23.8 ± 12.6 mg/day	Not reported	Not reported	Systemic corticosteroids could be initiated (for participants not receiving them at baseline) with the dose maintained, increased, or decreased until week 14 (maximum allowed dose, 40 mg per day) Participants who received systemic corticosteroid during the main study: Week 2: CG (AZA) - 48 participants at dose of 22.9 ± 12.5 mg/day; IG1 (IFX) - 50 participants at mean dose of 24.8 ± 16.2 mg/day; IG2 (Combo) - 49 participants at mean dose of 22.8 ± 11.9 mg/day Week 6: CG (AZA) - 53 participants at mean dose of 18.6 ± 11.6 mg/day; IG1 (IFX) - 52 participants at mean dose of 17.7 ± 11.0 mg/day; IG2 (Combo) - 51 participants at mean dose of 18.3 ± 11.6 mg/day Week 10: CG (AZA) - 56 participants at mean dose of 16.2 ± 11.2 mg/day; IG1 (IFX) - 56 participants at mean dose of 15.7 ± 14.9 mg/day; IG2 (Combo) - 52 participants at mean dose of 15.0 ± 11.1 mg/day Week 18: CG (AZA) - 59 participants at mean dose of 13.5 ± 10.9 mg/day; IG1 (IFX)

Table 2. Induction studies: intervention details (Continued)

							- 57 participants at mean dose of 13.2 ± 17.2 mg/day; IG2 (Combo) - 56 participants at mean dose of 11.6 ± 10.9 mg/day Week 26: CG (AZA) - 60 participants at mean dose of 11.6 ± 10.3 mg/day; IG1 (IFX) - 60 participants at mean dose of 11.0 ± 16.0 mg/day; IG2 (Combo) 58 participants at mean dose of 9.4 ± 10.1 mg/day
D'Haens 1999 (refractory CD)	IG-IFX5: Infliximab 5 mg/kg. Single 2-hour IV infusion. IG-IFX10: Infliximab 10 mg/kg. Single 2-hour IV infusion. IG-IFX20: Infliximab 20 mg/kg. Single 2-hour IV infusion	CG: 0.1% human serum albumin identical in appearance to the infliximab solution. Single 2-hour intravenous infusion (DOSE NOT REPORTED)	NOT REPORTED	Participants were eligible if CD was refractory to: mesalamine for 8 or more weeks (at least 4 weeks' stable dose); corticosteroids up to 40 mg prednisone-equivalent per day for 8 or more weeks with stable doses for at least 2 weeks; and mercaptopurine or azathioprine for 6 or more months, with stable doses for at least 8 weeks before screening Concomitant medication: (i) Azathioprine: CG: 3/8 (37.50%) IG-IFX5: 3/7 (42.86%) IG-IFX10: 1/7 (14.29%) IG-IFX20: 5/8 (62.50%) (ii) Steroids: CG: 5/8 (62.50%) IG-IFX5: 4/7 (57.14%) IG-IFX10: 3/7 (42.86%)	Participants were excluded if they underwent treatment with cyclosporine, methotrexate, or experimental agents within 3 months before inclusion	NOT REPORTED – No protocol found	Participants were eligible if CD was refractory to: mesalamine for 8 or more weeks (at least 4 weeks' stable dose); corticosteroids up to 40 mg prednisone-equivalent per day for 8 or more weeks with stable doses for at least 2 weeks; and mercaptopurine or azathioprine for 6 or more months, with stable doses for at least 8 weeks before screening Concomitant medication: (i) Azathioprine: CG: 3/8 (37.50%) IG-IFX5: 3/7 (42.86%) IG-IFX10: 1/7 (14.29%) IG-IFX20: 5/8 (62.50%) (ii) Steroids: CG: 5/8 (62.50%) IG-IFX5: 4/7 (57.14%)

Table 2. Induction studies: intervention details (Continued)

				IG-IFX20: 4/8 (50.0%)		IG-IFX10: 3/7 (42.86%)	IG-IFX20: 4/8 (50.0%)
D'Haens 2008	Early combined immunosuppression: participants assigned to early combined immunosuppression received 3 infusions of infliximab in doses of 5 mg/kg bodyweight at weeks 0, 2 and 6, in combination with azathioprine in doses of 2 to 2.5 mg/kg per day from day 0 onwards. If a participant responded to and tolerated both drugs, azathioprine was continued for the duration of the trial. Participants who were intolerant to azathioprine were given subcutaneous methotrexate at an initial dose of 25	Conventional management: participants received induction treatment with either methylprednisolone or budesonide. For those who responded to these treatments, the dose of corticosteroid was tapered. For methylprednisolone, an initial daily dose of 32 mg was prescribed for 3 weeks, followed by tapering by 4 mg per week to discontinuation. Participants who received budesonide were prescribed 9 mg per day for 8 weeks with tapering to discontinuation by 3 mg per week thereafter. Hence treat-	According to inclusion criteria, eligible participants included those who had not previously received biologic therapy. THEREFORE, CG - N, IG - N.	Mesalazine use: CG: 2/64 (3.1%) IG: 3/65 (4.6%)	NOT REPORTED	NOT REPORTED - No protocol	NOT REPORTED – may possibly be the same as medication up to study beginning but authors have not mentioned whether mesalazine was continued or discontinued when study began

Table 2. Induction studies: intervention details *(Continued)*

mg each week for 12 weeks with the dose reduced to 15 mg per week thereafter	ment with either drug lasted 10 weeks. If a participant's symptoms worsened during the course of corticosteroid tapering, the dose increased to the initial dose and repeated the induction treatment. If their symptoms continued to worsen despite this manoeuvre, azathioprine was introduced (2 to 2.5 mg/kg per day). Participants who relapsed after withdrawal of corticosteroids were given a second course of corticosteroids in combination with azathioprine. For participants who failed 4 weeks of corticosteroid treatment, the authors increased the methyl
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Table 2. Induction studies: intervention details *(Continued)*

prednisolone dose to 64 mg per day and added azathioprine. They gave 64 mg methyl-prednisolone per day for 2 weeks and then tapered this dose by 8 mg per week to a daily dose of 32 mg. Thereafter, methyl-prednisolone was tapered by 4 mg each week. Any participants who remained symptomatic after 16 weeks of azathioprine treatment received an induction course of infliximab (5 mg/kg body-weight at weeks 0, 2, and 6) and continued antimetabolite treatment. Participants who relapsed despite the use of methotrexate or those who were in-

Table 2. Induction studies: intervention details *(Continued)*

		tolerant to both azathioprine and methotrexate also received infliximab, without antimetabolite treatment. They repeated the infusion on relapse of symptoms in these participants.					
Duan 2013	IG1: infliximab 5 mg/kg at Week 0, Week 2 and Week 6 to induce remission, followed by infliximab 5 mg/kg every 8 weeks to maintain remission. Route NOT REPORTED. IG2: the combination treatment group was given azathioprine 2.5 mg/kg every day, and on the 0th and 2nd, infliximab 5 mg/kg was given for 6 weeks, and then infliximab 5 mg/	CG: azathioprine 2.5 mg/kg every day. Route NOT REPORTED	NOT REPORTED	NOT REPORTED	NOT REPORTED	No Protocol	NOT REPORTED

Table 2. Induction studies: intervention details (Continued)

	kg was given every 8 weeks. Route NOT REPORTED						
Feagan 2008c	IG1: MLN0002 (vedolizumab) 0.5 mg/kg IV at days 1 and 29 IG2: MLN0002 (vedolizumab) 2.0 mg/kg IV at days 1 and 29	CG: placebo IV at days 1 and 29 (AGENT and DOSE NOT REPORTED)	Participants previously treated with biologic therapy for Crohn's disease were ineligible. Therefore, it can be assumed that previous experience of biologic therapy: CG: N IG1: N IG2: N	Mesalamine use % CG: 47% IG1: 55% IG2: 42% Months on oral or continuous intravenous steroid therapy (mean (SD)): CG: 3.8 (6.72) IG1: 2.6 (5.59) IG2: 3.3 (6.08)	Participants requiring cyclosporine or immunosuppressives within 3 months or investigational drugs within 30 days before screening, systemic corticosteroids, heparin, nonsteroidal anti-inflammatory drugs, tube feeding, defined formula diets, or parenteral alimentation, were excluded. New treatments for Crohn's disease were not permitted during the study.	NOT REPORTED - No protocol	Participants could receive concomitant treatment for Crohn's disease with mesalamine or antibiotics provided they had been maintained on a stable dose for 2 weeks immediately before screening. Mesalamine use % CG: 47% IG1: 55% IG2: 42% Months on oral or continuous intravenous steroid therapy (mean(SD)) CG: 3.8(6.72) IG1: 2.6(5.59) IG2: 3.3(6.08)
Feagan 2015a (UNITI-1)	Interventions IG: to receive a single intravenous infusion of 130 mg of ustekinumab, a weight-range-based dose that approximated	CG: placebo (not specified)	History of TNF antagonist treatment failure — no. (%) Participants who received 1 drug: CG: 112 (45.3%) IG1: 124 (50.6%) IG2: 120 (48.2%)	Medications for Crohn's disease taken at baseline — no. (%) One or more medications: CG: 185 (74.9%) IG1: 178 (72.7%) IG2: 174 (69.9%) Immunosuppressant:	NR	NR	Stable doses of immunosuppressants, mesalamine, antibiotics, or oral glucocorticoids (≤ 40 mg of prednisone per day or ≤ 9 mg of budesonide per day) or a combination thereof were permitted.

Table 2. Induction studies: intervention details (Continued)

	6 mg of ustekinumab per kilogram of body weight, or placebo. (The administration of 6 mg of ustekinumab per kilogram meant that Participants weighing ≤ 55 kg received 260 mg, those weighing > 55 kg and ≤ 85 kg received 390 mg and those weighing > 85 kg received 520 mg)	Participants who received 2 or 3 drugs: CG: 134 (54.3%) IG1: 119 (48.6%) IG2: 126 (50.6%)	CG: 81 (32.8%) IG1: 74 (30.2%) IG2: 78 (31.3%) Aminosalicylate: CG: 54 (21.9%) IG1: 50 (20.4%) IG2: 50 (20.1%) Glucocorticoid: CG: 111 (44.9%) IG1: 121 (49.4%) IG2: 108 (43.4%)				
Feagan 2015b(UNITI-2)	IG: At week 0, participants in both induction trials were randomly assigned, in a 1:1:1 ratio, to receive a single intravenous infusion of 130 mg of ustekinumab, a weight-range-based dose that approximated 6 mg of ustekinumab per kilogram of body weight, or placebo.	CG: placebo (not specified)	In UNITI-2, 68.6% of participants had not received TNF antagonists (data not shown). NR	Medications for Crohn's disease taken at baseline — no. (%) One or more medications: CG: 185 (75.2%) IG1: 161 (77.0%) IG2: 170 (81.3%) Immunosuppressant: CG: 73 (34.8%) IG1: 74 (35.4%) IG2: 72 (34.4%) Aminosalicylate CG: 89 (42.4%) IG1: 89 (42.6%)	NR	NR	Stable doses of immunosuppressants, mesalamine, antibiotics, or oral glucocorticoids (≤ 40 mg of prednisone per day or ≤ 9 mg of budesonide per day) or a combination thereof were permitted.

Table 2. Induction studies: intervention details (Continued)

		(The administration of 6 mg of ustekinumab per kilogram meant that Participants weighing ≤ 55 kg received 260 mg, those weighing > 55 kg and ≤ 85 kg received 390 mg, and those weighing > 85 kg received 520 mg)		IG2: 93 (44.5%)			
				Glucocorticoid			
				CG: 75 (35.7%)			
				IG1: 80 (38.3%)			
				IG2: 92 (44.0%)			
Feagan 2017	IG-R200: risankizumab 200 mg intravenous (clear solution) at weeks 0, 4, and 8. IG-R600: risankizumab 600 mg intravenous (clear solution) at weeks 0, 4, and 8. IG-RP: this is the combination of IG-R200 and IG-R600 (risankizumab groups). This is not a separate study arm by itself.	Placebo (clear solution) intravenous at weeks 0, 4 and 8 (Agent and Dose NOT REPORTED)	Previous TNF antagonist use (number of participants/total participants in given group (percentage)): (i) 1: CG: 12/39 (31%) IG-R200: 9/41 (22%) IG-R600: 9/41 (22%) IG-RP: 18/82 (22%) (ii) 2: CG: 20/39 (51%) IG-R200: 23/41 (56%)	Corticosteroids only (number of participants/total participants in given group (percentage)): CG: 6/39 (15%) IG-R200: 7/41 (17%) IG-R600: 9/41 (22%) IG-RP: 16/82 (20%) Corticosteroids AND IM (number of participants/total participants in given group (percentage)): CG: 5/39 (13%) IG-R200: 3/41 (7%) IG-R600: 3/41 (7%) IG-RP: 6/82 (7%) IM only (number of participants/total participants in	NOT REPORTED	NOT REPORTED – No protocol	Note: These are the same as medications up to study beginning Corticosteroids only (number of participants/total participants in given group (percentage)): CG: 6/39 (15%) IG-R200: 7/41 (17%) IG-R600: 9/41 (22%) IG-RP: 16/82 (20%) Corticosteroids AND IM (number of participants/total participants in given group (percentage)): CG: 5/39 (13%) IG-R200: 3/41 (7%) IG-R600: 3/41 (7%) IG-RP: 6/82 (7%)

Table 2. Induction studies: intervention details *(Continued)*

IG-R600: 24/41
(59%) **IG-RP:**
47/82 (57%).

(iii) ≥ 3 :

CG: 5/39 (13%)

IG-R200: 7/41
(17%)

IG-R600: 4/41
(10%)

IG-RP: 11/82
(13%)

(iv) Missing:

CG: 2/39 (5%)

IG-R200: 2/41
(5%) **IG-R600:**
4/41 (10%)

IG-RP: 6/82 (7%)

**Worst outcome
of ≥ 1 previous
TNF antagonists
(number of par-
ticipants/to-
tal participants
in given group
(percentage)):**

**(i) Inadequate
response:**

CG: 10/39 (26%)

IG-R200: 16/41
(39%)

IG-R600: 11/41
(27%)

IG-RP: 27/82
(33%)

given group (percentage)):

CG: 8/39 (21%)

IG-R200: 7/41 (17%)

IG-R600: 5/41 (12%)

IG-RP: 12/82 (15%)

**IM only (number of partici-
pants/total participants in
given group (percentage)):**

CG: 8/39 (21%)

IG-R200: 7/41 (17%)

IG-R600: 5/41 (12%)

IG-RP: 12/82 (15%)

Table 2. Induction studies: intervention details *(Continued)*

(ii) Loss of response: CG:
21/39 (54%)

IG-R200: 13/41
(32%)

IG-R600: 17/41
(41%)

IG-RP: 30/82
(37%)

(iii) Unacceptable adverse events:

CG: 1/39 (3%)

IG-R200: 2/41
(5%) **IG-R600:**
5/41 (12%)

IG-RP: 7/82 (9%)

(iv) Other:

CG: 2/39 (5%)

IG-R200: 3/41
(7%)

IG-R600: 1/41
(2%)

IG-RP: 4/82 (5%)

(v) Unknown:

CG: 1/39 (3%)

IG-R200: 3/41
(7%)

IG-R600: 2/41
(5%)

IG-RP: 5/82 (6%)

Table 2. Induction studies: intervention details (Continued)

Data on non-TNF
antagonist bio-
logics is NOT RE-
PORTED

Ghosh 2003	IG1: one infusion of natalizumab dose 3 mg/kg (route NOT REPORTED) followed by one infusion of placebo (agent NOT REPORTED, route NOT REPORTED, dose NOT REPORTED) 4 weeks later. IG2: 2 infusions of natalizumab 3 mg/kg 4 weeks apart (route NOT REPORTED). IG3: two infusions of natalizumab 6 mg/kg 4 weeks apart (route NOT REPORTED)	CG: placebo. Two infusions 4 weeks apart. Agent NOT REPORTED. Route NOT REPORTED. Dose NOT REPORTED	Excluded participants with previous biologic treatment: "criteria for exclusion included prior treatment with any antibody agent"	At baseline, participants were receiving (listed as concomitant medications in paper but not made clear whether they were started at baseline or if participants were already on them before study): None for Crohn's disease*: CG: 12 (19%) IG1: 10 (15%) IG2: 9 (14%) IG3: 5 (10%) 5-aminosalicylate compounds: CG: 30 (48%) IG1: 41 (60%) IG2: 41 (62%) IG3: 30 (59%) Oral corticosteroids: CG: 31 (49%) IG1: 31 (46%) IG2: 37 (56%) IG3: 32 (63%) Azathioprine or mercaptopurine with or without corticosteroids:	NOT REPORTED but paper states: "Patients who had received methotrexate, cyclosporine, or any investigational agents within three months before randomization were excluded; patients who were receiving azathioprine or mercaptopurine were required to have been taking a stable dose for at least four months before randomization. Other criteria for exclusion included prior treatment with any antibody agent, current use of oral prednisolone at a dose of more than 25 mg per day or another corticosteroid at an equivalent dose".	NOT REPORTED/No Protocol	None for Crohn's disease*: CG: 12 (19%) IG1: 10 (15%) IG2: 9 (14%) IG3: 5 (10%) 5-aminosalicylate compounds: CG: 30 (48%) IG1: 41 (60%) IG2: 41 (62%) IG3: 30 (59%) Oral corticosteroids: CG: 31 (49%) IG1: 31 (46%) IG2: 37 (56%) IG3: 32 (63%) Azathioprine or mercaptopurine with or without corticosteroids: CG: 22 (35%) IG1: 26 (38%) IG2: 17 (26%) IG3: 9 (18%) *The participants were not receiving 5-aminosalicy-
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Table 2. Induction studies: intervention details (Continued)

				CG: 22 (35%) IG1: 26 (38%) IG2: 17 (26%) IG3: 9 (18%) *The participants were not receiving 5-aminosalicylate compounds, corticosteroids or immunosuppressants.			late compounds, corticosteroids, or immunosuppressants
Gordon 2001	Natalizumab	Placebo	Previous exposure to infliximab IG: 5 (7%) CG: 7 (10%)	NR	NR	NR	NR
Hanauer 2002 (ACCENT I) - Induction (non-response)	At week 0, all eligible participants received a 5 mg/kg intravenous infusion of infliximab. IG1: 5 mg/kg infliximab infusions at week 2, 6 and every 8 weeks thereafter until week 46 IG2: 10 mg/kg infliximab infusions at week 2, 6 and every 8 weeks thereafter until week 46	At week 0, all eligible participants received a 5 mg/kg intravenous infusion of infliximab. CG: placebo infusions at week 2, 6 and every 8 weeks thereafter until week 46	"Patients were excluded from the study if they had received previous treatment with infliximab or any other agent targeted at TNF." THEREFORE, IG – N, CG – N	Participants receiving the following treatments were eligible: 5-aminosalicylates or antibiotics (if the dose remained constant for 4 weeks before the screening visit); corticosteroids (prednisone, prednisolone or budesonide) at the equivalent of 40 mg per day of prednisone or less (stable dose for 3 weeks); azathioprine and 6-mercaptopurine (stable dose for 8 weeks); or methotrexate (stable dose for 6 weeks) NOT REPORTED per IG/CG so the following data are based on non-responders as a whole 5-aminosalicylates: 129/238 (54%)	Participants not receiving medical therapy had to have discontinued treatment for at least 4 weeks before screening.	NOT REPORTED – No protocol	Note* - SAME AS medications up to study beginning Participants receiving the following treatments were eligible: 5-aminosalicylates or antibiotics (if the dose remained constant for 4 weeks before the screening visit); corticosteroids (prednisone, prednisolone, or budesonide) at the equivalent of 40 mg per day of prednisone or less (stable dose for 3 weeks); azathioprine and 6-mercaptopurine (stable dose for 8 weeks); or methotrexate (stable dose for 6 weeks) NOT REPORTED per IG/CG so the following data is based on non-responders as a whole.

Table 2. Induction studies: intervention details (Continued)

				6-mercaptopurine and azathioprine: 63/238 (27%) Methotrexate: 13/238 (6%) Corticosteroids of which (i) Any – 118/238 (50%) (ii) > 20 mg per day – 32/238 (13%)			5-aminosalicylates: 129/238 (54%) 6-mercaptopurine and azathioprine: 63/238 (27%) Methotrexate: 13/238 (6%) Corticosteroids of which: (i) Any: 118/238 (50%) (ii) > 20 mg per day: 32/238 (13%)
Hanauer 2006 (CLASSIC I)	IG1: adali-mumab 40 mg/20 mg week 0 and week 2 subcutaneous IG2: adali-mumab 80 mg/40 mg week 0 and week 2 subcutaneous IG3: adali-mumab 160 mg/80 mg subcutaneous week 0 and week 2	CG: placebo weeks 0 and 2	Participants are naive	Corticosteroid use: CG: 25 IG1: 17 IG2: 32 IG3: 24 Immunosuppressive agents: CG: 22 IG1: 23 IG2: 21 IG3: 22 Crohn's disease-related antibiotics: CG: 5 IG1: 10 IG2: 7 IG3: 4 5-aminosalicylates: CG: 37 IG1: 37	Not reported	Not reported	Corticosteroid use: CG: 25 IG1: 17 IG2: 32 IG3: 24 Immunosuppressive agents: CG: 22 IG1: 23 IG2: 21 IG3: 22 Crohn's disease-related antibiotics: CG: 5 IG1: 10 IG2: 7 IG3: 4 5-aminosalicylates: CG: 37 IG1: 37

Table 2. Induction studies: intervention details (Continued)

				IG2: 40			IG2: 40
				IG3: 39			IG3: 39
Hanauer 2021 (VOLTAIRE- CD)	BI 695501	Adalimumab	Previous expo- sure to infliximab IG: 5 (7%) CG: 7 (10%)	NR	NR	NR	NR
Lemann 2006	According to the randomi- sation, par- ticipants re- ceived at 0, 2 and 6 weeks either a 2- hour infusion of infliximab (Remicade) administered at a dose of 5 mg/kg or an infusion of an identical placebo	Placebo; ex- act name not mentioned. But given at same dosage and interval as infliximab.	31 participants on infliximab and 27 participants on placebo were naïve	Steroids. At baseline, all participants were treated with prednisone or pred- nisolone at a daily dose of 10 mg or more.	Concomitant treatment that has been shown to facilitate steroid withdraw- al in steroid-de- pendent par- ticipants such as mesalamine, budesonide, artificial nutri- tion, or other im- munosuppressive drugs were not allowed during the study.	All partici- pants were treated with AZA (2 to 3 mg/kg per day) or 6-MP (1 to 1.5 mg/ kg per day). Participants previously treated with AZA or 6-MP (failure stra- tum) con- tinued their treatment at the same dose; in the naïve stra- tum partic- ipants, AZA (2 to 2.5 mg/ kg per day) was started 1 week after the first infliximab infusion (to differentiate side effects related to in- fliximab from those related to AZA).	Not reported separately for IG/CG

Table 2. Induction studies: intervention details (Continued)

Present 1999	IG1: inflix-imab 5 mg/kg (at weeks 0, 2, 6) IG2: inflix-imab 10 mg/kg (at weeks 0, 2, 6)	CG: placebo	CG: Corticosteroids 11 (35%) Mercaptopurine or azathioprine 9 (29%) Aminosalicylates 19 (61%) Antibiotics 11 (35%) IG1: Corticosteroids 12 (39%) Mercaptopurine or azathioprine 12 (3%) Aminosalicylates 17 (55%) Antibiotics 6 (19%) IG2: Corticosteroids 10 (31%) Mercaptopurine or azathioprine 17 (53%) Aminosalicylates 16 (50%) Antibiotics 11 (34%)	Cyclosporin, steroids, methotrexate, azathioprine, mercaptopurine and antibiotics. If not already (and stable) on these medications, they had to have discontinued therapy at least 4 weeks before enrolment.	Not reported	Not reported	CG: Corticosteroids 11 (35%) Mercaptopurine or azathioprine 9 (29%) Aminosalicylates 19 (61%) Antibiotics 11 (35%) IG1: Corticosteroids 12 (39%) Mercaptopurine or azathioprine 12 (3%) Aminosalicylates 17 (55%) Antibiotics 6 (19%) IG2: Corticosteroids 10 (31%) Mercaptopurine or azathioprine 17 (53%) Aminosalicylates 16 (50%) Antibiotics 11 (34%)
Panes 2017 Induction	IG-T5: tofacitinib 5 mg twice daily for 8 weeks (ROUTE NOT REPORTED but likely to be ORAL given it is twice daily over a	CG: placebo twice daily for 8 weeks (ROUTE, DOSE, and AGENT NOT REPORTED but ROUTE is likely to be ORAL given it	Prior use of TN-Fi – Y: CG: 70/91 (76.9%) IG-T5: 68/86 (79.1%) IG-T10: 66/86 (76.7%)	Concomitant corticosteroids were allowed up to oral prednisone-equivalent 30 mg/day or budesonide 9 mg/day, provided a stable dose had been taken for at least 2 weeks prior to baseline. Concomitant oral 5-ASA or sulfasalazine (stable dose for at least 4 weeks	Participants receiving azathioprine, 6-mercaptopurine or methotrexate were required to stop these agents at least two weeks before randomisation.	NR – No protocol found	Note - Same as medications up to study beginning Concomitant corticosteroids were allowed up to oral prednisone-equivalent 30 mg/day or budesonide 9 mg/day, provided a stable dose had been taken for at least 2 weeks prior to base-

Table 2. Induction studies: intervention details (Continued)

period of 8 weeks). IG-T10: tofacitinib 10 mg twice daily for 8 weeks (ROUTE NOT REPORTED but likely to be ORAL given it is twice daily over a period of 8 weeks). IG-T15: tofacitinib 15 mg twice daily for 8 weeks (ROUTE NOT REPORTED but likely to be ORAL given it is twice daily over a period of 8 weeks)	is twice daily over a period of 8 weeks)	IG-T15: 11/16 (68.8%) Previous exposure to biologics as a whole (which includes JAK inhibitors) is NOT REPORTED
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prior to baseline), chronic treatment for CD with antibiotics (e.g. metronidazole, rifaximin; stable dose for at least 2 weeks prior to baseline), rectally administered formulation of corticosteroids or 5-ASA (stable dose for at least 2 weeks prior to baseline) was also permitted, but only data on corticosteroids are provided.

Use of corticosteroids at study entry:

CG: 35/91 (38.5%)

IG-T5: 32/86 (37.2%)

IG-T10: 28/86 (32.6%)

IG-T15: 4/16 (25.0%)

Concomitant tumour necrosis factor inhibitor (TNFi) was not allowed, and a wash-out period of at least eight weeks prior to randomisation was required.

Cyclosporine, mycophenolate or tacrolimus were required to be stopped at least 4 weeks prior to baseline.

Intravenous corticosteroids were required to be stopped at least 2 weeks prior to baseline.

Other investigational or marketed biologics with immunomodulatory properties were required to be stopped at least 3 months prior to baseline.

Treatment with natalizumab or any anti-adhesion molecule was to be stopped at least 1 year prior to baseline.

Moderate to potent CYP3A induc-

line. Concomitant oral 5-ASA or sulfasalazine (stable dose for at least 4 weeks prior to baseline), chronic treatment for CD with antibiotics (e.g. metronidazole, rifaximin; stable dose for at least 2 weeks prior to baseline), rectally administered formulation of corticosteroids or 5-ASA (stable dose for at least 2 weeks prior to baseline) was also permitted, but only data on corticosteroids is provided.

Use of corticosteroids at study entry:

CG: 35/91 (38.5%)

IG-T5: 32/86 (37.2%)

IG-T10: 28/86 (32.6%)

IG-T15: 4/16 (25.0%)

Table 2. Induction studies: intervention details (Continued)

					ers were prohibited within 28 days or 5 half-lives, whichever was longer, prior to first dose of study drug.		
					Moderate to potent CYP3A inhibitors were prohibited within 7 days or 5 half-lives, whichever was longer, prior to first dose of study drug.		
Sandborn 2001a	IG1: 2-hour IV infusion CDP571 (humicade) 10 mg/kg IG2: 2-hour IV infusion CDP571 20 mg/kg	CG: 2-hour IV infusion placebo: 5% (wt/vol) dextrose	Participants who had previously received anti-TNF therapies were ineligible	Participants receiving prednisone or budesonide for a least 4 weeks with a stable prednisone dose of 10 to 30 mg/day or a budesonide dose of 9 mg for at least 2 weeks were eligible. Participants receiving stable doses of azathioprine (at least 1.5 mg/kg/day) or 6-mercaptopurine (at least 0.75 mg/kg/day) for at least 4 months, or methotrexate (at least 15 mg/week) for at least 3 months were eligible.	NR	NR	Corticosteroids and/or immunosuppressive: CG (8 weekly): 18 CG (12 weekly): 20 IG1 (8 weekly): 20 IG1 (12 weekly): 17 IG2 (8 weekly): 19 IG2 (12 weekly): 20 Corticosteroids: CG (8 weekly): 9 CG (12 weekly): 18 IG1 (8 weekly): 18 IG1 (12 weekly): 13 IG2 (8 weekly): 14 IG2 (12 weekly): 17 Azathioprine or 6-mercaptopurine:

Table 2. Induction studies: intervention details (Continued)

Sandborn 2001b	IG: etanercept 25 mg subcutaneous TWICE WEEKLY for 8 weeks, injected by participants. Etaner-	CG: placebo subcutaneous TWICE WEEKLY for 8 weeks, injected by participants. Preparation	Prior treatment with infliximab within the last year: CG: 7/20 (35.00%) IG: 9/23 (39.13%)	Participants receiving prednisone (no upper limit on prednisone dose) or budesonide for at least 4 weeks with a stable dose for at least 2 weeks were eligible. Participants receiving stable doses of azathioprine or 6-	Participants who had received anti-TNF therapy with infliximab within 12 weeks were ineligible. No participant received other	NOT REPORTED – No protocol found	Participants receiving prednisone (no upper limit on prednisone dose) or budesonide for at least 4 weeks with a stable dose for at least 2 weeks were eligible. Participants receiving stable doses of azathioprine or 6-
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CG (8 weekly): 10
CG (12 weekly): 10
IG1 (8 weekly): 5
IG1 (12 weekly): 8
IG2 (8 weekly): 7
IG2 (12 weekly): 7

Methotrexate:

CG (8 weekly): 2
CG (12 weekly): 3
IG1 (8 weekly): 0
IG1 (12 weekly): 1
IG2 (8 weekly): 1
IG2 (12 weekly): 2

5-aminosalicylates (mesalamine, sulfasalazine, olsalazine):

CG (8 weekly): 17
CG (12 weekly): 16
IG1 (8 weekly): 17
IG1 (12 weekly): 13
IG2 (8 weekly): 17
IG2 (12 weekly): 12

Table 2. Induction studies: intervention details (Continued)

cept was supplied in single-dose vials containing a lyophilised powder consisting of 25 mg etanercept, 40 mg of mannitol, 10 mg of sucrose, and 1.2 mg of tromethamine. The study drug was reconstituted with 1.0 mL of bacteriostatic water and dispensed in pre-filled syringes by an unblinded study pharmacist at each centre. Etanercept preparation had a clear and colourless appearance similar to the placebo preparationN.	was supplied in single-dose vials containing a lyophilised powder consisting of 40 mg mannitol, 10 mg sucrose and 1.2 mg of tromethamine. The study drug was reconstituted with 1.0 mL of bacteriostatic water and dispensed in pre-filled syringes by an unblinded study pharmacist at each centre. Placebo preparation had a clear and colourless appearance similar to the etanercept preparation.	Prior treatment with infliximab at any time: CG: 8/20 (40.00%) IG: 12/23 (52.17%) Previous exposure to other biologics or biologics as a whole is NOT REPORTED	mercaptopurine for at least 12 weeks, or methotrexate or mycophenolate mofetil for at least 8 weeks were eligible. Continued treatment with oral or rectal 5-aminosalicylates, rectal corticosteroids, and oral antibiotics was allowed, provided the dose had been stable for at least 2 weeks. Corticosteroids: CG: 7/20 IG: 10/23 Azathioprine or 6-mercaptopurine: CG: 7/20 IG: 10/23 Methotrexate: CG: 1/20 IG: 1/20 Antibiotics: CG: 5/20 IG: 4/23 5-aminosalicylates*: CG: 9/20 IG: 11/23 *5-aminosalicylates indicates: mesalamine, sulfasalazine and olsalazine	investigational therapies within 28 days preceding randomisation. Participants who had received tacrolimus, cyclosporine, thalidomide, interleukin 10, or interleukin 11 within 4 weeks were not eligible. Nutritional therapy (parenteral nutrition or enteral nutrition with elemental or semi-elemental diet) was discontinued at the screening visit and was not permitted during the trial, with the exception of nocturnal tube feedings for adolescent participants with growth failure and enteral nutrition supplements for adult participants with malnutrition who were receiving a stable regimen for at least 2 weeks and continued that same regimen throughout the 8-week study.	mercaptopurine for at least 12 weeks, or methotrexate or mycophenolate mofetil for at least 8 weeks were eligible. Continued treatment with oral or rectal 5-aminosalicylates, rectal corticosteroids, and oral antibiotics was allowed, provided the dose had been stable for at least 2 weeks Corticosteroids: CG: 7/20 IG: 10/23 Azathioprine or 6-mercaptopurine: CG: 7/20 IG: 10/23 Methotrexate: CG: 1/20 IG: 1/20 Antibiotics: CG: 5/20 IG: 4/23 5-aminosalicylates*: CG: 9/20 IG: 11/23 *5-aminosalicylates indicates: mesalamine, sulfasalazine, and olsalazine
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Table 2. Induction studies: intervention details (Continued)

Sandborn 2005a (ENACT 1)	Natalizumab, 300 mg at week 0, 4 and 8, intravenous	Placebo (exact agent and dosage NOT REPORTED), dose at week 0, 4 and 8, intravenous	IG: Y 291 (40%), refractory to such therapy 196 (27%) CG: Y 69 (38%), refractory to such therapy 46 (25%)	NOT REPORTED for either IG or CG. May be possible that these are the same as concomitant medications (i.e. participants may already be on them before study), but it is not explicitly stated. Need to contact authors to confirm.	NOT REPORTED	No protocol	For both IG and CG, concurrent therapies, including stable doses of 5-aminosalicylates, prednisolone (25 mg per day or less) or equivalent, budesonide (6 mg per day or less), azathioprine, mercaptopurine, methotrexate, and antibiotics, were permitted. Corticosteroids: CG: 70 (39%) IG: 271 (37%) Prednisone CG: 53 (29%) IG: 197 (27%). Budesonide CG: 20 (11%) IG: 79 (11%) Immunosuppressants: CG: 52 (29%) IG: 247 (34%) Azathioprine CG: 38 (21%) IG: 167 (23%) Mercaptopurine: CG: 8 (4%) IG: 49 (7%) Methotrexate:
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Table 2. Induction studies: intervention details (Continued)

Sandborn 2007a	Subcutaneous adalimumab 160 mg at week 0, then subcutaneous adalimumab 80 mg at week 2	Placebo at week 0 then placebo at week 2. Dose, route and agent NOT REPORTED	The paper states "To be included, patients must have been intolerant of infliximab or must have previously responded to infliximab and then lost response." Therefore, number of participants with pre-	NOT REPORTED EXPLICITLY. Paper states: "Concurrent therapies, including stable dosages of 5-aminosalicylates, prednisone (≤ 40 mg/d), budesonide (≤ 9 mg/d), azathioprine, 6-mercaptopurine, methotrexate, and antibiotics, were permitted" "Dosages of all these medications were required to remain stable during the study". Not stated whether these participants were	Paper states: "We excluded patients who had a primary nonresponse to infliximab as defined by the investigator, received infliximab or another TNF antagonist within the past 8 weeks, previously received adali-	No protocol NOT REPORTED	Steroids*: CG: 73 (44%) IG: 55 (35%) Any immunosuppressive agent CG: 85 (51%) IG: 73 (46%) 5-aminosalicylates** CG: 60 (36%)
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CG: 6 (3%)

IG: 31 (4%)

Antibiotics (metronidazole, tinidazole, ciprofloxacin, and clarithromycin):

CG: 12 (7%)

IG: 43 (6%)

5-aminosalicylates (mesalamine, sulfasalazine, and olsalazine):

CG: 80 (44%)

IG: 343 (47%)

≥ 1 corticosteroids or immunosuppressants:

CG: 100 (55%)

IG: 407 (56%)

Corticosteroids and immunosuppressants:

CG: 22 (12%)

IG: 111 (15%)

Table 2. Induction studies: intervention details (Continued)

vious biologic experience:

CG: 166 (100%)

IG: 159 (100%)

Previous loss of response to infliximab:

CG: 87 (52%)

IG: 77 (48%)

Previous intolerance to infliximab:

CG: 95 (57%)

IG: 95 (60%). Of which **(i) Acute reaction CG:**

63 (38%), **IG:** 68 (43%). **(ii) Delayed reaction**

CG: 52 (31%), **IG:** 43 (27%). **(iii) Acute AND Delayed reaction**

CG: 20 (12%). **IG:** 15 (9%).

Previous loss of response to infliximab AND previous intolerance to infliximab:

CG: 21 (13%)

IG: 19 (12%)

Antibodies to infliximab: (i) Absent CG: 91 (57%)

already on these medications prior to study or were initiated at baseline during study. It is possible that the numbers of participants on each of the medications here is the same as for concomitant medications.

mumab (Humira, Abbott Laboratories, Abbott Park, Illinois), or participated in an adalimumab clinical trial' 'We excluded Participants who changed dosages or discontinued azathioprine, 6-mercaptopurine, or methotrexate treatment within 12 weeks of screening. Similarly, we excluded patients who changed dosages or discontinued 5-aminosalicylates, mesalamine, or sulfasalazine treatment within 4 weeks of screening. Prednisone (≤ 40 mg/d) and budesonide (≤ 9 mg/d) dosages must have been stable for 2 weeks or more before screening. Dosages of all these medications were required to remain stable during the study.' 'We al-

IG: 45 (28%)

*Budesonide, betamethasone, cortisone, cloprednol, corticosteroids, dexamethasone, deflazacort, flucortolone, glucocorticoids, glucocorticosteroids, hydrocortisone, methylprednisolone, paramethasone, prednisolone, prednisone, or, prednylidene.

**Balsalazide, mesalamine, olsalazine, or sulfasalazine.

Table 2. Induction studies: intervention details *(Continued)*

IG: 88 (59%). (ii) Indeterminate* CG: 8 (6%)** IG: 12 (8%). (iii) Present CG: 60 (38%)** IG: 50 (33%) *Results are indeterminate because of presence of infliximab in 8 placebo group participants and 12 adalimumab group participants. **One participant in the placebo group had measurable infliximab concentration and antibody to infliximab							
so excluded patients who had received antibiotic treatment for infections not related to Crohn disease within 3 weeks of the study 'We also excluded patients with a history of clinically significant drug or alcohol abuse in the past year'.							
Sandborn 2007c (PRECISE 1)	Injections of placebo Subcutaneous injections of certolizumab pegol at a dose of 400 mg	Subcutaneous injections of placebo (agent: NR)	Participants who had received any anti-TNF agent within the previous 3 months or who had had a severe hypersensitivity reaction or a lack of response to the first dose of another TNF antagonist were excluded	Glucocorticoids only: CG: 75 (23%) IG: 72 (22%) Immunosuppressive agents only: CG: 66 (20%) IG: 69 (21%) Glucocorticoids combined with immunosuppressive agents: CG: 55 (17%) IG: 57 (17%) Neither glucocorticoids nor immunosuppressive agents:	NR	NR	Participants could receive concomitant therapy with stable doses of 5-aminosalicylates, prednisolone or its equivalent (at a dose of 30 mg per day or less), azathioprine, 6-mercaptopurine, methotrexate, or antibiotics.

Table 2. Induction studies: intervention details (Continued)

				CG: 132 (40%)			
				IG: 133 (40%)			
Sandborn 2008 (Population 1)	IG SC-UST: ustekinumab 90 mg subcutaneous at weeks 0, 1, 2 and 3. IG IV-PBO: placebo (agent and dose NOT REPORTED). Intravenous placebo at week 0. IG IV-UST: ustekinumab 4.5 mg/kg intravenous at week 0. Note: Cross-over to alternative therapy occurred at week 8, from which point we are not considering the data in our NMA	CG SC-PBO: placebo (agent and dose NOT REPORTED). Subcutaneous placebo at weeks 0, 1, 2 and 3. Note: cross-over to ustekinumab 90 mg subcutaneous occurred at week 8, from which point we are not considering the data in our NMA	CG SC-PBO: Y 16/26 (62%) IG SC-UST: Y 14/25 (56%) IG IV-PBO: Y 11/27 (41%). IG IV-UST: Y 8/26 (31%) Note: These participants received infliximab more than 16 weeks before randomisation (see medications that had to be stopped prior to study)	Concomitant use of 5-aminosalicylates, antibiotics, prednisolone at a maximum daily dose of 20 mg, azathioprine, 6-mercaptopurine, or methotrexate was permitted. Concomitant medication doses remained constant, except corticosteroids, which could be tapered by 2.5 mg/week after week 8. The following are reported in baseline characteristics (Table 1 for the whole population and Table 2 for participants with previous infliximab exposure). A. Whole population Concomitant medications: CG SC-PBO: 21/26 (81%) IG SC-UST: 19/25 (76%) IG IV-PBO: 19/27 (70%) IG IV-UST: 20/26 (77%) Oral Corticosteroids: CG SC-PBO: 8/26 (31%) IG SC-UST: 10/25 (40%) IG IV-PBO: 8/27 (30%) IG IV-UST: 7/26 (27%) Of which: (i) ≤ 20 mg/day:	Participants were excluded from the study if recent treatment with any investigational agent or an anti-TNF agent, including infliximab, within the past 16 weeks. Therefore, these agents needed to have been stopped at least 16 weeks prior to the study.	NOT REPORTED. No protocol	Note*: Same as medications up to study beginning Concomitant use of 5-aminosalicylates, antibiotics, prednisolone at a maximum daily dose of 20 mg, azathioprine, 6-mercaptopurine, or methotrexate was permitted. Concomitant medication doses remained constant, except corticosteroids, which could be tapered by 2.5 mg/week after week 8. The following are reported in baseline characteristics (Table 1 for the whole population and table 2 for participants with previous infliximab exposure). A. Whole population Concomitant medications: CG SC-PBO: 21/26 (81%) IG SC-UST: 19/25 (76%) IG IV-PBO: 19/27 (70%) IG IV-UST: 20/26 (77%) Oral corticosteroids: CG SC-PBO: 8/26 (31%) IG SC-UST: 10/25 (40%) IG IV-PBO: 8/27 (30%) IG IV-UST: 7/26 (27%) Of which:

Table 2. Induction studies: intervention details *(Continued)*

CG SC-PBO: 7/26 (27%)

IG SC-UST: 9/25 (36%)

IG IV-PBO: 6/27 (22%)

IG IV-UST: 5/26 (19%)

(ii) > 20 mg/day: CG SC-PBO: 1/26 (4%)

IG SC-UST: 1/25 (4%)

IG IV-PBO: 2/27 (7%)

IG IV-UST: 2/26 (8%)

Immunosuppressants:

CG SC-PBO: 10/26 (39%)

IG SC-UST: 5/25 (20%)

IG IV-PBO: 10/27 (37%)

IG IV-UST: 10/26 (39%)

Of which:

(i) 6-mercaptopurine, azathioprine:

CG SC-PBO: 7/26 (27%)

IG SC-UST: 4/25 (16%)

IG IV-PBO: 8/27 (30%)

IG IV-UST: 9/26 (35%)

(ii) Methotrexate:

CG SC-PBO: 3/26 (12%)

IG SC-UST: 1/25 (4%)

IG IV-PBO: 2/27 (7%)

IG IV-UST: 1/26 (4%)

(i) ≤ 20 mg/day:

CG SC-PBO: 7/26 (27%)

IG SC-UST: 9/25 (36%)

IG IV-PBO: 6/27 (22%)

IG IV-UST: 5/26 (19%) **(ii) > 20 mg/day:**

CG SC-PBO: 1/26 (4%)

IG SC-UST: 1/25 (4%)

IG IV-PBO: 2/27 (7%)

IG IV-UST: 2/26 (8%)

Immunosuppressants: CG SC-PBO: 10/26 (39%)

IG SC-UST: 5/25 (20%)

IG IV-PBO: 10/27 (37%)

IG IV-UST: 10/26 (39%)

Of which:

(i) 6-mercaptopurine, azathioprine:

CG SC-PBO 7/26 (27%)

IG SC-UST 4/25 (16%)

IG IV-PBO 8/27 (30%)

IG IV-UST 9/26 (35%)

(ii) Methotrexate:

CG SC-PBO: 3/26 (12%)

IG SC-UST: 1/25 (4%)

IG IV-PBO: 2/27 (7%)

IG IV-UST: 1/26 (4%)

Table 2. Induction studies: intervention details *(Continued)*

Oral corticosteroids OR immunosuppressants:

CG SC-PBO: 14/26 (54%)

IG SC-UST: 13/ 25 (52%)

IG IV-PBO: 14/27 (52%)

IG IV-UST: 15/26 (58%)

Oral corticosteroids AND immunosuppressants:

CG SC-PBO: 4/26 (15%)

IG SC-UST: 2/25 (8%)

IG IV-PBO: 4/27 (15%)

IG IV-UST: 2/26 (8%)

Antibiotics:

CG SC-PBO: 2/26 (8%)

IG SC-UST: 2/25 (8%)

IG IV-PBO: 2/27 (7%)

IG IV-UST: 1/26 (4%)

Aminosalicylates:

CG SC-PBO: 13/26 (50%)

IG SC-UST: 9/25 (36%)

IG IV-PBO: 14/27 (52%)

IG IV-UST: 10/26 (39%)

B. Participants with previous infliximab use

Concomitant medications:

CG SC-PBO: 12/16 (75%)

IG SC-UST: 10/14 (71%)

Oral corticosteroids OR immunosuppressants: CG SC-PBO: 14/26 (54%)

IG SC-UST: 13/ 25 (52%)

IG IV-PBO: 14/27 (52%)

IG IV-UST: 15/26 (58%)

Oral corticosteroids AND immunosuppressants:CG SC-PBO: 4/26 (15%)

IG SC-UST: 2/25 (8%)

IG IV-PBO: 4/27 (15%)

IG IV-UST: 2/26 (8%)

Antibiotics:

CG SC-PBO: 2/26 (8%)

IG SC-UST: 2/25 (8%)

IG IV-PBO: 2/27 (7%)

IG IV-UST: 1/26 (4%)

Aminosalicylates:

CG SC-PBO: 13/26 (50%)

IG SC-UST: 9/25 (36%)

IG IV-PBO: 14/27 (52%)

IG IV-UST: 10/26 (39%)

B. Participants with previous infliximab use

concomitant medications:

CG SC-PBO: 12/16 (75%)

IG SC-UST: 10/14 (71%)

Table 2. Induction studies: intervention details (Continued)

IG IV-PBO: 7/11 (64%)

IG IV-UST: 6/8 (75%)

Oral Corticosteroids:

CG SC-PBO: 5/16 (31%)

IG SC-UST: 6/14 (43%)

IG IV-PBO: 3/11 (27%)

IG IV-UST: 3/8 (38%)

Of which:

(i) ≤ 20 mg/day: CG SC-PBO: 4/16 (25%)

IG SC-UST: 5/14 (36%)

IG IV-PBO: 3/11 (27%)

IG IV-UST: 1/8 (13%)

(ii) > 20 mg/day: CG SC-PBO: 1/16 (6%).

IG SC-UST: 1/14 (7%)

IG IV-PBO: 0/11 (0%)

IG IV-UST: 2/8 (25%)

Immunosuppressants:

CG SC-PBO: 8/16 (50%)

IG SC-UST: 3/14 (21%)

IG IV-PBO: 5/11 (46%)

IG IV-UST: 4/8 (50%).

Of which:

(i) 6-mercaptopurine, azathioprine:

CG SC-PBO: 5/16 (31%)

IG IV-PBO: 7/11 (64%)

IG IV-UST: 6/8 (75%)

Oral corticosteroids: CG

SC-PBO: 5/16 (31%)

IG SC-UST: 6/14 (43%)

IG IV-PBO: 3/11 (27%)

IG IV-UST: 3/8 (38%)

Of which:

(i) ≤ 20 mg/day:

CG SC-PBO: 4/16 (25%)

IG SC-UST: 5/14 (36%)

IG IV-PBO: 3/11 (27%)

IG IV-UST: 1/8 (13%)

(ii) > 20 mg/day:

CG SC-PBO: 1/16 (6%)

IG SC-UST: 1/14 (7%)

IG IV-PBO: 0/11 (0%)

IG IV-UST: 2/8 (25%).

Immunosuppressants: CG

SC-PBO: 8/16 (50%)

IG SC-UST: 3/14 (21%)

IG IV-PBO: 5/11 (46%)

IG IV-UST: 4/8 (50%)

Of which **(i) 6-mercaptopurine, azathioprine:**

CG SC-PBO: 5/16 (31%)

IG SC-UST: 2/14 (14%)

Table 2. Induction studies: intervention details *(Continued)*

IG SC-UST: 2/14 (14%)

IG IV-PBO: 3/11 (27%)

IG IV-UST: 4/8 (50%)

(ii) Methotrexate:

CG SC-PBO: 3/16 (19%)

IG SC-UST: 1/14 (7%)

IG IV-PBO: 2/11 (18%)

IG IV-UST: 0/8 (0%)

Oral corticosteroids OR immunosuppressants:

CG SC-PBO: 10/16 (63%)

IG SC-UST: 9/14 (64%)

IG IV-PBO: 6/11 (55%)

IG IV-UST: 5/8 (63%)

Oral corticosteroids AND immunosuppressants:

CG SC-PBO: 3/16 (19%)

IG SC-UST: 0/14 (0%)

IG IV-PBO: 2/11 (18%)

IG IV-UST: 2/8 (25%)

Antibiotics:

CG SC-PBO: 1/16 (6%)

IG SC-UST: 1/14 (7%)

IG IV-PBO: 2/11 (18%)

IG IV-UST: 1/8 (13%)

Aminosalicylates:CG SC-PBO: 6/16 (38%)

IG IV-PBO: 3/11 (27%)

IG IV-UST: 4/8 (50%)

(ii) Methotrexate:

CG SC-PBO: 3/16 (19%)

IG SC-UST: 1/14 (7%)

IG IV-PBO: 2/11 (18%)

IG IV-UST: 0/8 (0%)

Oral corticosteroids OR immunosuppressants: CG SC-PBO: 10/16 (63%)

IG SC-UST: 9/ 14 (64%) **IG IV-PBO:** 6/11 (55%) **IG IV-UST:** 5/8 (63%)

Oral corticosteroids AND immunosuppressants:CG SC-PBO: 3/16 (19%)

IG SC-UST: 0/14 (0%)

IG IV-PBO: 2/11 (18%)

IG IV-UST: 2/8 (25%)

Antibiotics:

CG SC-PBO: 1/16 (6%)

IG SC-UST: 1/14 (7%)

IG IV-PBO: 2/11 (18%)

IG IV-UST: 1/8 (13%)

Aminosalicylates:

CG SC-PBO: 6/16 (38%)

IG SC-UST: 3/14 (21%)

IG IV-PBO: 3/11 (27%)

Table 2. Induction studies: intervention details (Continued)

				IG SC-UST: 3/14 (21%)			IG IV-UST: 2/8 (25%)
				IG IV-PBO: 3/11 (27%)			
				IG IV-UST: 2/8 (25%)			
Sandborn 2011	Certolizumab 400 mg subcutaneous at weeks 0, 2 and 4	Placebo matched control	Biologic naive	Corticosteroid (with or without immunosuppressants) IG 97 CG 98 More than 20% of participants were receiving immunosuppressants	Not reported	Not reported	Corticosteroid (with or without immunosuppressants) IG 97 CG 98 More than 20% of participants were receiving immunosuppressants
Sandborn 2012 (CERTIFI) - Induction	IG-UST1: IV ustekinumab 1 mg/kg at week 0. IG-UST3: IV ustekinumab 3 mg/kg at week 0. IG-UST6: IV ustekinumab 6 mg/kg at week 0	CG-PBO: placebo IV at week 0. Placebo agent and dose NOT REPORTED	Previous experience of biologics*: CG-PBO: 132/132 (100%) IG-UST1: 131/131 (100%) IG-UST3: 132/132 (100%) IG-UST6: 131/131 (100%) Of which (i) 1 drug – CG-PBO: 71/132 (53.8%) IG-UST1: 64/131 (48.9%) IG-UST3: 66/132 (50.0%) IG-UST6: 66/131 (50.4%) (ii) 2 or 3 drugs: CG-PBO: 60/132 (45.5%)	Participants were permitted to continue receiving stable doses of drugs for the treatment of Crohn's disease, including oral prednisolone (≤ 40 mg per day) or budesonide (≤ 9 mg per day), immunomodulators (e.g. azathioprine, mercaptopurine, or methotrexate), mesalamine, and antibiotics, if they had been taking the drugs for at least the prespecified period before study entry of 3 weeks, 4 weeks, 3 weeks, and 3 weeks, respectively. Participants had not received previous therapies targeting interleukin-12 or interleukin-23. Crohn's disease drugs at baseline of which: (i) ≥ 1 medication: CG-PBO: 101/132 (76.5%) IG-UST1: 91/131 (69.5%) IG-UST3: 96/132 (72.7%)	IV glucocorticoids were not permitted at least 3 weeks before study entry. TNF antagonists were not permitted at least 8 weeks before study entry. Natalizumab was not permitted for at least 12 months before study entry.	NR – No protocol	Participants were permitted to continue receiving stable doses of drugs for the treatment of Crohn's disease, including oral prednisolone (≤ 40 mg per day) or budesonide (≤ 9 mg per day), immunomodulators (e.g. azathioprine, mercaptopurine, or methotrexate), mesalamine, and antibiotics, if they had been taking the drugs for at least the prespecified period before study entry of 3 weeks, 4 weeks, 3 weeks, and 3 weeks, respectively. Participants had not received previous therapies targeting interleukin-12 or interleukin-23. Crohn's disease drugs at baseline of which: (i) ≥ 1 medication: CG-PBO: 101/132 (76.5%) IG-UST1: 91/131 (69.5%) IG-UST3: 96/132 (72.7%)

Table 2. Induction studies: intervention details *(Continued)*

IG-UST1: 67/131 (51.1%). IG-UST3: 66/132 (50.0%)	IG-UST6: 92/131 (70.2%)	IG-UST6: 92/131 (70.2%)
	(ii) Immunomodulatory drug*:	(ii) Immunomodulatory drug*:
IG-UST6: 64/131 (48.9%)	CG-PBO: 30/132 (22.7%)	CG-PBO: 30/132 (22.7%)
Failure criteria met**. (i) Primary nonresponse:	IG-UST1: 33/131 (25.2%)	IG-UST1: 33/131 (25.2%)
CG-PBO: 44/132 (33.3%)	IG-UST3: 28/132 (21.2%)	IG-UST3: 28/132 (21.2%)
IG-UST1: 39/131 (29.8%)	IG-UST6: 35/131 (26.7%)	IG-UST6: 35/131 (26.7%)
IG-UST3: 41/132 (31.1%)	(iii) Aminosalicylate drug:	(iii) Aminosalicylate drug:
IG-UST6: 36/131 (27.5%)	CG-PBO: 24/132 (18.2%)	CG-PBO: 24/132 (18.2%)
(ii) Secondary nonresponse:	IG-UST1: 20/131 (15.3%)	IG-UST1: 20/131 (15.3%)
CG-PBO: 91/132 (68.9%)	IG-UST3: 22/132 (16.7%)	IG-UST3: 22/132 (16.7%)
IG-UST1: 96/131 (73.3%)	IG-UST6: 25/131 (19.1%)	IG-UST6: 25/131 (19.1%)
IG-UST3: 98/132 (74.2%)	(iv) Glucocorticoid**:	(iv) Glucocorticoid**:
IG-UST6: 95/131 (72.5%)	CG-PBO: 73/132 (55.3%)	CG-PBO: 73/132 (55.3%)
(iii) Unacceptable side effects:	IG-UST1: 59/131 (45.0%)	IG-UST1: 59/131 (45.0%)
CG-PBO: 41/132 (31.1%)	IG-UST3: 71/132 (53.8%)	IG-UST3: 71/132 (53.8%)
IG-UST1: 48/131 (36.6%)	IG-UST6: 59/131 (45.0%)	IG-UST6: 59/131 (45.0%)
IG-UST3: 40/132 (30.3%)	(ivA) Baseline corticosteroid dose, no., mg/day:	(ivA) Baseline corticosteroid dose, no., mg/day:
	CG-PBO: 60, 20.0	CG-PBO: 60, 20.0
	IG-UST1: 51, 15.0	IG-UST1: 51, 15.0
	IG-UST3: 62, 20.0	IG-UST3: 62, 20.0
	IG-UST6: 50, 20.0	IG-UST6: 50, 20.0
	(ivB) Baseline budesonide dose, no., mg/day:	(ivB) Baseline budesonide dose, no., mg/day:
	CG-PBO: 13, 9.0	CG-PBO: 13, 9.0
	IG-UST1: 9, 9.0	IG-UST1: 9, 9.0

Table 2. Induction studies: intervention details *(Continued)*

IG-UST6: 47/131 (35.9%)

*Although the proportion of participants with previous experience is not stated explicitly, it can be safely assumed to be 100% for all groups as it is a study looking at the efficacy of UST in CD refractory to TNF treatment. It is stated in the main paper: "In this 36-week, randomized, double-blind,

placebo-controlled phase 2b trial of ustekinumab

(comprising 8-week induction and 28-week maintenance

phases), we evaluated patients with moderate-

to-severe Crohn's disease that was resistant to TNF antagonists."

**These failure criteria seem to relate to all drugs

IG-UST3: 9, 9.0

IG-UST6: 10, 9.0 **(v) Antibiotics:**

CG-PBO: 7/132 (5.3%)

IG-UST1: 12/131 (9.2%)

IG-UST3: 11/132 (8.3%)

IG-UST6: 12/131 (9.2%)

Previous treatment failure of any immunomodulatory drug*:

CG-PBO: 113/132 (85.6%)

IG-UST1: 116/131 (88.5%)

IG-UST3: 119/132 (90.2%)

IG-UST6: 119/131 (90.8%)

Of which:

(i) Contraindication to drug developed:

CG-PBO: 101/113 (89.4%)

IG-UST1: 105/116 (90.5%)

IG-UST3: 98/119 (82.4%)

IG-UST6: 97/119 (81.5%)

Failure criteria met***

(i) Primary nonresponse:

CG-PBO: 44/132 (33.3%)

IG-UST1: 39/131 (29.8%)

IG-UST3: 41/132 (31.1%)

IG-UST6: 36/131 (27.5%)

IG-UST3: 9, 9.0

IG-UST6: 10, 9.0

(v) Antibiotics:

CG-PBO: 7/132 (5.3%)

IG-UST1: 12/131 (9.2%)

IG-UST3: 11/132 (8.3%)

IG-UST6: 12/131 (9.2%)

*Listed are all participants who received a full course of an immunomodulatory drug. Immunomodulatory drugs included azathioprine, mercaptopurine, and methotrexate

**Glucocorticoid formulations included budesonide

Table 2. Induction studies: intervention details *(Continued)*

(i.e. both TNF antagonists and also other drugs such as corticosteroids) combined. Separate data for TNF antagonists have not been reported. Participants may have reported more than one reason for treatment failure. Primary non-response refers to absence of an initial response. Secondary non-response refers to an initial response that was not maintained.

(ii) Secondary nonresponse:

CG-PBO: 91/132 (68.9%)

IG-UST1: 96/131 (73.3%)

IG-UST3: 98/132 (74.2%)

IG-UST6: 95/131 (72.5%)

(iii) Unacceptable side effects:

CG-PBO: 41/132 (31.1%)

IG-UST1: 48/131 (36.6%)

IG-UST3: 40/132 (30.3%)

IG-UST6: 47/131 (35.9%)

*Listed are all participants who received a full course of an immunomodulatory drug. Immunomodulatory drugs included azathioprine, mercaptopurine, and methotrexate

**Glucocorticoid formulations included budesonide

***These failure criteria seem to relate to all drugs (i.e. both TNF antagonists and also other drugs such as corticosteroids) combined. Separate data for TNF antagonists have not been reported. Participants may have reported more than one reason for treatment failure. Primary non-response refers to absence of an initial response. Secondary nonresponse refers

Table 2. Induction studies: intervention details (Continued)

Sandborn 2012 (CERTIFI) - Induction (non-response)	Subcutaneous ustekinumab 90 mg at weeks 8 and 16	Subcutaneous placebo at weeks 8 and 16. DOSE and AGENT NOT REPORTED	Previous experience of biologics*: CG: 110/110 (100%) IG: 109/109 (100%) *Although the proportion of participants with previous experience is not stated explicitly, it can be safely assumed to be 100% for all groups as it is a study looking at the efficacy of UST in CD refractory to TNF treatment. It is stated in the main paper: "In this 36-week, randomized, double-blind, placebo-controlled phase 2b trial of ustekinumab (comprising 8-week induction and 28-week maintenance phases), we evaluated Partici-	Participants were permitted to continue receiving stable doses of drugs for the treatment of Crohn's disease, including oral prednisolone (≤ 40 mg per day) or budesonide (≤ 9 mg per day), immunomodulators (e.g. azathioprine, mercaptopurine, or methotrexate), mesalamine, and antibiotics, if they had been taking the drugs for at least the prespecified period before study entry of 3 weeks, 4 weeks, 3 weeks, and 3 weeks, respectively. Participants had not received previous therapies targeting interleukin-12 or interleukin-23. Note: 'Study entry' here implies entry to the original induction phase of CERTIFI (i.e. week 0 to week 8 rather than week 8 to week 36) Proportions or numbers of participants per IG/CG per medication are NOT REPORTED for the week 8 to week 36 phase of CERTIFI (including non-responders induction study)	IV Glucocorticoids were not permitted at least 3 weeks before study entry. TNF antagonists were not permitted at least 8 weeks before study entry. Natalizumab was not permitted for at least 12 months before study entry. Note: 'Study entry' here relates week 0 to week 8 induction phase. So 8 weeks should be added to the times stated.	NR - No protocol	Participants were permitted to continue receiving stable doses of drugs for the treatment of Crohn's disease, including oral prednisolone (≤ 40 mg per day) or budesonide (≤ 9 mg per day), immunomodulators (e.g. azathioprine, mercaptopurine, or methotrexate), mesalamine, and antibiotics, if they had been taking the drugs for at least the prespecified period before study entry of 3 weeks, 4 weeks, 3 weeks, and 3 weeks, respectively. Participants had not received previous therapies targeting interleukin-12 or interleukin-23. Note - 'Study entry' here implies entry to the original induction phase of CERTIFI (i.e. week 0 to week 8 rather than week 8 to week 36) Proportions or numbers of participants per IG/CG per medication is NOT REPORTED for the week 8 to week 36 phase of CERTIFI (including non-responders induction study).
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to an initial response that was not maintained.

Table 2. Induction studies: intervention details (Continued)

			pants with moderate- to-severe Crohn's disease that was resistant to TNF antagonists."				
Sandborn 2013 (GEMINI II) - Induction	Vedolizumab 300 mg intravenously at weeks 0 and 2	Placebo matched control	About 50% equally biologic naive and not naive participants	Glucocorticoids only IG:336 CG: 45 Glucocorticoids and immunosuppressive agents IG 163 Less than 20% of participants were receiving immunosuppressants	Not reported	Not reported	Glucocorticoids only IG:336 CG: 45 Glucocorticoids and immunosuppressive agents IG 163 Less than 20% of participants were receiving immunosuppressants
Sandborn 2014	IGT1: tofacitinib 1 mg oral twice daily. IGT5: tofacitinib 5 mg oral twice daily. IGT15: tofacitinib 15 mg oral twice daily	CG: placebo oral twice daily (DOSE and AGENT NOT REPORTED)	Anti-TNF (alpha) use within the previous 12 months (n (%)): CG: 1/34 (2.9%) IGT1: 4/36 (11.1%) IGT5: 1/34 (2.9%) IGT15: 4/35 (11.4%) Anti-TNF (alpha) use before within 12 months of baseline – NOT REPORTED Use of other biologics – NOT REPORTED	The following at stable doses were permitted but not required: 5-acetylsalicylic acid or sulfasaline (≥ 3 weeks before baseline and during treatment); prednisolone 30 mg or less per day or budesonide 9 mg or less per day (≥ 2 weeks before baseline); chronic treatment with antibiotics and rectally administered corticosteroids or 5-acetylsalicylic acid (≥ 2 weeks before baseline and during treatment) Participants remaining on immunosuppressant at screening (n (%)): CG: 0/34 (0.0%) IGT1: 0/36 (0.0%) IGT5: 0/34 (0.0%) IGT15: 2/35 (5.7%)	Azathioprine, 6-mercaptopurine, methotrexate – within 7 days of baseline prohibited Cyclosporine, mycophenolate, tacrolimus – within 4 weeks prohibited Interferon and anti-TNF (alpha) therapy – within 8 weeks prohibited	NR – No protocol found	The following at stable doses were permitted but not required: 5-acetylsalicylic acid or sulfasaline (≥ 3 weeks before baseline and during treatment); prednisolone 30 mg or less per day, or budesonide 9 mg or less per day (≥ 2 weeks before baseline); Chronic treatment with antibiotics and rectally administered corticosteroids or 5-acetylsalicylic acid (≥ 2 weeks before baseline and during treatment) Participants remaining on immunosuppressant at screening (n (%)): CG: 0/34 (0.0%) IGT1: 0/36 (0.0%) IGT5: 0/34 (0.0%) IGT15: 2/35 (5.7%)

Table 2. Induction studies: intervention details *(Continued)*

				Number of participants on other medications at screening – NOT REPORTED	Number of participants on other medications at screening – NOT REPORTED
				Immunosuppressant use within the previous 12 months (n (%)):	
				CG: 7/34 (20.6%) IGT1: 8/36 (22.2%)	
				IGT5: 9/34 (26.5%)	
				IGT15: 15/35 (42.9%)	
				Steroid use within the previous 12 months (n (%))	
				CG: 15/34 (44.1%) IGT1: 16/36 (44.4%) IGT5: 14/34 (41.2%)	
				IGT15: 16/35 (45.7%)	
Sandborn 2020d (CELEST) - Induction	IG-U3B: immediate-release oral formulation of upadacitinib 3 mg TWICE daily for 16 weeks (Week 0 to Week 16 of CELEST). IG-U6B: immediate-release oral formulation of upadacitinib 6 mg TWICE daily for 16 weeks (Week 0 to Week 16 of CELEST). IG-U12B: immediate-release oral for-	CG-PBO: placebo for 16 weeks (Week 0 to Week 16 of CELEST). DOSE, REGIMEN, ROUTE and AGENT NOT REPORTED, but it can be safely assumed that the placebo was oral and ONCE or TWICE daily doses for the maintenance of blinding.	NOT REPORTED previous experience of biologic therapy as a whole but REPORTED previous experience of certain categories of biologics as follows: (i) Prior TNF antagonist (iA) 0: CG-PBO: 2/37 (5.4%) IG-U3B: 2/39 (5.1%) IG-U6B: 1/37 (2.7%)	Participants could enter the study while receiving stable doses of aminosalicylates, oral corticosteroids (equivalent prednisone dose of ≤ 30 mg/d or budesonide of ≤ 9 mg/d), methotrexate, and/or CD-related antibiotics. Starting at week 2, participants receiving oral corticosteroids initiated a mandatory corticosteroid tapering following a protocol-specified schedule until discontinuation Baseline corticosteroid use, n (%): CG-PBO: 15/37 (40.5%) IG-U3B: 21/39 (53.8%) IG-U6B: 18/37 (48.6%)	Participants had to discontinue azathioprine or mercaptopurine ≥ 10 days before baseline. Not reported – No protocol Note*: These are the same as medication up to study beginning Participants could enter the study while receiving stable doses of aminosalicylates, oral corticosteroids (equivalent prednisone dose of ≤ 30 mg/d or budesonide of ≤ 9 mg/d), methotrexate, and/or CD-related antibiotics. Starting at week 2, participants receiving oral corticosteroids initiated a mandatory corticosteroid tapering following a protocol-specified schedule until discontinuation Baseline corticosteroid use, n (%): CG-PBO: 15/37 (40.5%)

Table 2. Induction studies: intervention details *(Continued)*

mulation of upadacitinib 12 mg TWICE daily for 16 weeks (Week 0 to Week 16 of CELEST). IG-U24B: immediate-release oral formulation of upadacitinib 24 mg TWICE daily for 16 weeks (Week 0 to Week 16 of CELEST). IG-U24Q: immediate-release oral formulation of upadacitinib 24 mg ONCE daily for 16 weeks (Week 0 to Week 16 of CELEST)	IG-U12B: 2/36 (5.6%)	IG-U12B: 17/36 (47.2%)	IG-U3B: 21/39 (53.8%)
	IG-U24B: 0/36 (0%)	IG-U24B: 15/36 (41.7%)	IG-U6B: 18/37 (48.6%)
	IG-U24Q: 2/35 (5.7%)	IG-U24Q: 10/35 (28.6%)	IG-U12B: 17/36 (47.2%)
	(iB) 1:	Baseline daily corticosteroid dose, mg (median (range)):	IG-U24B: 15/36 (41.7%)
	CG-PBO: 15/37 (40.5%)	CG-PBO: 20.0 (5 to 45)	IG-U24Q: 10/35 (28.6%)
	IG-U3B: 17/39 (43.6%)	IG-U3B: 20.0 (5 to 45)	Baseline daily corticosteroid dose, mg (Median (Range)):
	IG-U6B: 12/37 (32.4%)	IG-U6B: 22.5 (10 to 45)	CG-PBO: 20.0 (5 to 45)
	IG-U12B: 6/36 (16.7%)	IG-U12B: 20.0 (5 to 45)	IG-U3B: 20.0 (5 to 45)
	IG-U24B: 10/36 (27.8%)	IG-U24B: 20.0 (10 to 45)	IG-U6B: 22.5 (10 to 45)
	IG-U24Q: 10/35 (28.6%)	IG-U24Q: 30.0 (10 to 45)	IG-U12B: 20.0 (5 to 45)
(iC) 2:	CG-PBO: 15/37 (40.5%)	NR	IG-U24B: 20.0 (10 to 45) IG-U24Q: 30.0 (10 to 45)
	IG-U3B: 16/39 (41.0%)		
	IG-U6B: 20/37 (54.1%)		
	IG-U12B: 24/36 (66.7%)		
	IG-U24B: 15/36 (41.7%)		
	IG-U24Q: 16/35 (45.7%)		

Table 2. Induction studies: intervention details (Continued)

(iD) ≥ 3:						
CG-PBO: 5/37 (13.5%)						
IG-U3B 4/39 (10.3%)						
IG-U6B 4/37 (10.8%)						
IG-U12B 4/36 (11.1%)						
IG-U24B 9/36 (25.0%)						
IG-U24Q 7/35 (20.0%)						
(ii) Prior non-TNF antagonist biologics:						
CG-PBO: 14/37 (37.8%)						
IG-U3B: 15/39 (38.5%)						
IG-U6B: 19/37 (51.4%)						
IG-U12B: 15/36 (41.7%)						
IG-U24B: 16/36 (44.4%)						
IG-U24Q: 14/35 (40.0%)						
Sandborn 2022 - GALAXI 1	IG-G200: guselkum- ab 200 mg intravenous at weeks 0, 4 and 8	CG: placebo (Agent, Dose, Regimen and Route are NOT REPORT- ED)	History of bio- logic use (num- ber of partici- pants/total par- ticipants in giv-	Crohn's disease medica- tion taken at baseline (YES) (Number of partici- pants/total participants in given group (percentage)): CG: 60/61 (98.4%)	NOT REPORTED	NOT REPORT- ED – No pro- tocol found Crohn's disease medica- tion taken at baseline (YES) (Number of partici- pants/total participants in given group (percentage)): CG: 60/61 (98.4%)

Table 2. Induction studies: intervention details (Continued)

IG-G600: guselkum- ab 600 mg intravenous at weeks 0, 4 and 8 IG-G1200: guselkum- ab 1200 mg intravenous at weeks 0, 4 and 8 IG-GC: this represents IG- G200, IG-G600 and IG-G1200 (guselkumab groups) com- bined. This is not a separate study arm by itself. IG-UST: ustekinum- ab 6 mg/kg in- travenous at week 0 AND ustekinum- ab 90 mg sub- cutaneous at week 8	en group (per- centage)):	IG-G200: 61/61 (100.0%)	IG-G200: 61/61 (100.0%)
	CG: 42/61 (68.9%)	IG-G600: 63/63 (100.0%)	IG-G600: 63/63 (100.0%)
	IG-G200: 36/61 (59.0%)	IG-G1200: 61/61 (100.0%)	IG-G1200: 61/61 (100.0%)
	IG-G600: 39/63 (61.9%)	IG-GC: 185/185 (100.0%)	IG-GC: 185/185 (100.0%)
	IG-G1200: 36/61 (59.0%)	IG-UST: 63/63 (100.0%)	IG-UST: 63/63 (100.0%)
	IG-GC: 111/185 (60.0%)	Of which (i) ≥ 1 medica- tions for Crohn's disease:	Of which:
	IG-UST: 44/63 (69.8%)	CG: 45/61 (73.8%)	(i) ≥ 1 medications for Crohn's disease:
	Participants with an inade- quate response to, or intoler- ance to biolog- ic therapy (num- ber of partici- pants/total parti- cipants in giv- en group (per- centage)):	IG-G200: 44/61 (72.1%)	CG: 45/61 (73.8%)
	CG: 30/61 (49.2%)	IG-G600: 47/63 (74.6%)	IG-G200: 44/61 (72.1%)
	IG-G200: 32/61 (52.5%)	IG-G1200: 46/61 (75.4%)	IG-G600: 47/63 (74.6%)
	IG-G600: 35/63 (55.6%)	IG-GC: 137/185 (74.1%)	IG-G1200: 46/61 (75.4%)
	IG-G1200: 34/61 (55.7%)	IG-UST: 53/63 (84.1%)	IG-GC: 137/185 (74.1%)
	IG-GC: 101/185 (54.6%)	(ii) Immunomodulatory therapy:	IG-UST: 53/63 (84.1%)
	IG-UST: 37/63 (58.7%)	CG: 26/61 (42.6%)	(ii) Immunomodulatory therapy:
		IG-G200: 15/61 (24.6%)	CG: 26/61 (42.6%)
		IG-G600: 18/63 (28.6%)	IG-G200: 15/61 (24.6%)
		IG-G1200: 25/61 (41.0%)	IG-G600: 18/63 (28.6%)
		IG-GC: 58/185 (31.4%)	IG-G1200: 25/61 (41.0%)
		IG-UST: 26/63 (41.3%)	IG-GC: 58/185 (31.4%)
		(iiia) Corticosteroids (Number of partici- pants/total partici- pants in given group (percentage)):	IG-UST: 26/63 (41.3%)
		CG: 24/61 (39.3%)	(iiia) Corticosteroids (Number of partici- pants/total partici- pants in given group (percentage)):
		IG-G200: 24/61 (39.3%)	CG: 24/61 (39.3%)

Table 2. Induction studies: intervention details (Continued)

Of which:

(i) Anti-TNF only: CG: 25/61 (41.0%)
IG-G200: 26/61 (42.6%)
IG-G600: 27/63 (42.9%)
IG-G1200: 31/61 (50.8%)
IG-GC: 84/185 (45.4%)
IG-UST: 32/63 (50.8%)

(ii) ≥ 1 anti-TNF:
CG: 29/61 (47.5%)

IG-G200: 30/61 (49.2%)

IG-G600: 35/63 (55.6%)

IG-G1200: 33/61 (54.1%)

IG-GC: 98/185 (53.0%)

IG-UST: 37/63 (58.7%) **(iii)**

Vedolizumab:

CG: 5/61 (8.2%)

IG-G200: 6/61 (9.8%)

IG-G600: 8/63 (12.7%)
IG-G1200: 3/61 (4.9%)

IG-GC: 17/185 (9.2%)

IG-G600: 19/63 (30.2%)

IG-G1200: 20/61 (32.8%)

IG-GC: 63/185 (34.1%)

IG-UST: 26/63 (41.3%)

(iiib) Corticosteroid dose (mg/day) (Median (IQR)):
CG: 20.0 (10.0 to 30.0)

IG-G200: 20.0 (20.0 to 25.0)

IG-G600: 20.0 (10.0 to 25.0)

IG-G1200: 20.0 (15.0 to 22.5)

IG-GC: 20.0 (15.0 to 25.0)

IG-UST: 20.0 (15.0 to 32.5)

Note: ‘Immunomodulatory therapy’ included azathioprine, mercaptopurine, and methotrexate. ‘Corticosteroids’ included budesonide and beclomethasone dipropionate.

IG-G200: 24/61 (39.3%)

IG-G600: 19/63 (30.2%)

IG-G1200: 20/61 (32.8%)

IG-GC: 63/185 (34.1%)

IG-UST: 26/63 (41.3%)

(iiib) Corticosteroid dose (mg/day) (Median (IQR)):

CG: 20.0 (10.0 to 30.0)

IG-G200: 20.0 (20.0 to 25.0)

IG-G600: 20.0 (10.0 to 25.0)

IG-G1200: 20.0 (15.0 to 22.5)

IG-GC: 20.0 (15.0 to 25.0)

IG-UST: 20.0 (15.0 to 32.5)

Note: ‘Immunomodulatory therapy’ included azathioprine, mercaptopurine, and methotrexate. ‘Corticosteroids’ included budesonide and beclomethasone dipropionate.

Table 2. Induction studies: intervention details *(Continued)*

IG-UST: 5/63
(7.9%). **(iv)**

**Vedolizum-
ab AND > 1 an-
ti-TNF:**

CG: 4/61 (6.6%)

IG-G200: 4/61
(6.6%)

IG-G600: 8/63
(12.7%)

IG-G1200: 2/61
(3.3%)

IG-GC: 14/185
(7.6%)

IG-UST: 5/63
(7.9%)

**Biologic-naïve
(number of par-
ticipants/to-
tal participants
in given group
(percentage):**

CG: 19/61 (31.1%)

IG-G200: 25/61
(41.0%)

IG-G600: 24/63
(38.1%)

IG-G1200: 25/61
(41.0%)

IG-GC: 74/185
(40.0%)

IG-UST: 19/63
(30.2%)

Note: 'Biolog-
ics' included in-

Table 2. Induction studies: intervention details *(Continued)*
 fliximab, adal-
 imumab, cer-
 tolizumab pe-
 gol, or vedolizum-
 ab

Sands 2004 (ACCENT II) Induction (Non-responders)	Infliximab 5 mg/kg (Route NOT REPORT-ED) at weeks 14, 22, 30, 38 and 46. Beginning at week 22, participants who had a loss of response (defined by recrudescence of draining fistulas, the need for additional therapy for persistent or worsening luminal disease activity, the need for a surgical procedure for Crohn's disease, or the discontinuation of the study medication owing to a perceived lack of efficacy) were eligible to cross over to maintenance treatment with infliximab 10 mg/kg	Placebo (Agent, Dose, and Route NOT REPORT-ED) at weeks 14, 22, 30, 38 and 46. Beginning at week 22, participants who had a loss of response (defined by recrudescence of draining fistulas, the need for additional therapy for persistent or worsening luminal disease activity, the need for a surgical procedure for Crohn's disease, or the discontinuation of the study medication owing to a perceived lack of efficacy) were eligible to cross over to maintenance treatment with in-	NR	NR	NR	NR	NR
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Table 2. Induction studies: intervention details (Continued)

		fliximab 5 mg/ kg					
Sands 2007	IG: natalizumab 300 mg at weeks 0,4, 8 + intravenous infliximab infusion (5 mg/kg) in week 6	CG: placebo + intravenous infliximab infusion (5 mg/kg) in week 6	NR (other than infliximab that both groups needed to have)	NR Permitted CD medications included oral 5-aminosalicylic acid (5-ASA) compounds, if on a stable dose for at least 4 weeks; oral antibiotics if on a stable dose for at least 4 weeks; oral corticosteroids not exceeding 25 mg prednisolone or its equivalent for at least 4 weeks and at a stable dose for at least 2 weeks; and immunosuppressants (azathioprine, 6-mercaptopurine, or methotrexate) for at least 4 months before natalizumab and at a stable dose for at least 8 weeks.	Concurrent use of herbal preparations, rectal steroids, rectal 5-ASAs, tacrolimus, cyclosporine, mycophenolate mofetil, or any investigational or experimental agents was prohibited, as was prior treatment with natalizumab.	NR	Concomitant CD-related medications, n (%) 5-Aminosalicylate compounds: CG: 10 (37) IG: 24 (46) Azathioprine, mercaptopurine, or methotrexate: CG: 15 (56) IG: 26 (50) Oral steroids: CG: 8 (30) IG: 14 (27) Antibiotics: CG: 5 (19) IG: 10 (19)
Sands 2010	IG1: apilimod mesylate 50 mg oral every day IG2: apilimod mesylate 100 mg	CG: placebo oral daily (Agent, Dose, and Frequency NOT REPORTED)	Prior medication use (n (%)) of which: (i) Biologic agents: CG: 38/73 (52%) IG1: 47/73 (64%) IG2: 45/74 (61%) (ii) Infliximab: CG: 29/73 (40%) IG1: 43/73 (59%)	(i) Any (n (%)): CG: 53/73 (73%) IG1: 50/73 (68%) IG2: 55/74 (74%) (ii) Aminosalicylates (mesalamine) (n (%)): CG: 4/73 (5%) IG1: 2/73 (3%) IG2: 1/74 (1%) (iii) Antibiotics (n (%)): CG: 5/73 (7%)	NOT REPORTED	NOT REPORTED – No protocol. No mention of mandatory medications in the paper	Note – these are the same as medications up to study beginning (i) Any (n (%)): CG: 53/73 (73%) IG1: 50/73 (68%) IG2: 55/74 (74%) (ii) Aminosalicylates (mesalamine) (n (%)): CG: 4/73 (5%) IG1: 2/73 (3%)

Table 2. Induction studies: intervention details (Continued)

Study		Intervention group		Control group		Outcome	
Sands 2014 (GEMINI III)	Vedolizumab 300 mg IV in 250 mL of 0.9% sodium chloride	Placebo administered intravenously in 250 mL	IG2: 38/74 (41%)	IG1: 6/73 (8%)	NR	NR	IG2: 1/74 (1%)
			(iii) Adalimumab:	IG2: 9/74 (12%)			(iii) Antibiotics (n (%)):
			CG: 9/73 (12%)	(iv) Systemic corticosteroids (n (%)):			CG: 5/73 (7%)
			IG1: 8/73 (11%)	CG: 14/73 (19%)			IG1: 6/73 (8%)
			IG2: 10/74 (14%)	IG1: 8/73 (11%)			IG2: 9/74 (12%)
			(iv) Certolizumab:	IG2: 9/74 (12%)			(iv) Systemic corticosteroids (n (%)):
			CG: 2/73 (3%)	(v) Mercaptopurine/azathioprine (n (%)):			CG: 14/73 (19%)
			IG1: 0/73 (0%)	CG: 1/73 (1%)			IG1: 8/73 (11%)
			IG2: 1/74 (1%)	IG1: 2/73 (3%)			IG2: 9/74 (12%)
			(v) Natalizumab:	IG2: 4/74 (5%)			(v) Mercaptopurine/azathioprine (n (%)):
			CG: 9/73 (12%)	(vi) Methotrexate (n (%)):			CG: 1/73 (1%)
			IG1: 13/73 (18%)	CG: 0/73 (0%)			IG1: 2/73 (3%)
			IG2: 12/74 (16%)	IG1: 1/73 (1%)			IG2: 4/74 (5%)
			(vi) CDP-571:	IG2: 2/74 (3%)			(vi) Methotrexate (n (%))
			CG: 1/73 (1%)	(vii) Biologic agents (n (%)):			CG: 0/73 (0%)
			IG1: 0/73 (0%)	CG: 2/73 (3%)			IG1: 1/73 (1%)
			IG2: 0/74 (0%)	IG1: 2/73 (3%)			IG2: 2/74 (3%)
			(vii) Onercept:	IG2: 1/74 (1%)			(vii) Biologic agents (n (%)):
			CG: 0/73 (0%)				CG: 2/73 (3%) IG1: 2/73 (3%)
			IG1: 1/73 (1%)				IG2: 1/74 (1%)
			IG2: 0/74% (0%)				
			Other biologics are NOT REPORTED				
Sands 2014 (GEMINI III)	Vedolizumab 300 mg IV in 250 mL of 0.9% sodium	Placebo administered intravenously in 250 mL	Prior TNF antagonist failure, n	NR	NR	NR	Corticosteroid use, n (%)
			CG: 157				CG: 108 (52)

Table 2. Induction studies: intervention details (Continued)

	chloride at weeks 0, 2, 6	of 0.9% sodium chloride at weeks 0, 2 and 6	IG: 158				IG: 110 (53) Immunosuppressive use, n (%) CG: 69 (33) IG: 71 (34) Mesalamine use, n (%) CG: 61 (29) IG: 68 (33)
Sands 2022 (SEAVUE)	Ustekinumab 6 mg/kg IV at baseline then 90 mg SC every 8 weeks	Adalimumab 160/80 mg SC at baseline/week 2, then 40 mg SC every 2 weeks	Biologic naive	IG: steroids including budesonide 75, steroids excluding budesonide 46, steroid dose 20mg/day (10-20) mean (SD), budesonide 29, budesonide dose 9mg/day (6-9) mean (SD) CG: steroids including budesonide 70, steroids excluding budesonide 42, steroid dose 20mg/day (10-20) mean (SD), budesonide 28, budesonide dose 9mg/day (6-9) mean (SD) More than 20% of participants were receiving immunosuppressants	Not reported	Not reported	IG: steroids including budesonide 75, steroids excluding budesonide 46, steroid dose 20mg/day (10-20) mean (SD), budesonide 29, budesonide dose 9mg/day (6-9) mean (SD) CG: steroids including budesonide 70, steroids excluding budesonide 42, steroid dose 20mg/day (10-20) mean (SD), budesonide 28, budesonide dose 9mg/day (6-9) mean (SD) More than 20% of participants were receiving immunosuppressants
Schreiber 2005	IG1: subcutaneous injection certolizumab 100 mg IG2: subcutaneous injection certolizumab 200 mg	Subcutaneous injection placebo (saline)	Biologic naive	IG1: overall 24, systemic steroids 15, budesonide 9 IG2: overall 29, systemic steroids 22, budesonide 8 IG3: overall 22, systemic steroids 16, budesonide 6 CG: overall 29, systemic steroids 20, budesonide 9 More than 20% of participants were receiving immunosuppressants	Not reported	Not reported	IG1: overall 24, systemic steroids 15, budesonide 9 IG2: overall 29, systemic steroids 22, budesonide 8 IG3: overall 22, systemic steroids 16, budesonide 6 CG: overall 29, systemic steroids 20, budesonide 9 More than 20% of participants were receiving immunosuppressants

Table 2. Induction studies: intervention details (Continued)

IG3: subcutaneous injection certolizumab 400 mg							
Schreiber 2018a	Received subcutaneous injections of placebo weekly, an-decaliximab 150 mg every 2 weeks, an-decaliximab 150 mg or an-decaliximab 300 mg weekly	Placebo	Prior exposure to TNF- α antagonist IG1: 46 (86.8%) IG2: 45 (84.9%) IG3: 43 (81.1%) CG: 24 (85.7%)	Concomitant use of oral corticosteroid at randomisation, n (%) IG1: 24 (45.3%) IG2: 26 (49.1%) IG3: 24 (45.3%) CG: 9 (32.1%) Concomitant use of immunomodulator IG1: 19 (35.8%) IG2: 17 (32.1%) IG3: 8 (15.1%) IG3: 8 (28.6%)	NR	NR	Concomitant administration of the following was permitted: oral 5-aminosalicylate compounds; prednisone $\leq$ 30 mg/day or budesonide $\leq$ 9 mg/day at a stable dose for $\geq$ 2 weeks prior to screening; anti-diarrhoeals; azathioprine, 6-mercaptopurine or methotrexate at a stable dose for 4 weeks prior to screening; and antibiotics for treatment of CD at stable doses for $\geq$ 2 weeks prior to screening or doses consistent with the subject's standard low-dose regimen. Thiopurines or methotrexate could be initiated up to 4 weeks prior to screening.
Schwartz 2021 - ENTERPRISE	IG: vedolizumab 300 mg intravenous at weeks 0, 2, 6, 10, 14 and 22. If a seton was placed as part of the standard of care, it had to be removed by week 14. IGP: pooled group, which is a combination of IG	CG: vedolizumab 300 mg intravenous at weeks 0, 2, 6, 14 and 22. At week 10, participants received placebo intravenous (AGENT NOT REPORTED). If a seton was placed as part of the standard of care,	Previous anti-TNF treatment (number of participants/total participants in group (percentage)): CG: 12/16 (75.0%) IG: 13/16 (81.3%) IGP: 25/32 (78.1%) Other biologics are NOT REPORTED	Concomitant treatment at baseline (number of participants/total participants in group (percentage)) of which: (i) Corticosteroids: CG: 3/16 (18.8%) IG: 3/16 (18.8%) IGP: 6/32 (18.8%) (ii) Immunosuppressants: CG: 5/16 (31.3%) IG: 3/16 (18.8%)	NOT REPORTED	No protocol found. All participants were prescribed companion antibiotics , per local label, from day 1 until week 6 . Approved antibiotics were metronidazole, ciprofloxacin,	Note: These are the same as medications up to study beginning Concomitant treatment at baseline (number of participants/total participants in group (percentage)) of which: (i) Corticosteroids: CG: 3/16 (18.8%) IG: 3/16 (18.8%) IGP: 6/32 (18.8%) (ii) Immunosuppressants:

Table 2. Induction studies: intervention details *(Continued)*

	and CG. It is not a separate study arm by itself.	it had to be removed by week 14.		IGP: 8/32 (25.0%)		amoxicillin clavulanate, and tinidazole per local label.	CG: 5/16 (31.3%) IG: 3/16 (18.8%) IGP: 8/32 (25.0%)
Targan 1997	IG1: IV infusion 5 mg of infliximab/kg IG2: IV infusion 10 mg of infliximab/kg IG3: IV infusion 20 mg of infliximab2/kg	IV infusion placebo (agent: NR)	Participants were excluded if prior treatment with murine, chimeric or humanised monoclonal antibodies within 4 weeks before screening.	Participants who were receiving mesalamine, corticosteroids, azathioprine, or mercaptopurine before the study continued to receive a stable dose during the trial period.	NR	NR	Prednisone equivalent < 20 mg/day orally: CG: 10 (40%) IG1: 8 (30%) IG2: 8 (29%) IG3: 10 (36%) Prednisone equivalent > 20 mg/day orally: CG: 6 (24%) IG1: 7 (26%) IG2: 8 (29%) IG3: 7 (25%) Mercaptopurine: CG: 4 (16%) IG1: 4 (15%) IG2: 4 (14%) IG3: 4 (14%) Azathioprine CG: 7 (28%) IG1: 5 (19%) IG2: 4 (14%) IG3: 8 (29%) Oral aminosalicylates

Table 2. Induction studies: intervention details *(Continued)*

CG: 17 (68%) IG1: 16 (59%) IG2: 18 (64%) IG3: 13 (46%)							
Targan 2007 (ENCORE)	IV natalizumab 300 mg	IV infusion placebo (exact agent and dosage: NR)	Participants who had received anti-tumour necrosis factor therapy within the previous 12 weeks, or had ever previously been treated with natalizumab were excluded	NR	Prior use of anti-tumour necrosis factor agents	NR	Concomitant Crohn's disease medication or diet:
							CG: 227 (91%)
							IG: 247 (95%)
							5-ASA compounds:
							CG: 120 (48%)
							IG: 128 (49%)
							Immunosuppressants:
							CG: 96 (38%)
							IG: 97 (37%)
							Corticosteroids:
							CG: 94 (38%)
							IG: 109 (42%)
							Antibiotics:
							CG: 13 (5%)
							IG: 17 (7)%
							Diet:
							CG: 4 (2%)
							IG: 3 (1%)
Loftus 2023 (U-EXCEL)	Upadacitinib 45 mg oral daily (frequency NOT REPORTED –	Placebo oral daily for 12 weeks (Agent, Dose, and Frequency NOT	NOT REPORTED – Need full paper for this information	NOT REPORTED – Need full paper for this information	NOT REPORTED – Need full paper for this information	According to the trial registration, female participants were	NOT REPORTED – Need full paper for this information

Table 2. Induction studies: intervention details (Continued)

	Need full paper for this information) for 12 weeks	REPORTED – need full paper for this information)				required to have contraception .	
Vermeire 2017 FITZROY - induction	Filgotinib 200 mg oral once daily	Matching placebo oral once daily (AGENT NOT REPORTED)	One of the exclusion criteria is as follows: "Previous treatment with biological investigational medicinal products including murine, chimeric or humanized monoclonal antibodies or a chemokine receptor blocker within less than 5 half-lives prior to baseline. Previous treatment with a Janus kinase inhibitor is prohibited." Therefore, previous biologic (any biologic) exposure* (reported as 'previous anti-TNF exposure'): CG: 28/44 (64%) IG: 73/130 (56%) *The anti-TNF naïve group in-	Concurrent treatment with oral steroids (≤ 30 mg prednisolone or budesonide ≤ 9 mg per day) or probiotics was allowed if dosages had been stable for at least 2 weeks before the first dose of study drug. Concurrent treatment with Crohn's disease-related antibiotics, or with mesalazine or olsalazine, was permitted if dosages had been stable for at least 4 weeks before the first dose of study drug. Previous exposure to sulfasalazine was allowed but had to be discontinued at least 4 weeks before screening in male participants, whereas immunomodulators (e.g. thiopurines and methotrexate) must have been discontinued at least 25 days before the first dose of the study drug. Oral corticosteroids: CG: 23/44 (52%) IG: 65/130 (50%) Daily dose (mg/day) (Mean/Median (SD/IQR)): CG*: 21.3/20.0 (8.6/15.0 to 30.0) IG**: 20.7/20.0 (8.6/15.0 to 30.0)	Previous exposure to sulfasalazine was allowed but had to be discontinued at least 4 weeks before screening in male participants. Immunomodulators must have been discontinued at least 25 days before the first dose of the study drug. Tube feeding, defined formula diets, or total parenteral alimentation was not allowed during the study. IV corticosteroids were to be discontinued 14 days prior to screening. Enemas or suppositories other than those required for colonoscopy were to be discontinued 7 days prior to screening.	If using paracetamol at baseline, it should not be discontinued Contraception	Concurrent treatment with oral steroids (≤ 30 mg prednisolone or budesonide ≤ 9 mg per day) or probiotics was allowed if dosages had been stable for at least 2 weeks before the first dose of study drug. Concurrent treatment with Crohn's disease-related antibiotics, or with mesalazine or olsalazine, was permitted if dosages had been stable for at least 4 weeks before the first dose of study drug. Previous exposure to sulfasalazine was allowed but had to be discontinued at least 4 weeks before screening in male participants, whereas immunomodulators (e.g. thiopurines and methotrexate) must have been discontinued at least 25 days before the first dose of the study drug. Oral corticosteroids: CG: 23/44 (52%) IG: 65/130 (50%) Daily dose (mg/day) (Mean/Median (SD/IQR)): CG*: 21.3/20.0 (8.6/15.0 to 30.0). IG**: 20.7/20.0 (8.6/15.0 to 30.0) *n = 22

*n = 63

Cyclosporine, mycophenolate, tacrolimus, or interferon were to be discontinued at least **10 weeks prior** to screening.

Any **prior lymphocyte depleting agent** (e.g. alemtuzumab) was **prohibited**. **Lymphocyte apheresis or selective monocyte granulocyte apheresis** was to be discontinued **12 months prior** to screening.

Previous faecal microbiota transplant and stem cell transplants were **prohibited**.

Investigational chemical agents were to be discontinued **4 weeks before** screening.

Previous treatment with biological investigational medicinal products including murine, chimeric or humanised monoclonal antibodies or a

*n = 22

*n = 63

cluded 4 participants who were anti-TNF responders but who discontinued for financial reasons

Table 2. Induction studies: intervention details *(Continued)*

Table 2. Induction studies: intervention details *(Continued)*

Vermeire 2017 FITZROY - in- duction non- responders	Filgotinib 200 mg oral once daily	Matching placebo oral once daily (Agent NOT REPORTED). Placebo was presented as orally admin-	NOT REPORTED However, one of the exclusion criteria states that previous JAK inhibitor ther- apy was not al-	Note – the time points here apply to the original induc- tion study (i.e. screening means week 0 rather than week 10, where the latter is where the maintenance phase began)	Note – the time points here ap- ply to the orig- inal induction study (i.e. screen- ing means week 0 rather than week 10, where the	If using paracetamol at baseline , it should not be discontinued Contracep- tion	Note: the time points here apply to the original induc- tion study (i.e. screening means week 0 rather than week 10, where the latter is where the maintenance phase began)
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**chemokine re-
ceptor blocker
within less than
5 half-lives prior
to baseline. Pre-
vious treatment
with a janus ki-
nase inhibitor is
prohibited.**

Live vaccines
must not have
been adminis-
tered **within 90
days** or an **atten-
uated vaccine**
within 30 days of
first dose of study
drug as well as
during the study
duration **until 6
weeks follow-
ing last intake** of
GLPG0634.

Participants **cur-
rently on any
therapy for
chronic infection**
(such as pneu-
mocystis, CMV,
herpes simplex,
herpes zoster,
or atypical my-
cobacteria) were
excluded from
the study.

Table 2. Induction studies: intervention details (Continued)

istered brown film-coated tablets that were identical in appearance to filgotinib and contained the same excipients.	lowed. Therefore, no participant had previous JAK inhibitors (CG: 0/13, IG: 0/39) but unable to comment on TNF inhibitors and hence on biologics as a whole	Concurrent treatment with oral steroids (≤ 30 mg prednisolone or budesonide ≤ 9 mg per day) or probiotics was allowed if dosages had been stable for at least 2 weeks before the first dose of study drug . Concurrent treatment with Crohn's disease-related antibiotics , or with mesalazine or olsalazine , was permitted if dosages had been stable for at least 4 weeks before the first dose of study drug . Previous exposure to sulfasalazine was allowed but had to be discontinued at least 4 weeks before screening in male participants, whereas immunomodulators (e.g. thiopurines and methotrexate) must have been discontinued at least 25 days before the first dose of the study drug.	latter is where the maintenance phase began)	Previous exposure to sulfasalazine was allowed but had to be discontinued at least 4 weeks before screening in male participants.	Concurrent treatment with oral steroids (≤ 30 mg prednisolone or budesonide ≤ 9 mg per day) or probiotics was allowed if dosages had been stable for at least 2 weeks before the first dose of study drug . Concurrent treatment with Crohn's disease-related antibiotics , or with mesalazine or olsalazine , was permitted if dosages had been stable for at least 4 weeks before the first dose of study drug . Previous exposure to sulfasalazine was allowed but had to be discontinued at least 4 weeks before screening in male participants, whereas immunomodulators (e.g. thiopurines and methotrexate) must have been discontinued at least 25 days before the first dose of the study drug
		Proportion of participants on each medication – NOT REPORTED for the maintenance phase		Immunomodulators must have been discontinued at least 25 days before the first dose of the study drug.	Proportion of participants on each medication – NOT REPORTED for the maintenance phase
				Tube feeding, defined formula diets, or total parenteral alimentation was not allowed during the study.	
				IV corticosteroids were to be discontinued 14 days prior to screening.	
				Enemas or suppositories other than those required for colonoscopy were to be discontinued 7 days prior to screening.	

Table 2. Induction studies: intervention details *(Continued)*

Cyclosporine, mycophenolate, tacrolimus, or interferon were to be discontinued at least **10 weeks prior** to screening.

Any **prior lymphocyte depleting agent** (e.g. alemtuzumab) was **prohibited**. **Lymphocyte apheresis or selective monocyte granulocyte apheresis** was to be discontinued **12 months prior** to screening.

Previous faecal microbiota transplant and stem cell transplants were **prohibited**

Investigational chemical agents were to be discontinued **4 weeks before** screening.

Previous treatment with biological investigational medicinal products including murine, chimeric or humanised monoclonal antibodies or a

Table 2. Induction studies: intervention details *(Continued)*

Watanabe 2012 - Induction	Adalimumab	Placebo	REPORTED under the heading 'Crohn's disease-related medication prior to or at baseline of induction and	REPORTED under the heading 'Crohn's disease-related medication prior to or at baseline of induction and maintenance trials'.	Previous use of infliximab or biologic, if any, had to be discontinued at least 8 weeks before baseline.	NR	Concomitant use of immunomodulators (azathioprine and 6-mercaptopurine), aminosalicylates, and Crohn's-related antibiotics, was permitted in the induction and maintenance trials, provided that
	Adalimumab 80/40 mg						
	Adalimumab 160/80 mg						

chemokine receptor blocker within less than 5 half-lives prior to baseline. Previous treatment with a Janus kinase inhibitor is prohibited.

Live vaccines must not have been administered **within 90 days** or an **attenuated vaccine within 30 days** of first dose of study drug as well as **during the study** duration **until 6 weeks following last intake** of GLPG0634.

Participants **currently on any therapy for chronic infection** (such as pneumocystis, CMV, herpes simplex, herpes zoster, or atypical mycobacteria) were **excluded** from the study.

the doses remained stable. Stable doses of corticosteroids (≤ 40 mg/day of prednisolone or equivalent), and stable enteral nutrition ≤ 1200 kcal/day were permitted. No changes of Crohn's-related concomitant therapies were allowed during the induction trial.

Table 2. Induction studies: intervention details (Continued)

			maintenance trials'. Anti-TNF n (%) CG: 13 (56.5%) IG1: 20 (58.8%) IG2: 19 (57.6%)	Baseline medication use induction trial full analysis set n (%) Aminosalicylates: CG: 23 (100) IG1: 27 (79.4) IG2: 32 (97.0) Immunosuppressants: CG: 8 (34.8) IG1: 11 (32.4) IG2: 10 (30.3) Corticosteroids: CG: 5 (21.7) IG1: 6 (17.6) IG2: 8 (24.2) CD-related antibiotics: CG: 2 (8.7) IG1: 1 (2.9) IG2: 2 (6.1) Enteral nutrition: CG: 16 (69.6) IG1: 22 (64.7) IG2: 23 (69.7)			
Watanabe 2020 - Induction	Vedolizumab 300 mg intravenous infusion at Weeks 0, 2 and 6	Placebo by intravenous infusion at Weeks 0, 2 and 6 (DOSE and AGENT NOT REPORTED)	Reported as 'Prior anti-TNF (alpha) treatment' (numbers (percentage)): (i) No: CG: 16/78 (20.5%) IG: 18/79 (22.8%) (ii) Yes: CG: 62/78 (79.5%)	Participants were allowed concomitant treatment with oral 5-aminosalicylic acid, probiotics, antibiotics for CD, oral corticosteroids, immunomodulators and enteral nutrients (≤ 900 kcal/day) under specific conditions. Concomitant medication for CD at baseline (numbers (percentage)): (i) Enteral nutrient:	These participants were excluded from the study (otherwise discontinued medication NOT REPORTED): (i) Participants who started oral 5-ASAs, probiotics, antibiotics for CD treatment, or oral corticos-	NOT REPORTED – No protocol found	Note: These were the same as medications up to study beginning Participants were allowed concomitant treatment with oral 5-aminosalicylic acid, probiotics, antibiotics for CD, oral corticosteroids, immunomodulators and enteral nutrients (≤ 900 kcal/day) under specific conditions. Concomitant medication for CD at baseline (num-

Table 2. Induction studies: intervention details *(Continued)*

IG: 61/79 (77.2%)	CG: 43/78 (55.1%)	teroids (≤ 30 mg/day) within 13 days before the first dose of the study drug. Or participants who changed dosage of or discontinued these drugs within 13 days before the first dose of the study drug if the participant had used these drugs for more than 14 days before the first dose of the study drug	bers (percentage)): (i) Enteral nutrient:
Prior anti-TNF (alpha) failure (numbers (percentage)):	IG: 38/79 (48.1%)		CG: 43/78 (55.1%)
(i) Overall:	CG: 59/78 (75.6%)	(ii) 5-aminosalicylic acid:	IG: 38/79 (48.1%)
CG: 61/78 (78.2%)	IG: 64/79 (81.0%)		(ii) 5-aminosalicylic acid:
IG: 60/79 (75.9%)	(iii) Oral corticosteroids AND no immunomodulators:		CG: 59/78 (75.6%)
(ii) Inadequate response:	CG: 7/78 (9.0%)		IG: 64/79 (81.0%)
CG: 10/78 (12.8%)	IG: 13/79 (16.5%)	(iv) No oral corticosteroids OR immunomodulators:	(iii) Oral corticosteroids AND no immunomodulators:
IG: 14/79 (17.7%)	(iv) No oral corticosteroids OR immunomodulators:		CG: 7/78 (9.0%)
(iii) Loss of response:	CG: 31/78 (39.7%)	(ii) Participants who had received 5-ASA, corticosteroid enemas/suppositories, corticosteroid IV infusion, oral corticosteroid at > 30 mg/day, drugs for diarrhoea-predominant irritable bowel syndrome, or Chinese herbal medicine for the CD treatment (e.g. Daikenchuto) within 13 days before the first dose of the study drug	IG: 13/79 16.5%) (iv) No oral corticosteroids OR immunomodulators:
CG: 46/78 (59.0%)	IG: 30/79 (38.0%)		CG: 31/78 (39.7%)
IG: 45/79 (57.0%)	(v) Oral corticosteroids AND immunomodulators:		IG: 30/79 (38.0%)
(iv) Intolerance:	CG: 11/78 (14.1%)		(v) Oral corticosteroids AND immunomodulators:
CG: 5/78 (6.4%)	IG: 9/79 (11.4%)		CG: 11/78 (14.1%)
IG: 1/79 (1.3%)			IG: 9/79 (11.4%)
Number of drugs of anti-TNF (alpha) failure (numbers (percentage)):			
(i) None:			
CG: 17/78 (21.8%)			
IG: 19/79 (24.1%).			
(ii) One:			
CG: 29/78 (37.2%)			
IG: 29/79 (36.7%)			
(iii) Two:			

Table 2. Induction studies: intervention details *(Continued)*

CG: 32/78 (41.0%)

IG: 31/79 (39.2%)

Worst prior treatment failure (numbers (percentage)):

(i) Prior anti-TNF (alpha) failure:

CG: 61/78 (78.2%)

IG: 60/79 (75.9%)

(ii) Prior immunomodulator failure but not anti-TNF (alpha) failure:

CG: 9/78 (11.5%)

IG: 12/79 (15.2%)

(iii) Participants who had received **AZA, 6-MP, or MTX within 27 days before the first dose of the study drug**. However, this did **not apply** to participants **who had used these drugs for more than 83 days before the first dose of the study drug and continued the steady dose of the drugs for more than 27 days before the first dose of the study drug**

(iv) Participants who had received **cyclosporine, tacrolimus, tofacitinib or any study drugs of low molecular compound for CD treatment within 27 days before the first dose of the study drug**

(v) Participants who had received **adalimumab within 27 days before the first dose of the study drug or any biologics other than adal-**

Table 2. Induction studies: intervention details *(Continued)*

imumab within 55 days before the first dose of the study drug.

However, this did **not apply** to participants **who had received localised injections of these drugs** (e.g. intraocular injection for treatment of age-related macular degeneration)

(vi) Participants who had received any **live-vaccinations within 27 days before the first dose of the study drug**

(vii) Participants who had a **history or a complication of alcohol dependence or illicit drug use within one year before the first dose of the study drug**

(viii) Participants who had received **central venous nutrition therapy or enteral total nutrition therapy, or fasted within 20 days before the first dose of the study drug**

Table 2. Induction studies: intervention details (Continued)

Winter 2004	Certolizumab (CDP870) single IV dose	Placebo Normal saline 100 mL	Previous anti-TNF therapy, n (%) CG: 3 (13%) IG1: 1 (50%) IG2: 6 (23%) IG3: 5 (29%) IG4: 7 (30%)	Participants receiving aminosalicylate therapy, n (%) CG: 10 (40) IG1: 0 (0) IG2: 12 (48) IG3: 9 (53) IG4: 9 (39)	NR	NR	Participants receiving aminosalicylate therapy, n (%) CG: 10 (40) IG1: 0 (0) IG2: 12 (48) IG3: 9 (53) IG4: 9 (39)
	IG1 dose: 1.25 mg/kg (n = 2)						
	IG2 dose: 5 mg/kg (n = 26)						
	IG3 dose: 10 mg/kg (n = 17)						
	IG4 dose 20 mg/kg (n = 23)			Participants receiving steroid therapy, n (%): CG: 7 (28)			Participants receiving steroid therapy, n (%): CG: 7 (28)

(ix) Participants who had received an **enteral nutrient of > 900 kcal/day** or who had started an **enteral nutrient of ≤ 900 kcal/day** within **20 days** before the **first dose of the study drug**. Or participants who **changed dosage of or discontinued the enteral nutrient** within **20 days** before the **first dose of the study drug** if the participant had received an **enteral nutrient of ≤ 900 kcal/day** more than **21 days** before the **first dose of the study drug**

Table 2. Induction studies: intervention details *(Continued)*

				IG1: 1 (50)			IG1: 1 (50)
				IG2: 6 (24)			IG2: 6 (24)
				IG3: 6 (35)			IG3: 6 (35)
				IG4: 6 (26)			IG4: 6 (26)
				Participants receiving immunosuppressive therapy, n (%):			Participants receiving immunosuppressive therapy, n (%)
				CG: 11 (44)			CG: 11 (44)
				IG1: 0 (0)			IG1: 0 (0)
				IG2: 11 (44)			IG2: 11 (44)
				IG3: 9 (53)			IG3: 9 (53)
				IG4: 10 (44)			IG4: 10 (44)
Ye 2019	Cross-over at week 30	Cross-over at week 30	IG: 84 (76%) CG: 80 (73%)	Corticosteroid use: IG: 37 (33%) CG: 33 (30%)	NR	NR	IG: 111 randomly assigned to initiate CT-P13, 56 to CT-P13–CT-P13, 55 to CT-P13–infliximab CG: 109 randomly assigned to initiate infliximab, 54 to infliximab–infliximab, 55 to infliximab–CT-P13
	Before week 30	Before week 30					
	IG: infliximab	CG: CT-P13					

5-ASA: 5-aminosalicylic acid
 AZA: azathioprine
 CD: Crohn's disease
 CG: control group
 Eow: every other week
 IFX: infliximab
 IG: intervention group
 IV: intravenous
 JAK: Janus kinase
 NMA: network meta-analysis
 NR: not reported
 PBO: placebo
 SC: subcutaneous
 SD: standard deviation
 TNF: tumour necrosis factor

UST: ustekinumab

Table 3. Maintenance studies: characteristics

Study ID	Numbers randomised	Purine analogues use	Biologic naive or not	Mixed disease state or remission only
Buhl 2022 (STOP-IT)	CG (placebo) 56 IG (ifx) 59	Thiopurines 55% of all patients > 50%	Not	Remission
Colombel 2007 (CHARM) - Maintenance (responders)	CG (placebo) 170 IG1 (ada eow) 172 IG2 (ada weekly) 157	AZA 165/499 6-MP 38/499 < 50%	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Colombel 2023 - LIBERTY-CD	CG (placebo) 112 IG (CT-P13 SC) 231	NR	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Feagan 2015c (IM-UNITI)	CG (placebo) 133 IG (ust 8 weeks) 132 IG (ust 12 weeks) 131	Immunosuppressants including azathioprine, mercaptopurine, and methotrexate, n (%) CG (placebo) 47 (35.3) IG (ust 8 weeks) 44 (33.3) IG (ust 12 weeks) 52 (39.4) < 50%	Not	Mixed responders and remission
Ferrante 2022 - FORTIFY	CG (placebo) 184 IG1 (ris 180 mg) 179 IG2 (ris 360 mg) 179	Immunomodulator use, n (%) CG 40 (24) IG1 41 (26) IG2 41 (28) < 50%	Not (the study recruited from two induction studies with the same biologics)	Mixed responders and remission
Hanauer 2002 (ACCENT I) - Maintenance (Responders)	CG (placebo) 110 IG1 (ifx 5 mg) 113 IG2 (ifx 10 mg) 112	6CMP and AZA 81/335 (24%) < 50%	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Jorgensen 2017 (NORSWITCH)	CG (ifx) 78 IG (CT-P13) 77	Use of immunosuppressives CG 30/78 (38.5%) IG 36/77 (46.8%) < 50%	Mix 60/77 and 61/78 naive	Mixed responders and remission
Louis 2023 - SPARE	Group 1 (combination) 71 Group 2 (antimetabolite) 71	Part of the intervention	Not	Remission

Table 3. Maintenance studies: characteristics (Continued)

Group 3 (ifx): 69

NCT00349752 (COSPAR1)	CG (placebo) 87 IG (certolizumab) 87	NR	Not (extension of Sandborn 2010b (WEL-COME))	Remission
Panes 2017 Maintenance	CG (placebo) 59 IG (tof 5 mg) 60 IG (tof 10 mg) 61	Patients receiving azathioprine, 6-mercaptopurine or methotrexate were required to stop these agents at least two weeks before randomisation < 50%	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Rutgeerts 1999	CG (placebo) 36 IG (ifx) 37	6-MP or AZA > 6 months' duration at a stable dosage for 8 weeks was acceptable	Not (The study included patients that had responded to an initial infusion of infliximab or placebo)	Mixed responders and remission
Rutgeerts 2012 (EXTEND)	CG (placebo) 65 IG (ada) 64	Concomitant immunomodulators, n (%) CG 25 (38.5) IG 28 (43.8) < 50%	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Sandborn 2005a (ENACT 2)	CG (placebo) 179 responders IG (nat) 175 responders	Concomitant AZA and 6MP, n (%) AZA CG 45 (26) IG 42 (25) 6MP CG 7 (4) IG 12 (7) < 50%	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Sandborn 2007b (CLASSIC II)	CG (placebo): 18 IG1 (Ada eow): 19 IG2 (Ada weekly): 18	Concomitant AZA and 6MP, n (%) AZA CG 1 (6) IG1 4 (21) IG2 2 (11) 6MP CG 1 (6) IG1 0 (0)	Not (The study had an induction phase with the biologic)	Remission

Table 3. Maintenance studies: characteristics (Continued)

IG2 3 (17)				
< 50%				
Sandborn 2012 (CERTIFI) - Maintenance	CG (placebo) 73 IG (ust) 72	Not given for the maintenance responder randomised population. For the total (n = 526) induction randomised population, 126 (24.0%) were receiving an immunomodulatory drug at baseline.	Not	Mixed responders and remission
< 50%				
Sandborn 2013 (GEMINI II) - Maintenance	CG (placebo) 153 IG1 (Ved E8w) 154 IG2 (Ved E4w) 154	Stable doses of azathioprine or 6-mercaptopurine, n (%) CG (placebo) 152 (99.3) IG1 (Ved E8w) 153 (99.4) IG2 (Ved E4w) 154 (100.0)	Not (The study had an induction phase with the biologic)	Mixed responders and remission
> 50%				
Sandborn 2020d (CELEST) - Maintenance	IG (Upa 3 mg) 61 IG (Upa 6 mg) 23 IG (Upa 12 mg) 59 IG (Upa 24 mg) 37	Included patients had failed or inadequately responded to immunosuppressants	Not	Mixed responders and remission
Sandborn 2023 - BERGAMOT Maintenance	CG (placebo): 217 IG (etro) 217 (One patient was randomised but left the study before receiving allocated medication, and the authors have not included them in these numbers)	Baseline use of immunosuppressants CG (placebo) 71 (33%) IG (etro) 67 (31%)	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Sands 2004 (ACCENT II) Maintenance (Responders)	CG (placebo) 99 IG (ifx) 96	6CMP or AZA CG 35/99 (35%) IG 29/96 (30%)	Not (The study had an induction phase with the biologic)	Mixed responders and remission
< 50%				
Schreiber 2007 (PRECISE 2)	CG (placebo) 212 IG (cert) 216	Immunosuppressive agents only, n (%) CG 52 (25) IG 59 (27) Glucocorticoids plus immunosuppressive agents, n (%) CG 34 (16) IG 28 (13)	Not (The study had an induction phase with the biologic)	Mixed responders and remission

Table 3. Maintenance studies: characteristics (Continued)
< 50%

VanAssche 2012 (SWITCH)	CG (ifx) 37 IG (ADA) 36	AZA IG 6/36 (16.7%) CG 0/37 (0%) < 50%	Not Previous exposure to ADA, and receiving IFX doses > 5 mg/kg intravenously were excluded	Mixed responders and remission
Vermeire 2017 FITZROY - maintenance responders	CG1 (fil/pl) 14 IG1 (fil 100 mg) 30 IG2 (fil 200 mg) 30	Immunomodulators (e.g. thiopurines and methotrexate) must have been discontinued at least 25 days before the first dose of the study drug < 50%	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Vermeire 2021 (VISIBLE 2)	CG (placebo): 135 IG (ved): 275	Immunosuppressive agents only, n (%) CG 34 (25.4) IG 51 (18.5) Corticosteroids plus immunosuppressive agents, n (%) CG 13 (9.7) IG 31 (11.3) < 50%	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Vermeire 2025 - DIVERSITY (maintenance)	IG1 filgotinib 100 mg (n = 105) IG 2 filgotinib 200 mg (n = 118) CG1 placebo (filgotinib 100 mg at induction) (n = 56) CG1 placebo (filgotinib 100 mg at induction) (n = 56)	IG1/2 more than 20% CG1/2 less than 20% < 50%	Not (The study had an induction phase with the biologic)	Mixed clinical remission or endoscopic response at end of induction
Volkers 2017 - SIMILAR	IG (biosimilar) NR CG (ifx) NR (total 47 participants)	NR	Not	Remission
Watanabe 2012 - Maintenance	CG (placebo) 25 IG (ada) 25	Immunosuppressants, n (%) CG 7 (28.0) IG 1 (44.0) < 50%	Not (The study had an induction phase with the biologic)	Mixed responders and remission
Watanabe 2020 - Maintenance	CG (placebo) 12 IG (ved) 12	Immunomodulators, n (%) CG 3 (25.0%)	Not	Mixed responders and remission

Table 3. Maintenance studies: characteristics (Continued)

		IG 2 (16.7%)		(The study had an induction phase with the biologic)
		Immunomodulators and corticosteroids, n (%)		
		CG 3 (25.0)		
		IG 1 (8.3)		
		< 50%		
Young 2024 - iBaSS	IG: reference adalimumab (cross-over to SB5) (n = 57)	IG n = 24 (53.3%)	Not	Remission
		CG n = 20 (46.5%)		(CD in remission or with mild disease activity (modified Harvey-Bradshaw Index (mHBI) score < 8))
	CG: SB5 adalimumab biosimilar (cross-over to reference adalimumab) (n = 55)	~50%		

6-MP: 6-mercaptopurine

ada: adalimumab

AZA: azathioprine

CD: Crohn's disease

CG: control group

eow: every other week

E8w/E4w: every 8 weeks/every 4 weeks

fil: filgotinib

etro: etrolizumab

IG: intervention group

ifx: infliximab

IV: intravenous

nat: natalizumab

NR: not reported

pl: placebo

SC: subcutaneous

Upa: upadacitinib

ust: ustekinumab

Ved/ved: vedolizumab

Table 4. Maintenance studies: intervention details

Study ID	Intervention medications IG	Intervention medications CG	Previous experience of biologic therapy	Medications up to study beginning	Medications that had to be discontinued prior to starting study	Mandatory medications per study protocol	Concomitant medications during study
Buhl 2022 (STOP-IT)	Continue infliximab therapy at an unchanged dose for a duration of 48 weeks	Matching placebo for a duration of 48 weeks	Yes. All patients had been treated with standard IFX maintenance therapy for at least 1 year	Concomitant therapy with immune suppressants, except steroids, was allowed. The dosage and frequency must have been stable 3 months prior to inclusion, and must remain stable throughout the study period. Continued concomitant therapy with stable doses of immunomodulators (thiopurines, methotrexate) was allowed (55% of patients).	No use of oral steroids within 3 months prior to inclusion	None	Thiopurines; methotrexate was allowed (55% of patients)
Colombel 2007 (CHARM) - Maintenance (responders)	IG1 (eow): adalimumab 40 mg once every other week, subcutaneous IG2 (weekly): adalimumab 40 mg weekly, subcutaneous	Placebo (no further details)	Previous TNF-antagonist exposure: n (%) 238 (47.7) out of 499 participants	Concurrent therapies for CD, including stable dosages (for at least 4 weeks before screening) of azathioprine, 6-mercaptopurine, methotrexate, 5-aminosalicylates, sulfasalazine, oral mesalamine, and CD-related antibiotics, were permitted, as were stable dosages (for at least 2 weeks before screening) of prednisone (≤ 30 mg/day or equivalent) or budesonide (≤ 9 mg/day) (patients could not be on both prednisone and budesonide). Patients who had received infliximab or any TNF antagonist other than	Patients could not be enrolled if: currently, on parenteral nutrition; had received investigational chemical agents within 30 days or investigational biologic therapy within 3 months before screening; had received antibiotic treatment for non-CD-related infections within 3 weeks before screening; had received treatment with adalimumab or participated in an adalimumab clinical study; had received enema therapy within 2 weeks before screening; had received cyclosporine, mycophenolate mofetil,	All patients entered a 2-week screening period before their baseline assessments. At the baseline visit (week 0), all eligible patients received open-label adalimumab 80 mg subcutaneously followed by a 40 mg dose at week 2.	From all three groups n (%) out of 499 participants Any corticosteroid: 210; prednisone: 130; budesonide: 48 Any immunosuppressive: 240; azathioprine: 165; 6-mercaptopurine: 38; methotrexate: 49 5-aminosalicylates: 206

Table 4. Maintenance studies: intervention details (Continued)

				adalimumab more than 12 weeks before screening could be enrolled provided that they did not exhibit initial nonresponse to the agent (i.e. no clinical response to first injection as judged by the investigator). Female patients of child-bearing potential were required to use an effective form of birth control.	or tacrolimus within 8 weeks of screening		
Colombel 2023 - LIBERTY-CD	Subcutaneous injection of CT-P13 SC	Subcutaneous injection of placebo	NR	NR	NR	NR	NR
Feagan 2015c (IM-UNITI)	IG1 (90 mg/12 weeks) IG2 (90 mg/8 weeks)	Placebo	No history of TNF antagonist treatment, no. (%) CG 52 (39.1) IG1 53 (40.2) IG252 (39.4)	NR	NR	NR	NR
Ferrante 2022 - FORTIFY	IG1 SC risankizumab 180 mg IG2 SC risankizumab 360 mg	SC placebo	Induction phase experience (induction population characteristics in Table 1)	(Induction population characteristics in Table 1)	(Induction population characteristics in Table 1)	None other	Corticosteroid use IG1 32%, IG2 30%, CG 32% Immunomodulator use IG1 26%, IG2 28%, CG 24%
Hanauer 2002 (ACCENT I) - Maintenance (Responders)	IG1 Infliximab infusion, each infusion 5 mg/kg. Infusions at week 2, week 6, and every 8 weeks	CG Placebo – identical in appearance to infliximab infusion. Infusions at week 2 (start point	Patients were excluded from the study if they had received Infliximab in the past or any other agent targeting TNF	"Patients receiving the following treatments were eligible: 5-aminosalicylates or antibiotics (if the dose remained constant for 4 weeks before the screening visit); corticosteroids (prednisone, prednisolone,	"Patients not receiving medical therapy had to have discontinued treatment for at least 4 weeks before screening"	None	"Patients receiving the following treatments were eligible: 5-aminosalicylates or antibiotics (if the dose remained constant for 4 weeks before the screening visit); corticosteroids (prednisone, prednisolone, or budesonide)

Table 4. Maintenance studies: intervention details (Continued)

	thereafter until week 46	of maintenance study), week 6, and every 8 weeks thereafter until week 46.		or budesonide) at the equivalent of 40 mg per day of prednisone or less (stable dose for 3 weeks); azathioprine and 6-mercaptopurine (stable dose for 8 weeks); or methotrexate (stable dose for 6 weeks)."			at the equivalent of 40 mg per day of prednisone or less (stable dose for 3 weeks); azathioprine and 6-mercaptopurine (stable dose for 8 weeks); or methotrexate (stable dose for 6 weeks)."
	IG2						NOT REPORTED per IG/CG , the following data is based on responders as a whole.
	Infliximab infusion, 5 mg/kg infusions at week 2 and week 6; 10 mg/kg every 8 weeks thereafter until week 46			NOT REPORTED per IG/CG , the following data are based on responders as a whole.			
				5-aminosalicylates			5-aminosalicylates
				159/335 (47%)			159/335 (47%)
				6-mercaptopurine and azathioprine			6-mercaptopurine and azathioprine
				81/335 (24%)			81/335 (24%)
				Methotrexate			Methotrexate
				10/335 (3%)			10/335 (3%)
				Corticosteroids of which			Corticosteroids of which
				(i) Any			(i) Any
				175/335 (52%)			175/335 (52%)
				(ii) > 20 mg per day			(ii) > 20 mg per day
				61/335 (18%)			61/335 (18%)
Jorgensen 2017 (NORSWITCH)	IG Patients were switched to biosimilar CT-P13 infusions on the same dose	CG The dose and infusion intervals of patients' infliximab treatment regi-	IG 60/77 and CG 61/78 naïve	Immunosuppressive therapy: IG 36/77, CG 30/78 Prednisolone: IG 0/77, CG 2/78 Pre-infusion steroids IV: IG 6/77, CG 8/78	Initiation of systemic corticosteroids or an immunosuppressant or other medication, which according to the investigator would interfere with the stability of the disease, was not allowed	None	Immunosuppressive therapy: IG 36/77, CG 30/78 Prednisolone: IG 0/77, CG 2/78 Pre-infusion steroids IV: IG 6/77, CG 8/78

Table 4. Maintenance studies: intervention details (Continued)

	and intervals as their pre-randomisation infliximab regimen	mens were kept unchanged from those before randomisation					
NCT00349752 (COSPAR1)	IG Certolizumab pegol 400 mg provided in a solution for subcutaneous injection (2 x 200 mg/mL) in single-use vials for administration at Weeks 0, 2, 4 and then every 4 weeks until Week 36	CG Placebo provided for subcutaneous injection in 2 single-use vials for administration at Weeks 0, 2, 4 and then every 4 weeks until Week 36	All patients not naive	NR	Patients had to be receiving corticosteroids at a dose no higher than 30 mg/day prednisone or equivalent during the week prior to randomisation. Patients followed a corticosteroid forced-taper regimen after randomisation.	None	NR
Panes 2017 Maintenance	IG-T5 Tofacitinib 5 mg twice daily for 26 weeks (ROUTE NOT REPORTED but likely to be ORAL as it is twice daily over 26 weeks) IG-T10 Tofacitinib 10 mg twice daily for 26 weeks (ROUTE NOT REPORTED but likely to be ORAL as it	CG Placebo twice daily for 26 weeks (ROUTE, AGENT, and DOSE NOT REPORTED but ROUTE is likely to be ORAL as it is twice daily over 26 weeks)	Previous experience of biologic therapy as a whole (which includes JAKi) is NOT REPORTED Previous experience of TNFi CG 39/59 (66.1%) IG-T5 48/60 (80.0%) IG-T10 48/61 (78.7%)	For the preceding induction study Concomitant corticosteroids were allowed up to oral prednisone-equivalent 30 mg/day or budesonide 9 mg/day, provided a stable dose had been taken for at least 2 weeks prior to baseline. Concomitant oral 5-ASA or sulfasalazine (stable dose for at least 4 weeks prior to baseline), chronic treatment for CD with antibiotics (e.g. metronidazole, rifaximin; stable dose for at least 2 weeks prior to baseline), rectal-	For the preceding induction study Patients receiving azathioprine, 6-mercaptopurine, or methotrexate were required to stop these agents at least 2 weeks before randomisation. Concomitant tumour necrosis factor inhibitors (TNFi) were not allowed, and a wash-out period of at least 8 weeks prior to randomisation was required. Cyclosporine, mycophenolate or tacrolimus were required to be	N/A – No protocol found	For the preceding induction study Concomitant corticosteroids were allowed up to oral prednisone-equivalent 30 mg/day or budesonide 9 mg/day, provided a stable dose had been taken for at least 2 weeks prior to baseline. Concomitant oral 5-ASA or sulfasalazine (stable dose for at least 4 weeks prior to baseline), chronic treatment for CD with antibiotics (e.g. metronidazole, rifaximin; stable dose for at least 2 weeks prior to baseline), rectally administered formu-

Table 4. Maintenance studies: intervention details (Continued)

	is twice daily over 26 weeks)			ly administered formulation of corticosteroids or 5-ASA (stable dose for at least 2 weeks prior to baseline) was also permitted, but only data on corticosteroids are provided.	stopped at least 4 weeks prior to baseline.		lation of corticosteroids or 5-ASA (stable dose for at least 2 weeks prior to baseline) was also permitted, but only data on corticosteroids are provided.
				Use of corticosteroids at entry to the maintenance phase	Intravenous corticosteroids were required to be stopped at least 2 weeks prior to baseline.		In the maintenance study, a mandatory steroid taper algorithm was applied for patients receiving oral corticosteroids starting at baseline of the maintenance period with the dose decreased by 5 mg prednisolone equivalent per week until the dose reached 20 mg/day and then by 2.5 mg/week to 10 mg/day. Further taper was at the discretion of the investigator.
				CG 18/59 (30.5%)	Other investigational or marketed biologics with immunomodulatory properties were required to be stopped at least 3 months prior to baseline.		Use of corticosteroids at entry to the maintenance phase
				IG-T5 18/60 (30.0%)	Treatment with natalizumab or any anti-adhesion molecule was to be stopped at least 1 year prior to baseline.		CG 18/59 (30.5%)
				IG-T10 23/61 (37.7%)	Moderate to potent CYP3A inducers were prohibited within 28 days or 5 half-lives, whichever was longer, prior to first dose of study drug.		IG-T5 18/60 (30.0%)
					Moderate to potent CYP3A inhibitors were prohibited within 7 days or 5 half-lives, whichever was longer, prior to first dose of study drug.		IG-T10 23/61 (37.7%)
Rutgeerts 1999	IV 5, 10, or 20 mg/kg infliximab	IV infusion placebo	No (patients were induced with infliximab in a previous trial, one of the first on infliximab)	For patients undergoing concomitant therapy for Crohn's disease, acceptable regimens for inclusion in the study were as follows: mesalamine 8 weeks' duration and at a stable dosage for 4 weeks before screening; oral corticosteroids, 8	NR	None	For patients undergoing concomitant therapy for Crohn's disease, acceptable regimens for inclusion in the study were as follows: mesalamine 8 weeks' duration and at a stable dosage for 4 weeks before screening; oral corticosteroids, 8 weeks du-

Table 4. Maintenance studies: intervention details (Continued)

				weeks duration at a stable dosage for 2 weeks, with a maximum dosage of 40 mg/day; and 6-mercaptopurine or azathioprine, 6 months duration at a stable dosage for 8 weeks			ration at a stable dosage for 2 weeks, with a maximum dosage of 40 mg/day; and 6-mercaptopurine or azathioprine, 6 months duration at a stable dosage for 8 weeks
Rutgeerts 2012 (EXTEND)	Adalimumab subcutaneous 40 mg every other week (n = 64)	Placebo (n = 65)	Yes (patients undergoing previous treatment with infliximab or other anti-TNF agents)	More than 20% of patients on IMM IG: 9 on steroids CG: 25 on steroids	Not reported	Not reported	More than 20% of patients on IMM IG: 9 on steroids CG: 25 on steroids
Sandborn 2005a (ENACT 2)	IV 300 mg natalizumab every 4 weeks	IV infusion placebo (exact agent and dosage: NR)	IG Y 55 (33%), refractory to such therapy 32 (58%) CG Y 68 (40%), refractory to such therapy 40 (23%)	NR	NR	No protocol	For both IG and CG, concurrent therapies, including stable doses of 5-aminosalicylates, prednisolone (25 mg per day or less) or equivalent, budesonide (6 mg per day or less), azathioprine, mercaptopurine, methotrexate, and antibiotics, were permitted. Corticosteroids CG 76 (44%), IG 64 (38%) Prednisone CG 54 (32%), IG 44 (26%) Budesonide CG 24 (14%), IG 21 (13%) Immunosuppressants CG 60 (35%), IG 62 (37%) Azathioprine CG 45 (26%), IG 42 (25%)

Table 4. Maintenance studies: intervention details (Continued)

							Mercaptopurine CG 7 (4%), IG 12 (7%) Methotrexate CG 8 (5%), IG 8 (5%) Antibiotics (metronidazole, tinidazole, ciprofloxacin, and clarithromycin) CG 10 (6%), IG 15 (9%) 5-aminosalicylates (mesalamine, sulfasalazine, and olsalazine) CG 93 (54%), IG 76 (45%) ≥ 1 Corticosteroids or immunosuppressants CG 103 (60%), IG 96 (57%) Corticosteroids and immunosuppressants CG 33 (19%), IG 30 (18%)
Sandborn 2007b (CLASSIC II)	IG1 (subcutaneous adalimumab 40 mg every other week) IG2 (subcutaneous adalimumab 40 mg weekly)	Placebo not specified	All patients had received biologics during induction	NR		Patients' dosages of all concurrent drugs were required to remain constant, with the exception of corticosteroids. Steroid tapering was mandated for randomised patients at week 8 and was permitted in the open-label cohort for those patients who were responders (that is, who experienced a reduction of > 70 points in CDAI score from week 0 in induction). After week 8, dai-	NR Concomitant drug treatment (n (%)): Any corticosteroid CG 10 (56), IG1 8 (47), IG2 9 (50) Systemic corticosteroid CG 16 (33), IG1 4 (21), IG2 5 (28) Budesonide CG 4 (22), IG1 4 (21), IG2 4 (22) Any immunosuppressive agent

Table 4. Maintenance studies: intervention details (Continued)

					ly doses for randomised patients receiving prednisone 10 mg were reduced by 5 mg weekly until a dosage of 10 mg/day was reached. Thereafter, dosage was reduced by 2.5 mg weekly to the point of discontinuation. Similarly, budesonide dosage was decreased by 3 mg every week until discontinuation.	CG 3 (17), IG1 4 (21), IG2 5 (28) Azathioprine CG 1 (6), IG1 4 (21), IG2 2 (11) 6-mercaptopurine CG 1 (6), IG1 0 (0), IG2 3 (17) Methotrexate CG 1 (6), IG1 0 (0), IG2 0 (0) Crohn's-related antibiotics CG 1 (6), IG1 0 (0), IG2 (0) 5-aminosalicylates CG 8 (44), IG1 14 (74), IG2 12 (67)	
Sandborn 2012 (CERTIFI) - Maintenance	Subcutaneous ustekinumab 90 mg at weeks 8 and 16	Subcutaneous placebo at weeks 8 and 16. DOSE and AGENT NOT REPORTED	Previous experience of biologics* CG 110/110 (100%) IG 109/109 (100%) *Although the proportion of patients with previous experience not stated explicitly, it can be safely assumed to be 100% for all groups as it is a study looking at efficacy of UST in CD refractory to TNF treatment	Patients were permitted to continue receiving stable doses of drugs for the treatment of Crohn's disease, including oral prednisolone (≤ 40 mg per day) or budesonide (≤ 9 mg per day), immunomodulators (e.g. azathioprine, mercaptopurine, or methotrexate), mesalamine, and antibiotics, if they had been taking the drugs for at least the prespecified period before study entry of 3 weeks, 4 weeks, 3 weeks, and 3 weeks, respectively. Patients had not received previous therapies targeting interleukin-12 or interleukin-23.	IV Glucocorticoids were not permitted at least 3 weeks before study entry. TNF antagonists were not permitted at least 8 weeks before study entry. Natalizumab was not permitted for at least 12 months before study entry. Note – 'Study entry' here relates week 0 to week 8 induction phase. So, 8 weeks should be added to the times stated.	Not reported – no protocol	Patients were permitted to continue receiving stable doses of drugs for the treatment of Crohn's disease, including oral prednisolone (≤ 40 mg per day) or budesonide (≤ 9 mg per day), immunomodulators (e.g. azathioprine, mercaptopurine, or methotrexate), mesalamine, and antibiotics, if they had been taking the drugs for at least the prespecified period before study entry of 3 weeks, 4 weeks, 3 weeks, and 3 weeks, respectively. Patients had not received previous therapies targeting interleukin-12 or interleukin-23.

Note – ‘Study entry’ here implies entry to the original induction phase of CERTIFI (i.e. week 0 to week 8 rather than week 8 to week 36)

Proportions or numbers of patients per IG/CG per medication is NOT REPORTED for the week 8 to week 36 phase of CERTIFI (including responders maintenance study).

Note – ‘Study entry’ here implies entry to the original induction phase of CERTIFI (i.e. week 0 to week 8 rather than week 8 to week 36)

Proportions or numbers of patients per IG/CG per medication is NOT REPORTED for the week 8 to week 36 phase of CERTIFI (including responders maintenance study).

ment. It is stated in the main paper "In this 36-week, randomized, double-blind,

placebo-controlled phase 2b trial of ustekinumab

(comprising 8-week induction and 28-week maintenance

phases), we evaluated patients with moderate-to-severe Crohn's disease that was resistant to TNF antagonists."

Table 4. Maintenance studies: intervention details *(Continued)*

Sandborn 2013 (GEMINI II) - Maintenance	VCG Vedolizumab responders placebo VIG1: vedolizumab responders vedolizumab every 8 weeks (n = 154) VIG2: vedolizumab every 4 weeks (n = 154)	CG placebo + concomitant meds	Prior Crohn's disease medications, n (%) Prior anti-TNF use VCG 82 (54) VIG1 88 (57) VIG2 83 (54) CG 72 (49) IG3 364 (72) Prior anti-TNF failure ≥ 1 failure	Glucocorticoids were tapered in patients who had a response at week 6, according to a prespecified regimen; immunosuppressive agents were discontinued in patients at U.S. sites (see the Supplementary Appendix). The use of other concomitant medications remained constant.	Glucocorticoids were tapered in patients who had a response at week 6, according to a prespecified regimen; immunosuppressive agents were discontinued in patients at U.S. sites (see the Supplementary Appendix). The use of other concomitant medications remained constant.	NR	Concomitant CG placebo VCG vedolizumab responders placebo VIG1: vedolizumab responders Vedolizumab Every 8 weeks (n = 154) VIG2: vedolizumab every 4 weeks (n = 154) IG3: vedolizumab induction non-responders every 4 weeks (n = 506) Crohn's 52 disease medications, n (%)
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Table 4. Maintenance studies: intervention details (Continued)

IG3: vedolizum- ab induction non-respon- ders Every 4 weeks (n = 506)	VCG 78 (51)	Corticosteroids only
	VIG1 82 (55)	VCG 56 (37)
	VIG2 77 (50)	VIG1 59 (38)
	CG 70 (48) IG3 338 (67)	VIG2 58 (38)
	Inadequate	CG 45 (30)
	VCG 35 (45)	IG3 163 (32)
	VIG1 37 (45)	Immunosuppressives on-ly**
	VIG2 31 (40)	VCG 23 (15)
	CG 41 (59)	VIG1 27 (18)
	IG3 176 (52)	VIG2 31 (20)
	Loss of re- sponse	CG 25 (17)
	VCG 29 (37)	IG3 75 (15)
	VIG1 35 (43)	Corticosteroids and im- munosuppressives
	VIG2 33 (43)	VCG 26 (17)
	CG 22 (31)	VIG1 23 (15)
	IG3 132 (39)	VIG2 22 (14)
	Intolerance	CG 26 (18)
	VCG 14 (18)	IG3 92 (18)
	VIG1 10 (12)	No corticosteroids or im- munosuppressives
	VIG2 13 (17)	VCG 48 (31)
	CG 7 (10)	VIG1 45 (29)
	IG3 30 (9)	VIG2 43 (28)
	≥ 2 failures	CG 52 (35)
	VCG 53 (35)	IG3 176 (35)
	VIG1 46 (30)	

Table 4. Maintenance studies: intervention details (Continued)

VIG2 49 (32)

CG 42 (28)

IG3 208 (41)

Sandborn 2020d (CELEST) - Maintenance	IG-U6B Upadacitinib 6 mg immediate-release twice daily for 36 weeks (ROUTE NOT REPORTED but likely to be ORAL as it is twice daily for 36 weeks) IG-U12B Upadacitinib 12 mg immediate-release twice daily for 36 weeks (ROUTE NOT REPORTED but likely to be ORAL as it is twice daily for 36 weeks) IG-U24Q Upadacitinib 24 mg immediate-release once daily for 36 weeks (ROUTE NOT REPORTED but likely to be ORAL as it	CG-U3B Upadacitinib 3 mg immediate-release twice daily for 36 weeks (ROUTE NOT REPORTED but likely to be ORAL as it is twice daily for 36 weeks)	NOT REPORTED	NOT REPORTED per IG/CG for the maintenance phase For the preceding induction phase Patients could enter the study while receiving stable doses of aminosalicylates, oral corticosteroids (equivalent prednisone dose of ≤ 30 mg/d or budesonide of ≤ 9 mg/d), methotrexate, and/or CD-related antibiotics. Starting at week 2, patients receiving oral corticosteroids initiated a mandatory corticosteroid tapering following a protocol-specified schedule until discontinuation	Patients had to discontinue azathioprine or mercaptopurine ≥ 10 days before baseline in the induction study	Not reported – no protocol	NOT REPORTED per IG/CG for the maintenance phase For the preceding induction phase Patients could enter the study while receiving stable doses of aminosalicylates, oral corticosteroids (equivalent prednisone dose of ≤ 30 mg/d or budesonide of ≤ 9 mg/d), methotrexate, and/or CD-related antibiotics. Starting at week 2, patients receiving oral corticosteroids initiated a mandatory corticosteroid tapering following a protocol-specified schedule until discontinuation

Table 4. Maintenance studies: intervention details (Continued)

	is once daily for 36 weeks)						
Sandborn 2023 - BERG- AMOT Mainte- nance	IG Subcu- taneous etrolizumab 105 mg every 4 weeks for 52 weeks	CG Placebo every 4 weeks for 52 weeks Patients ran- domly as- signed to placebo dur- ing induc- tion who achieved CDAI-70 re- sponses re- ceived blinded place- bo during the maintenance phase after sham re-ran- domisation; these patients were not in- cluded in maintenance efficacy re- sults	All patients went through an induction phase with etrolizumab Experience with biologics at in- duction base- line is men- tioned in the in- duction tables for this study.	Patients must have had intolerance, inade- quate response, or no response to one or more of corticosteroids, im- munosuppressants, or anti-TNF therapy with- in 5 years of screening. Patients receiving stable doses of oral corticos- teroids (≤ 20 mg per day prednisone or equiva- lent or ≤ 6 mg per day budesonide) for at least 2 weeks before screen- ing or immunosuppres- sants (e.g. azathioprine, 6- mercaptopurine, and methotrexate) for at least 8 weeks before screening were includ- ed.	Patients with previous exposure to antiadhe- sion molecule therapy were not allowed. Pa- tients were excluded if they had taken one or more of adalimum- ab, certolizumab pe- gol, or infliximab with- in 8 weeks before ran- domisation or had taken ustekinumab within 14 weeks before randomi- sation.	Only the ones given as inter- ventions	Baseline use of oral cor- ticosteroids 42% in both groups. (Immunosuppres- sant use in Table 3).
Sands 2004 (ACCENT II) Maintenance (Responders)	Infliximab 5 mg/kg at weeks 14, 22, 30, 38, and 46. Beginning at week 22, patients who had a loss of response (de- fined by re- crudescence of draining fistulas, the need for addi-	Placebo at weeks 14, 22, 30, 38, and 46. Beginning at week 22, patients who had a loss of response (de- fined by re- crudescence of draining fistulas, the need for addi- tional thera-	NR However, ex- clusion crite- ria state that if patients had received inflix- imab previous- ly, they were ex- cluded from the study. No men- tion of other bi- ologics	Concurrent therapies for Crohn's disease, includ- ing stable doses of 5- aminosalicylates, oral corticosteroids, azathio- prine, mercaptopurine, mycophenolate mofetil, methotrexate, and an- tibiotics, were permitted (i) 5-Aminosalicylates (oral or rectal) (n (%))	NR	NR	Concurrent therapies for Crohn's disease, includ- ing stable doses of 5- aminosalicylates, oral corticosteroids, azathio- prine, mercaptopurine, mycophenolate mofetil, methotrexate, and antibi- otics, were permitted (i) 5-Aminosalicylates (oral or rectal) (n (%))

Table 4. Maintenance studies: intervention details (Continued)

	tional therapy for persistent or worsening luminal disease activity, the need for a surgical procedure for Crohn's disease, or the discontinuation of the study medication owing to a perceived lack of efficacy) were eligible to cross over to maintenance treatment with infliximab 10 mg/kg	py for persistent or worsening luminal disease activity, the need for a surgical procedure for Crohn's disease, or the discontinuation of the study medication owing to a perceived lack of efficacy) were eligible to cross over to maintenance treatment with infliximab 5 mg/kg		CG 49/99 (49%) IG 41/96 (43%) (ii) Mercaptopurine or azathioprine (n (%)) CG 35/99 (35%) IG 29/96 (30%) (iii) Methotrexate (n (%)) CG 2/99 (2%) IG 1/96 (1%) (iv) Antibiotics (n (%)) CG 26/99 (26%) IG 28/96 (29%) (v) Corticosteroids (va) Any (n (%)) CG 30/99 (30%) IG 25/96 (26%) (vb) > 20 mg per day (n (%)) CG 8/99 (8%) IG 8/96 (8%)			CG 49/99 (49%) IG 41/96 (43%) (ii) Mercaptopurine or azathioprine (n (%)) CG 35/99 (35%) IG 29/96 (30%) (iii) Methotrexate (n (%)) CG 2/99 (2%) IG 1/96 (1%) (iv) Antibiotics (n (%)) CG 26/99 (26%) IG 28/96 (29%) (v) Corticosteroids (va) Any (n (%)) CG 30/99 (30%) IG 25/96 (26%) (vb) > 20 mg per day (n (%)) CG 8/99 (8%) IG 8/96 (8%)
Schreiber 2007 (PRECISE 2)	Subcutaneous injection certolizumab 400 mg	Subcutaneous injection placebo (agent: NR)	Patients who had received any certolizumab pegol, who had received an anti-TNF agent or other biologic therapy within 3 months before enrollment, or who had a severe	Glucocorticoids only: CG 44 (21%) IG 47 (22%) Immunosuppressive agents only: CG 52 (25%) IG 59 (27%)	NR	NR	Permitted concomitant therapies for Crohn's disease were stable doses of 5-aminosalicylates, 30 mg or less of prednisolone per day (or equivalent), azathioprine, 6-mercaptopurine, methotrexate, and antibiotics

Table 4. Maintenance studies: intervention details (Continued)

			hypersensitivity reaction or no clinical response after initial dosing with an anti-TNF were excluded	Glucocorticoids plus immunosuppressive agents: CG 34 (16%) IG 28 (13%) Neither glucocorticoids nor immunosuppressive agents CG 80 (38%) IG 81 (38%)			
VanAssche 2012 (SWITCH)	IG-ADA 80 mg subcutaneous adalimumab at inclusion and 40 mg subcutaneous adalimumab every other week for 54 weeks. Patients with disease flare were allowed dose intensification – step-up to 40 mg every week. If complete loss of response, patients were allowed to switch over to the alternative treatment. Short (4-week) courses of steroids were	CG-IFX Continue infliximab 5 mg/kg intravenously at 8-week intervals for 56 weeks. Patients with disease flare were allowed dose intensification – decrease of dosing interval with 2-week decrements. Short (4-week) courses of steroids were also allowed per protocol	Patients with previous exposure to ADA, and receiving IFX doses > 5 mg/kg intravenously were excluded	Azathioprine CG-IFX 0/37 (0.00%) IG-ADA 6/36 (16.67%) Methotrexate CG-IFX 2/37 (5.41%) IG-ADA 0/36 (0.00%)	IFX doses > 5 mg/kg intravenously were excluded	None	Azathioprine CG-IFX 0/37 (0.00%) IG-ADA 6/36 (16.67%) Methotrexate CG-IFX 2/37 (5.41%) IG-ADA 0/36 (0.00%)

Table 4. Maintenance studies: intervention details (Continued)

		also allowed per protocol					
Vermeire 2017 FITZROY - maintenance responders	IG1 Filgotinib 100 mg oral once daily IG2 Filgotinib 200 mg oral once daily	Matching placebo oral once daily (AGENT NOT REPORTED). Placebo was presented as orally admin- istered brown film-coated tablets that were identi- cal in appear- ance to fil- gotinib and contained the same excipi- ents	NOT REPORTED However, one of the exclusion criterion states that previous JAK inhibitor therapy was not allowed. There- fore, no patient had previous JAK inhibitors (CG 0/15, IG1 0/30, IG2 0/31) but unable to comment on TNF inhibitors and hence on biologics as a whole	Note – the time points here apply to the origi- nal induction study (i.e. screening means week 0 rather than week 10, where the latter is where the maintenance phase began) Concurrent treatment with oral steroids ($\leq$ 30 mg prednisolone or budesonide $\leq$ 9 mg per day) or probiotics was allowed if dosages had been stable for at least 2 weeks be- fore the first dose of study drug . Concur- rent treatment with Crohn's disease-re- lated antibiotics , or with mesalazine or ol- salazine , was permit- ted if dosages had been stable for at least 4 weeks before the first dose of study drug . Previous exposure to sulfasalazine was al- lowed but had to be discontinued at least 4 weeks before screening in male patients, where- as immunomodula- tors (e.g. thiopurines and methotrexate) must have been discontin- ued at least 25 days be- fore the first dose of the study drug	Note – the time points here apply to the origi- nal induction study (i.e. screening means week 0 rather than week 10, where the latter is where the maintenance phase began) Previous exposure to sulfasalazine was al- lowed but had to be discontinued at least 4 weeks before screening in male patients. Immunomodulators must have been discon- tinued at least 25 days before the first dose of the study drug. Tube feeding, defined formula diets, or to- tal parenteral alimen- tation was not allowed during the study. IV corticosteroids were to be discontinued 14 days prior to screening. Enemas or suppos- itories other than those required for colonoscopy were to be discontinued 7 days pri- or to screening. Cyclosporine, mycophenolate, tacrolimus, or interfer- on were to be discontin-	If using paracetamol at baseline , it should not be discontinued Contracep- tion	Note – the time points here apply to the origi- nal induction study (i.e. screening means week 0 rather than week 10, where the latter is where the maintenance phase began) Concurrent treatment with oral steroids ($\leq$ 30 mg prednisolone or budesonide $\leq$ 9 mg per day) or probiotics was allowed if dosages had been stable for at least 2 weeks before the first dose of study drug . Con- current treatment with Crohn's disease-relat- ed antibiotics , or with mesalazine or olsalazine , was permitted if dosages had been stable for at least 4 weeks before the first dose of study drug . Previous expo- sure to sulfasalazine was allowed but had to be discontinued at least 4 weeks before screening in male patients, where- as immunomodulators (e.g. thiopurines and methotrexate) must have been discontinued at least 25 days before the first dose of the study drug. Proportion of patients on each medication – NOT REPORTED for the mainte- nance phase

Table 4. Maintenance studies: intervention details *(Continued)*

Proportion of patients on each medication – NOT REPORTED for the maintenance phase

ued at least **10 weeks prior** to screening.

Any **prior lymphocyte depleting agent** (e.g. alemtuzumab) was **prohibited**. **Lymphocyte apheresis or selective monocyte granulocyte apheresis** was to be discontinued **12 months prior** to screening.

Previous faecal microbiota transplant and stem cell transplants were **prohibited**.

Investigational chemical agents were to be discontinued **4 weeks before** screening.

Previous treatment with biological investigational medicinal products including murine, chimeric or humanised monoclonal antibodies or a chemokine receptor blocker within less than **5 half-lives prior** to baseline. **Previous treatment with a Janus kinase inhibitor** is **prohibited**.

Live vaccines must not have been administered **within 90 days** or **an attenuated vaccine within 30 days** of first dose of study drug as well as **during the study** duration **until 6 weeks fol-**

Table 4. Maintenance studies: intervention details (Continued)

				lowing last intake of GLPG0634.			
				Patients currently on any therapy for chronic infection (such as pneumocystis, CMV, herpes simplex, herpes zoster, or atypical mycobacteria) were excluded from the study.			
Vermeire 2021 (VISIBLE 2)	Vedolizumab 108 mg SC	SC placebo (agent: NR)	Patients with previous exposure to approved or investigational anti-integrin antibodies (eg, natalizumab, efalizumab, etrolizumab, abrilumab (A mg 181)), anti-mucosal addressing cell adhesion molecule-1 antibodies, or rituximab and vedolizumab were excluded	Anti-TNF naïve CG 64 (47.8%) IG 110 (40.0%) Prior anti-TNF use: CG 71 (53.0%) IG 168 (61.1%) Prior use of IMM (only): CG 4 (3.0%) IG 16 (5.8%) Prior use of oral CS (only): CG 23 (17.2%) IG 67 (24.4%) Prior use of oral CS and IMM: CG 103 (76.9%) IG 189 (68.7%) Prior anti-TNF use: CG 71 (53.0%) IG 168 (61.1%)	NR	NR	Only IMM: CG 34 (25.4%) IG 51 (18.5%) Only CS: CG 31 (23.1%) IG 64 (23.3%) IMM and CS: CG 13 (9.7%) IG 31 (11.3%)
Vermeire 2025 - DIVERSI-	Filgotinib 100 mg	Placebo (after receiving	All patients induced with the	About 50% of all patients had concomitant	None	Vaccines, contraceptives,	Only IMM (n):



Table 4. Maintenance studies: intervention details *(Continued)*

TY (maintenance)	Filgotinib 200 mg	filgotinib 100 mg at induction)	filgotinib at induction phase	use of systemic corticosteroids, immunosuppressants or both	fertility medication	CG1 9 CG2 10 IG1 23 IG2 24 Only CS: CG1 14 CG2 15 IG1 27 IG2 34 IMM and CS: CG1 4 CG2 4 IG1 8 IG2 8
	Patients received their induction dose or placebo orally, once daily	Placebo (after receiving filgotinib 200 mg at induction)				
		Patients received their induction dose or placebo orally, once daily				
Volkers 2017 - SIMILAR	Infliximab biosimilar – patients switched from infliximab-biological and received 4 to 6 doses of 5 mg/kg to 10 mg/kg of infliximab-biosimilar during the 30-week study period (Route and Regimen NOT REPORTED)	Infliximab biological – Patients continued infliximab-biological and received 4 to 6 doses of 5 mg/kg to 10 mg/kg of infliximab-biosimilar during the 30-week study period (Route and Regimen NOT REPORTED)	NR	NR	NR	NR

Table 4. Maintenance studies: intervention details (Continued)

Watanabe 2012 - Maintenance	Adalimumab 40 mg, every other week (n = 25)	Placebo (n = 25)	Yes (all exposed at induction)	More than 20% of patients on IMM IG: 3 on steroids CG: 5 on steroids	Not reported	Not reported	More than 20% of patients on IMM IG: 3 on steroids CG: 5 on steroids
Watanabe 2020 - Maintenance	Vedolizumab 300 mg intravenously at week 14, then every 8 weeks until week 54	Placebo intravenously at week 14, then every 8 weeks until week 54 (AGENT and DOSE NOT REPORTED)	Reported as 'Prior anti-TNF (alpha) treatment' (numbers (percentage)) : (i) No CG 5/12 (41.7%) IG 4/12 (33.3%) (ii) Yes CG 7/12 (58.3%) IG 8/12 (66.7%) Prior anti-TNF (alpha) failure (numbers (percentage)) : (i) Overall CG 7/12 (58.3%) IG 8/12 (66.7%) (ii) Inadequate response CG 1/12 (8.3%) IG 3/12 (25.0%) (iii) Loss of response CG 6/12 (50.0%) IG 5/12 (41.7%)	Patients were allowed concomitant treatment with oral 5-aminosalicylic acid, probiotics, antibiotics for CD, oral corticosteroids, immunomodulators and enteral nutrients (≤ 900 kcal/day) under specific conditions Concomitant medication for CD at baseline (numbers (percentage)) : (i) Enteral nutrient CG 5/12 (41.7%) IG 8/12 (66.7%) (ii) 5-Aminosalicylic acid CG 11/12 (91.7%) IG 8/12 (66.7%) (iii) Oral corticosteroids AND no immunomodulator CG 3/12 (25.0%) IG 2/12 (16.7%) (iv) No oral corticosteroids OR immunomodulator	These patients were excluded from the study (otherwise discontinued medication NOT REPORTED): (i) Patients who started oral 5-ASAs, probiotics, antibiotics for CD treatment, or oral corticosteroids (≤ 30 mg/day) within 13 days before the first dose of the study drug. Or patients who changed dosage of or discontinued these drugs within 13 days before the first dose of the study drug if the patient had used these drugs for more than 14 days before the first dose of the study drug (ii) Patients who had received 5-ASA, corticosteroid enemas/suppositories, corticosteroid IV infusion, oral corticosteroid at > 30 mg/day, drugs for diarrhoea-predominant irritable bowel syndrome, or Chinese herbal medicine for the CD treatment (e.g. Daikenchuto) within	NOT REPORTED – no protocol found	NOTE – These are the same as medications up to study beginning Patients were allowed concomitant treatment with oral 5-aminosalicylic acid, probiotics, antibiotics for CD, oral corticosteroids, immunomodulators and enteral nutrients (≤ 900 kcal/day) under specific conditions Concomitant medication for CD at baseline (numbers (percentage)) : (i) Enteral nutrient CG 5/12 (41.7%) IG 8/12 (66.7%) (ii) 5-Aminosalicylic acid CG 11/12 (91.7%) IG 8/12 (66.7%) (iii) Oral corticosteroids AND no immunomodulators CG 3/12 (25.0%) IG 2/12 (16.7%)

Table 4. Maintenance studies: intervention details *(Continued)*

(iv) Intolerance	CG 3/12 (25.0%)	13 days before the first dose of the study drug	(iv) No oral corticosteroids OR immunomodulators
	IG 1/12 (8.3%)		
CG 0/12 (0.0%)	(v) Oral corticosteroids AND immunomodulator	(iii) Patients who had received AZA, 6-MP, or MTX within 27 days before the first dose of the study drug. However, this did not apply to patients who had used these drugs for more than 83 days before the first dose of the study drug and continued the steady dose of the drugs for more than 27 days before the first dose of the study drug	CG 3/12 (25.0%)
IG 0/12 (0.0%)	CG 0/12 (0.0%)		IG 1/12 (8.3%)
Number of drugs of anti-TNF (alpha) failure (numbers (percentage)):	IG 3/12 (25.0%)	(iv) Patients who had received cyclosporine, tacrolimus, tofacitinib or any study drugs of low molecular compound for CD treatment within 27 days before the first dose of the study drug	(v) Oral corticosteroids AND immunomodulators
(i) None			CG 0/12 (0.0%)
CG 5/12 (41.7%)			IG 3/12 (25.0%)
IG 4/12 (33.3%)			
(ii) One			
CG 5/12 (41.7%)			
IG 2/12 (16.7%)			
(iii) Two			
CG 2/12 (16.7%)			
IG 6/12 (50.0%)			
Worst prior treatment failure (numbers (percentage)):		(v) Patients who had received adalimumab within 27 days before the first dose of the study drug or any biologics other than adalimumab within 55 days before the first dose of the study drug. However, this did not apply to patients who had received localised injections of these drugs (e.g. intraocular injection for treatment of age-related macular degeneration)	
(i) Prior anti-TNF (alpha) failure			
CG 7/12 (58.3%)			
IG 8/12 (66.7%)			
(ii) Prior immunomodula-			

Table 4. Maintenance studies: intervention details *(Continued)*

**tors failure but
not anti-TNF
(alpha) failure**

CG 2/12 (16.7%)

IG 3/12 (25.0%)

vi) Patients who had received any **live-vaccinations within 27 days before the first dose of the study drug**

(vii) Patients who had a **history or a complication of alcohol dependence or illicit drug use within one year before the first dose of the study drug**

(viii) Patients who had received **central venous nutrition therapy or enteral total nutrition therapy, or fasted within 20 days before the first dose of the study drug**

(ix) Patients who had received an **enteral nutrient of > 900 kcal/day or who had started an enteral nutrient of ≤ 900 kcal/day within 20 days before the first dose of the study drug**. Or patients who **changed dosage of or discontinued the enteral nutrient within 20 days before the first dose of the study drug if the patient had received an enteral nutrient of ≤ 900 kcal/day more than 21 days before the first dose of the study drug**

NOTE – First dose of the study drug refers to Week 0 dose during the induction phase (i.e. not



Table 4. Maintenance studies: intervention details (Continued)

					Week 0 of maintenance phase, which is Week 14)		
Young 2024 - iBaSS	Reference adalimumab according to existing regimen	SB5 adalimumab biosimilar according to existing regimen	All patients were induced with adalimumab or biosimilar	Currently taking concomitant immunosuppressive therapy at baseline, n (%)	NR	None	NR
			Other biologics:	IG 24 (53.3)			
			IG n = 14 (31.1%),	CG 20 (46.5)			
			CG n = 11 (25.6%)				

5-ASA: 5-aminosalicylic acid
6-MP: 6-mercaptopurine
ada: adalimumab
AZA: azathioprine
CD: Crohn's disease
CDAI: Crohn's Disease Activity Index
CG: control group
CS: corticosteroids
eow: every other week
ifx: infliximab
IG: intervention group
IMM: immunomodulators
IV: intravenous
JAK: Janus kinase
N/A: not applicable
nat: natalizumab
NR: not reported
pl: placebo
SC: subcutaneous
TNF: tumour necrosis factor
UC: ulcerative colitis
Upa: upadacitinib
ust: ustekinumab
Ved/ved: vedolizumab

Table 5. Induction studies: critical and important outcomes

Study ID	I.a Clinical remission	I.b Clinical response	I.c Endoscopic remission	I.d Withdrawals due to adverse events	I.e Serious adverse events	I.f Total numbers of participants with adverse events
Chen 2020	IG 38/102, CG 7/103 at 4 weeks	Clinical response (decrease in CDAI $\geq$ 70 from baseline): CG 28/103, IG 69/102	NR	IG 3/102, CG 5/103	IG 2/102, CG 1/103	IG 38/102, CG 38/103
Colombel 2007 (CHARM) - Induction (non-responders)	Week 12: CG (Plb) NR/91, IG1 (Ada eow) 25/88 (calculated from 28%), IG2 (Ada weekly) NR/100 <i>Need to contact authors to get further details on non-responders clinical remission.</i>	<i>Need to contact authors for more details on clinical response CR-70 in non-responders</i>	NR	<i>Need to contact authors to separate information for non-responders and responders</i>	<i>Need to contact authors to separate information for non-responders and responders</i>	<i>Need to contact authors to separate information for non-responders and responders</i>
Colombel 2010 (SONIC)	By week 26 CG (AZA) 54/170, IG1 (IFX) 81/169, IG2 (Combo) 102/169 By week 50 CG 41/170, IG1 64/169, IG2 80/169	<i>CDAI 70 Response (wk 26):</i> CG 71/170, IG1 95/169, IG2 113/169 <i>CDAI 70 Response (wk 50):</i> CG 56/170, IG1 75/169, IG2 88/169 <i>CDAI 100 response (wk 26):</i> CG 64/170, IG1 92/169, IG2 105/169 <i>CDAI 100 Response (wk 50):</i> CG 47/170, IG1 70/169, IG2 85/169	Mucosal healing (wk 26): CG 18/170, IG1 28/169, IG2 47/169 <i>Ran-domised participants not assessed with endoscopy were counted as failures</i>	At week 54: CG 42/170, IG1 29/169, IG2 37/169	At week 54: CG 43/170, IG1 39/169, IG2 27/169	Through week 30: CG 63/170, IG1 41/169, IG2 39/169 Through week 50: CG 69/170, IG1 53/169, IG2 53/169
D'Haens 1999 (refractory CD)	NR	NR as dichotomous CDAI mean (SEM) at week 4: CG 261.3 (33.3) IG1 (5 mg): 122.8 (26.1) IG2 (10 mg): 220.5 (63.4) IG3 (20 mg): 161.9 (34.5)	NR as dichotomous CDEIS mean (SEM) at week 4: CG 7.5 (5.4) IG1 (5 mg): 6.4 (5.1) IG2 (10 mg): 4.3 (5.4)	The follow-up ileo-colonoscopy could not be performed in 2 CG participants: 1 due to surgery, 1 due to abdominal pain. In 3 participants with ileitis, the ileocecal valve	NR (Except for 1 infliximab-treated participant with exclusive rectosigmoidal Crohn's disease who developed a new rectal stricture at the site of earlier severe ulceration. This stricture needed	NR

Table 5. Induction studies: critical and important outcomes (Continued)

Table 3: Induction studies: critical and important outcomes				(continued)	IG3 (20 mg): 5.2 (2.8)	could not be intubated so that the ileum could not directly be visualised, but it is not explained in which groups they belonged.	multiple endoscopic dilations in combination with stool softeners.)
D'Haens 2008	Week 14 CG (corticosteroids) 21/66 IG (combined) 43/67 (taken from graph) After this time, infliximab was given to the control group and therefore no further data have been reported	NR	NR	At end of study CG 14/66, IG 14/67	CG 19/66, IG 20/67 Not known whether at end of the trial – we have assumed that these occurred at 26 weeks.	Need to contact authors for number of participants with AEs as Table 2 only provides number of adverse events. UNCLEAR	
D'Haens 2022 (ADVANCE)	CG (pl) 44/186 IG1 (ris600) 152/373 IG2 (ris1200) 141/372	NR	CG (pl) 16/186 IG1 (ris600) 81/373 IG2 (ris1200) 81/372	CG 14/186 IG1 9/373 IG2 7/372	Combined severe and serious, serious infections and death CG 55/186 IG1 52/373 IG2 34/372	CG 105/186 IG1 210/373 IG2 191/372	
Duan 2013	Clinical remission at week 26 – decrease in CDAI < 70 points but a total CDAI score of < 150 CG (AZA) 3/8, IG1 (IFX) 4/8, IG2 (Combo - AZA + IFX) 5/8	Clinical efficacy at week 26 – decrease in CDAI > 70 points or a decrease of ≥ 25% of the total CDAI score CG (AZA) 2/8, IG1 (IFX) 2/8, IG2 (Combo - AZA + IFX) 2/8	"Fully healed": CG (AZA) 3/8, IG1 (IFX) 1/8, IG2 (Combo - AZA + IFX) 4/8 Basically healing: CG (AZA) 1/8, IG1 (IFX) 1/8, IG2 (Combo - AZA + IFX) 1/8	CG (AZA) 1/8, IG1 (IFX) 1/8, IG2 (Combo - AZA + IFX) 0/8	NR	NR	

Table 5. Induction studies: critical and important outcomes (Continued)

Feagan 2008c	Day 57 (CDAI $\leq$ 150) CG 21%, n = 12 IG1 (0.5 mg): 30%, n = 19 IG2 (2 mg): 37%, n = 24	Day 57 $\geq$ 70-point decrement in CDAI from baseline CG 41%, n = 24 IG1 (0.5 mg): 49%, n = 30 IG2 (2 mg): 53%, n = 34 Day 57 $\geq$ 100-point decrement in CDAI from baseline CG 31%, n = 18 IG1 (0.5 mg): 43%, n = 27 IG2 (2 mg): 47%, n = 31 Day 180 $\geq$ 100-point decrement in CDAI from baseline CG 27%, n = 16 IG1 (0.5 mg): 32%, n = 20 IG2 (2 mg): 43%, n = 28 (Days 8, 15, 29, 43 illustrated in a graph)	NR	CG 0 IG1 (0.5 mg): 1 (aggravated CD) IG2 (2 mg): 1 (aggravated CD)	SAEs: CG 10 IG1 (0.5 mg): 6 IG2 (2 mg): 10 BUT "Severe AEs were experienced by 37(29%) MLN0002-treated and 19 (33%) placebo-treated patients"	CG 50 IG1 (0.5 mg): 58 IG2 (2 mg): 59
Feagan 2015a (UNITI-1)	Clinical remission (CDAI < 150) at week 8: CG (Plb) 18/247, IG1 (Ust 130) 39/245, IG2 (Ust 6) 52/249	100-point clinical response at week 6: CG (Plb) 53/247, IG1 (Ust 130) 84/245, IG2 (Ust 6) 84/249 100-point clinical response at week 8: CG (Plb) 50/247, IG1 (Ust 130) 82/245, IG2 (Ust 6) 94/249 70-point clinical response at week 3: CG (Plb) 67/247, IG1 (Ust 130) 94/245, IG2 (Ust 6) 101/249 70-point clinical response at week 6: CG (Plb) 75/247, IG1 (Ust 130) 113/245, IG2 (Ust 6) 109/249	NR	CG (Plb) 10/247, IG1 (Ust 130) 9/245, IG2 (Ust 6) 14/249 <i>Need to contact authors to describe reasons for withdrawal and loss to follow-up further.</i>	CG (Plb) 15/247, IG1 (Ust 130) 12/245, IG2 (Ust 6) 18/249	CG (Plb) 159/247, IG1 (Ust 130) 159/245, IG2 (Ust 6) 164/249
Feagan 2015b(UNITI-2)	Clinical remission (CDAI < 150) at week 8: CG (Plb) 41/210, IG1 (Ust 130)	100-point clinical response at week 6: CG (Plb) 60/210, IG1 (Ust 130) 108/209, IG2 (Ust 6) 116/209	NR	CG (Plb) 12/210, IG1 (Ust 130) 9/209, IG2 (Ust 6) 2/209	CG (Plb) 12/210, IG1 (Ust 130) 10/209, IG2 (Ust 6) 6/209	CG (Plb) 113/210, IG1 (Ust 130) 106/209,

Table 5. Induction studies: critical and important outcomes (Continued)

64/209, IG2 (Ust 6) 84/209		100-point clinical response at week 8:	<i>Need to contact authors to describe reasons for withdrawal and loss to follow-up further.</i>	IG2 (Ust 6) 115/209		
		CG (Plb) 67/210, IG1 (Ust 130) 99/209, IG2 (Ust 6) 121/209				
		70-point clinical response at week 3:				
		CG (Plb) 66/210, IG1 (Ust 130) 103/209, IG2 (Ust 6) 106/209				
		70-point clinical response at week 6:				
		CG (Plb) 81/210, IG1 (Ust 130) 123/209, IG2 (Ust 6) 135/209				
Feagan 2017	Clinical remission defined as 'CDAI score < 150'	Clinical response defined as "either CDAI score < 150 or a CDAI reduction of >= 100 from baseline"	Endoscopic remission defined as "CDEIS score of <=4 at week 12 (<=2 for participants with initial isolated ileitis)"	Through week 12	Through week 12	Through week 12
	Week 4	Week 4		Withdrawals per IG/CG (number of participants withdrawing/total participants in given group (percentage)): CG 6/39 (15.38%), IG-R200 6/41 (14.63%), IG-R600 1/41 (2.44%)	Serious adverse events (number of participants/total participants in given group (percentage)):	Total number of participants with adverse events (number of participants/total participants in given group (percentage)):
	In clinical remission (number of participants/total participants in given group (percentage)): CG 3/39 (7.7%), IG-R200 4/41 (9.8%), IG-R600 8/41 (19.5%)	In clinical response CG 6/39, IG-R200 10/41, IG-R600 13/41			CG 12/39 (31%), IG-R200 9/41 (22%), IG-R600 3/41 (7%)	
	Week 8	Week 8	Week 12		A serious adverse event was defined as any that results in death, is immediately life-threatening, results in persistent or significant disability or incapacity, requires or prolongs patient hospital stay, is a congenital anomaly or birth defect, or is an important medical event, based upon appropriate medical judgement that might jeopard-	
	In clinical remission (number of participants/total participants in given group (percentage)): CG 1/39 (2.6%), IG-R200 7/41 (17.1%), IG-R600 10/41 (24.4%)	In clinical response CG 5/39, IG-R200 13/41, IG-R600 13/41	In endoscopic remission (number of participants/total participants in given group (percentage)) CG 1/39 (3%), IG-R200 6/41 (15%), IG-R600 8/41 (20%)			CG 32/39 (82%), IG-R200 32/41 (78%), IG-R600 31/41 (76%)
	Week 12 (primary)	Week 12				
	In clinical remission (number of patients/total patients in given group (percentage), Difference vs placebo (95% CI, P value)):	In clinical response CG 8/39, IG-R200 15/41, IG-R600 17/41				

Table 5. Induction studies: critical and important outcomes (Continued)

	CG 6/39, IG-R200 10/41, IG-R600 15/41				dise the patient and might require medical or surgical intervention
					Note: Severe adverse events were reported as a subcategory of adverse events whereas serious adverse events were a category of their own. It is unclear how many participants had both severe and serious adverse events.
Ghosh 2003	Clinical remission (CDAI < 150) Week 6 (PRIMARY END-POINT): CG (Plb) 17/63, IG1 (Nat 1 of 3) 20/68, IG2 (Nat 2 of 3) 29/66, IG3 (Nat 2 of 6) 16/51	70-point clinical response Week 2 CG (Plb) 19/63, IG1 (Nat 1 of 3) 31/68, IG2 (Nat 2 of 3) 36/66, IG3 (Nat 2 of 6) 22/51 Week 4 CG (Plb) 18/63, IG1 (Nat 1 of 3) 32/68, IG2 (Nat 2 of 3) 41/66, IG3 (Nat 2 of 6) 27/51 Week 6 CG (Plb) 24/63, IG1 (Nat 1 of 3) 40/68, IG2 (Nat 2 of 3) 47/66, IG3 (Nat 2 of 6) 29/51 Week 8 CG (Plb) 22/63, IG1 (Nat 1 of 3) 38/68, IG2 (Nat 2 of 3) 44/66, IG3 (Nat 2 of 6) 28/51 Week 12 CG (Plb) 27/63, IG1 (Nat 1 of 3) 34/68, IG2 (Nat 2 of 3) 40/66, IG3 (Nat 2 of 6) 33/51	NR	CG (Plb) 7/63, IG1 (Nat 1 of 3) 1/68, IG2 (Nat 2 of 3) 5/66, IG3 (Nat 2 of 6) 4/51 <i>These numbers were based on withdrawal due to lack of efficacy and adverse events. Authors need to be contacted to provide clear numbers on withdrawal numbers and reasons.</i>	CG (Plb) 7/63, IG1 (Nat 1 of 3) 7/68, IG2 (Nat 2 of 3) 6/66, IG3 (Nat 2 of 6) 6/51 CG (Plb) 51/63, IG1 (Nat 1 of 3) 50/68, IG2 (Nat 2 of 3) 57/66, IG3 (Nat 2 of 6) 40/51
Gordon 2001	Clinical remission (CDAI < 150) at week 2 (primary endpoint): CG (Plb) 1/12, IG (Nat) 7/18	At week 2: CG mean CDAI reduction of 11 IG mean CDAI reduction of 45 (unknown SD) from 258	NR	CG (Plb) 2/12, IG (Nat) 1/18 <i>Need to contact the authors for further clarity on</i>	<i>Need to contact authors for full clarity on what serious adverse events occurred.</i> <i>Need to contact the author as information provided only speaks about most common</i>

adverse events and not all events.

Need to contact authors for more information on the clinical response observed in each group.

Table 5. Induction studies: critical and important outcomes (Continued)

	IG1 (BI 695501) 46/72, CG (Ada) 54/75	Week 24 (secondary end-point): IG1 (BI 695501) 55/72, CG (Ada) 59/75 <i>Exploratory endpoints included other weeks – 12 and 48</i>		<i>Need to contact authors for further clarity on 'withdrawal by patient', 'lost to follow-up' and 'physician decision' reasons for discontinuation of study.</i>	Week 24 to 56: IG1 (BI 695501) 2/72, CG (Ada) 9/75	45/72, CG (Ada) 42/75 Week 24 to 56: IG1 (BI 695501) 31/72, CG (Ada) 34/75
Lemann 2006	Week 12: CG (Plb) 22/58 IG (IFX) 43/57 Week 24 (primary): CG (Plb) 17/58 IG (IFX) 32/57 Week 52: CG (Plb) 13/58 IG (IFX) 23/57	NR	CDEIS decrease from baseline (median (interquartile range)): Week 24: CG (Plb) 1.2 (-1.5 to 4.4) IG (IFX) 6.9 (4.1 to 9.5)	CG (Plb) 5/58, IG (IFX) 2/57	CG (Plb) 3/58, IG (IFX) 3/57 Table 2 describes "Serious adverse event in 5 cases (infliximab, n=3; placebo, n=3)"	CG (Plb) 28/58, IG (IFX) 29/57
Allez 2023 (TRIDENT-Part 1)	Week 12 CG (placebo) 7/72, IG (tesnatilimab) 15/73	> 100 CDAI reduction Week 12 CG (placebo) 11/72 IG (tesnatilimab) 21/73	Week 12 CG (placebo) 5/72 IG (tesnatilimab) 10/73	CG (placebo) 14/72 IG (tesnatilimab) 25/73 (of which one was death)	CG (placebo) 1/72 IG (tesnatilimab) 8/73 (of which one was death)	CG (placebo) 24/72 IG (tesnatilimab) 45/73 (of which one was death)
Allez 2023 (TRIDENT-Part 2)	Week 12 CG (placebo) 7/48, IG1 (tesnatilimab) 35/148 IG2 (ustekinumab) 25/47	> 100 CDAI reduction Week 12 CG (placebo) 11/48 IG1 (tesnatilimab) 56/148 IG2 (ustekinumab) 34/47	Week 12 CG (placebo) 6/48 IG1 (tesnatilimab) 6/148 IG2 (ustekinumab) NR	G (placebo) 8/48 IG1 (tesnatilimab) 31/148 IG2 (ustekinumab) 5/47	CG (placebo) 3/48 IG1 (tesnatilimab) 9/148 IG2 (ustekinumab) 2/47	CG (placebo) 19/72 IG1 (tesnatilimab) 88/73 IG2 (ustekinumab) 22/47
Panac-cione 2022 (MOTIVATE)	CG (pl) 37/187 IG1 (ris600) 81/191 IG2 (ris1200) 77/191	NR	CG (pl) 8/187 IG (ris) 76/382	CG (pl) 2/187 IG1 (ris600) 5/191 IG2 (ris1200) 16/191	CG (pl) 24/187 IG1 (ris600) 9/191 IG2 (ris1200) 24/191	CG (pl) 124/187 IG1 (ris600) 90/191

Table 5. Induction studies: critical and important outcomes (Continued)

						IG2 (ris1200) 113/191
Panes 2017 Induction	Week 8: CG (Plb) 33/92, IG1 (Tof 5) 37/86, IG2 (Tof 10) 37/86, IG3 (Tof 15) 7/16	<u>≥ 70-point decrease clinical response</u> Week 8: CG (Plb) 56/92 IG1 (Tof 5) 65/86 IG2 (Tof 10) 64/86 IG3 (Tof 15) 13/16 <u>≥ 100-point decrease clinical response</u> Week 8: CG (Plb) 49/92 IG1 (Tof 5) 60/86 IG2 (Tof 10) 59/86 IG3 (Tof 15) 12/16	NR	<i>Need to contact authors to:</i> - confirm contents of reasons such as 'no longer willing to participate', 'study terminated by sponsor', 'other reasons'; and - confirm discontinuation due to AEs as there are discrepancies between Table 3 and Figure 2 flowchart.	Participants with serious and severe adverse events CG (Plb) 11/92, IG1 (Tof 5) 8/86, IG2 (Tof 10) 17/86, IG3 (Tof 15) 1/16 <i>Need to contact authors for total number of participants that experienced serious and severe adverse events.</i>	CG (Plb) 55/92, IG1 (Tof 5) 50/86, IG2 (Tof 10) 52/86, IG3 (Tof 15) 9/16
Present 1999	Complete response (defined as the absence of any draining fistulas at two consecutive visits) CG 4/31 IG1 17/31 IG2 12/32	A reduction of 50 percent or more from baseline in the number of draining fistulas observed at two or more consecutive study visits CG 8/31 IG1 21/31 IG2 18/32	NR	CG 0 IG1 1 IG2 1	5 in total IG1 1/31 IG2 4/31 CG 0/32	CG 12/31 IG1 10/31 IG2 26/32
Sandborn 2001a	Clinical remission (CDAI < 150) at each visit Week 24 Placebo 8 weekly 1/27 Placebo 12 weekly 1/31 CDP57 8 weekly 6/57 CDP57 12 weekly 6/54	Week 2 CDAI70 response 15/58 50/111 Week 2 CDAI100 response 8/58 33/111	NR	Placebo 8 weekly, 10 mg/kg CDP571 8 weekly, placebo 12 weekly, and 10 mg/kg CDP57112 weekly <i>Withdrawal due to adverse events information was combined in Table 2. Need to contact au-</i>	Placebo 8 weekly, 10 mg/kg CDP571 8 weekly, placebo 12 weekly, and 10 mg/kg CDP571 12 weekly <i>Need to contact authors for no. of patients experiencing serious adverse events per group as information was</i>	Placebo 8 weekly, 10 mg/kg CDP571 8 weekly, placebo 12 weekly, and 10 mg/kg CDP57112 weekly <i>Need to contact the authors for no. of patients ex-</i>

Table 5. Induction studies: critical and important outcomes (Continued)

				thors for separation of data per group.	not provided in tables or texts.	periencing adverse events as information was not presented in tables or text.
Sandborn 2001b	CDAI < 150 at week 4	≥ 70-point decrease clinical response at week 4	NR	CG 9/20 IG 14/23	SAEs: CG 2/20 IG 1/23 Severe: CG 3/20 IG 1/23	CG 10/20 IG 17/23
	CG 4/20	CG 9/20				
	IG 2/ 23	IG 9/ 23				
	Week 2:	Week 2:				
	CG 3/20	CG 5/20				
	IG 2/ 23	IG 11/ 23				
	Week 8:	Week 8:				
	CG 5/20	CG 6/20				
	IG 3/ 23	IG 7/ 23				
Sandborn 2005a (EN-ACT 1)	CDAI < 150 at week 10	≥ 70-point decrease clinical response at week 10	NR	CG 13/181 IG 59/724	CG 12/181 IG 52/724	CG 153/181 IG 626/724
	CG 55/181	CG 88/181				
	IG 267/724	IG 408/724				
	Week 2 %	Week 2 %				
	CG 10	CG 33				
	IG 14	IG 40				
	Week 4 %	Week 4 %				
	CG 19	CG 45				
	IG 23	IG 51				
	Week 6 %	Week 6 %				
	CG 23	CG 53				
	IG 33	IG 58				
	Week 8 %	Week 8 %				
	CG 28	CG 50				
	IG 34	IG 57				
	Week 12 %	Week 12 %				
	CG 31	CG 51				
	IG 40	IG 61				

Table 5. Induction studies: critical and important outcomes (Continued)

Sandborn 2007a	CDAI < 150 at week 4	<u>≥ 70-point decrease clinical response at week 4</u>	NR	CG 5/166	CG 8/166	CG 121/166
	CG 12/166	CG 56/166		IG 3/159	IG 2/159	IG 91/159
	IG 34/159	IG 82/159				
	Week 1	Week 1				
	CG 6/166	CG 34/166				
	IG 10/159	IG 55/159				
	Week 2	Week 2				
	CG 10/166	CG 54/166				
	IG 34/159	IG 83/159				
		<u>≥ 100-point decrease clini- cal response at week 4</u>				
		CG 41/166				
		IG 61/159				
		Week 1				
		CG 20/166				
		IG 31/159				
		Week 2				
		CG 30/166				
		IG 58/159				
Sandborn 2007c (PRECISE 1)	CDAI < 150 at week 6	<u>≥ 100-point decrease clini- cal response at week 6</u>	NR	CG 151/329	CG 23/329	CG 260/329
	CG 57/329	CG 87/329		IG 126/333	IG 34/333	IG 269/333
	IG 71/333	IG 115/333				
	CDAI < 150 at week 26	<u>≥ 100-point decrease clini- cal response at week 26</u>				
	CG 32/329	CG 52/329				
	IG 47/333	IG 75/333				
Sandborn 2008 (Pop- ulation 1)	CDAI < 150 at week 8	<u>A reduction of 25% or more and 70 points or more from the baseline CDAI score at week 8</u>	NR	CGsub 2/26	CGsub 2/26	CGsub 22/26
	CGsub 6/26			IGsub 0/25	IGsub 0/25	IGsub 17/25
	IGsub 6/25	CGsub 13/26		GCiv 3/27	GCiv 1/27	GCiv 19/27
	GCiv 3/27	IGsub 12/25		IGiv 2/26	IGiv 2/26	IGiv 20/26
	IGiv 7/26	GCiv 8/27				
	Week 2	IGiv 13/26				
	CGsub 3/26	Week 2				

Table 5. Induction studies: critical and important outcomes (Continued)

IGsub 6/25	CGsub 11/26
GCiv 2/27	IGsub 11/25
IGiv 5/26	GCiv 6/27
Week 4	IGiv 10/26
CGsub 4/26	Week 4
IGsub 6/25	CGsub 8/26
GCiv 4/27	IGsub 13/25
IGiv 7/26	GCiv 8/27
Week 6	IGiv 14/26
CGsub 6/26	Week 6
IGsub 7/25	CGsub 10/26
GCiv 3/27	IGsub 13/25
IGiv 6/26	GCiv 6/27
	IGiv 14/26
	<u>≥ 100-point decrease clinical response at week 8</u>
	CGsub 9/26
	IGsub 12/25
	GCiv 7/27
	IGiv 13/26
	Week 2
	CGsub 6/26
	IGsub 9/25
	GCiv 6/27
	IGiv 8/26
	Week 4
	CGsub 7/26
	IGsub 11/25
	GCiv 7/27
	IGiv 11/26
	Week 6
	CGsub 9/26
	IGsub 12/25
	GCiv 4/27

Table 5. Induction studies: critical and important outcomes (Continued)

IGiv 13/26

Sandborn 2011	CDAI < 150 at week 6	≥ 100-point decrease clinical response at week 8	NR	CG 24/216 IG 16/223	CG 8/216 IG 12/223	CG 100/216 IG 114/223
	CG 53/216	CG 71/216				
	IG 68/223	IG 87/223				
	Week 2	Week 2				
	CG 33/216	CG 42/216				
	IG 50/223	IG 71/223				
	Week 4	Week 4				
	CG 40/216	CG 55/216				
	IG 57/223	IG 76/223				
Sandborn 2012 (CERTIFI) - Induction	Secondary outcome is remission at week 6 and week 22	Primary outcome is CDAI100 6 weeks and secondary outcome at week 4	NR	CG (Plb) 19/132	CG (Plb) 11/132	CG (Plb) 94/132
	Week 4	Week 4		IG1 (Ust 1) 10/131	IG1 (Ust 1) 6/131	IG1 (Ust 1) 89/131
	CG (Plb) 12/132, IG1 (Ust 1) 14/131, IG2 (Ust 3) 25/132, IG3 (Ust 6) 16/131	CG (Plb) 33/132, IG1 (Ust 1) 54/131, IG2 (Ust 3) 61/132, IG3 (Ust 6) 53/131		IG2 (Ust 3) 12/132	IG2 (Ust 3) 8/132	IG2 (Ust 3) 88/132
	Week 6 (primary)	Week 6		IG3 (Ust 6) 8/131	IG3 (Ust 6) 9/131	IG3 (Ust 6) 80/131
	CG (Plb) 14/132, IG1 (Ust 1) 21/131, IG2 (Ust 3) 21/132, IG3 (Ust 6) 16/131	CG (Plb) 38/132, IG1 (Ust 1) 57/131, IG2 (Ust 3) 57/132, IG3 (Ust 6) 62/131				
	Week 8	Week 8				
	CG (Plb) 14/132, IG1 (Ust 1) 23/131, IG2 (Ust 3) 24/132, IG3 (Ust 6) 24/131	CG (Plb) 33/132, IG1 (Ust 1) 56/131, IG2 (Ust 3) 59/132, IG3 (Ust 6) 73/131				
Sandborn 2012 (CERTIFI) - Induction (non-response)	Week 8	Week 8	NR	CG (Plb) 22/110	CG (Plb) 21/110	CG (Plb) 89/110
	CG (Plb)	CG (combined Plb)		IG1 (Ust) 17/109	IG1 (Ust 90) 22/109	IG1 (Ust) 86/109
	7/110	15/110				
	IG (Ust)	IG (combined Ust)		IG2 (ust270/90) 10/85		
	3/109	16/109				
	Week 22 (primary)	Week 22				
	CG (Plb)	CG (combined Plb)				
	7/110	20/110				
		IG (combined Ust)				

Table 5. Induction studies: critical and important outcomes (Continued)

	IG (Ust)	22/109				
	12/109	Week 36				
	Week 36	CG (combined Plb)				
	CG (Plb)	18/110				
	10/110	IG (combined Ust)				
	IG (Ust)	16/109				
	7/109					
Sandborn 2013 (GEMINI II) - Induction	CDAI < 150 at week 6	≥ 100-point decrease clinical response at week 6	NR	CG 11/148	CG 9/148	CG 88/148
	CG 10/148	CG 38/148		IG 21/220	IG 20/220	IG 124/220
	IG 32/220	IG 69/220				
Sandborn 2014	CDAI < 150 at week 4	≥ 70-point decrease clinical response at week 4	NR	CG 8/34	CG 5/34	CG 22/34
	CG 7/34	CG 16/34		IG1 (1 mg) 2/36	IG1 (1 mg) 4/36	IG1 (1 mg) 18/36
	IG1 (1 mg) 11/36	IG1 (1 mg) 13/36		IG2 (5 mg) 2/34	IG2 (5 mg) 4/34	IG2 (5 mg) 21/34
	IG2 (5 mg) 8/34	IG2 (5 mg) 19/34		IG3 (15 mg) 1/35	IG3 (15 mg) 1/35	IG3 (15 mg) 22/35
	IG3 (15 mg) 5/35	IG3 (15 mg) 16/35				
		≥ 100-point decrease clinical response at week 4				
		CG 10/34				
		IG1 (1 mg) 11/36				
		IG2 (5 mg) 15/34				
		IG3 (15 mg) 13/35				
Sandborn 2023 - BERGAMOT (cohort 1)	CDAI < 150 at week 14	NR	NR	NR	NR	NR
	CG (pl) 10/59					
	IG1 (etr105) 28/120					
	IG2 (etr210) 35/121					
Sandborn 2020d (CELEST) - Induction	Clinical remission 1.5/1.0 (defined as average daily stool frequency of ≤ 1.5 and AP score of ≤ 1.0, with neither worse than the baseline value) at week 16	Clinical response (CDAI < 150 and decrease in CDAI from baseline ≥ 70 points (CR-70)) at week 16	Endoscopic remission at week 12/16 (defined as SES-CD of ≤ 4 and a ≥ 2-point reduction)	CG 9/37	CG 2/37	CG 27/37
				IG1 (3 mg) 5/39	IG1 (3 mg) 5/39	IG1 (3 mg) 34/39
		CG 2/37		IG2 (6 mg) 4/37	IG2 (6 mg) 2/37	IG2 (6 mg) 29/37
		IG1 (3 mg) 8/39		IG3 (12 mg) 9/36	IG3 (12 mg) 10/36	IG3 (12 mg) 29/36
		IG2 (6 mg) 11/37			IG4 (24 mg) 3/36	

Table 5. Induction studies: critical and important outcomes (Continued)

	CG 4/37	IG3 (12 mg) 14/36	(Continued) from base- line, with no sub- score > 1) at week 12/16	IG4 (24 mg) 6/36	IG5 (24 mg once daily) 7/35	IG4 (24 mg) 30/36
	IG1 (3 mg) 5/39	IG4 (24 mg) 11/36				
	IG2 (6 mg) 10/37	IG5 (24 mg once daily) 7/35		IG5 (24 mg once daily) 4/35		IG5 (24 mg once daily) 29/35
	IG3 (12 mg) 4/36					
	IG4 (24 mg) 8/36		CG 0/37			
	IG5 (24 mg once daily) 5/35		IG1 (3 mg) 4/39			
			IG2 (6 mg) 3/37			
			IG3 (12 mg) 3/36			
			IG4 (24 mg) 8/36			
			IG5 (24 mg once daily) 5/35			
Sandborn 2022 - GALAXI 1	Clinical remission defined as "CDAI score <150" Week 12 A. In clinical remission: CG 10/61, IG-G200 35/61, IG- G600 35/63, IG- G1200 28/61, IG-UST 29/63	Clinical response defined as ">=100-point reduction from baseline in CDAI score or CDAI score <150" Week 12 A. In clinical response CG 15/61, IG-G200 43/61, IG-G600 42/63, IG-G1200 37/61, IG- UST 42/63	NR	Through week 12 Patients with adverse event(s) (AE) leading to withdrawal CG 11/70 (15.71%), IG- G200 14/73 (19.18%), IG- G600 12/73 (16.44%), IG- G1200 14/73 (19.18%), IG- GC 40/219 (18.26%), IG-UST 8/71 (11.27%)	Through week 12 Patients with ≥ 1 serious ad- verse event (SAE) CG 4/70 (5.7%), IG-G200 3/73 (4.1%), IG-G600 4/73 (5.5%), IG-G1200 1/73 (1.4%), IG-GC 8/219 (3.7%), IG-UST 4/71 (5.6%)	Through week 12 Patients with ≥ 1 adverse event (AE) CG 42/70 (60.0%), IG- G200 32/73 (43.8%), IG- G600 37/73 (50.7%), IG-G1200 31/73 (42.5%), IG- GC 100/219 (45.7%), IG- UST 36/71 (50.7%)
Sands 2004 (ACCENT II) Induction (Non-re- sponders)	NOT REPORTED	A response was defined as a reduction of at least 50% from baseline in the number of draining fistulas at con- secutive visits four or more weeks apart. Time point of response is NOT REPORTED Patients who achieved re- sponse (n (%)) CG 7/44 (16%), IG 9/43 (21%)	NOT RE- PORTED	NOT REPORT- ED – They provide com- bined data for the respon- ders and non- responders but no sepa- rate informa- tion. NEED TO CON- TACT AU- THORS TO	NOT REPORTED – They provide combined data for the respon- ders and non- responders but no separate in- formation. NEED TO CON- TACT AUTHORS TO REQUEST SEPARATE DATA FOR RESPON-	NOT RE- PORTED – They provide combined data for the respon- ders and non-re- sponders but no sepa- rate infor- mation.

Table 5. Induction studies: critical and important outcomes (Continued)

				REQUEST SEPARATE DATA FOR RESPONDERS AND NON- RESPONDERS PER IG/CG	DERS AND NON-RESPON- DERS PER IG/ CG Serious ad- verse events (combined da- ta for both the responders and non-re- sponders) (n (%)) CG 33/144 (23%), IG1 19/138 (14%)	NEED TO CONTACT AUTHORS TO RE- QUEST SEPARATE DATA FOR RESPON- DERS AND NON- RESPON- DERS PER IG/CG Adverse events (combined data for both the respon- ders and non-re- sponders) (n (%)) CG 132/144 (92%), IG 123/138 (89%)
				Adverse events lead- ing to discon- tinuation of study agent (combined data for both the respon- ders and non- responders) (n (%)) CG 12/144 (8%), IG 5/138 (4%)		
Sands 2007	CDAI < 150 at week 10	≥ 70-point decrease clinical response at week 10	NR	CG 4/27 IG 7/52	CG 1/27 IG 1/52	CG 27/27 IG 48/52
	CG 8/27	CG UNCLEAR/27				
	IG 19/52	IG UNCLEAR/52				
	Week 8					
	CG 8/27					
	IG 17/52					
	Week 6					
	CG 7/27					
	IG 12/52					
	Week 4					
	CG 3/27					
	IG 9/52					
	Week 2					
	CG 2/27					
	IG 8/52					
Sands 2010	Clinical remis- sion defined as	Clinical response defined as "decrease in CDAI score of ≥100 points from baseline"	NR	Time points for adverse events, se-	Time points for adverse events, serious adverse	Time points for adverse events,

Table 5. Induction studies: critical and important outcomes (Continued)

"CDAI score of <150 points"	Day 29	rious adverse events, and withdrawals due to adverse events are NOT REPORTED (NEED TO CONTACT AUTHORS FOR MORE INFORMATION). They report this data for induction (randomised) and maintenance (non-randomised) phases combined per IG/CG, which makes it difficult to know how many had adverse events, serious adverse events, or had withdrawn due to adverse events by day 43.	events, and withdrawals due to adverse events are NOT REPORTED (NEED TO CONTACT AUTHORS FOR MORE INFORMATION). They report this data for induction (randomised) and maintenance (non-randomised) phases combined per IG/CG, which makes it difficult to know how many had adverse events, or had withdrawn due to adverse events by day 43.	serious adverse events, and withdrawals due to adverse events are NOT REPORTED (NEED TO CONTACT AUTHORS FOR MORE INFORMATION). They report this data for induction (randomised) and maintenance (non-randomised) phases combined per IG/CG, which makes it difficult to know how many had adverse events, serious adverse events, or had withdrawn due to adverse events by day 43.
Day 29	Patients in clinical response (n (%))			
Patients in clinical remission (n (%))	CG 21/73 (28.8%)			
CG 10/73 (13.7%)	IG1 18/73 (24.7%)			
IG1 10/73 (13.7%)	IG2 19/74 (25.7%)			
IG2 13/74 (17.6%)	Day 43			
Day 43	Patients in clinical response (n (%))			
Patients in clinical remission (n (%))	CG 22/73 (30.1%) IG1 10/73 (13.7%) IG2 16/74 (21.6%)			
CG 14/73 (19.2%)				
IG1 8/73 (11%)				
IG2 11/74 (14.9%)				
		However, they do report overall how many patients completed the induction phase, but they do not specify why those patients who did not complete induction failed to do so.	Number of patients experiencing serious adverse events (n (%))	
			CG 8/73 (11%)	
			IG1 6/73 (8%)	
			IG2 4/74 (5%)	
		Patients not completing induction phase (n (%))		
		CG 19/73 (26%)		
		IG1 16/73 (22%)		
		IG2 16/74 (22%)		
			They do not specify what time point these figures apply to but they are likely to represent the induction and maintenance phases combined, so these can be considered as figures for the worst-case scenario.	They report the following for the study overall:
				Number of patients experiencing adverse events (n (%))
				CG 59/73 (81%)

Table 5. Induction studies: critical and important outcomes (Continued)

				As they have not specified the reasons for failure to complete induction, it should be assumed the above figures represent withdrawals due to adverse events as a worst-case scenario for the induction phase.	nario for the induction phase alone.	IG1 53/73 (73%) IG2 59/74 (80%) They do not specify what time point these figures apply to but they are likely to represent the induction and maintenance phases combined, so these can be considered as figures for the worst-case scenario for the induction phase alone.
Sands 2014 (GEMINI III)	CDAI < 150 at week 10	≥ 100-point decrease clinical response at week 10	NR	CG 15/207 IG 13/209	CG 16/207 IG 13/209	CG 124/207 IG 117/209
	CG (fail) 19/157	CG (fail) 39/157				
	IG (fail) 42/158	IG (fail) 74/158				
	CG (naive) 8/50	CG (naive) 11/50				
	IG (naive) 18/51	IG (naive) 26/51				
	Week 6 (primary)	Week 6				
	CG (fail) 19/157	CG (fail) 35/157				
	IG (fail) 24/158	IG (fail) 62/158				
	CG (naive) 6/50	CG (naive) 12/50				
	IG (naive) 16/51	IG (naive) 20/51				
Sands 2022 (SEAVUE)	CDAI < 150 at week 52 (primary, but not used in our analysis)	≥ 100-point decrease clinical response at week 52	SES-CD score ≤ 3/0 for pts with BL score = 3 at week 52	CG 22/195 IG 12/191	CG 32/195 IG 25/191	CG 152/195 IG 153/191
	CG 119/195	CG 129/195				
	IG 124/191	IG 138/191				
			CG 55/195			
			IG 51/191			

Table 5. Induction studies: critical and important outcomes (Continued)

Week 16 (used in our analysis)						
CG 117/195						
IG 109/191						
Schreiber 2005	CDAI < 150 at week 12	≥ 100-point decrease clinical response at week 12	NR	CG 32/73	CG 6/73	CG 51/73
				IG1 36/74	IG1 7/74	IG1 57/74
	CG 17/73	CG 26/73		IG2 32/72	IG2 10/72	IG2 55/72
	IG1 20/74	IG1 27/74		IG3 32/73	IG3 6/73	IG3 48/73
	IG2 14/72	IG2 26/72				
	IG3 19/73	IG3 32/73				
Schreiber 2018a	CDAI < 150 at week 8	Liquid or very soft stool frequency and abdominal pain composite score ≤ 8 at week 8	NR	CG 1/28	CG 3/28	CG 19/28
	CG 6/28			IG1 (150 x 2) 1/53	IG1 (150 x 2) 1/53	IG1 (150 x 2) 32/53
	IG1 (150 x 2) 11/53	CG 4/28		IG2 (150) 5/53	IG2 (150) 6/53	IG2 (150) 33/53
	IG2 (150) 9/53	IG1 (150 x 2) 9/53		IG3 (300) 6/53	IG3 (300) 8/53	IG3 (300) 37/53
	IG3 (300) 6/53	IG2 (150) 7/53				
		IG3 (300) 6/53				
Schwartz 2021 - ENTERPRISE	NR	NOT REPORTED. They report "fistulae response" Fistulae response defined as "≥50% decrease from baseline in the number of draining perianal fistulae at the visit, assessed by gentle finger compression during clinical examinations. Patients with missing data at the visit were classed as nonresponder" Week 30 Fistulae response (number of patients/total patients in group (percentage)) CG 9/14 (64.3%) IG 6/14 (42.9%) IGP 15/28 (53.6%) Week 22 and 30 Fistulae response (number of patients/total patients in group (percentage)) CG 8/14 (57.1%)	NOT REPORTED	Adverse events leading to discontinuation (number of patients/total patients in group (percentage)) CG 1/16 (6.3%) IG 3/18 (16.7%) IGP 4/34 (11.8%)	Serious adverse events (number of patients/total patients in group (percentage)) CG 4/16 (25.0%) IG 3/18 (16.7%) IGP 7/34 (20.6%)	Adverse events (number of patients/total patients in group (percentage)) CG 15/16 (93.8%) IG 17/18 (94.4%) IGP 32/34 (94.1%)

Table 5. Induction studies: critical and important outcomes (Continued)

IG 5/14 (35.7%)						
IGP 13/28 (46.4%)						
Targan 1997	CDAI < 150 at week 4	<u>≥ 70-point decrease clinical response at week 4</u>	NR	UNCLEAR	UNCLEAR	UNCLEAR
	CG 1/25	CG 4/25				
	IG (5) 19/27	IG (5) 22/27				
	IG (10) 11/28	IG (10) 14/28				
	IG (20) 11/28	IG (20) 18/28				
	Week 2					
	CG 1/25					
	IG (5) 16/27					
	IG (10) 10/28					
	IG (20) 10/28					
	(taken from graph)					
Targan 2007 (EN- CORE)	CDAI < 150 at week 12	<u>≥ 70-point decrease clinical response at week 12</u>	NR	CG 42/250	CG 24/250	CG 206/250
	CG 62/250	CG 110/250		IG 36/259	IG 13/259	IG 222/259
	IG 98/259	IG 155/259				
	Week 8	Week 8				
	CG 52/250	CG 100/250				
	IG 83/259	IG 145/259				
	Week 4	Week 4				
	CG 40/250	CG 92/250				
	IG 62/259	IG 132/259				
		<u>≥ 100-point decrease clini- cal response at week 12</u>				
		CG 77/250				
		IG 127/259				
		Week 8				
		CG 82/250				
		IG 124/259				
		Week 4				
		CG 54/250				
		IG 101/259				

Table 5. Induction studies: critical and important outcomes (Continued)

Vermeire 2017 FITZROY - induction	CDAI < 150 at week 10	<u>≥ 100-point decrease clinical response at week 10</u>	SES-CD ≤ 4, ulcerated surface subscore of 1 or less in all five segments at week 10	CG 7/44 IG 31/152 (more people because 22 were re-ran- domised from placebo to fil- gotinib)	CG 8/44 IG 33/152 (more people because 22 were re-ran- domised from placebo to fil- gotinib)	CG 31/44 IG 128/152 (more people because 22 were re-ran- domised from place- bo to filgo- tinib)
	CG 10/44 IG 60/130 Week 6 CG 12/44 IG 53/130 Week 4 CG 8/44 IG 44/130 Week 2 CG 9/44 IG 29/130	CG 18/44 IG 76/130	CG 3/44 IG 18/130			
Vermeire 2017 FITZROY - induction non-re- sponders	Unclear	Unclear	Unclear	Unclear	Unclear	Unclear
Watanabe 2012 - In- duction	CDAI < 150 at week 4	<u>≥ 70-point decrease clinical response at week 4</u>	NR	CG 1/23 IG1 2/34 IG2 1/33	CG 2/23 IG1 3/34 IG2 1/33	CG 12/23 IG1 20/34 IG2 17/33
	CG 3/23 IG1 6/34 IG2 11/33 Week 2 CG 1/23 IG1 5/34 IG2 6/33	CG 7/23 IG1 20/34 IG2 23/33 <u>≥ 100-point decrease clinical response at week 4</u> CG 4/23 IG1 17/34 IG2 15/33				
Watanabe 2020 - In- duction	CDAI < 150 at week 10	<u>≥ 100-point decrease clinical response at week 10</u>	NR	CG 12/78 IG 3/79	CG 10/78 IG 8/79	CG 42/78 IG 49/79
	CG 8/78 IG 14/79 Week 6 CG 10/78 IG 11/79 Week 2	CG 13/78 IG 21/79 Week 6 CG 10/78 IG 19/79 Week 2				

Table 5. Induction studies: critical and important outcomes (Continued)

CG 4/78		CG 5/78				
IG 5/79		IG 13/79				
Winter 2004	CDAI < 150 at week 4	≥ 100-point decrease clinical response at week 4	NR	CG 5/25	CG 2/25	CG 15/25
				IG 31/67	IG 7/67	IG 43/67
	CG 10/25	CG 14/25				
	IG1 (5) 9/25	IG1 (5) 15/25				
	IG2 (10) 5/19	IG2 (10) 11/19				
	IG3 (20) 4/23	IG3 (20) 10/23				
Ye 2019	Clinical remission at weeks 6, 14, and 30	Week 6	NR	Up to 30 weeks	Up to 30 weeks	Up to 30 weeks
		CDAI-70		CG 21/109	CG 9/109	CG 70/109
	Week 6 (primary)	IG 77		IG 17/111	IG 6/111	IG 63/111
	IG 47	CG 81				
	CG 49	CDAI-100				
	Week 14	IG 67				
	IG 59	CG 70				
	CG 60	Week 14				
	Week 30	CDAI-70				
	IG 61	IG 96				
	CG 62	CG 96				
		CDAI-100				
		IG 78				
		CG 83				
		Week 30				
		CDAI-70				
		IG 85				
		CG 82				
		CDAI-100				
		IG 80				
		CG 80				
Loftus 2023 (U-EXCEL)	Clinical remission defined as 'CDAI < 150' OR "Clinical remission per SF (stool frequency)/AP (abdominal pain) (also referred to as PRO-2) is defined	Clinical response is NOT REPORTED in the press release version. Full paper may provide this information.	Endoscopic remission is NOT REPORTED in the press release version. Full paper may provide this	NOT REPORTED – Need full paper for this information	Serious adverse events (%) CG 6.8% IG 6.9%	NOT REPORTED – Need full paper for this information

Table 5. Induction studies: critical and important outcomes *(Continued)*

as average daily very soft or liquid stool frequency ≤ 2.8 AND average daily abdominal pain score ≤ 1.0 , and both not greater than baseline"	information.	Note – Time points of measurement are NOT REPORTED and numbers randomised per IG/CG are NOT REPORTED, therefore it is not possible at this stage to convert the percentage to numbers. The full paper may provide this information.
Week 12	The press release version does report endoscopic response	
Clinical remission per CDAI (n (%))	Endoscopic response is defined as "a decrease in simple endoscopic score for Crohn's disease (SES-CD) of >50 percent from baseline (or at least a 2-point reduction from baseline for subjects with a baseline SES-CD of 4), as scored by central reviewer."	
CG 51/176 (29%)		
IG 172/350 (49%)		
Clinical remission per SF/AP (n (%))		
CG 39/176 (22%)		
IG 178/350 (51%)		
	Week 12	
	Endoscopic response (n (%))	
	CG 23/176 (13%)	
	IG 161/350 (46%)	

Ada: adalimumab
 AE: adverse event
 AZA: azathioprine
 CD: Crohn's disease
 CDAI: Crohn's Disease Activity Index
 CDEIS: Crohn's Disease Endoscopic Index of Severity
 CG: control group
 CI: confidence interval
 eow: every other week
 IFX: infliximab
 IG: intervention group
 IGiv: IG intravenous
 IGsub: IG subcutaneous

IV: intravenous
Nat: natalizumab
NR: not reported
Plb/pl: placebo
ris: risenkizumab
SAE: serious adverse event
SD: standard deviation
SEM: standard error of the mean
SES-CD: simple endoscopic score for Crohn's disease
TNF: tumour necrosis factor
Tof: tofacitinib
Ust: ustekinumab

Table 6. Maintenance studies: critical and important outcomes

Study ID	M.aClinical relapse	M.bClinical loss of response	M.c Endoscopic relapse	M.d Withdrawals due to adverse events	M.e Serious adverse events	M.f Total number of patients with adverse events
Buhl 2022 (STOP-IT)	Week 48 (reported + withdrawals counted as treatment failures) CG 31/56 IG 5/59	NR	CG 42/56 IG 17/59	CG 8/56 IG 4/59	CG 2/56 IG 2/59	NR
Colombel 2007 (CHARM) - Maintenance (responders)	(Data collected from Figure 2 and calculated from clinical remission) Week 26 CG (Plb) 141/170, IG1 (Ada eow) 104/172, IG2 (Ada weekly) 84/157 Week 56 CG (Plb) 150/170, IG1 (Ada eow) 110/172, IG2 (Ada weekly) 92/157	< 70-point decrease CDAI Score Week 26 CG 122/170, IG1 79/172, IG2 69/157 Week 56 CG 140/170, IG1 98/172, IG2 80/157 < 100-point decrease CDAI Score Week 26 CG 125/170, IG1 83/172, IG2 75/157 Week 56 CG 142/170, IG1 101/172, IG2 82/157	NR	<i>Need to contact authors for separation of safety data from Table 4 for responders and non-responders of each group.</i>	<i>Need to contact authors for separation of safety data from Table 4 for responders and non-responders of each group.</i>	<i>Need to contact authors for separation of safety data from Table 4 for responders and non-responders of each group.</i>
Colombel 2023 - LIBERTY-CD	Week 54 CG (placebo) 87/231, IG (CT-P13 SC) 76/112	Week 54 CG (placebo) 79/231, IG (CT-P13 SC) 69/112	Week 54 CG (placebo) 151/231, IG	NR	CG (placebo) 8/105, IG (CT-P13 SC) 16/238	CG (placebo) 65/105, IG (CT-P13 SC) 172/238

Table 6. Maintenance studies: critical and important outcomes (Continued)
(CT-P13 SC)
100/112

Feagan 2015c (IM-UNITI)	Calculated from clinical remission information in table Week 44: CG (Plb) 86/133, IG1 (Ust q12w) 69/132, IG2 (q8w) 64/132	Calculated from sustained 100-point clinical response at week 44 from table Week 44: CG (Plb) 75/133, IG1 (Ust q12w) 57/132, IG2 (q8w) 56/132	NR	CG (Plb) 13/133, IG1 (Ust q12w) 9/132, IG2 (q8w) 14/132 <i>Need to contact the authors for further clarity on 'terminated study' individuals described in the supplemental figure.</i>	CG (Plb) 20/133, IG1 (Ust q12w) 16/132, IG2 (q8w) 13/132	CG (Plb) 111/133, IG1 (Ust q12w) 106/132, IG2 (q8w) 107/132
Ferrante 2022 - FORTIFY	Week 52 CG (placebo) 117/184 IG1 (ris 180 mg) 105/179 IG2 (ris 360 mg) 92/179	NR	Week 52 CG (placebo) 163/184 IG1 (ris 180 mg) 132/179 IG2 (ris 360 mg) 124/179	CG (placebo) 10/184 IG1 (ris 180 mg) 10/179 IG2 (ris 360 mg) 17/179	CG (placebo) 23/184 IG1 (ris 180 mg) 24/179 IG2 (ris 360 mg) 22/179	CG (placebo) 135/184 IG1 (ris 180 mg) 129/179 IG2 (ris 360 mg) 128/179
Hanauer 2002 (AC-CENT I) - Maintenance (Responders)	Week 30: CG (Placebo) 87/110, IG1 (IFX 5 mg) 69/113, IG2 (IFX 5-10 mg) 62/112	<i>Need to contact authors for exact proportion of patients with 70-point clinical response at week 30 and 54 to calculate clinical loss of response.</i>	NR	<i>Need to contact the authors to provide separate information for the safety information for responders and non-responders.</i>	<i>Need to contact the authors to provide separate information for the safety information for responders and non-responders.</i>	<i>Need to contact the authors to provide separate information for the safety information for responders and non-responders.</i>
Jorgensen 2017 (NORSWITCH)	Harvey-Bradshaw Index remission (score less than or equal to 4): CG (IFX) 37/78, IG1 (CT-P13) 31/77	Disease worsening was defined by worsening in Harvey-Bradshaw Index (HBI) for CD (a change from baseline of 4 points or more and a total score of 7 points or greater) CG (IFX) 38/78, IG1 (CT-P13) 25/77	NR	CG (IFX) 2/78 IG1 (CT-P13) 1/77	CG (IFX) 8/78 IG1 (CT-P13) 8/77	CG (IFX) 57/78 IG1 (CT-P13) 49/77
Louis 2023 - SPARE	39 in total before retreatment, but NR per group	NR	NR	NR	Group 1: 10 Group 2: 8 Group 3: 13	NR

Table 6. Maintenance studies: critical and important outcomes (Continued)

(2/23/3 patients
were retreated)

NCT00349752 (COSPAR1)	CG (placebo): 68/87 IG (certolizumab): 64/87	NR	NR	NR	CG (placebo): 8/87 IG (certolizumab): 9/87	NR
Panes 2017 Maintenance	CG (Plb) 47/59, IG1 (Tof 5) 44/60, IG2 (Tof 10) 43/61	Loss of clinical response - 100 Week 26: CG (Plb) 44/59, IG1 (Tof5) 44/60, IG2 (Tof10) 37/61	NR	CG (Plb) 28/59, IG1 (Tof 5) 26/60, IG2 (Tof 10) 19/61	Patients with serious and severe AEs: CG (Plb) 13/59, IG1 (Tof 5) 11/60, IG2 (Tof 10) 14/61 <i>Need to contact authors for combined data on patients experiencing serious and severe AEs.</i>	CG (Plb) 44/59, IG1 (Tof 5) 50/60, IG2 (Tof 10) 48/61
Rutgeerts 1999	Week 44 (randomisation at week 12) CG (placebo) 23/36 IG (ifx) 15/37	Week 44 (randomisation at week 12) CG (placebo) 23/36 IG (ifx) 14/37	NR	CG (Plb) 12/36 IG (Ifx) 10/37	<i>Need to contact authors for numbers of patients with serious adverse events.</i>	CG (Plb) 35/36 IG (Ifx) 35/37
Rutgeerts 2012 (EXTEND)	Week 12 (8 weeks after the randomisation): CG (Plb) 47/65, IG (Ada) 34/64 Week 52: CG (Plb) 59/65, IG (Ada) 43/64	NR	Patients without mucosal healing (mucosal ulceration was considered present if the final outcome of the ileo-colonoscopy review process determined that 1 or more of the 5 segments of the ileo-	From flow-chart: CG (Plb) 4/65, IG (Ada) 8/64 From table: CG (Plb) 5/65, IG (Ada) 7/64 <i>Discrepancies in the numbers of discontinuations due to adverse events between table and flowchart.</i>	CG (Plb) 5/65, IG (Ada) 4/64	CG (Plb) 55/65, IG (Ada) 61/64

Table 6. Maintenance studies: critical and important outcomes (Continued)

		colon had ulceration):				
		Week 12:				
		CG (Plb) 57/65, IG (Ada) 47/64				
		Week 52:				
		CG (Plb) 65/65, IG (Ada) 49/64				
		Patients without CDEIS Remission (remission defined as score of 4 or less)				
		Week 12:				
		CG (Plb) 47/65, IG (Ada) 31/64				
		Week 52:				
		CG (Plb) 63/65, IG (Ada) 46/64				
Sandborn 2005a (EN-ACT 2)	Week 36 (randomisation at week 12): CG (Plb) 89/214 (26% of 120 had remission), IG (Nat) 73/214 (44% of 130 had remission) Week 60: CG (Plb) 102/214 (15% of 120 had remission), IG (Nat) 79/214 (39% of 130 had remission)	Patients without CDAI-70 response at week 36 : CG (Plb) 122/214 (28% of 170 had sustained response), IG (Nat) 66/214 (61% of 168 had sustained response) Patients without CDAI-70 response at week 60 : CG (Plb) 136/214 (20% of 170 had sustained response), IG (Nat) 77/214 (54% of 168 had sustained response) <i>This may be a loss of response as the values reported are sustained response. What do you think?</i>	NR	CG (Plb) 61/214, IG (Nat) 29/214	CG (Plb) 21/214, IG (Nat) 18/214	CG (Plb) 207/214, IG (Nat) 194/214
Sandborn 2007b (CLASSIC II)	Week 24: CG (Plb) 9/18 (50%), IG1 (Ada eow) 3/19 (16%), IG2 (Ada weekly) 1/18 (6%)	Patients with < 70-point clinical response Week 24: CG (Plb) 3/18, IG1 (Ada eow) 1/19, IG2 (Ada weekly) 1/18	NR	CG (Plb) 2/18, IG1 (Ada eow) 2/19, IG2 (Ada weekly) 1/18	CG (Plb) 2/18, IG1 (Ada eow) 1/19, IG2 (Ada weekly) 0/18	CG (Plb) 18/18, IG1 (Ada eow) 15/19, IG2 (Ada weekly) 14/18

Table 6. Maintenance studies: critical and important outcomes (Continued)

	Week 56:	Week 56:				
	CG (Plb) 10/18 (56%), IG1 (Ada eow) 4/19 (21%), IG2 (Ada weekly) 3/18 (17%)	CG (Plb) 5/18, IG1 (Ada eow) 4/19, IG2 (Ada weekly) 2/18 Patients with < 100-point clinical response Week 24: CG (Plb) 7/18, IG1 (Ada eow) 3/19, IG2 (Ada weekly) 1/18 Week 56: CG (Plb) 2/18, IG1 (Ada eow) 4/19, IG2 (Ada weekly) 8/18				
Sand-born 2012 (CERTIFI) - Maintenance	(Results collected and calculated from clinical remission in patients in remission at week 6 in Table S6 - ustekinumab responders in maintenance phase) Week 22 - CG (Plb) 57/73, IG1 (Ust 90) 50/72	(Results collected and calculated from sustained 70-point clinical response in Table S6 - ustekinumab responders in maintenance phase) Week 22 - CG (Plb) 49/73, IG1 (Ust 90) 32/72	NR	Maintenance only CG (Plb) 10/73, IG1 (Ust 90) 5/72 <i>Will need to contact authors to clarify "unsatisfactory therapeutic effect" but considering worse-case scenario these numbers were included)</i>	Induction and maintenance phases added up CG (Plb) 33/183 IG (Ust) 31/181	Induction and maintenance phases added up CG (Plb) 151/183 IG (Ust) 140/181
Sand-born 2013 (GEMINI II) - Maintenance	Week 52 (Randomisation occurred at week 6): CG (Plb) 120/153 (21.6% maintained remission from week 6) IG1 (Ved q4w) 98/154 (36.4% maintained remission) IG2 (Ved q8w) 94/154 (39.0% maintained remission) <i>(reported as proportion of patients at remission at week 6)</i>	Patients without CDAI-100 response Week 52: CG (Plb) 107/153 (30.1% with response) IG1 (Ved q4w) 84/154 (45.5% with response) IG2 (Ved q8w) 87/154 (43.5% with response)	NR	CG (Plb) 81/153, IG1 (Ved q4w) 60/154, IG2 (Ved q8w) 73/154	CG (Plb) 23/153, IG1 (Ved q4w) 25/154, IG2 (Ved q8w) 28/154	CG (Plb) 128/153, IG1 (Ved q4w) 130/154, IG2 (Ved q8w) 135/154

Table 6. Maintenance studies: critical and important outcomes (Continued)

Sandborn 2020d (CELEST) - Maintenance	Authors need to be contacted to separate data for remission and clinical responders.	Authors need to be contacted to separate data for remission and clinical responders.	Authors need to be contacted to separate data for remission and clinical responders.	Authors need to be contacted to separate data for remission and clinical responders.	Authors need to be contacted to separate data for remission and clinical responders.	Authors need to be contacted to separate data for remission and clinical responders.
Sandborn 2023 - BERGAMOT Maintenance	CG (placebo) 151/217 IG (etrolizumab) 134/217	NR	CG (placebo) 204/217 IG (etrolizumab) 191/217	CG 19/217 IG 27/217	CG 9/217 IG 9/217	CG 190/217 IG 189/217
Sands 2004 (ACCENT II) Maintenance (Responders)	NR	A response was defined as a reduction of at least 50 percent from base line (week 0) in the number of draining fistulas at consecutive visits four or more weeks apart A complete response was defined by – the absence of draining fistulas Loss of response (n (%)) as read from Figure 2A Week 14 (for reference) CG: 0/98 (0%), IG: 0/91 (0%) Week 22 CG: 31/98 (32%), IG: 14/91 (15%) Week 30 CG: 56/98 (57%), IG: 26/91 (29%) Week 38 CG: 65/98 (66%), IG: 32/91 (35%) Week 46 CG: 66/98 (67%), IG: 35/91 (38%) Week 54 CG: 68/98 (69%), IG: 39/91 (43%) Loss of response as stated in text (n (%)) Week 54 CG: 75/98 (77%), IG: 49/91 (54%) Loss of complete response (n (%)) as read from Figure 2B	NR	NEED TO CONTACT AUTHORS TO REQUEST SEPARATE DATA FOR RESPONDERS AND NON-RESPONDERS PER IG/CG Adverse events leading to discontinuation of study agent (combined data for both the responders and non-responders) (n (%)) CG 12/144 (8%), IG 5/138 (4%)	NEED TO CONTACT AUTHORS TO REQUEST SEPARATE DATA FOR RESPONDERS AND NON-RESPONDERS PER IG/CG Serious adverse events (combined data for both the responders and non-responders) (n (%)) CG 33/144 (23%), IG 19/138 (14%)	NEED TO CONTACT AUTHORS TO REQUEST SEPARATE DATA FOR RESPONDERS AND NON-RESPONDERS PER IG/CG Adverse events (combined data for both the responders and non-responders) (n (%)) CG 132/144 (92%) IG 123/138 (89%)

Table 6. Maintenance studies: critical and important outcomes (Continued)

Week 14 (for reference) CG: 28/98 (29%), IG: 23/91 (25%) Week 22 CG: 58/98 (59%), IG: 40/91 (44%). Week 30 CG: 71/98 (72%), IG: 47/91 (52%) Week 38 CG: 76/98 (78%), IG: 49/91 (54%) Week 46 CG: 78/98 (80%), IG: 52/91 (57%) Week 54 CG: 79/98 (81%), IG: 57/91 (63%)						
Schreiber 2007 (PRECISE 2)	Week 26 (randomisation at week 6): CG (Plb) 152/212, IG (Cert) 113/216	Patients without CDAI-100 response who had response at week 6 Week 26: CG (Plb) 140/212 (34% had response), IG (Cert) 82/216 (62% had response) Patients without CDAI-100 response Week 26: CG (Plb) 136/212 (36% had response), IG (Cert) 80/216 (63% had response) <i>Need to clarify</i>	NR	From AEs in induction and maintenance phases: CG (Plb) 112/212, IG (Cert) 69/216 From AEs in maintenance phase: CG (Plb) 111/212, IG (Cert) 66/216 <i>Numbers of patients with AEs that withdrew differ between flow-chart and table. Need to contact authors for clarity on these numbers.</i>	CG (Plb) 14/212, IG (Cert) 12/216	CG (Plb) 143/212, IG (Cert) 140/216
VanAssche 2012 (SWITCH)	NR	Patients without CDAI-100 response Week 54 (randomisation occurred at week 0): CG (Ada) 10/36, IG (Ifx) 7/37	NR	CG (Ada) 10/36, IG (Ifx) 1/37	CG (Ada) 5/36, IG (Ifx) 0/37	CG (Ada) 30/36, IG (Ifx) 27/37

Table 6. Maintenance studies: critical and important outcomes (Continued)

Vermeire 2017 FITZROY - maintenance responders	Placebo: /15 FIL 200 to PL: /14 FIL 200 to 100: /30 FIL 200: /30 NR have asked authors	PL to FIL 100: /22 FIL 200 to PL: /9 FIL 200: /25 NR have asked authors	NR have asked authors	NR	NR	NR
Vermeire 2021 (VISIBLE 2)	Week 52 (randomisation at week 6): CG (Plb) 88/135, IG (Ved) 143/275	Patients without CDAI-100 clinical response at Week 52 (randomisation at week 6): CG (Plb) 75/135, IG (Ved) 132/275	NR	CG (Plb) 56/135, IG (Ved) 106/275	CG (Plb) 14/135, IG (Ved) 23/275	CG (Plb) 102/135, IG (Ved) 202/275
Vermeire 2025 - DIVERSITY (maintenance)	FIL 100 82/105 FIL 200 70/118 Placebo 85/112	FIL 100 72/105 FIL 200 67/118 Placebo 78/112	NR	FIL 100 17/105 FIL 200 18/118 Placebo 7/112	FIL 100 14/104 FIL 200 13/118 Placebo 8/111	FIL 100 75/104 FIL 200 80/118 Placebo 71/111
Volkers 2017 - SIMILAR	NR	NR	NR	NR	NR	NR
Watanabe 2012 - Maintenance	<i>Estimated from graphs in Figure 5</i> Week 52 (randomisation at week 0): CG (Plb) 23/25 (9% with remission), IG (ADA 40) 16/25 (37% with remission)	<i>Estimated from graphs in Figure 5</i> Patients without CDAI-70 response at Week 52: CG (Plb) 23/25 (11%), IG1 (ADA 40) 14/25 (43% with response) Patients without CDAI-100 response at week 52: CG (Plb) 23/25 (10% with response), IG (ADA 40) 16/25 (38% with response)	NR	CG (Plb) 23/25, IG1 (ADA 40) 15/25, IG2 (This is taking into account the patients with flares that were moved to the open-label phase)	CG (Plb) 6/25, IG1 (ADA 40) 2/25	CG (Plb) 23/25, IG (ADA 40) 20/25 As the authors have not counted flares as adverse events and reported 21 (CG) and 20 (IG) total AEs, we've gone with the worst case scenario in each group
Watanabe 2020 - Maintenance	Week 60 (randomisation at week 14): CG (Plb) 10/12 (16.7% had remission), IG (Ved)	Patients without CDAI-100 response at week 60: CG (Plb) 11/12 (8.3% had response), IG (Ved) 5/12 (58.3% had response)	NR	CG (Plb) 8/12, IG (Ved) 5/12	CG (Plb) 4/12, IG (Ved) 2/12	CG (Plb) 10/12, IG (Ved) 9/12

Table 6. Maintenance studies: critical and important outcomes (Continued)

7/12 (41.7% had
remission)

Young 2024 - iBaSS	NR (cross-over study without dif- ferentiation be- tween phase 1 and 2 results)	NR (cross-over study without dif- ferentiation between phase 1 and 2 results)	NR (cross- over study without dif- ferentiation between phase 1 and 2 re- sults)	NR (cross-over study without differentia- tion between phase 1 and 2 results)	NR (cross- over study without dif- ferentiation between phase 1 and 2 re- sults)	NR (cross- over study without dif- ferentiation between phase 1 and 2 re- sults)
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ada: adalimumab

AE: adverse events

CD: Crohn's disease

CDAI: Crohn's Disease Activity Index

CG: control group

eow: every other week

fil: filgotinib

ifx/IFX: infliximab

IG: intervention group

IV: intravenous

N/A: not applicable

nat: natalizumab

NR: not reported

plb/PL: placebo

q12w/q8w: every 12 weeks/every 8 weeks

ris: risankizumab

SC: subcutaneous

tof: tofacitinib

ust: ustekinumab

Ved/ved: vedolizumab